

ZlibValidation

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Chapter 1

ZlibValidation

1.1 Description

Command line tool to validate standard cell libraries in .lib format.

1.2 Usage

```
ZlibValidation
Usage: ./zlibvalidation [OPTIONS] [SUBCOMMAND]

Options:
  -h,--help          Print this help message and exit
  -v,--version        Display program version information and exit

Subcommands:
  parse              Parse the Liberty file and write JSON to a file
  mono              Check the monotonicity of timing arc values
  compare            Compare the comparison library against the reference library and report differences
  supercell         Generate supercells for the given Liberty file
  zlibboost          ZlibBoost - Multi-threaded Library Processing Tool
  clear              Clear the log, JSON, map, markdown files in this directory
  verilog            Generate Verilog file for given Liberty file
  spice             Generate SPICE file for given Liberty file
  func              Check functional equivalence of two Liberty files or Verilog files
```

1.3 Development Diary

1.3.1 2025-01-27

- 创建了一个C++空项目，使用CMake管理编译链，并手动添加了外部库 libsi2dr_liberty.a。
- 添加了 CLI11 库以处理命令行参数解析。
- 添加了 spdlog 日志库，实现高效的日志格式化输出。

1.3.2 2025-01-28

- 添加 `nlohmann/json` 库用于 JSON 数据存储和检索。
- 在 `version.h` 中存储项目作者、版本、构建日期等信息，由 CMake 自动维护。
- 简化了 `--version` 参数解析的代码实现，并添加了 `--mode` 选项以选择工作模式。
- 使用 `spdlog` 库的日志记录器将 debug 级别及以上的日志输出到文件，将 info 级别及以上的日志输出到控制台。
- 将库文件顶层封装为 `LibFile` 对象，提供了读取、解析和修改库文件的方法，包含名称和 PVT 信息作为公共属性。
 - 但仍然使用过程化方法循环迭代读取库文件名称和 PVT 信息。
- 按照 C++ 标准库、第三方库和本地库的顺序重新排列了 `#include` 语句。

1.3.3 2025-02-01

- 使用面向对象编程 (OOP) 重构了原先用 C 语言循环遍历的代码，将简单属性的读取封装为 `LibAttribute` 类，
 - 并将函数返回的 `char *` 类型的属性值转换为 `std::string` 类型，便于外部使用。
- 将属性迭代过程封装为 `AttributesIterator` 类，在析构函数中处理迭代退出，实现资源获取即初始化 (RAII)。

1.3.4 2025-02-10

- 将组的迭代过程封装为 `GroupsIterator` 类，同样在析构函数中处理迭代退出。
- 封装了 `LibGroup` 类，提供了读取组名、类别、属性、子组的方法。
- 将类的定义和实现分离，将类的实现放在 `src` 目录下，将类的定义放在 `include` 目录下，修改了 `CMakeLists.txt` 自动收集文件，便于模块化编译。

1.3.5 2025-02-11

- 类的头文件改为 `hpp` 后缀，修改了 `CMakeLists.txt` 中的文件收集规则，CMake 编译时间戳更新。
- 为 `LibFile` 类添加了输出同名 json 格式文件的方法 `LibFile::writeJsonToFile`，目前能输出 PVT、`cell_name`、`cell_footprint`、`cell_area` 信息。
- [x] voltage 浮点数误差未解决。

1.3.6 2025-02-14

- 在 `json_utils.cpp` 中，实现了以 JSON 数据结构循环迭代存储库 (lib) 信息的功能，具体包括 `generateCellJson`, `generatePinJson`, `generatePowerJson`, `generateLutJson` 等函数，并添加了同名的 `hpp` 头文件。
- [x] 时序 (timing) 信息解析功能待完成。
- 新增辅助函数 `parseStringToVector`，该函数可将以逗号分隔的字符串解析为浮点数向量，以便于读取 Look Up Table (LUT)。
- 为 `LibFile` 对象添加了私有属性 `lib_json_`，用于存储库 (lib) 的 JSON 对象。
- 封装了用于迭代复杂属性值的 `ValuesIterator` 类。
- 修改了 `LibAttribute::isComplex()` 函数的返回值类型，由原类型变更为 `bool` 类型。
- 重写了 `LibGroup::getName()` 函数，使其直接返回 `std::string` 类型的组名字。
- 重构了迭代器使用代码。移除了中间变量类似 `si2drGroupsIdT sub_groups` 的声明步骤。直接使用 `lib_group.getGroups()` 的返回值进行初始化 `GroupsIterator`，提升了代码的可读性和简洁性。

1.3.7 2025-02-15

- 实现 `generateTimingJson` 函数，用于全面解析时序信息。

1.3.8 2025-02-17

- 简化了迭代器在外部的调用流程，删除所有迭代器的 `begin()` 方法，改为在构造函数内部初始化。

1.3.9 2025-02-18

- 新建 `LibFile::mono()` 方法，准备实现库文件输出引脚的 `values` 单调递增性检查。
- 修改了 `setupLogger()` 函数，将日志文件名可作为参数传入，便于根据不同工作模式切换日志文件。

1.3.10 2025-02-19

- 修改了 `mode == "mono"` 情况下的逻辑，使用 C++17 引入的 `std::filesystem::exists()` 函数检查同名 JSON 文件是否存在，如果 JSON 文件不存在，则先调用库文件解析功能，生成 JSON 文件，然后再执行单调性检查。
- 实现了 `LibFile::mono()` 方法，检查 `cell_rise`、`cell_fall`、`rise_transition`、`fall_transition` 这四种 timing 信息，针对其 LUT 数据结构，检查其 `value` 值在表格的每一行是否保持单调递增。

1.3.11 2025-02-20

- `LibFile::mono()` 方法增加结果统计功能，在程序运行的最后输出单调性检查中 passed (通过) 和 failed (失败) 的 cell 数量，输出格式参考 `liberate_lv` 工具的风格。
- 以 `sl018_ff_3.96_-40.lib` 为例，与 `liberate_lv` 工具对比测试，结果均为 205 out of 559 cells failed。
- 对于非单调信息警告，增加输出 when 信息，方便用户查看具体位置。

1.3.12 2025-02-25

- 重构命令行参数解析，使用 `CLI11` 库的子命令，在每个回调函数中实现 `parse`、`mono` 功能。

1.3.13 2025-02-26

- 子命令可自定义输出文件名、日志文件名，并设置了相应的默认值。
- 改为在 `Logger` 初始化时打印版本信息，避免在 `-h`、`-v` 时多余打印。
- `LibFile::mono()` 方法增加对 `input_slew` 的单调性检查，如果在输入时有指定，就检查 value 矩阵的每一列是否单调。经过测试，`tcbn65lpbc.lib` 输出与 `liberate_lv` 工具一致。

1.3.14 2025-02-27

- 多线程并行尝试，采用 GDB 调试：
 - `si2dr_liberty` 库在解析 `Liberty` 文件时，可能使用了共享的数据结构（例如哈希表、字符串表）来存储解析结果或中间数据。
 - 在多线程并行解析多个文件时，不同的线程同时调用 `si2dr_liberty` 库的函数，并发地访问和操作这些共享数据结构。
 - `si2dr_liberty` 库可能没有采取足够的线程同步措施来保护这些共享数据结构，导致了数据竞争。
 - 数据竞争最终导致了内存损坏，使得 `strcmp` 函数在后续操作中访问了无效内存，触发了段错误。

1.3.15 2025-02-28

- 修改 `LibFile` 的构造函数与析构函数，`read` 方法改为私有，移入 `parse` 方法。将 `si2dr` 初始化与销毁放在 `parse` 方法中，使得 `mono` 方法与 `si2dr` 库解耦，从而实现多线程并行验证单调性。
- [x] 顶层 `si2drPIGetGroups` 未退出的 bug: `WARNING: si2drPIQuit: GetGroups called 1 more times than IterQuit`，内存泄漏？
- 学到了可以在 `CMakeLists.txt` 中使用 `set(CMAKE_BUILD_TYPE Release)` 减少编译器优化程度，配合 GDB 等调试工具检查堆栈信息。

1.3.16 2025-03-01

- `sub_groups_iter` 是在每次 `for` 循环的迭代中声明的局部变量，在每次迭代结束后其作用域结束，析构函数被调用，释放相关资源。所以 `si2drPIQuit` 未发现内存泄漏。
- 顶层 `group_iter` 的析构函数发生在 `parse` 方法结束时，所以 `parse` 方法末尾的 `si2drPIQuit` 检测到了顶层 `groups` 未退出的情况。
- 综上，`si2drPIQuit` 的警告是由于当时顶层 `groups` 未退出导致的，在 `parse` 方法结束后能正确释放，不会造成实际上的内存泄漏。
- [] 顺序执行 `parse`，`log` 输出仍存在问题：解析第二第三个库的时候会多余打印之前库的PVT信息，时间上也比单独解析一个库的总是要慢 (`parse` 6s 左右，`mono` 1s)。并行执行 `mono`，`log` 输出会集中到最后一个库的文件。
- 给 `LibFile` 类的私有属性添加了初始值，避免了未初始化的问题，吗？

1.3.17 2025-03-07

- 实现了 `compare` 子命令的解析，能够对两个库文件依次进行解析，并生成对应的 JSON 文件。
- [] 仍然存在日志重复输出的问题，多文件处理时数据可能会相互干扰。

1.3.18 2025-03-10

- 新增 `supercell` 子命令和 `LibFile::supercell` 方法，用于生成超级单元以供验证。
- 集成了 `zlibboost` 子命令，支持调用开源库特征化工具 `zlibboost`。已与开发者取得联系。
- 调整了子命令输出文件的位置为当前目录，避免文件混淆。
- 添加了 `clear` 子命令，方便清理生成的 JSON 文件和日志文件。

1.3.19 2025-03-14

- 添加了 `tabulate/table` 库，通过 CMake FetchContent 进行管理，用于生成格式化的表格输出。
- 新建了 `compare.cpp` 和 `compare.hpp` 文件，创建了 `LibraryComparator` 类，用于比较两个库文件的差异。目前实现了读取参考库 JSON 文件的功能。

1.3.20 2025-03-15

- 使用 `std::filesystem::path` 重构了 `LibFile` 类的文件路径管理，方便文件路径的拼接和处理。`LibFile` 类添加了成员变量 `basename_`、`filename_`、`libname_`、`jsonname_` 和 `loggername_`，用于存储文件名、库名、JSON 文件名和日志文件名。
- 通过作用域 (scope) 管理 `LibFile::parse` 方法中的顶层迭代器的生命周期，提前结束并调用 `si2drPIQuit` 保证资源释放，解决了分析多个库时日志重复输出和数据保存错误的问题。
- `clear` 子命令增加了删除 `.map` 和 `.md` 文件的功能。
- 实现了 `mono` 子命令的多线程并行。首先检查是否是多文件输入，如果是，就顺序检查是否准备好了每个文件的 JSON 文件，否则顺序进行多文件库解析。在所有文件的 JSON 准备就绪后，根据文件数量创建线程池，每个线程负责一个库的单调性检查。
- 重构了 `spdlog` 日志初始化，返回 `logger` 对象而不是设置全局默认 `logger`。保留名为 `APP_NAME` 的全局 `logger`，用于输出总体提示信息。
- 为 `LibFile` 类添加了独立的成员变量 `logger`，用于记录每个库文件的日志信息。
- 完成了 `supercell` 子命令的多线程并行，处理逻辑与 `mono` 子命令类似。
- `LibraryComparator::generateReport()` 方法测试了 `tabulate` 库的 markdown 表格输出功能，完善了报告的头部信息输出，包括参考库、比较库、相对容差、软件信息、报告生成时间和图例说明。

1.3.21 2025-03-16

- 针对 `compare` 子命令，增加了对报告文件名的检查，确保其以 `.md` 或 `.txt` 结尾。若不符合，则给出警告并自动添加 `.md` 后缀。
- 进一步完善了 `LibraryComparator::generateReport()` 方法。现在，该方法能够遍历比较库中的所有 `cell`，并在参考库中查找对应的 `cell`。如果找到，则调用 `LibraryComparator::compareCell()` 方法进行比较；否则，会记录 `cell` 未找到的警告信息。
- 新增了 `LibraryComparator::compareCell()` 方法，用于比较两个库中同名 `cell` 的输出引脚。该方法会遍历比较库 `cell` 的每个输出引脚，在参考库 `cell` 中寻找同名引脚。如果找到，则调用 `comparePin` 函数比较引脚；如果未找到，则记录警告信息。如果比较库 `cell` 没有输出引脚，则会记录一条信息级别的日志。
- 新增了 `LibraryComparator::comparePin()` 方法，用于比较两个库中同名 `cell` 的同名引脚。它会遍历比较库 `pin` 的每个 `timing arc`，并在参考库 `pin` 中查找相同类型的 `timing arc`。如果找到，则调用 `compareTimingArc` 函数进行比较；如果未找到，则记录警告信息。如果比较库 `pin` 没有 `timing arc`，则记录一条信息级别的日志。

- 修改了 `LibraryComparator::compareTimingArc()` 方法，用于比较两个 timing arc JSON 对象。该方法首先记录正在比较的 timing arc 的类型，然后提取 "related_pin" 属性。接着，它遍历预定义的 timing arc 名称列表 ("cell_rise", "cell_fall", "rise_transition", "fall_transition")，并在比较库中查找这些 timing arc。如果找到，则在参考库中查找对应的 timing arc，并调用 `compareLut` 方法进行比较。如果参考库中未找到对应的 timing arc，则记录警告信息。
- 修改了 `LibraryComparator::compareLut()` 方法（原 `compareValue` 方法，已更正命名），用于比较两个 JSON 对象中具有相同名称的 value。它首先提取两个 JSON 对象中的 "index_1" 和 "index_2" 数组，如果这两个数组不相等，则记录错误信息并返回。如果索引匹配，则比较 LUT 中的实际 value 值，并根据相对容差 (reltol) 和绝对容差 (abstol) 标记差异。
- 重新设计了层级比较方法，现在可以逐层级地传入 `cell_name`、`pin_name`、`timing_type`、`related_pin` 和 `arc_name`，以便在最内层制表或提示时能够准确输出层级信息。
- 修复了 `LibraryComparator::compareValue()` 方法的命名错误，已更正为 `LibraryComparator::compareLut`
- 已经可以输出表格，数据数量和商用工具结果一致，但格式还需要进一步调整。
- 新增了 `abstol` 参数，默认值为 0.002ns，用于设置绝对容差，与库验证工具保持一致。

1.3.22 2025-03-17

- 新增 `verilog` 和 `spice` 子命令，用于生成 Verilog 和 SPICE 代码。
- 完善 `LibraryComparator` 报告生成：
 - 为每个 cell 增加表头，仅在存在表格数据时输出表格，避免冗余输出。
 - 增加 cell 结果统计功能，输出结果统计表格，目前仅支持 Timing 中 Delay 的比较。

1.3.23 2025-03-18

- 优化统计表格，包含 | Cell Name | Data Type | Failed Count | Avg Diff | Avg Diff% | Max Diff | Max Diff% | Outliers |。
- 优化报告对比表格，包含 | Pin Name | Reference | Comparison | Diff | Diff % | Type | Arc Name | Row # | Index_1 | Column # | Index_2 | Note |
- 报告表格的 stdout 输出增加表头加粗、黄色的格式化。
- 增加 `si2drSimpleAttrGetBooleanValue` 封装，用于获取布尔类型的属性值，以判断引脚是否是时钟引脚。
- `LibFile::mono()` 增加对 `min_pulse_width` 的单调性检查：
 - 针对包含 "input_pins" 的 cell，遍历每个输入引脚。
 - 如果引脚是时钟引脚，则检查其 "timing_arcs"。

- 针对 "timing_type" 为 "min_pulse_width" 的 timing arc, 对 "rise_constraint" 和 "fall_constraint" 调用 checkTimingArcMonotonicity 进行单调性检查。
- checkTimingArcMonotonicity 函数日志区分有无 when 信息, 输出不同日志。
- checkTimingArcMonotonicity 核心比较功能取消了等于的情况, 相等情况默认为通过。
- checkTimingArcMonotonicity 核心比较功能增加了判断 related_pin 是否等于当前 pin, 不等于则跳过。最终改为: 如果矩阵中当前的值小于前一个值, 并且当前引脚不是相关引脚, 或者当前值和前一个值都为零, 那么就认为这些值不是单调递增的。

1.3.24 2025-03-19

- 完善 supercell 方法, 如果有有时钟信号引脚, 链式长度被设为 1, 再生成超级单元。
- 解决 voltage 浮点数误差问题, `std::round(voltage_ * 100) / 100.0` 保留两位小数。

1.3.25 2025-03-21

- 新增 func 子命令, 用于检查库或者 Verilog 文件的逻辑等价性。

1.3.26 2025-03-24

- 调研了 C++ 操作 Verilog 相关的库, 找到了一个实用的 Verilog 库: `slang`, 它可以解析出抽象语法树 (AST)。已修改 CMakeLists 文件, 将 slang 集成到项目中, 目前正在研究其 API。

1.3.27 2025-03-25

- 新增 `verilog_utils.cpp` 和 `verilog_utils.hpp` 文件, 并创建了 `VerilogVisitor` 类, 用于自定义遍历 Verilog AST。
- 验证了 slang 对不同类型逻辑单元的解析能力:
 - 简单组合逻辑 (如 AND2D0)、复杂时序逻辑 (如 CMPE42D1) 和基本时序逻辑 (如 DFQD1) 均能被正确解析。
 - 用户自定义原语 (UDP), 例如 `tsmc_dff`, 会被错误地识别为模块实例化, 并产生 "Invalid instance declaration" 警告。
 - 成功提取了模块名、端口方向 (input、output) 及名称。
 - 能够解析命名端口连接的层次化实例化, 并获取模块名、实例名称和端口映射关系。
 - 能够解析门级原语实例化, 并获取门类型、实例名称、输出端口和输入端口。

1.3.28 2025-03-26

- 进一步完善了对 UDP（如 tsmc_dff）端口映射关系的解析，但其内部 Entry 尚未处理。
- 尝试了通过继承 slang::syntax::SyntaxRewriter 类创建自定义类 CellExtractor，实现了只保留目标模块、删除其他模块，并将修改后的语法树另存为新文件。使用 slang::syntax::SyntaxPrinter::printFile() 方法可以将 Verilog 代码输出到文件。
- 创建了 CellPrinter 类，继承自 slang::syntax::SyntaxVisitor 类。当访问到目标模块时，使用 module.toString() 方法也能将 Verilog 代码输出到文件，但发现输出结果中空行会被删除。
- 修改了 LibFile::supercell() 方法，该方法可以根据输入的 cell_names 生成指定的 supercell，并在遇到未找到的单元时提示警告信息。
- 优化了时钟引脚的处理方式，现在时钟引脚不再被视为 supercell 的输入引脚，而是仅记录为时序单元，并跳过存入 input_pins 集合的步骤。
- 引入了 Doxygen 文档生成工具，用于可视化分析项目，并生成了 Doxygen HTML 文档和 LaTeX 参考手册。
- [x] 尚未解决手册中文显示不正确的问题，可能需要自定义 LaTeX 头文件。

1.3.29 2025-03-27

- **重大失误！** 误操作 Git 分支，导致部分代码修改丢失。务必牢记：** 及时提交代码！ **
- 实现了 LibFile::verilog 方法的核心流程：
 - 首先调用 this->supercell() 生成超级单元。
 - 读取 .map 文件，提取单元映射关系到 std::pair<std::string, std::string> 中（原始单元名 -> 超级单元名/模块名）。
 - 从 lib_json_ 中提取单元的输入/输出端口信息。
 - 根据是否存在时钟引脚，确定 Verilog 模块中 instance 的生成数量（时序逻辑为 1，组合逻辑等于 chain length）。
 - 通过字符串操作生成 ANSI 风格的端口列表，并与模块名组合成完整的 fullModuleText。
 - 使用 slang::syntax::SyntaxTree::fromText 从 fullModuleText 生成语法树。
 - 将语法树传递给 ModuleRewriter 类，使用其 transform 方法遍历模块成员，并通过 handle() 方法进行处理。
 - 最后，使用 slang::syntax::SyntaxPrinter::printFile(*tree) 将处理后的语法树输出到文件。
- LibFile::verilog 目前能够正确输出模块名和 ANSI 风格的端口声明。

1.3.30 2025-03-28

- 实现了在模块成员列表中插入新节点的功能。具体而言，在 `ModuleRewriter::handle(const slang::syntax::ModuleDeclarationSyntax &module)` 方法中，通过 `insertAtBack(module.members, newDataNode)` 函数，成功地在模块成员的末尾插入了链式单元所需的中间变量声明，例如 `wire OP_i;`。
- 成功地在模块内部添加了关键连接网络变量。通过解析 `CELLNAME__X#__CRITICALPORT__OUTPUTPORT` 格式的字符串，提取出关键输入端口 (critical input port) 和输出端口 (OUTPUTPORT)，用于后续的实例化端口连接。
- 增加了用于处理多输出接口的中间网络变量 `P__UNUSEOUTPUTPORT`，用于连接不需要考察的输出端口。
- 成功地在模块内添加了模块实例化所需的模块名和实例名称，完成了端口映射关系的连接处理。
- 验证了代码对简单组合逻辑单元（如 `INVD0` 反相器）和复杂组合逻辑单元（如 `FA1D0` 全加器）的输出能力，结果均符合预期。
- 为 `ModuleRewriter` 类添加了私有属性 `std::map<std::string, std::string> portInfoMap_`，用于存储端口映射关系，将端口名映射到其方向 (input/output)。
- 完成了实例化端口连接的细节处理：
 - 如果是关键输入端口且为第一个实例，则直接连接到输入端口；否则，连接到 `OP_(i-1)` 这样的中间网络变量。
 - 如果是关键输出端口，且为最后一个实例，则连接到输出端口；否则，连接到 `OP_i` 中间网络变量。
 - 如果是其他输出端口且不是最后一个实例，则连接到 `P_i__portName` 这样的中间网络变量；否则，连接名等于端口名。
 - 其余输入端口，连接名等于端口名。
- 文档方面，GitHub 远程仓库已开放为公开访问，并配置了 GitHub Pages。可以通过 <https://cedar17.github.io/ZlibValidation/> 访问项目文档，该页面包含 Doxygen 生成的 HTML 文档和仓库地址的链接。
- 配置了 GitHub Actions，实现了 Doxygen 文档和 Graphviz 图的自动化构建，以及 LaTeX 参考手册的自动编译（暂不支持中文）。每次在 `dev` 和 `main` 分支的提交都会触发文档和参考手册的生成，并发布到 `gh-pages` 分支。
- 将第三方库放置到 `include_3rd_party` 目录下，方便管理。修改了 `CMakeLists` 文件，将第三方库的头文件路径添加到 `include_directories` 中。修改了 `Doxyfile` 文件，将第三方库的头文件路径添加到 `INPUT` 中，便于文档生成工具理解第三方库函数、类的关系及使用方法。

1.3.31 2025-03-29

- 修复了 Doxygen 生成 LaTeX 参考手册时中文显示不正确的问题。通过修改 Doxyfile 文件，添加 `EXTRA_PACKAGES = ctex` 选项，并指定 `lualatex` 作为编译器，成功解决了中文显示问题。同时，从 `include_3rd_party` 目录中移除了 `Allsyntax.h` 文件，避免了由于 LaTeX 文档过大导致的 "TeX capacity exceeded, sorry [main memory size=5000000]" 内存不足问题。
- 实现了 Verilog 文件的顶层模块生成功能，采用非 ANSI 风格的端口声明方式，格式与 `liberate_lv` 工具生成的完全一致。实现过程如下：
 - 首先将各模块的 Verilog 代码存放到临时文件中。
 - 在处理模块的过程中，使用 `std::vector<std::string>` 收集每个模块的 `input_pins` 和 `output_pins`。
 - 遍历所有模块名称列表，拼接模块名和引脚名，生成顶层模块的 `all_input_ports` 和 `all_output_ports`。
 - 使用字符串拼接方式构建 `validate_top` 模块的完整代码。
 - 最后将临时文件内容与顶层模块代码合并，输出到最终的 Verilog 文件中。
- 更改 `deploy-docs.yml`，修复了 GitHub Action 生成的 LaTeX 参考手册中的图像显示问题。具体步骤如下：
 - 设置系统时区为 Asia/Shanghai (东八区)。
 - 安装 miniconda 并更新 conda。
 - 通过 conda 安装 1.9.5 版本的 Doxygen，并安装 graphviz 以解决依赖关系，确保每个 dot 图都能正确编译出 pdf 文件。
 - 使用 doxygen 命令生成 HTML 文档和 tex 文件。
 - 采用 action marketplace 提供的 `xu-cheng/texlive-action@v2` 安装全量的 texlive2024。
 - 进入 `./doc_doxygen/latex` 目录，运行 `make` 命令使用 `lualatex` 编译 tex 文件，生成 pdf 文件。
 - 使用 `JamesIves/github-pages-deploy-action@v4` 将生成的 pdf 文件上传到 GitHub Pages 发布，并保留 `README.md` 和 `index.html` 文件。

1.3.32 2025-03-30

- 完成了 `LibFile::logic(const std::string &cell_name)` 方法，该方法返回指定 `cell_name` 所有输出引脚对应的逻辑表达式，以 `std::map<std::string, std::string>` 的形式返回，存储输出引脚名称到其逻辑表达式字符串的映射关系。
- 实现了 `func` 子命令的基本框架，用于检查库或 Verilog 文件的逻辑等价性。
 - 完善了 `funcLibFile` 函数，能够处理 Liberty 文件和 Verilog 文件作为参考和比较对象。
 - 增加了文件类型判断，根据文件扩展名选择不同的处理方式。

- 实现了对 .lib 文件的解析，并提取指定 cell 的逻辑表达式。
- 目前仅支持 Liberty 文件的逻辑表达式提取，Verilog 文件的逻辑表达式提取功能尚未完成 (TODO)。
- 增加了对 report 文件名的检查，确保其以 .md 或 .txt 结尾。若不符合，则给出警告并自动添加 .md 后缀。
- 实现了从 Liberty 文件中提取逻辑表达式的功能，使用了 `LibFile::logic` 方法。
- 增加了日志输出，记录每个 cell 的逻辑表达式提取和比较过程。
- 逻辑比较部分目前为空，待实现 (TODO)。

1.3.33 2025-03-31

- 实现了从 Verilog 结构网表中提取逻辑表达式的功能，使用了 `LogicExtractor` 类。
 - 设计了如何使用 Slang 库从 Verilog 结构网表中提取逻辑表达式的方案：
 - * 使用 Slang 解析 Verilog 文件获取语法树 (AST)。
 - * 遍历 AST 定位目标模块。
 - * 在目标模块内构建网表的内部表示，识别输入、输出、线网及门级连接关系。
 - * 对每个输出端口，递归反向追踪门和线网直至到达主输入端口。
 - * 在回溯过程中根据遇到的门类型构建逻辑表达式。
 - * 应用记忆化技术避免对相同线网/信号的重复计算。
 - 创建了 `LogicExtractor` 类，用于从 Verilog 代码中提取逻辑表达式。
 - * `LogicExtractor` 类使用 Slang 库解析 Verilog 文件，并构建目标模块的网表表示。
 - * 实现了 `handle(const slang::syntax::ModuleDeclarationSyntax &module)` 方法，用于定位目标模块，并初始化内部数据结构。
 - * 实现了 `handle(const slang::syntax::PortDeclarationSyntax &portDecl)` 方法，用于提取端口信息，包括端口方向和名称。
 - * 实现了 `handle(const slang::syntax::NetDeclarationSyntax &netDecl)` 方法，用于提取线网信息。
 - * 实现了 `handle(const slang::syntax::PrimitiveInstantiationSyntax &primitiveInst)` 方法，用于提取门级单元信息，包括门类型、输入和输出信号。
 - * 实现了 `getLogicExpressions()` 方法，用于根据提取的网表信息，递归地推导每个输出端口的逻辑表达式。
 - * 实现了 `deriveLogicRecursive(const std::string &signalName)` 方法，用于递归地推导指定信号的逻辑表达式。
 - * 实现了 `formatExpression(const GateInfo &gateInfo, const std::vector<std::string> &inputExprs)` 方法，用于根据门类型和输入表达式，生成逻辑表达式字符串。
 - 实现了 `extractAndPrintNetlistInfo(const std::string &verilog_file, const std::string &cell)` 函数，用于提取并打印网表信息，包括输入、输出、线网和门级单元。

- 实现了 `extractLogicFromVerilog(const std::string &verilog_file, const std::string &cell)` 函数，用于从 Verilog 文件中提取指定 cell 的逻辑表达式，并以 `std::map<std::string, std::string>` 的形式返回。
- 为了更好地组织代码结构，将 `LogicExtractor` 类从 `verilog_utils.hpp` 和 `verilog_utils.cpp` 中分离出来，并创建了独立的 `LogicExtractor.hpp` 和 `LogicExtractor.cpp` 文件。同时，统一了所有 `hpp` 和 `cpp` 文件的命名规范，使其与对应的类名保持一致。
- 新增 `LogicComparator` 类及其头文件、源文件，用于比较两个逻辑表达式的等价性，并生成比较报告。

1.3.34 2025-04-01

- 整理头文件引用，删除了不必要的引用，提高编译效率。
- 引入 `exprtk.cpp` 到第三方头文件目录，为后续的逻辑表达式解析做准备。
- 完善了 `LogicComparator` 类的实现，增加了其示例代码模板 `template <typename T> void logic();`，该模板能够对一个表达式遍历输入变量的所有组合，计算出逻辑值，并打印真值表，为逻辑等价性验证提供基础。
- 实现了 `LogicComparator::preprocessExpression()` 方法，用于预处理逻辑表达式，目的是简化表达式，使其更易于比较。预处理的逻辑如下：
 - ** 统一 AND 处理 **：将 `*` 和空格隐式 AND 都转换为带空格的 `and`，保证 AND 运算符的显式性。
 - ** 显式符号处理 **：
 - * 在所有括号 `()` 周围添加空格，方便后续分词。
 - * 将 `+` 替换为 `or`。
 - * 将 `^` 替换为 `xor`。
 - * 将 `!` (非 `!=` 中的 `!`) 替换为 `not`。
 - * 将 `*` 替换为 `and`。
 - ** 分词 **：基于空格将处理后的字符串分割为 `token` 列表。
 - ** 精确插入隐式 AND **：
 - * 遍历 `token` 列表。
 - * 检查当前 `token` 和下一个 `token`，如果满足以下条件，则在当前 `token` 之后插入 `and`：
 - 当前 `token` 是一个标识符 (如 `A`, `CIX`) 或右括号 `)`。
 - 下一个 `token` 是一个标识符或左括号 `(`。
 - 当前 `token` 不是一个显式逻辑运算符 (`and`, `or`, `xor`, `not`) 或左括号 `(`。
 - 下一个 `token` 不是一个显式逻辑运算符或右括号 `)`。
 - ** 重组与清理 **：将处理后的 `token` 列表用单个空格连接起来，并进行最终的空格清理 (去除多余空格，修剪首尾空格)，得到最终的预处理表达式。

- 在 CMakeLists.txt 中，强制启用了 lld 链接器和 ccache，以提升链接和编译速度。
- 为了保持 main.cpp 的整洁，将 LibFile() 函数的相关操作剥离出来，并放入 LibFileOperations.cpp 文件中。
- 实现了 LogicComparator::extractVariables() 方法，用于从两个表达式中提取并验证变量名，确保它们一致。
 - 该方法使用正则表达式初步提取所有可能的标识符。
 - 对每个提取出的标识符，先用 isIdentifier 函数（检查大写开头等）进行验证。
 - 通过 isIdentifier 验证后，仍然将其转换为小写并与 keywords 集合比较，以过滤掉可能的关键词（这是一个保守但安全的操作）。
 - 只有通过这两步验证的标识符才被认为是有效变量，并以原始大小写形式添加到各自的 std::set 中。
 - 最后比较两个 std::set 是否相等，如果不等则报告差异并返回 false，如果相等则将变量列表（保留原始大小写）排序后存入 sorted_vars 并返回 true。
- 修复了 LogicComparator::preprocessExpression() 方法中对 not 操作符的处理。exprtk 库要求 not 关键字后必须跟着括号，如 not(A) 而非 not A。为此增加了专门的处理步骤：
 - 遍历预处理后的 token 列表，识别所有 not 关键字。
 - 当发现 not 后，检查其后是否跟随标识符（通过 isIdentifier() 函数判断）。
 - 如果后接标识符，自动将 not A 形式转换为 not(A)，具体做法是依次添加 not、(、标识符和)，并跳过已处理的标识符。
 - 如果后接其他元素（如 (、其他运算符或表达式结尾），保留原样，让 ExprTk 处理 not(表达式) 形式。
 - 对其他 token，直接添加到最终 token 列表，不做特殊处理。
- 实现了 LogicComparator::compareSingleExpressionPair() 方法，用于比较两个逻辑表达式的等价性：
 - 接收预处理后的表达式字符串和已排序的变量列表作为输入参数。
 - 为参考表达式和比较表达式分别创建 ExprTk 环境（符号表、表达式对象和解析器）。
 - 将所有变量添加到两个符号表中，确保变量一致性和顺序性。
 - 使用 ExprTk 解析器编译两个表达式，并检查可能的语法错误。
 - 通过二进制计数法，遍历所有可能的输入变量组合 (2^N 种)，其中 N 为变量数量。
 - 对每种组合：
 - * 设置两个符号表中的变量值（0 或 1）。
 - * 计算两个表达式的值，并处理可能的运行时错误。
 - * 将计算结果转换为布尔值并存入结果向量。
 - * 同步构建真值表，记录输入组合和对应的输出值。
 - 比较两个结果向量判断表达式是否等价。

- 将生成的真值表存入 `PinComparisonResult` 对象的 `std::optional` 成员中，以便后续报告生成。
- 实现了 `template <typename T> std::map<std::string, PinComparisonResult> LogicComparator::compareCellLogic();`
 - 该方法用于比较两个 cell 的逻辑表达式，返回一个 `std::map`，键为 cell 名称，值为 `PinComparisonResult` 结构体。
 - 遍历在类私有属性里的整个独立输出引脚键，提取其值作为逻辑字符串表达式，首先使用 `preprocessExpression` 进行预处理，然后使用 `extractVariables` 提取变量向量。
 - 调用 `LogicComparator::compareSingleExpressionPair()` 方法进行比较，结果存入 `PinComparisonResult` 对象。
 - 最后将 `PinComparisonResult` 对象存入 `std::map` 中，键为 cell 名称，值为 `PinComparisonResult` 对象，返回。
- 实现了 `LogicComparator::generateReport()` 方法，报告格式仍需要调整。

1.3.35 2025-04-02

- 取消所有模板函数的 `template <typename T>` 声明，改为使用 `double` 作为参数类型，加快编译速度。
- 完善了 `LogicComparator::generateReport()` 方法，增加了对逻辑比较结果的详细报告输出，包括：
 - 元数据，例如：**Performed by ZlibValidation v0.1.0 from Song Zixuan. on: Wed Apr 2 11:11:39 2025**
 - 图例，包括参考 pin 名称、pin function，参考 pin 名称、pin function，符号解释表。
 - 参考 pin 的真值表
 - 比较 pin 的真值表
 - 比较结果表格，例如：

Property	Value
Status	[OK]
Reference (Raw)	$(!(\neg(A1+A2))+B)$
Comparison (Raw)	$!(\neg A1 * \neg A2) + B$
Reference (Processed)	$(\neg(\neg(A1 \text{ or } A2)) \text{ or } B)$
Comparison (Processed)	$\neg(\neg(A1) \text{ and } \neg(A2)) \text{ or } B$
Ref Expression Compiled	Yes
Comp Expression Compiled	Yes
Logically Equivalent	Yes

1.3.36 2025-04-05

- 新增了 `LibFile::spice()` 方法，为 `spice` 子命令添加了 SPICE 网表生成功能。
 - 通过执行 `which v2lvs` 命令检测系统中是否安装了 V2LVS 工具。如果已安装，则调用该工具生成基本的 SPICE 网表。
 - 添加了 `--vl` 和 `--sl` 命令行选项，允许用户指定 Verilog 库文件和 SPICE 库文件的路径。
 - 目前已能生成基本的 SPICE 网表，后续需要进一步完善输出格式。

Generate SPICE file for given Liberty file

Usage: ./zlibvalidation spice [OPTIONS] library_path...

Positionals:

library_path TEXT:FILE ... REQUIRED

Specify the library file to process

Options:

-h,--help

Print this help message and exit

-l,--log TEXT

Specify the log file name. Default: <basename>.spice.log

-c,--chain INT

Specify the chain length for SPICE generation. Default: 1

--cells TEXT ...

Specify the cell names to generate SPICE for

--vl TEXT

Specify the location of the Verilog primitive library file

--sl TEXT

Specify the location of the SPICE library file to be included in the output

Chapter 2

Hierarchical Index

2.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

AttributesIterator	23
GateInfo	31
GroupsIterator	33
LibAttribute	36
LibFile	40
LibGroup	60
LibraryComparator	63
LogicComparator	73
PinComparisonResult	94
slang::syntax::SyntaxRewriter	
CellExtractor	26
ModuleRewriter	90
slang::syntax::SyntaxVisitor	
CellPrinter	29
LogicExtractor	82
VerilogVisitor	101
ValuesIterator	98

Chapter 3

Class Index

3.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

AttributesIterator	23
CellExtractor	26
CellPrinter	29
GateInfo	31
GroupsIterator	33
LibAttribute	36
LibFile	40
LibGroup	60
LibraryComparator	63
LogicComparator	73
LogicExtractor	82
ModuleRewriter	90
PinComparisonResult	94
ValuesIterator	98
VerilogVisitor	101

Chapter 4

File Index

4.1 File List

Here is a list of all files with brief descriptions:

include/Iterators.hpp	107
include/json_utils.hpp	109
include/LibAttribute.hpp	119
include/LibFile.hpp	121
include/LibFileOperations.hpp	123
include/LibGroup.hpp	138
include/LibraryComparator.hpp	140
include/LogicComparator.hpp	142
include/LogicExtractor.hpp	144
include/verilog_utils.hpp	147
include/version.h	150
src/Iterators.cpp	153
src/json_utils.cpp	154
src/LibAttribute.cpp	168
src/LibFile.cpp	169
src/LibFileOperations.cpp	182
src/LibGroup.cpp	200
src/LibraryComparator.cpp	201
src/LogicComparator.cpp	206
src/LogicExtractor.cpp	223
src/main.cpp	234
src/verilog_utils.cpp	240

Chapter 5

Class Documentation

5.1 AttributesIterator Class Reference

```
#include <Iterators.hpp>
```

Public Member Functions

- [AttributesIterator](#) (si2drAttrSldT attrs, si2drErrorT &err)
- [~AttributesIterator](#) ()
- void [next](#) ()
- bool [end](#) ()
- [LibAttribute](#) [get](#) ()

Private Attributes

- si2drAttrSldT [attrs_](#)
- si2drAttrIdT [attr_](#)
- si2drErrorT & [err_](#)

5.1.1 Detailed Description

Definition at line 23 of file [lterators.hpp](#).

5.1.2 Constructor & Destructor Documentation

5.1.2.1 AttributesIterator()

```
AttributesIterator::AttributesIterator (
    si2drAttrsIdT attrs,
    si2drErrorT & err )
```

Definition at line 13 of file [lterators.cpp](#).

5.1.2.2 ~AttributesIterator()

```
AttributesIterator::~AttributesIterator ( )
```

Definition at line 17 of file [lterators.cpp](#).

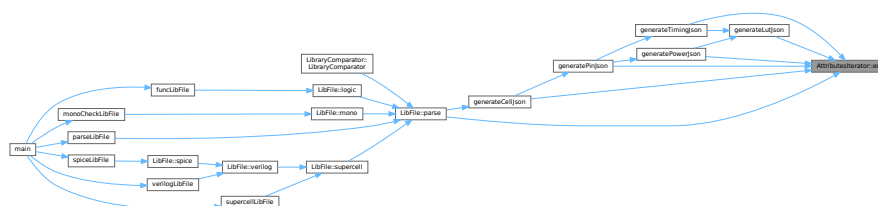
5.1.3 Member Function Documentation

5.1.3.1 end()

```
bool AttributesIterator::end ( )
```

Definition at line 20 of file [lterators.cpp](#).

Here is the caller graph for this function:

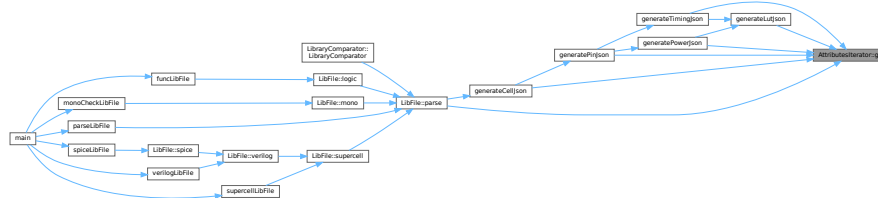


5.1.3.2 get()

LibAttribute AttributesIterator::get ()

Definition at line 22 of file [Iterators.cpp](#).

Here is the caller graph for this function:

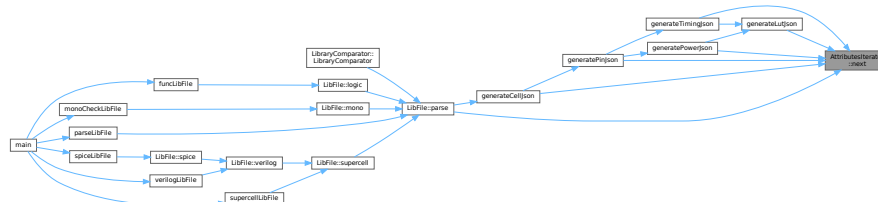


5.1.3.3 next()

void AttributesIterator::next ()

Definition at line 19 of file [Iterators.cpp](#).

Here is the caller graph for this function:



5.1.4 Member Data Documentation

5.1.4.1 attr_

si2drAttrIdT AttributesIterator::attr_ [private]

Definition at line 33 of file [Iterators.hpp](#).

5.1.4.2 attrs_

```
si2drAttrIdT AttributesIterator::attrs_ [private]
```

Definition at line 32 of file [lterators.hpp](#).

5.1.4.3 err_

```
si2drErrorT& AttributesIterator::err_ [private]
```

Definition at line 34 of file [lterators.hpp](#).

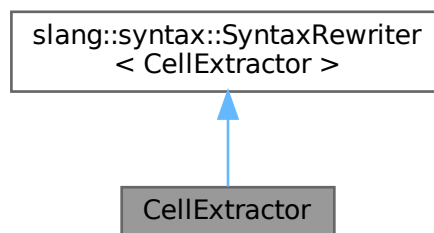
The documentation for this class was generated from the following files:

- [include/lterators.hpp](#)
- [src/lterators.cpp](#)

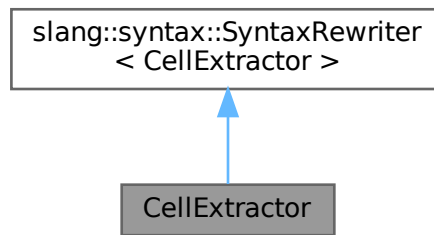
5.2 CellExtractor Class Reference

```
#include <verilog_utils.hpp>
```

Inheritance diagram for CellExtractor:



Collaboration diagram for CellExtractor:



Public Member Functions

- `CellExtractor` (`const std::string &targetCell`)
- `void handle` (`const slang::syntax::ModuleDeclarationSyntax &module`)
- `bool foundTargetCell` (`() const`)

Private Attributes

- `const std::string & targetCell_`
- `bool foundTarget_`

5.2.1 Detailed Description

Definition at line 33 of file [verilog_utils.hpp](#).

5.2.2 Constructor & Destructor Documentation

5.2.2.1 CellExtractor()

```

CellExtractor::CellExtractor (
    const std::string & targetCell ) [inline], [explicit]

```

Definition at line 35 of file [verilog_utils.hpp](#).

5.2.3 Member Function Documentation

5.2.3.1 foundTargetCell()

```
bool CellExtractor::foundTargetCell ( ) const
```

Definition at line 364 of file [verilog_utils.cpp](#).

Here is the caller graph for this function:



5.2.3.2 handle()

```
void CellExtractor::handle (
    const slang::syntax::ModuleDeclarationSyntax & module )
```

Definition at line 348 of file [verilog_utils.cpp](#).

5.2.4 Member Data Documentation

5.2.4.1 foundTarget_

```
bool CellExtractor::foundTarget_ [private]
```

Definition at line 42 of file [verilog_utils.hpp](#).

5.2.4.2 targetCell_

```
const std::string& CellExtractor::targetCell_ [private]
```

Definition at line 41 of file [verilog_utils.hpp](#).

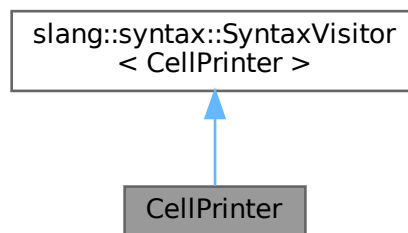
The documentation for this class was generated from the following files:

- include/[verilog_utils.hpp](#)
- src/[verilog_utils.cpp](#)

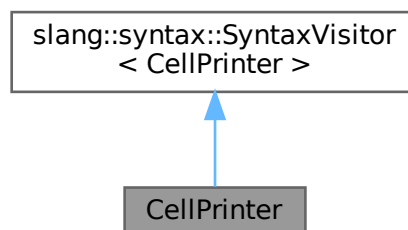
5.3 CellPrinter Class Reference

```
#include <verilog_utils.hpp>
```

Inheritance diagram for CellPrinter:



Collaboration diagram for CellPrinter:



Public Member Functions

- [CellPrinter](#) (const std::string &targetCell, std::ostream &out)
- void [handle](#) (const slang::syntax::ModuleDeclarationSyntax &module)

Private Attributes

- const std::string & [targetCell_](#)
- std::ostream & [out_](#)
- bool [foundTarget_](#)

5.3.1 Detailed Description

Definition at line [46](#) of file [verilog_utils.hpp](#).

5.3.2 Constructor & Destructor Documentation

5.3.2.1 CellPrinter()

```
CellPrinter::CellPrinter (  
    const std::string & targetCell,  
    std::ostream & out ) [inline], [explicit]
```

Definition at line [48](#) of file [verilog_utils.hpp](#).

5.3.3 Member Function Documentation

5.3.3.1 handle()

```
void CellPrinter::handle (  
    const slang::syntax::ModuleDeclarationSyntax & module )
```

Definition at line [367](#) of file [verilog_utils.cpp](#).

5.3.4 Member Data Documentation

5.3.4.1 foundTarget_

```
bool CellPrinter::foundTarget_ [private]
```

Definition at line 55 of file [verilog_utils.hpp](#).

5.3.4.2 out_

```
std::ostream& CellPrinter::out_ [private]
```

Definition at line 54 of file [verilog_utils.hpp](#).

5.3.4.3 targetCell_

```
const std::string& CellPrinter::targetCell_ [private]
```

Definition at line 53 of file [verilog_utils.hpp](#).

The documentation for this class was generated from the following files:

- include/[verilog_utils.hpp](#)
- src/[verilog_utils.cpp](#)

5.4 GateInfo Struct Reference

```
#include <LogicExtractor.hpp>
```

Public Attributes

- slang::parsing::TokenKind [kind](#)
- std::string [gateTypeName](#)
- std::vector< std::string > [inputSignals](#)
- std::string [outputSignal](#)

5.4.1 Detailed Description

Definition at line 9 of file [LogicExtractor.hpp](#).

5.4.2 Member Data Documentation

5.4.2.1 gateTypeName

```
std::string GateInfo::gateTypeName
```

Definition at line 11 of file [LogicExtractor.hpp](#).

5.4.2.2 inputSignals

```
std::vector<std::string> GateInfo::inputSignals
```

Definition at line 12 of file [LogicExtractor.hpp](#).

5.4.2.3 kind

```
slang::parsing::TokenKind GateInfo::kind
```

Definition at line 10 of file [LogicExtractor.hpp](#).

5.4.2.4 outputSignal

```
std::string GateInfo::outputSignal
```

Definition at line 13 of file [LogicExtractor.hpp](#).

The documentation for this struct was generated from the following file:

- include/[LogicExtractor.hpp](#)

5.5 GroupsIterator Class Reference

```
#include <Iterators.hpp>
```

Public Member Functions

- [GroupsIterator](#) (si2drGroupsIdT groups, si2drErrorT &err)
- [~GroupsIterator](#) ()
- void [next](#) ()
- bool [end](#) ()
- [LibGroup](#) [get](#) ()

Private Attributes

- si2drGroupsIdT [groups_](#)
- si2drGroupIdT [group_](#)
- si2drErrorT & [err_](#)

5.5.1 Detailed Description

Definition at line 9 of file [l iterators.hpp](#).

5.5.2 Constructor & Destructor Documentation

5.5.2.1 GroupsIterator()

```
GroupsIterator::GroupsIterator (  
    si2drGroupsIdT groups,  
    si2drErrorT & err )
```

Definition at line 3 of file [l iterators.cpp](#).

5.5.2.2 ~GroupsIterator()

```
GroupsIterator::~GroupsIterator ( )
```

Definition at line 6 of file [l iterators.cpp](#).

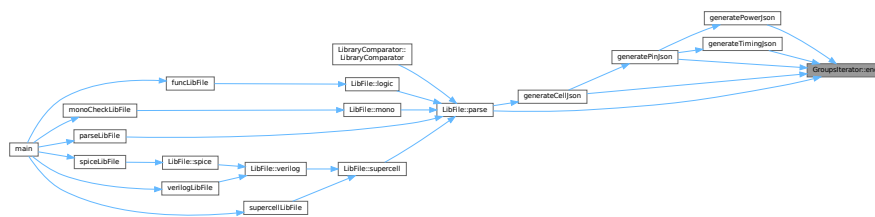
5.5.3 Member Function Documentation

5.5.3.1 end()

```
bool GroupsIterator::end ( )
```

Definition at line 9 of file [Iterators.cpp](#).

Here is the caller graph for this function:

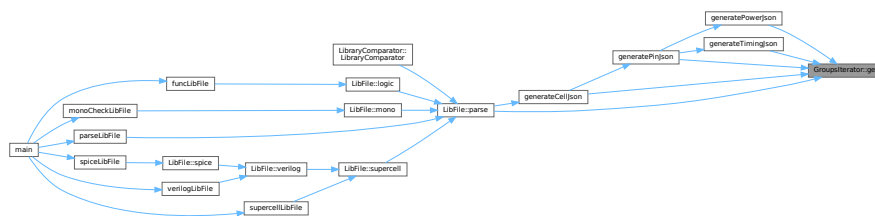


5.5.3.2 get()

```
LibGroup GroupsIterator::get ( )
```

Definition at line 11 of file [Iterators.cpp](#).

Here is the caller graph for this function:

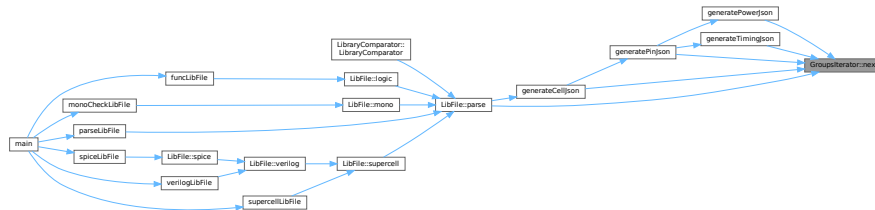


5.5.3.3 next()

```
void GroupsIterator::next ( )
```

Definition at line 8 of file [lterators.cpp](#).

Here is the caller graph for this function:



5.5.4 Member Data Documentation

5.5.4.1 err_

```
si2drErrorT& GroupsIterator::err_ [private]
```

Definition at line 20 of file [lterators.hpp](#).

5.5.4.2 group_

```
si2drGroupIdT GroupsIterator::group_ [private]
```

Definition at line 19 of file [lterators.hpp](#).

5.5.4.3 groups_

```
si2drGroupsIdT GroupsIterator::groups_ [private]
```

Definition at line 18 of file [lterators.hpp](#).

The documentation for this class was generated from the following files:

- include/[lterators.hpp](#)
- src/[lterators.cpp](#)

5.6 LibAttribute Class Reference

```
#include <LibAttribute.hpp>
```

Public Member Functions

- [LibAttribute](#) (si2drAttrIdT attr, si2drErrorT &err)
- [~LibAttribute](#) ()
- std::string [getName](#) ()
- bool [isComplex](#) ()
- si2drValuesIdT [getValues](#) ()
- long int [getInt](#) ()
- double [getFloat](#) ()
- std::string [getString](#) ()
- bool [getBoolean](#) ()

Private Attributes

- si2drAttrIdT [attr_](#)
- si2drErrorT & [err_](#)

5.6.1 Detailed Description

Definition at line 8 of file [LibAttribute.hpp](#).

5.6.2 Constructor & Destructor Documentation

5.6.2.1 LibAttribute()

```
LibAttribute::LibAttribute (  
    si2drAttrIdT attr,  
    si2drErrorT & err )
```

Definition at line 3 of file [LibAttribute.cpp](#).

5.6.2.2 ~LibAttribute()

```
LibAttribute::~~LibAttribute ( )
```

Definition at line 5 of file [LibAttribute.cpp](#).

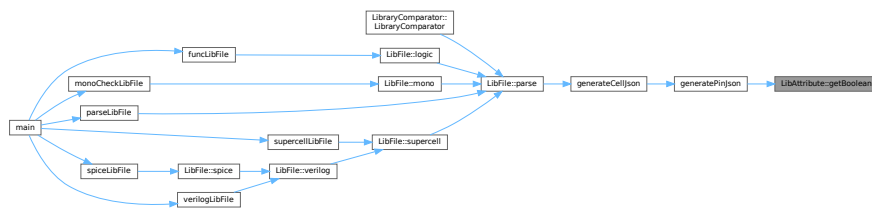
5.6.3 Member Function Documentation

5.6.3.1 getBoolean()

```
bool LibAttribute::getBoolean ( )
```

Definition at line 25 of file [LibAttribute.cpp](#).

Here is the caller graph for this function:

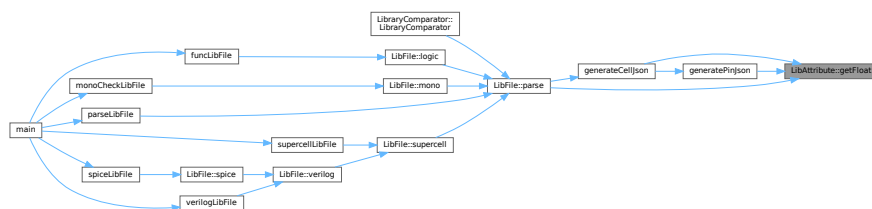


5.6.3.2 getFloat()

```
double LibAttribute::getFloat ( )
```

Definition at line 18 of file [LibAttribute.cpp](#).

Here is the caller graph for this function:

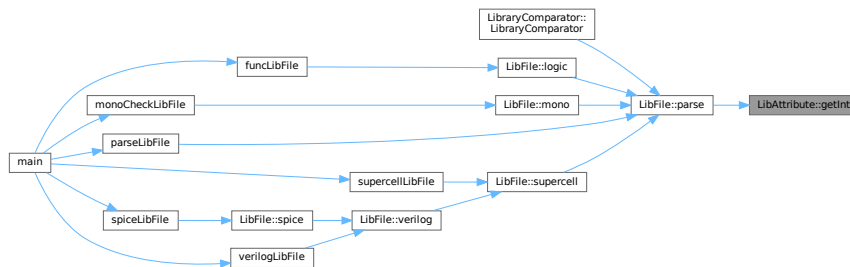


5.6.3.3 getInt()

```
long int LibAttribute::getInt ( )
```

Definition at line 16 of file [LibAttribute.cpp](#).

Here is the caller graph for this function:

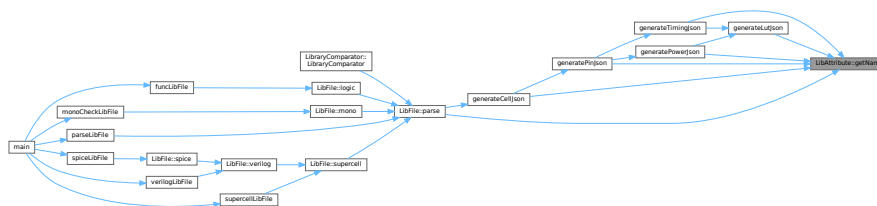


5.6.3.4 getName()

```
std::string LibAttribute::getName ( )
```

Definition at line 7 of file [LibAttribute.cpp](#).

Here is the caller graph for this function:

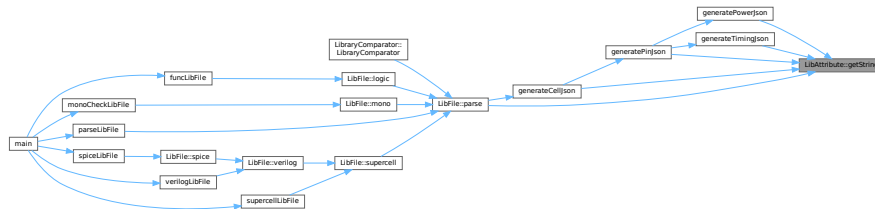


5.6.3.5 getString()

```
std::string LibAttribute::getString ( )
```

Definition at line 20 of file [LibAttribute.cpp](#).

Here is the caller graph for this function:

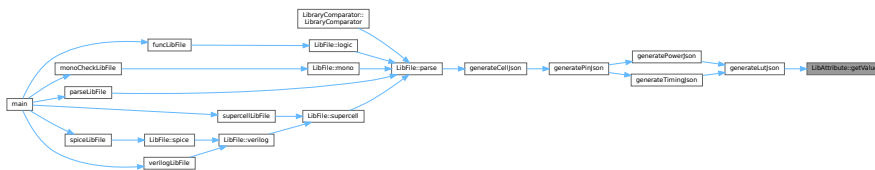


5.6.3.6 getValues()

```
si2drValuesIdT LibAttribute::getValues ( )
```

Definition at line 14 of file [LibAttribute.cpp](#).

Here is the caller graph for this function:



5.6.3.7 isComplex()

```
bool LibAttribute::isComplex ( )
```

Definition at line 12 of file [LibAttribute.cpp](#).

5.6.4 Member Data Documentation

5.6.4.1 attr_

```
si2drAttrIdT LibAttribute::attr_ [private]
```

Definition at line 21 of file [LibAttribute.hpp](#).

5.6.4.2 err_

```
si2drErrorT& LibAttribute::err_ [private]
```

Definition at line 22 of file [LibAttribute.hpp](#).

The documentation for this class was generated from the following files:

- include/[LibAttribute.hpp](#)
- src/[LibAttribute.cpp](#)

5.7 LibFile Class Reference

```
#include <LibFile.hpp>
```

Public Member Functions

- [LibFile](#) (const std::string &filepath, const std::string &loggername)
Constructs a [LibFile](#) object with specified file path and logger name.
- [~LibFile](#) ()
- void [writeJsonToFile](#) ()
Writes the JSON data stored in the object to a file.
- void [parse](#) ()
Parses the Liberty file and extracts library information into a JSON structure.
- void [modify](#) ()
- void [mono](#) (const bool is_slew)
Validates that lookup tables are monotonically increasing with respect to the output load.
- void [supercell](#) (const int chain_length, const std::vector< std::string > &cell_names)

Creates supercells based on standard cells in the library file.

- void [verilog](#) (const int chain_length, const std::vector< std::string > &cell_names)

Generates Verilog files for cell validation based on the library file.

- void [spice](#) (const int chain_length, const std::vector< std::string > &cell_names, const std::string &verilog_lib_file, const std::string &spice_lib_file)
- std::map< std::string, std::string > [logic](#) (const std::string &cell_name)

Retrieves the logic functions for the output pins of a specified cell within the Liberty file.

Public Attributes

- std::shared_ptr< spdlog::logger > [logger_](#)
- std::filesystem::path [filepath_](#)
- std::string [basename_](#)
- std::string [filename_](#)
- std::string [libname_](#) = ""
- std::string [jsonname_](#) = ""
- std::string [loggername_](#) = ""
- [json lib_json_](#) = json::object()

Private Member Functions

- void [read](#) ()
- Reads the Liberty file specified by the filename_ member variable.*
- bool [checkTimingArcMonotonicity](#) (const [json](#) &cell, const [json](#) &pin, const [json](#) &arc, const std::string &type, bool is_slew)
- Checks if the values in a timing arc are monotonically increasing.*

Private Attributes

- si2drErrorT [err_](#)
- int [process_](#) = 0
- float [voltage_](#) = 0.0
- int [temperature_](#) = 0

5.7.1 Detailed Description

Definition at line 22 of file [LibFile.hpp](#).

5.7.2 Constructor & Destructor Documentation

5.7.2.1 LibFile()

```
LibFile::LibFile (
    const std::string & filepath,
    const std::string & loggername )
```

Constructs a [LibFile](#) object with specified file path and logger name.

This constructor initializes a [LibFile](#) object with the given file path and creates a logger with the specified name. It sets up:

- Two logging sinks: a colored console sink and a file sink
- File path information including filename, basename, and a corresponding JSON filename
- Log levels (info for console, debug for file)

Parameters

<i>filepath</i>	The path to the file to be processed
<i>loggername</i>	The name for the logger and the log file

Definition at line [15](#) of file [LibFile.cpp](#).

5.7.2.2 ~LibFile()

```
LibFile::~LibFile ( )
```

Definition at line [35](#) of file [LibFile.cpp](#).

5.7.3 Member Function Documentation

5.7.3.1 checkTimingArcMonotonicity()

```
bool LibFile::checkTimingArcMonotonicity (
    const json & cell,
    const json & pin,
    const json & arc,
    const std::string & timing_arc_name,
    bool is_slew ) [private]
```

Checks if the values in a timing arc are monotonically increasing.

This function examines the values in the specified timing arc to ensure they are monotonically increasing. It checks monotonicity in two dimensions:

1. Across rows (by output load capacitance)
2. Across columns (by input slew, only if `is_slew` is true)

The function logs detailed information about any non-monotonic values found, including cell name, pin names, and conditional statements (when clause) if present.

Parameters

<i>cell</i>	The JSON object representing the cell being checked
<i>pin</i>	The JSON object representing the pin being checked
<i>arc</i>	The JSON object representing the timing arc being checked
<i>timing_arc_name</i>	The name of the timing arc to check (e.g., "cell_rise", "cell_fall")
<i>is_slew</i>	Boolean flag indicating whether to check monotonicity across input slew values

Returns

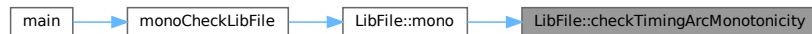
true if all values in the timing arc are monotonic, false otherwise

Exceptions

<i>None, but</i>	logs errors or warnings for invalid data formats or non-monotonic values
------------------	--

Definition at line 205 of file [LibFile.cpp](#).

Here is the caller graph for this function:



5.7.3.2 logic()

```
std::map< std::string, std::string > LibFile::logic (  
    const std::string & cell_name )
```

Retrieves the logic functions for the output pins of a specified cell within the Liberty file.

This method searches for a cell with the given name in the parsed Liberty file (either by parsing the file directly or reading from a JSON representation). It then iterates through the output pins of the cell, extracting the logic function associated with each pin. The logic functions are stored in a map, where the key is the pin name and the value is the logic function string.

If the JSON file does not exist, the Liberty file is parsed, and a JSON file is created. If the JSON file exists but cannot be opened or parsed, an error is logged, and an empty map is returned. If the specified cell is not found or if no logic functions are found for the cell, a warning is logged.

Parameters

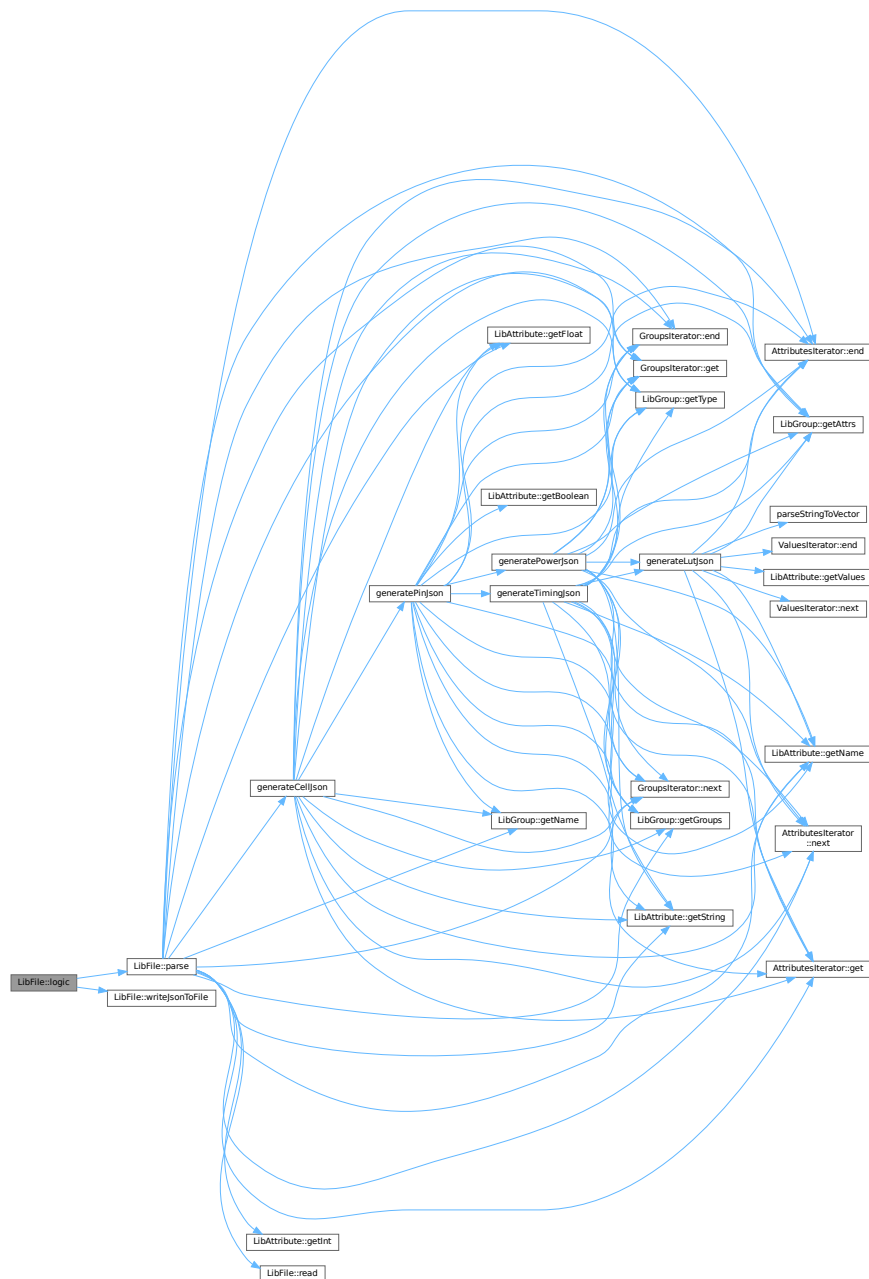
<i>cell_name</i>	The name of the cell for which to retrieve the logic functions.
------------------	---

Returns

A map containing the logic functions for the output pins of the specified cell. The keys of the map are the pin names, and the values are the corresponding logic function strings. Returns an empty map if no logic functions are found or if an error occurs during file processing.

Definition at line 950 of file LibFile.cpp.

Here is the call graph for this function:



Here is the caller graph for this function:



5.7.3.3 modify()

```
void LibFile::modify ( )
```

Definition at line 182 of file [LibFile.cpp](#).

5.7.3.4 mono()

```
void LibFile::mono (
    const bool is_slew )
```

Validates that lookup tables are monotonically increasing with respect to the output load.

This function checks the following timing data in the current library to ensure monotonicity:

- cell_rise and retaining_rise
- cell_fall and retaining_fall
- rise_transition and retain_rise_slew
- fall_transition and retain_fall_slew
- min_pulse_width constraints (rise_constraint, fall_constraint)

The function first checks if a parsed JSON representation exists. If not, it parses the Liberty file and creates the JSON file. It then evaluates each timing arc in all cells and reports the pass/fail status at the end.

Here is the caller graph for this function:



5.7.3.5 parse()

```
void LibFile::parse ( )
```

Parses the Liberty file and extracts library information into a JSON structure.

This method handles the parsing of a Liberty file by:

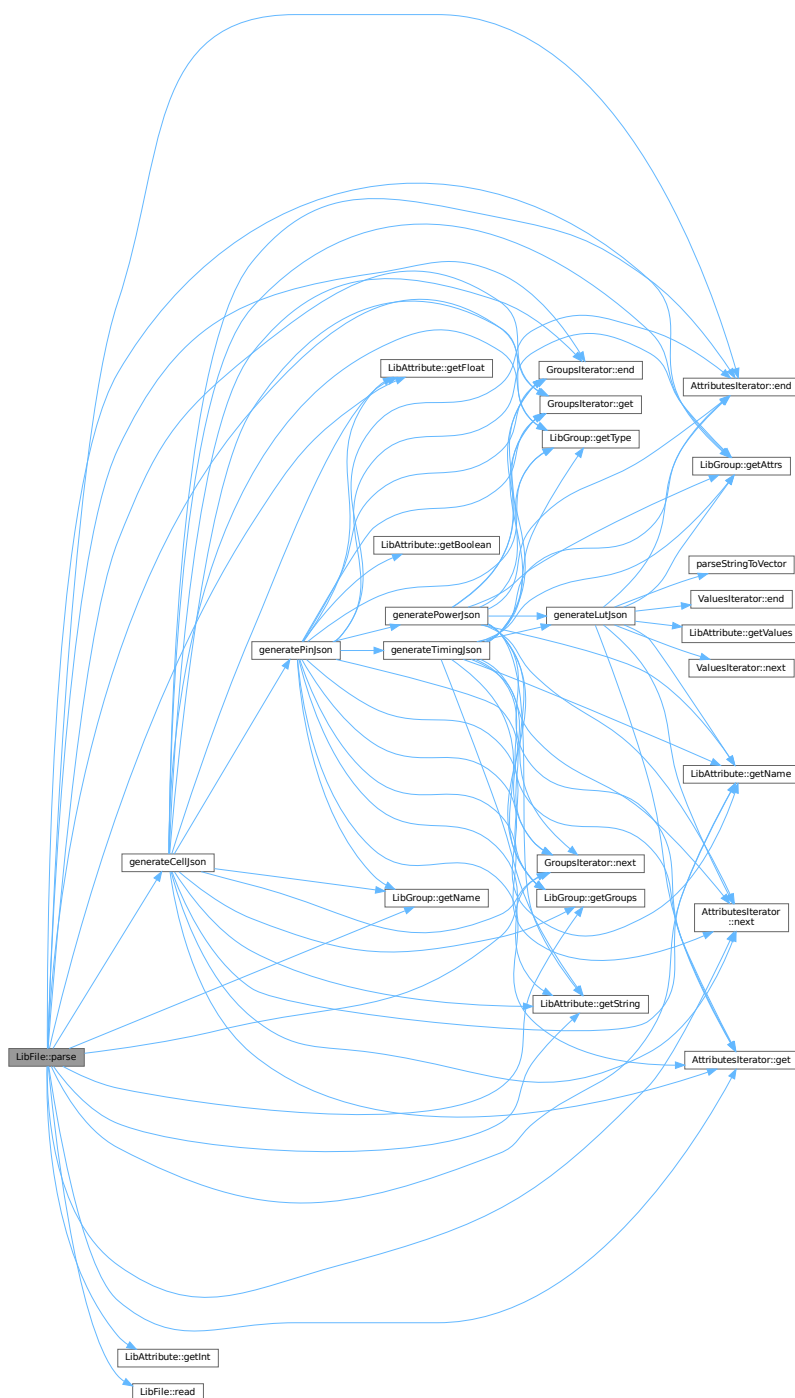
1. Initializing the SI2 parser interface error handler
2. Reading the Liberty file contents
3. Extracting the library name
4. Processing second-level groups including:
 - Cell definitions, which are added to the JSON structure
 - Operating conditions, extracting PVT (Process, Voltage, Temperature) values

The parsed data is stored in the `lib_json_` member variable, with the following structure:

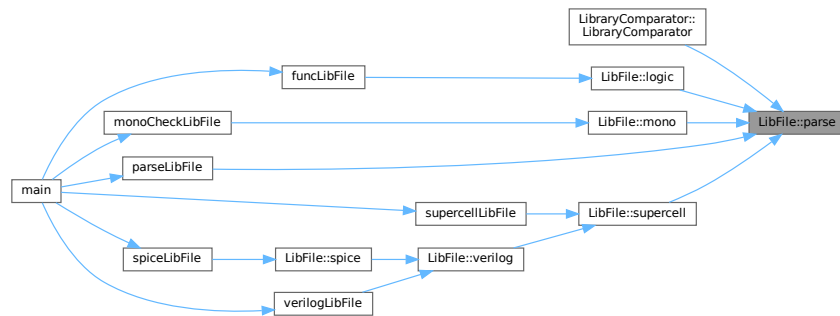
- "library_name": Name of the library
- "cells": Array of cell definitions
- "process": Process value
- "voltage": Voltage value (rounded to 2 decimal places)
- "temperature": Temperature value

This method creates local scopes to ensure proper cleanup of SI2 Iterators.

Here is the call graph for this function:



Here is the caller graph for this function:



5.7.3.6 read()

```
void LibFile::read ( ) [private]
```

Reads the Liberty file specified by the filename_ member variable.

This function attempts to read a Liberty file and logs the process. It measures the time taken to read the file and logs any errors encountered during the read operation. If an error occurs, the function logs the error and terminates the program.

- Logs the start of the read operation.
- Measures the duration of the read operation.
- Checks for specific errors (invalid name or syntax error) and logs them.
- Terminates the program with specific exit codes if errors are detected.
- Logs the completion of the read operation and the time taken.

Note

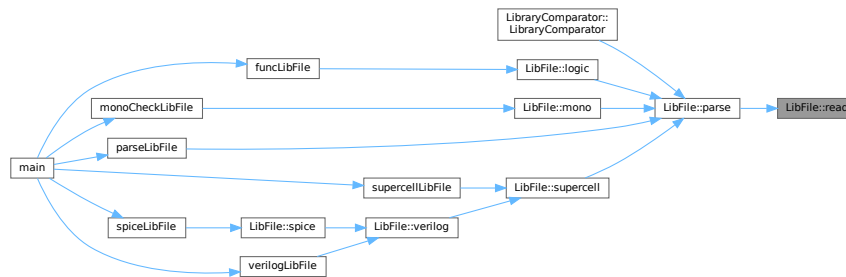
This function uses the spdlog library for logging and the si2drReadLibertyFile function for reading the Liberty file.

Exceptions

<i>This</i>	function will terminate the program with exit codes 301 or 401 if specific errors are encountered.
-------------	--

Definition at line 77 of file [LibFile.cpp](#).

Here is the caller graph for this function:



5.7.3.7 spice()

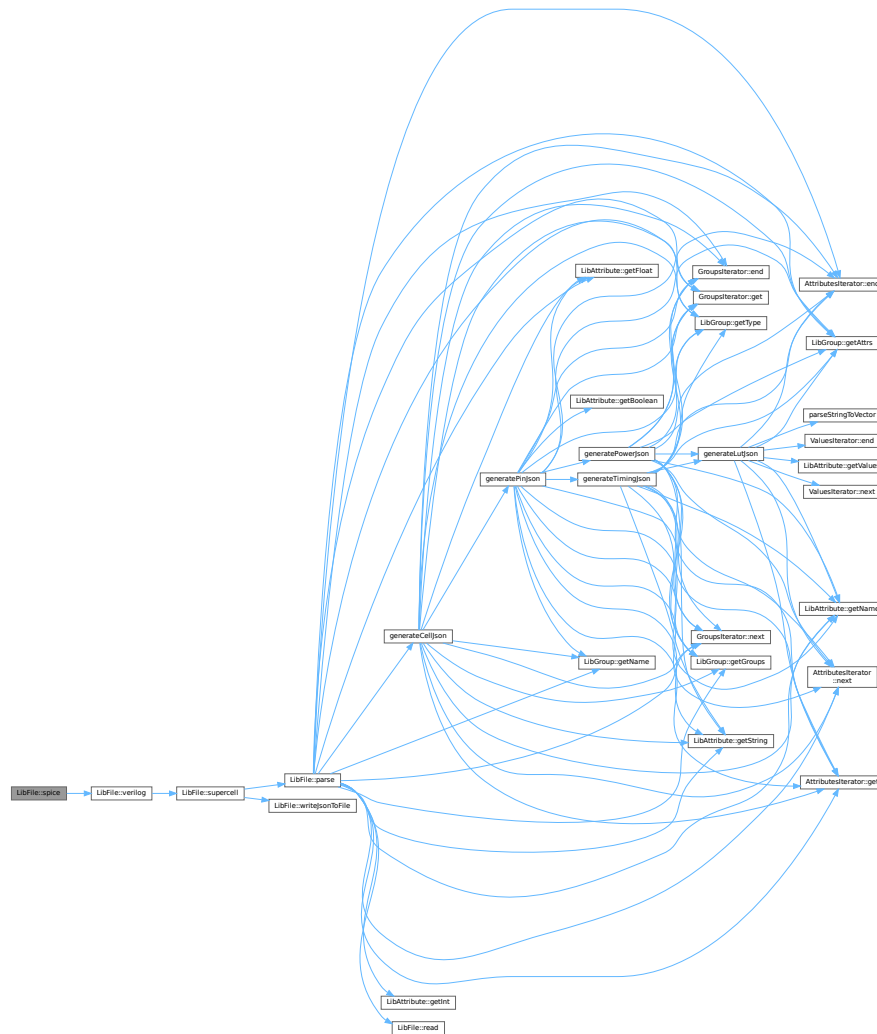
```

void LibFile::spice (
    const int chain_length,
    const std::vector< std::string > & cell_names,
    const std::string & verilog_lib_file,
    const std::string & spice_lib_file )

```

Definition at line 878 of file [LibFile.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



5.7.3.8 supercell()

```
void LibFile::supercell (
```

```
const int chain_length,  
const std::vector< std::string > & cell_names )
```

Creates supercells based on standard cells in the library file.

This method creates supercells by combining standard cells in chains. For each input-output pin pair of a cell, it creates a supercell entry with naming format: <cellname>**X**<chain_length><input_pin>↔____<output_pin>. The results are written to a .map file.

For sequential cells (with clock pins), the chain length is always set to 1 regardless of the requested chain length.

Parameters

<i>chain_length</i>	The length of the chain for combinational cells
<i>cell_names</i>	Vector of specific cell names to process. If empty, all cells will be processed.

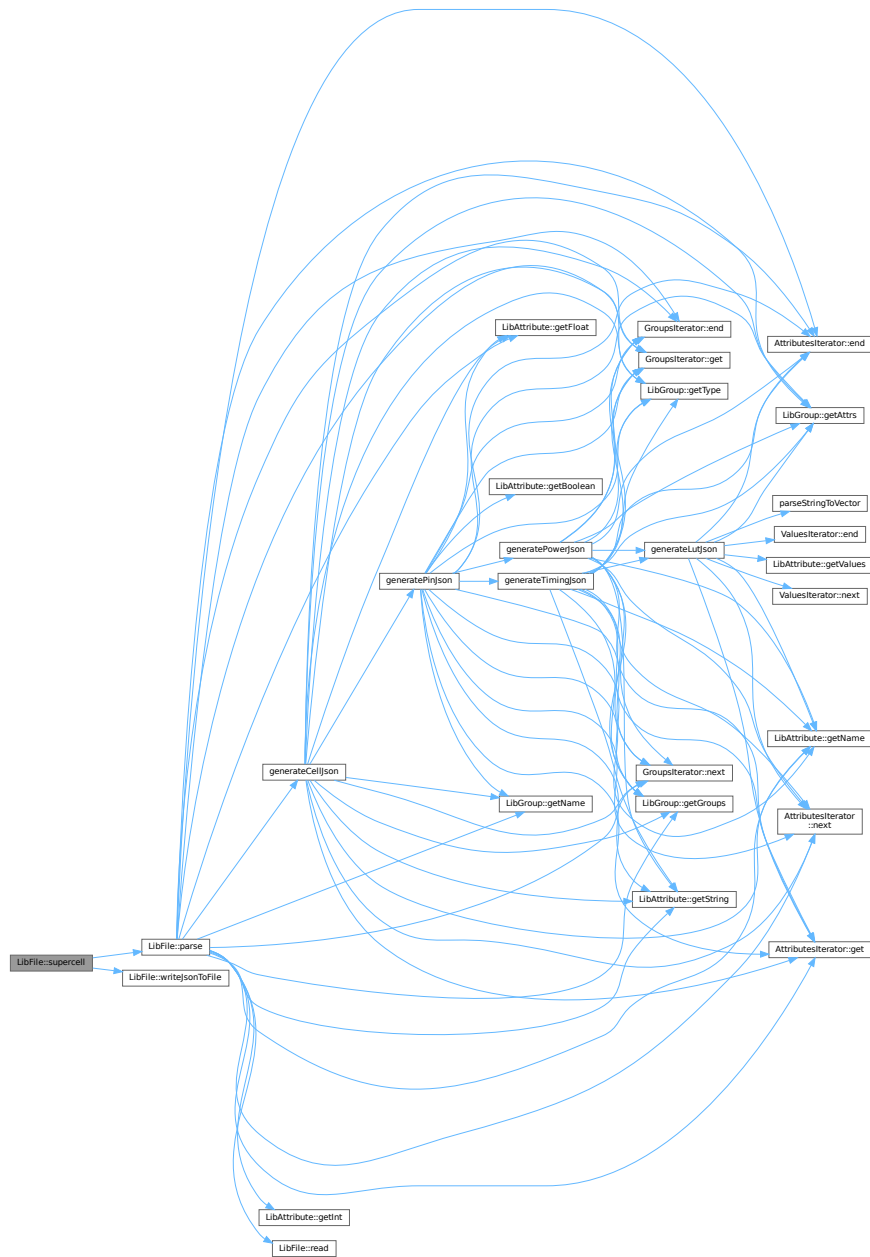
Note

Before creating supercells, this method checks for the existence of a JSON representation of the Liberty file and parses it if not found.

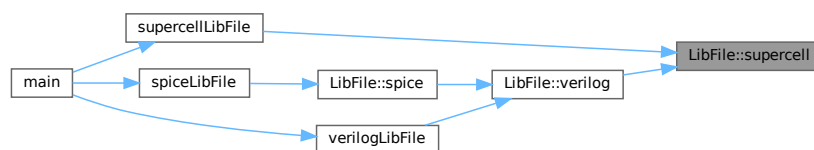
The method will report any requested cells that were not found in the library.

Definition at line 460 of file [LibFile.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



5.7.3.9 verilog()

```
void LibFile::verilog (
    const int chain_length,
    const std::vector< std::string > & cell_names )
```

Generates Verilog files for cell validation based on the library file.

This function creates Verilog modules for each cell in the specified chain and generates a top-level module that connects them all together. The process involves:

1. Creating supercells based on the specified chain length and cell names
2. Reading the mapping file to identify supercell name relationships
3. Generating individual Verilog modules for each supercell, handling both sequential and combinational cells differently
4. Creating ANSI-style port definitions for each module
5. Using the slang library to build proper syntax trees for Verilog modules
6. Creating a top-level module that instantiates all the individual modules
7. Writing the complete Verilog output to the output file

Sequential cells are instantiated once, while combinational cells are instantiated multiple times based on the specified chain length.

Parameters

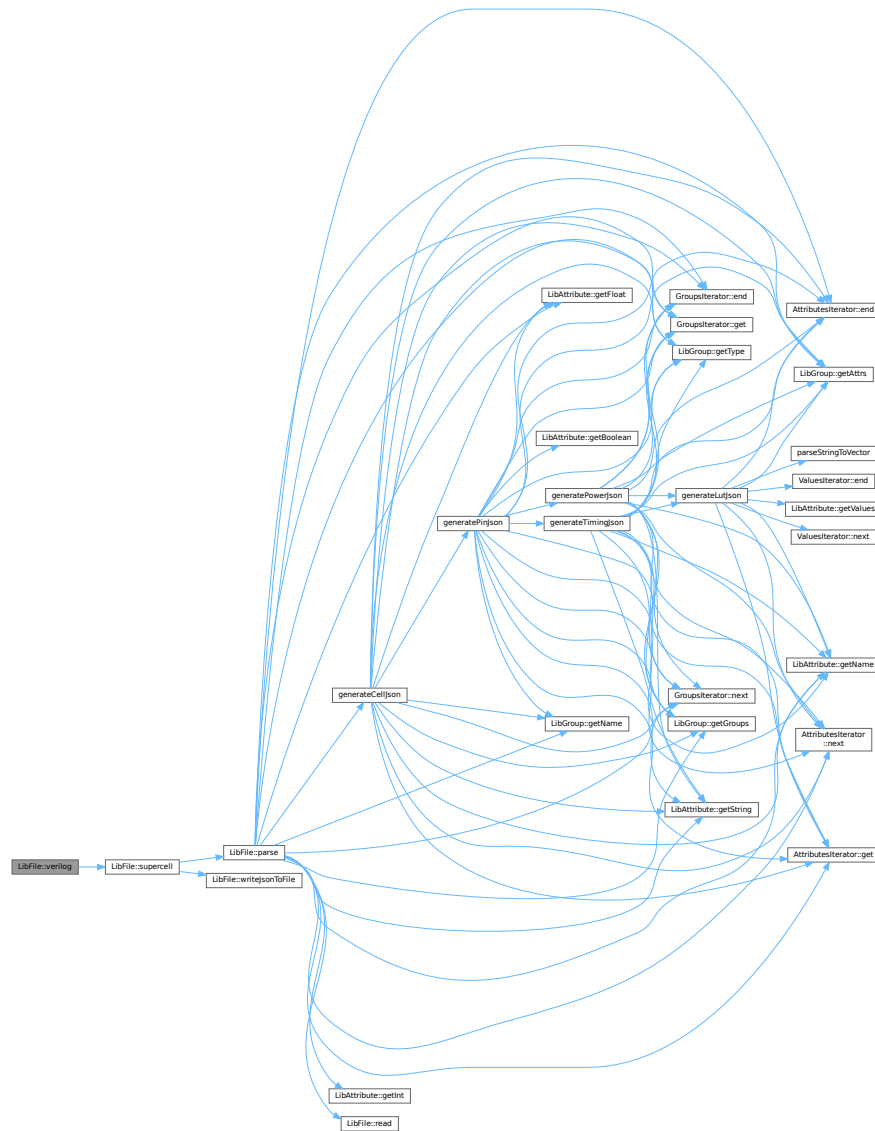
<i>chain_length</i>	The number of times to chain combinational cells together
<i>cell_names</i>	Vector of cell names to include in the Verilog generation

Note

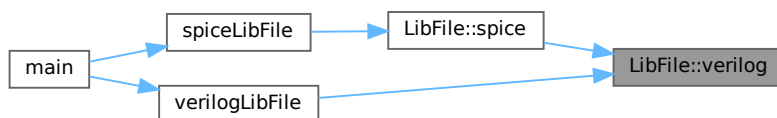
The function writes a temporary file during processing that might not be removed when the function completes (commented out cleanup code).

Definition at line 614 of file [LibFile.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



5.7.3.10 writeJsonToFile()

```
void LibFile::writeJsonToFile ( )
```

Writes the JSON data stored in the object to a file.

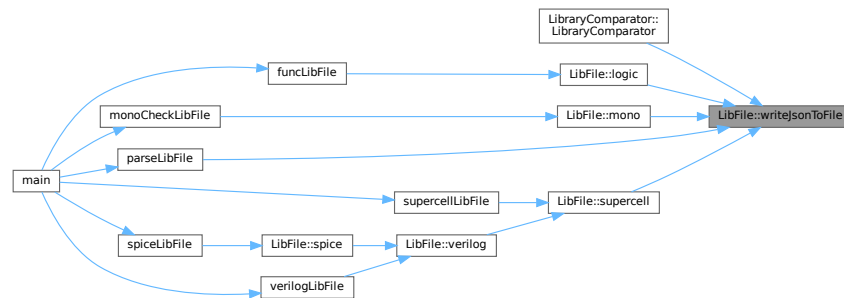
This method opens the file specified by `jsonname_` and writes the content of `lib_json_` with an indentation of 2 spaces. If the file cannot be opened, an error message is logged. Upon successful writing, an informational message is logged.

Exceptions

None	directly, but <code>std::ofstream</code> operations may throw exceptions
------	--

Definition at line 46 of file [LibFile.cpp](#).

Here is the caller graph for this function:



5.7.4 Member Data Documentation

5.7.4.1 basename_

```
std::string LibFile::basename_
```

Definition at line 28 of file [LibFile.hpp](#).

5.7.4.2 err_

```
si2drErrorT LibFile::err_ [private]
```

Definition at line 45 of file [LibFile.hpp](#).

5.7.4.3 filename_

```
std::string LibFile::filename_
```

Definition at line 29 of file [LibFile.hpp](#).

5.7.4.4 filepath_

```
std::filesystem::path LibFile::filepath_
```

Definition at line 27 of file [LibFile.hpp](#).

5.7.4.5 jsonname_

```
std::string LibFile::jsonname_ = ""
```

Definition at line 31 of file [LibFile.hpp](#).

5.7.4.6 lib_json_

```
json LibFile::lib_json_ = json::object()
```

Definition at line 33 of file [LibFile.hpp](#).

5.7.4.7 libname_

```
std::string LibFile::libname_ = ""
```

Definition at line 30 of file [LibFile.hpp](#).

5.7.4.8 logger_

```
std::shared_ptr<spdlog::logger> LibFile::logger_
```

Definition at line 26 of file [LibFile.hpp](#).

5.7.4.9 loggername_

```
std::string LibFile::loggername_ = ""
```

Definition at line 32 of file [LibFile.hpp](#).

5.7.4.10 process_

```
int LibFile::process_ = 0 [private]
```

Definition at line 46 of file [LibFile.hpp](#).

5.7.4.11 temperature_

```
int LibFile::temperature_ = 0 [private]
```

Definition at line 48 of file [LibFile.hpp](#).

5.7.4.12 voltage_

```
float LibFile::voltage_ = 0.0 [private]
```

Definition at line 47 of file [LibFile.hpp](#).

The documentation for this class was generated from the following files:

- include/[LibFile.hpp](#)
- src/[LibFile.cpp](#)

5.8 LibGroup Class Reference

```
#include <LibGroup.hpp>
```

Public Member Functions

- [LibGroup](#) (si2drGroupIdT group, si2drErrorT &err)
- [~LibGroup](#) ()
- std::string [getName](#) ()
- std::string [getType](#) ()
- si2drAttrIdT [getAttrs](#) ()
- si2drGroupIdT [getGroups](#) ()

Private Attributes

- si2drGroupIdT [group_](#)
- si2drErrorT & [err_](#)

5.8.1 Detailed Description

Definition at line 8 of file [LibGroup.hpp](#).

5.8.2 Constructor & Destructor Documentation

5.8.2.1 LibGroup()

```
LibGroup::LibGroup (
    si2drGroupIdT group,
    si2drErrorT & err )
```

Definition at line 3 of file [LibGroup.cpp](#).

5.8.2.2 ~LibGroup()

```
LibGroup::~~LibGroup ( )
```

Definition at line 5 of file [LibGroup.cpp](#).

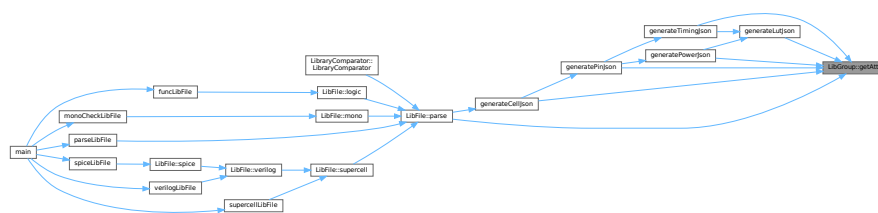
5.8.3 Member Function Documentation

5.8.3.1 getAttrs()

```
si2drAttrsIdT LibGroup::getAttrs ( )
```

Definition at line 19 of file [LibGroup.cpp](#).

Here is the caller graph for this function:

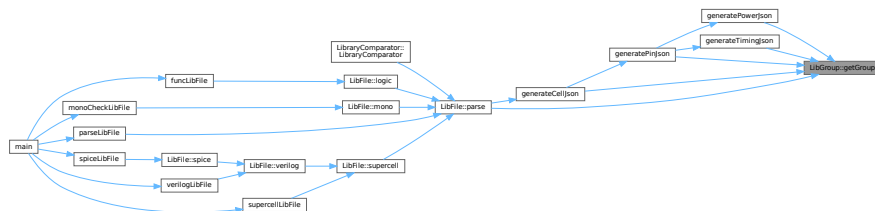


5.8.3.2 getGroups()

```
si2drGroupsIdT LibGroup::getGroups ( )
```

Definition at line 21 of file [LibGroup.cpp](#).

Here is the caller graph for this function:

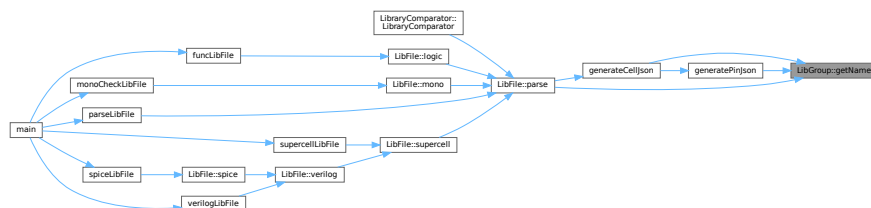


5.8.3.3 getName()

```
std::string LibGroup::getName ( )
```

Definition at line 7 of file [LibGroup.cpp](#).

Here is the caller graph for this function:

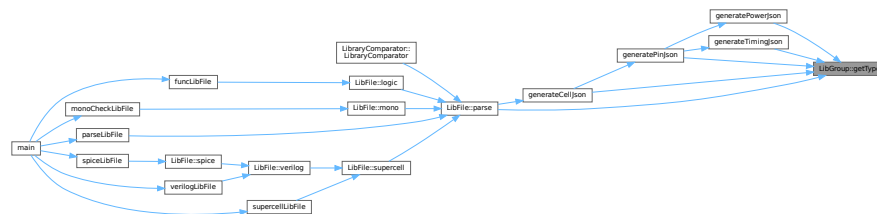


5.8.3.4 getType()

```
std::string LibGroup::getType ( )
```

Definition at line 14 of file [LibGroup.cpp](#).

Here is the caller graph for this function:



5.8.4 Member Data Documentation

5.8.4.1 err_

```
si2drErrorT& LibGroup::err_ [private]
```

Definition at line 19 of file [LibGroup.hpp](#).

5.8.4.2 group_

```
si2drGroupIdT LibGroup::group_ [private]
```

Definition at line 18 of file [LibGroup.hpp](#).

The documentation for this class was generated from the following files:

- include/[LibGroup.hpp](#)
- src/[LibGroup.cpp](#)

5.9 LibraryComparator Class Reference

```
#include <LibraryComparator.hpp>
```

Public Member Functions

- [LibraryComparator](#) ([LibFile](#) &ref_libfile, [LibFile](#) &comp_libfile, double reltol, double abstol)
Constructor for the [LibraryComparator](#) class.
- void [generateReport](#) (const std::string &output_file)
Generates a comparison report between the reference and comparison libraries.

Public Attributes

- std::filesystem::path [ref_lib_path_](#)
- std::filesystem::path [comp_lib_path_](#)

Private Member Functions

- void [compareCell](#) (const std::string &cell_name, const [json](#) &ref_cell, const [json](#) &comp_cell, Table &table)
Compares the output pins of a cell between a reference JSON and a comparison JSON.
- void [comparePin](#) (const std::string &cell_name, const std::string &pin_name, const [json](#) &ref_pin, const [json](#) &comp_pin, Table &table)
Compares the timing arcs of a given pin between a reference JSON and a comparison JSON.
- void [compareTimingArc](#) (const std::string &cell_name, const std::string &pin_name, const std::string &timing_type, const [json](#) &ref_timing_arc, const [json](#) &comp_timing_arc, Table &table)
Compares a timing arc between two JSON objects.
- void [compareLut](#) (const std::string &cell_name, const std::string &pin_name, const std::string &timing_type, const std::string &related_pin, const std::string &arc_name, const [json](#) &ref_lut, const [json](#) &comp_lut, Table &table)
Compares lookup tables (LUTs) between reference and comparison libraries for a specific timing arc.

Private Attributes

- [json](#) [ref_json_](#)
- [json](#) [comp_json_](#)
- double [reltol_](#)
- double [abstol_](#)

5.9.1 Detailed Description

Definition at line 19 of file [LibraryComparator.hpp](#).

5.9.2 Constructor & Destructor Documentation

5.9.2.1 LibraryComparator()

```
LibraryComparator::LibraryComparator (
    LibFile & ref_libfile,
    LibFile & comp_libfile,
    double reltol,
    double abstol )
```

Constructor for the [LibraryComparator](#) class.

Initializes a [LibraryComparator](#) object by loading JSON data from reference and comparison library files. If JSON files don't exist, parses the library files first and creates the JSON files.

The constructor performs the following steps:

1. Looks for reference JSON file, parses library file if not found
2. Loads reference JSON data into `ref_json_` member
3. Looks for comparison JSON file, parses library file if not found
4. Loads comparison JSON data into `comp_json_` member
5. Stores file paths and initializes tolerance values for comparison

Parameters

<i>ref_libfile</i>	Reference library file object
<i>comp_libfile</i>	Comparison library file object
<i>reltol</i>	Relative tolerance for numerical comparisons (default defined elsewhere)
<i>abstol</i>	Absolute tolerance for numerical comparisons (default defined elsewhere)

Definition at line 21 of file [LibraryComparator.cpp](#).

[illegible]

5.9.3.1 compareCell()

Generated on Sat Apr 5 2025 14:07:22 for ZlibValidation by Doxygen


```

const json & ref_cell,
const json & comp_cell,
Table & table ) [private]

```

Compares the output pins of a cell between a reference JSON and a comparison JSON.

This function iterates through the output pins of the comparison cell and checks if each pin exists in the reference cell. If a pin is found in both cells, it calls the comparePin function to compare the details of the pin. If a pin is not found in the reference cell, a warning is logged. If the comparison cell does not contain any output pins, an informational message is logged.

Parameters

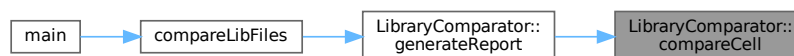
<i>cell_name</i>	The name of the cell being compared.
<i>ref_cell</i>	The reference JSON object containing the cell data.
<i>comp_cell</i>	The comparison JSON object containing the cell data.
<i>table</i>	The table object where the comparison results are stored.

Definition at line 301 of file [LibraryComparator.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



5.9.3.2 compareLut()

```

void LibraryComparator::compareLut (
    const std::string & cell_name,
    const std::string & pin_name,

```

```

const std::string & timing_type,
const std::string & related_pin,
const std::string & arc_name,
const json & ref_lut,
const json & comp_lut,
Table & table ) [private]

```

Compares lookup tables (LUTs) between reference and comparison libraries for a specific timing arc.

This function compares the lookup table values between reference and comparison libraries for a given timing arc, checking the consistency of indices and values. It verifies that the indices match between libraries and then compares each value in the LUT against relative and absolute tolerance thresholds. Any discrepancies that exceed the tolerance limits are recorded in the provided table.

Parameters

<i>cell_name</i>	The name of the cell containing the LUT.
<i>pin_name</i>	The name of the pin associated with the LUT.
<i>timing_type</i>	The timing type of the arc (e.g., "rise_transition", "fall_transition").
<i>related_pin</i>	The related pin for the timing arc.
<i>arc_name</i>	The name of the specific timing arc being compared.
<i>ref_lut</i>	The lookup table from the reference library (in JSON format).
<i>comp_lut</i>	The lookup table from the comparison library (in JSON format).
<i>table</i>	Table object to store comparison results for any mismatches found.

Note

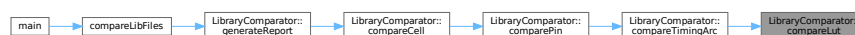
The function logs various information/warning/error messages:

- Logs index mismatches as warnings
- Logs value mismatches as debug messages
- Logs format errors in LUT structure as errors
- Records values exceeding tolerance in the provided table

The comparison uses both relative tolerance (`reltol_`) and absolute tolerance (`abstol_`) to determine if values differ significantly.

Definition at line 98 of file [LibraryComparator.cpp](#).

Here is the caller graph for this function:



5.9.3.3 comparePin()

```
void LibraryComparator::comparePin (
    const std::string & cell_name,
    const std::string & pin_name,
    const json & ref_pin,
    const json & comp_pin,
    Table & table ) [private]
```

Compares the timing arcs of a given pin between a reference JSON and a comparison JSON.

This function iterates through the timing arcs of the comparison pin and looks for matching timing types in the reference pin. If a matching timing type is found, it calls the compareTimingArc function to compare the timing arcs. If a timing type is not found in the reference JSON, a warning is logged.

Parameters

<i>cell_name</i>	The name of the cell containing the pin.
<i>pin_name</i>	The name of the pin to compare.
<i>ref_pin</i>	The reference JSON object containing the pin's data.
<i>comp_pin</i>	The comparison JSON object containing the pin's data.
<i>table</i>	A reference to a Table object where comparison results are stored.

Definition at line 261 of file [LibraryComparator.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



5.9.3.4 compareTimingArc()

```
void LibraryComparator::compareTimingArc (
    const std::string & cell_name,
    const std::string & pin_name,
    const std::string & timing_type,
    const json & ref_timing_arc,
    const json & comp_timing_arc,
    Table & table ) [private]
```

Compares a timing arc between two JSON objects.

This function compares a specific timing arc (e.g., cell_rise, cell_fall, rise_transition, fall_transition) between a reference JSON object and a comparison JSON object. It extracts the related pin from the reference timing arc and iterates through a predefined list of arc names. If an arc name is found in the comparison timing arc, it checks if the same arc name exists in the reference timing arc. If both exist, it calls the compareLut function to compare the LUT data associated with the arc. If the arc name is found in the comparison JSON but not in the reference JSON, a warning message is logged.

Parameters

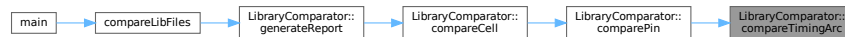
<i>cell_name</i>	The name of the cell.
<i>pin_name</i>	The name of the pin.
<i>timing_type</i>	The type of timing (e.g., "combinational", "sequential").
<i>ref_timing_arc</i>	The reference JSON object containing the timing arc information.
<i>comp_timing_arc</i>	The comparison JSON object containing the timing arc information.
<i>table</i>	The table to store comparison results.

Definition at line 220 of file [LibraryComparator.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



5.9.3.5 generateReport()

```
void LibraryComparator::generateReport (
    const std::string & output_file )
```

Generates a comparison report between the reference and comparison libraries.

This function generates a detailed comparison report between the reference library and the comparison library. The report includes metadata such as the reference and comparison library paths, absolute and relative tolerances, and the application details. It also includes a legend explaining various symbols used in the report.

The function iterates through each cell in the comparison library, compares it with the corresponding cell in the reference library, and generates a table of differences. If there are any differences, it calculates summary statistics such as the average and maximum differences, and the number of outliers. The results are written to the specified output file in either Markdown or plain text format.

Parameters

<i>output_file</i>	The path to the output file where the report will be written.
--------------------	---

Definition at line 343 of file [LibraryComparator.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



5.9.4 Member Data Documentation

5.9.4.1 abstol_

```
double LibraryComparator::abstol_ [private]
```

Definition at line 30 of file [LibraryComparator.hpp](#).

5.9.4.2 comp_json_

```
json LibraryComparator::comp_json_ [private]
```

Definition at line 28 of file [LibraryComparator.hpp](#).

5.9.4.3 comp_lib_path_

```
std::filesystem::path LibraryComparator::comp_lib_path_
```

Definition at line 23 of file [LibraryComparator.hpp](#).

5.9.4.4 ref_json_

```
json LibraryComparator::ref_json_ [private]
```

Definition at line 27 of file [LibraryComparator.hpp](#).

5.9.4.5 ref_lib_path_

std::filesystem::path LibraryComparator::ref_lib_path_

Definition at line 22 of file [LibraryComparator.hpp](#).

5.9.4.6 retol_

double LibraryComparator::retol_ [private]

Definition at line 29 of file [LibraryComparator.hpp](#).

The documentation for this class was generated from the following files:

- include/[LibraryComparator.hpp](#)
- src/[LibraryComparator.cpp](#)

5.10 LogicComparator Class Reference

```
#include <LogicComparator.hpp>
```

Public Member Functions

- [LogicComparator](#) (const std::map< std::string, std::string > &ref_outpin_map, const std::map< std::string, std::string > &comp_outpin_map, const std::string &cell_name)
- void [logic](#) ()

This function demonstrates the usage of the exprtk library to evaluate a boolean logic expression.
- std::string [preprocessExpression](#) (const std::string &input_expr)
- bool [extractVariables](#) (const std::string &expr1, const std::string &expr2, std::vector< std::string > &sorted_vars)

Extracts and validates variables from two expressions, ensuring they match.
- void [compareSingleExpressionPair](#) (const std::string &ref_expression_processed, const std::string &comp_expression_processed, const std::vector< std::string > &sorted_vars, [PinComparisonResult](#) &result)

Compares two preprocessed logic expression strings for equivalence. Internal helper function.
- void [compareCellLogic](#) ()

Compares logic for all output pins defined in the input maps, and stores results in all_pin_results_.
- void [generateReport](#) (const std::string &output_file)

Generates a comparison report file based on cell logic comparison results.

Private Attributes

- `std::map< std::string, std::string >` [ref_outpin_map_](#)
- `std::map< std::string, std::string >` [comp_outpin_map_](#)
- `std::string` [cell_name_](#)
- `std::map< std::string, PinComparisonResult >` [all_pin_results_](#)

5.10.1 Detailed Description

Definition at line 38 of file [LogicComparator.hpp](#).

5.10.2 Constructor & Destructor Documentation

5.10.2.1 LogicComparator()

```
LogicComparator::LogicComparator (
    const std::map< std::string, std::string > & ref_outpin_map,
    const std::map< std::string, std::string > & comp_outpin_map,
    const std::string & cell_name )
```

Definition at line 3 of file [LogicComparator.cpp](#).

5.10.3 Member Function Documentation

5.10.3.1 compareCellLogic()

```
void LogicComparator::compareCellLogic ( )
```

Compares logic for all output pins defined in the input maps, and stores results in `all_pin_results_`.

Compares the logic expressions for each output pin of a cell between a reference and a comparison design.

This method performs a detailed comparison of the logic expressions associated with each output pin of a cell. It iterates through each unique pin name found in both the reference and comparison designs, retrieves their corresponding logic expressions, preprocesses them, extracts and validates the variables used, and then compares the processed expressions. The results of each pin comparison, including any errors or discrepancies, are stored in the `all_pin_results_` map.

The comparison process includes the following steps:

1. **Pin Collection:** Collects all unique pin names from both the reference (`ref_outpin_map_`) and comparison (`comp_outpin_map_`) maps.
2. **Iteration:** Iterates through each unique pin name.
3. **Expression Retrieval:** Retrieves the raw logic expressions for the current pin from both the reference and comparison maps. If a pin is missing in either map, an error is logged, and the comparison is marked as impossible for that pin.
4. **Preprocessing:** Preprocesses the raw expressions to simplify them and remove any irrelevant characters or formatting. If preprocessing results in an empty expression, an error is logged, and the comparison is marked as impossible.
5. **Variable Extraction and Validation:** Extracts the variables used in both expressions and validates that the sets of variables match. If the variable sets do not match, an error is logged, and the comparison is marked as impossible.
6. **Expression Comparison:** Compares the preprocessed expressions using the extracted variables. This step determines whether the logic functions represented by the expressions are equivalent.
7. **Result Storage:** Stores the detailed results of the pin comparison, including the raw and processed expressions, any errors encountered, and the comparison result, in the `all_pin_results_` map.

Note

The `spdlog` library is used for logging information, warnings, and errors throughout the comparison process.

The `preprocessExpression` method is used to simplify the raw logic expressions before comparison.

The `extractVariables` method is used to extract and validate the variables used in the logic expressions.

The `compareSingleExpressionPair` method is used to compare the preprocessed expressions and determine whether they are equivalent.

See also

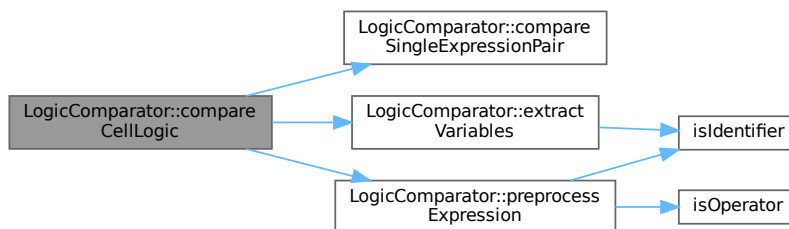
[preprocessExpression](#)

[extractVariables](#)

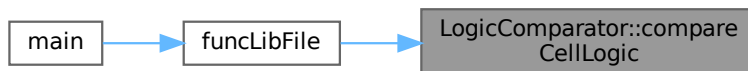
[compareSingleExpressionPair](#)

Definition at line 788 of file [LogicComparator.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



5.10.3.2 compareSingleExpressionPair()

```

void LogicComparator::compareSingleExpressionPair (
    const std::string & ref_expression_processed,
    const std::string & comp_expression_processed,
    const std::vector< std::string > & sorted_vars,
    PinComparisonResult & result )
  
```

Compares two preprocessed logic expression strings for equivalence. Internal helper function.

Compares two boolean expressions for logical equivalence using a truth table.

Template Parameters

<i>T</i>	The numeric type used by ExprTk (e.g., double, float).
----------	--

Parameters

<i>ref_expression_processed</i>	The preprocessed reference logic expression.
---------------------------------	--

Parameters

<i>comp_expression_processed</i>	The preprocessed comparison logic expression.
<i>sorted_vars</i>	A sorted vector of unique variable names common to both expressions.
<i>result</i>	Reference to a PinComparisonResult object to store detailed results.

This function takes two processed boolean expressions (*ref_expression_processed* and *comp_expression_processed*), a sorted list of variable names (*sorted_vars*), and a [PinComparisonResult](#) struct to store the results. It evaluates both expressions for all possible combinations of variable values and compares the resulting truth tables.

The function uses the ExprTk library to parse and evaluate the boolean expressions. It generates a truth table for each expression and compares the results. If the truth tables are identical, the expressions are considered logically equivalent.

Parameters

<i>ref_expression_processed</i>	The processed reference boolean expression.
<i>comp_expression_processed</i>	The processed comparison boolean expression.
<i>sorted_vars</i>	A sorted vector of variable names used in the expressions.
<i>result</i>	A PinComparisonResult struct to store the comparison results, including processed expressions, compilation status, equivalence, error messages, and the generated truth tables.

Note

The function limits the number of variables to 20 to prevent excessively large truth tables. It also performs overflow checking on the number of combinations.

The function uses spdlog for logging debug, warning, and error messages.

The function assumes that the input expressions are valid boolean expressions containing only the variables listed in *sorted_vars*.

Definition at line 529 of file [LogicComparator.cpp](#).

Here is the caller graph for this function:



5.10.3.3 extractVariables()

```
bool LogicComparator::extractVariables (
    const std::string & expr1_raw,
    const std::string & expr2_raw,
    std::vector< std::string > & sorted_vars )
```

Extracts and validates variables from two expressions, ensuring they match.

This function parses two raw expression strings (*expr1_raw* and *expr2_raw*) to identify potential variable names. It uses a regular expression to find identifiers and then validates them against a set of rules:

1. The identifier must be a valid identifier as determined by the `isIdentifier` function.
2. The identifier must not be a keyword or function name defined in the `keywords` set.

The function compares the sets of validated variables from both expressions. If the sets are identical, the variables are extracted, sorted alphabetically, and stored in the *sorted_vars* vector. If the sets differ, an error is reported, and the differences between the sets are logged.

Parameters

<i>expr1_raw</i>	The raw string representation of the first expression.
<i>expr2_raw</i>	The raw string representation of the second expression.
<i>sorted_vars</i>	A vector to store the sorted list of unique variable names if the variable sets from both expressions match. This vector is cleared if the sets do not match.

Returns

`true` if the variable sets from both expressions match, indicating that the *sorted_vars* vector contains the sorted list of unique variable names. `false` if the variable sets do not match, indicating an error.

Note

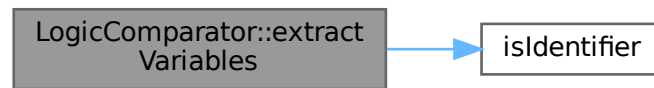
The `isIdentifier` function is used to validate potential variable names.

The `keywords` set contains a list of reserved words that cannot be used as variable names.

The function uses `spdlog` for logging debug, trace, info, and warning messages.

Definition at line 303 of file [LogicComparator.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



5.10.3.4 generateReport()

```
void LogicComparator::generateReport (
    const std::string & output_file )
```

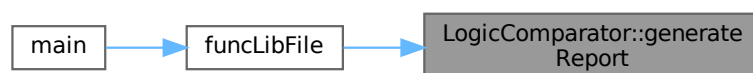
Generates a comparison report file based on cell logic comparison results.

Parameters

<code>output_file</code>	Path to the output report file.
--------------------------	---------------------------------

Definition at line 911 of file [LogicComparator.cpp](#).

Here is the caller graph for this function:



5.10.3.5 logic()

```
void LogicComparator::logic ( )
```

This function demonstrates the usage of the `exprtk` library to evaluate a boolean logic expression.

It defines a boolean expression "not(A and B) or C" and evaluates it for all possible combinations of boolean values for the variables A, B, and C. The results are then printed to the console in a tabular format.

The function utilizes the `exprtk` library for:

- Defining a symbol table to hold the variables A, B, and C.
- Creating an expression object and registering the symbol table with it.
- Compiling the boolean expression string into the expression object.
- Iterating through all possible boolean combinations for A, B, and C.
- Assigning the boolean values to the variables in the symbol table.
- Evaluating the expression using `expression.value()`.
- Printing the input values and the result to the console.

Definition at line [24](#) of file [LogicComparator.cpp](#).

Here is the caller graph for this function:

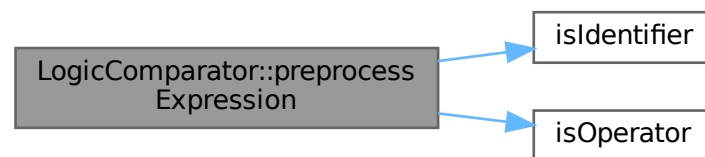


5.10.3.6 preprocessExpression()

```
std::string LogicComparator::preprocessExpression (
    const std::string & input_expr )
```

Definition at line 104 of file [LogicComparator.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



5.10.4 Member Data Documentation

5.10.4.1 all_pin_results_

```
std::map<std::string, PinComparisonResult> LogicComparator::all_pin_results_ [private]
```

Definition at line 86 of file [LogicComparator.hpp](#).

5.10.4.2 cell_name_

```
std::string LogicComparator::cell_name_ [private]
```

Definition at line 85 of file [LogicComparator.hpp](#).

5.10.4.3 comp_outpin_map_

```
std::map<std::string, std::string> LogicComparator::comp_outpin_map_ [private]
```

Definition at line 84 of file [LogicComparator.hpp](#).

5.10.4.4 ref_outpin_map_

```
std::map<std::string, std::string> LogicComparator::ref_outpin_map_ [private]
```

Definition at line 83 of file [LogicComparator.hpp](#).

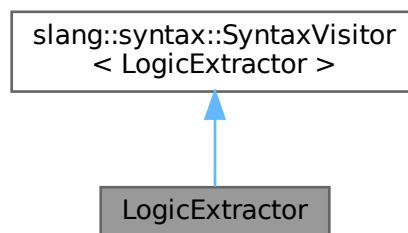
The documentation for this class was generated from the following files:

- include/[LogicComparator.hpp](#)
- src/[LogicComparator.cpp](#)

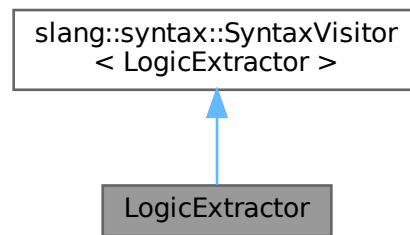
5.11 LogicExtractor Class Reference

```
#include <LogicExtractor.hpp>
```

Inheritance diagram for LogicExtractor:



Collaboration diagram for LogicExtractor:



Public Member Functions

- `LogicExtractor` (const std::string &targetCell)
- void `handle` (const slang::syntax::ModuleDeclarationSyntax &module)
- void `handle` (const slang::syntax::PortDeclarationSyntax &portDecl)
- void `handle` (const slang::syntax::NonAnsiPortListSyntax &portList)
- void `handle` (const slang::syntax::NetDeclarationSyntax &netDecl)
- void `handle` (const slang::syntax::PrimitiveInstantiationSyntax &primitiveInst)
- const std::unordered_map< std::string, `GateInfo` > & `getExtractedGates` () const
- const std::unordered_set< std::string > & `getPrimaryInputs` () const
- const std::unordered_set< std::string > & `getPrimaryOutputs` () const
- const std::unordered_set< std::string > & `getInternalWires` () const
- std::map< std::string, std::string > `getLogicExpressions` ()

Private Member Functions

- std::string `deriveLogicRecursive` (const std::string &signalName)
- std::string `formatExpression` (const `GateInfo` &gateInfo, const std::vector< std::string > &inputExprs)

Private Attributes

- const std::string & `targetCell_`
- bool `inTargetModule_`
- bool `parsingComplete_`
- std::unordered_set< std::string > `primaryInputs_`
- std::unordered_set< std::string > `primaryOutputs_`
- std::unordered_set< std::string > `internalWires_`
- std::unordered_map< std::string, std::string > `portDirections_`
- std::unordered_map< std::string, `GateInfo` > `gateOutputDrivers_`
- std::unordered_map< std::string, std::string > `logicCache_`

5.11.1 Detailed Description

Definition at line 17 of file [LogicExtractor.hpp](#).

5.11.2 Constructor & Destructor Documentation

5.11.2.1 LogicExtractor()

```
LogicExtractor::LogicExtractor (
    const std::string & targetCell ) [inline], [explicit]
```

Definition at line 20 of file [LogicExtractor.hpp](#).

5.11.3 Member Function Documentation

5.11.3.1 deriveLogicRecursive()

```
std::string LogicExtractor::deriveLogicRecursive (
    const std::string & signalName ) [private]
```

Definition at line 380 of file [LogicExtractor.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



5.11.3.2 formatExpression()

```
std::string LogicExtractor::formatExpression (
    const GateInfo & gateInfo,
    const std::vector< std::string > & inputExprs ) [private]
```

Definition at line 437 of file [LogicExtractor.cpp](#).

Here is the caller graph for this function:



5.11.3.3 getExtractedGates()

```
const std::unordered_map< std::string, GateInfo > & LogicExtractor::getExtractedGates ( ) const [inline]
```

Definition at line 37 of file [LogicExtractor.hpp](#).

Here is the caller graph for this function:

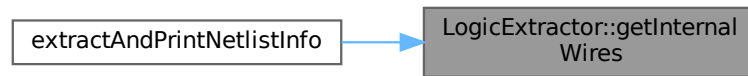


5.11.3.4 getInternalWires()

```
const std::unordered_set< std::string > & LogicExtractor::getInternalWires ( ) const [inline]
```

Definition at line 42 of file [LogicExtractor.hpp](#).

Here is the caller graph for this function:



5.11.3.5 getLogicExpressions()

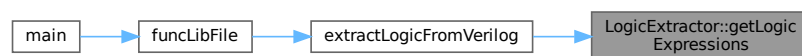
```
std::map< std::string, std::string > LogicExtractor::getLogicExpressions ( )
```

Definition at line 350 of file [LogicExtractor.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:

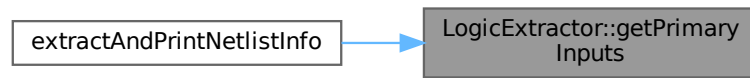


5.11.3.6 getPrimaryInputs()

```
const std::unordered_set< std::string > & LogicExtractor::getPrimaryInputs ( ) const [inline]
```

Definition at line 40 of file [LogicExtractor.hpp](#).

Here is the caller graph for this function:

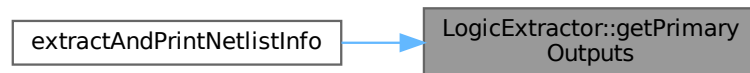


5.11.3.7 getPrimaryOutputs()

```
const std::unordered_set< std::string > & LogicExtractor::getPrimaryOutputs ( ) const [inline]
```

Definition at line 41 of file [LogicExtractor.hpp](#).

Here is the caller graph for this function:



5.11.3.8 handle() [1/5]

```
void LogicExtractor::handle (
    const slang::syntax::ModuleDeclarationSyntax & module )
```

Definition at line 5 of file [LogicExtractor.cpp](#).

5.11.3.9 handle() [2/5]

```
void LogicExtractor::handle (
    const slang::syntax::NetDeclarationSyntax & netDecl )
```

Definition at line 201 of file [LogicExtractor.cpp](#).

5.11.3.10 `handle()` [3/5]

```
void LogicExtractor::handle (
    const slang::syntax::NonAnsiPortListSyntax & portList )
```

Definition at line 112 of file [LogicExtractor.cpp](#).

5.11.3.11 `handle()` [4/5]

```
void LogicExtractor::handle (
    const slang::syntax::PortDeclarationSyntax & portDecl )
```

Definition at line 49 of file [LogicExtractor.cpp](#).

5.11.3.12 `handle()` [5/5]

```
void LogicExtractor::handle (
    const slang::syntax::PrimitiveInstantiationSyntax & primitiveInst )
```

Definition at line 221 of file [LogicExtractor.cpp](#).

5.11.4 Member Data Documentation

5.11.4.1 `gateOutputDrivers_`

```
std::unordered_map<std::string, GateInfo> LogicExtractor::gateOutputDrivers_ [private]
```

Definition at line 63 of file [LogicExtractor.hpp](#).

5.11.4.2 `inTargetModule_`

```
bool LogicExtractor::inTargetModule_ [private]
```

Definition at line 50 of file [LogicExtractor.hpp](#).

5.11.4.3 internalWires_

```
std::unordered_set<std::string> LogicExtractor::internalWires_ [private]
```

Definition at line 56 of file [LogicExtractor.hpp](#).

5.11.4.4 logicCache_

```
std::unordered_map<std::string, std::string> LogicExtractor::logicCache_ [private]
```

Definition at line 66 of file [LogicExtractor.hpp](#).

5.11.4.5 parsingComplete_

```
bool LogicExtractor::parsingComplete_ [private]
```

Definition at line 51 of file [LogicExtractor.hpp](#).

5.11.4.6 portDirections_

```
std::unordered_map<std::string, std::string> LogicExtractor::portDirections_ [private]
```

Definition at line 60 of file [LogicExtractor.hpp](#).

5.11.4.7 primaryInputs_

```
std::unordered_set<std::string> LogicExtractor::primaryInputs_ [private]
```

Definition at line 54 of file [LogicExtractor.hpp](#).

5.11.4.8 primaryOutputs_

```
std::unordered_set<std::string> LogicExtractor::primaryOutputs_ [private]
```

Definition at line 55 of file [LogicExtractor.hpp](#).

5.11.4.9 targetCell_

```
const std::string& LogicExtractor::targetCell_ [private]
```

Definition at line 49 of file [LogicExtractor.hpp](#).

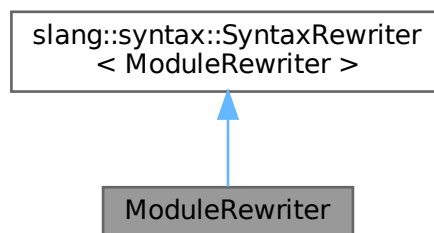
The documentation for this class was generated from the following files:

- include/[LogicExtractor.hpp](#)
- src/[LogicExtractor.cpp](#)

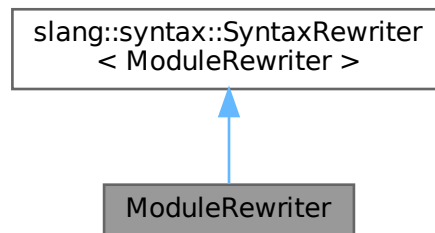
5.12 ModuleRewriter Class Reference

```
#include <verilog_utils.hpp>
```

Inheritance diagram for ModuleRewriter:



Collaboration diagram for ModuleRewriter:



Public Member Functions

- [ModuleRewriter](#) (const std::vector< std::string > &inputPins, const std::vector< std::string > &outputPins, const std::pair< std::string, std::string > &supercell_entry, int instance_count, std::shared_ptr< spdlog::logger > logger)
- void [handle](#) (const slang::syntax::SyntaxNode &node)
Processes a syntax node in the module's AST.
- void [handle](#) (const slang::syntax::ModuleDeclarationSyntax &module)

Public Attributes

- std::shared_ptr< spdlog::logger > [logger_](#)

Private Attributes

- const std::vector< std::string > & [inputPins_](#)
- const std::vector< std::string > & [outputPins_](#)
- const std::string [cellName_](#)
- const std::string [moduleName_](#)
- std::map< std::string, std::string > [portInfoMap_](#)
- int [depth_](#)
- int [instance_count_](#)

5.12.1 Detailed Description

Definition at line 59 of file [verilog_utils.hpp](#).

5.12.2 Constructor & Destructor Documentation

5.12.2.1 ModuleRewriter()

```
ModuleRewriter::ModuleRewriter (
    const std::vector< std::string > & inputPins,
    const std::vector< std::string > & outputPins,
    const std::pair< std::string, std::string > & supercell_entry,
    int instance_count,
    std::shared_ptr< spdlog::logger > logger ) [inline], [explicit]
```

Definition at line 61 of file [verilog_utils.hpp](#).

5.12.3 Member Function Documentation

5.12.3.1 handle() [1/2]

```
void ModuleRewriter::handle (
    const slang::syntax::ModuleDeclarationSyntax & module )
```

Definition at line 406 of file [verilog_utils.cpp](#).

5.12.3.2 handle() [2/2]

```
void ModuleRewriter::handle (
    const slang::syntax::SyntaxNode & node )
```

Processes a syntax node in the module's AST.

This method is called for each syntax node during the traversal of the AST. It logs debug information about the current node and continues processing its child nodes by recursively calling the appropriate visit methods.

Parameters

<i>node</i>	The syntax node to process
-------------	----------------------------

Definition at line 396 of file [verilog_utils.cpp](#).

5.12.4 Member Data Documentation

5.12.4.1 cellName_

```
const std::string ModuleRewriter::cellName_ [private]
```

Definition at line 75 of file [verilog_utils.hpp](#).

5.12.4.2 depth_

```
int ModuleRewriter::depth_ [private]
```

Definition at line 78 of file [verilog_utils.hpp](#).

5.12.4.3 inputPins_

```
const std::vector<std::string>& ModuleRewriter::inputPins_ [private]
```

Definition at line 73 of file [verilog_utils.hpp](#).

5.12.4.4 instance_count_

```
int ModuleRewriter::instance_count_ [private]
```

Definition at line 79 of file [verilog_utils.hpp](#).

5.12.4.5 logger_

```
std::shared_ptr<spdlog::logger> ModuleRewriter::logger_
```

Definition at line 68 of file [verilog_utils.hpp](#).

5.12.4.6 moduleName_

```
const std::string ModuleRewriter::moduleName_ [private]
```

Definition at line 76 of file [verilog_utils.hpp](#).

5.12.4.7 outputPins_

```
const std::vector<std::string>& ModuleRewriter::outputPins_ [private]
```

Definition at line 74 of file [verilog_utils.hpp](#).

5.12.4.8 portInfoMap_

```
std::map<std::string, std::string> ModuleRewriter::portInfoMap_ [private]
```

Definition at line 77 of file [verilog_utils.hpp](#).

The documentation for this class was generated from the following files:

- [include/verilog_utils.hpp](#)
- [src/verilog_utils.cpp](#)

5.13 PinComparisonResult Struct Reference

```
#include <LogicComparator.hpp>
```

Public Attributes

- std::string [pin_name](#)
- std::string [ref_expr_raw](#)
- std::string [comp_expr_raw](#)
- std::string [ref_expr_processed](#)
- std::string [comp_expr_processed](#)
- bool [comparison_possible](#) = false
- bool [are_equivalent](#) = false
- bool [ref_compiles](#) = false
- bool [comp_compiles](#) = false
- std::optional< Table > [ref_truth_table](#)
- std::optional< Table > [comp_truth_table](#)
- std::string [error_message](#)

5.13.1 Detailed Description

Definition at line 23 of file [LogicComparator.hpp](#).

5.13.2 Member Data Documentation

5.13.2.1 [are_equivalent](#)

```
bool PinComparisonResult::are_equivalent = false
```

Definition at line 30 of file [LogicComparator.hpp](#).

5.13.2.2 [comp_compiles](#)

```
bool PinComparisonResult::comp_compiles = false
```

Definition at line 32 of file [LogicComparator.hpp](#).

5.13.2.3 comp_expr_processed

```
std::string PinComparisonResult::comp_expr_processed
```

Definition at line 28 of file [LogicComparator.hpp](#).

5.13.2.4 comp_expr_raw

```
std::string PinComparisonResult::comp_expr_raw
```

Definition at line 26 of file [LogicComparator.hpp](#).

5.13.2.5 comp_truth_table

```
std::optional<Table> PinComparisonResult::comp_truth_table
```

Definition at line 34 of file [LogicComparator.hpp](#).

5.13.2.6 comparison_possible

```
bool PinComparisonResult::comparison_possible = false
```

Definition at line 29 of file [LogicComparator.hpp](#).

5.13.2.7 error_message

```
std::string PinComparisonResult::error_message
```

Definition at line 35 of file [LogicComparator.hpp](#).

5.13.2.8 pin_name

```
std::string PinComparisonResult::pin_name
```

Definition at line 24 of file [LogicComparator.hpp](#).

5.13.2.9 ref_compiles

```
bool PinComparisonResult::ref_compiles = false
```

Definition at line 31 of file [LogicComparator.hpp](#).

5.13.2.10 ref_expr_processed

```
std::string PinComparisonResult::ref_expr_processed
```

Definition at line 27 of file [LogicComparator.hpp](#).

5.13.2.11 ref_expr_raw

```
std::string PinComparisonResult::ref_expr_raw
```

Definition at line 25 of file [LogicComparator.hpp](#).

5.13.2.12 ref_truth_table

```
std::optional<Table> PinComparisonResult::ref_truth_table
```

Definition at line 33 of file [LogicComparator.hpp](#).

The documentation for this struct was generated from the following file:

- [include/LogicComparator.hpp](#)

5.14 ValuesIterator Class Reference

```
#include <Iterators.hpp>
```

Public Member Functions

- [ValuesIterator](#) (si2drValuesIdT values, si2drErrorT &err)
- [~ValuesIterator](#) ()
- void [next](#) ()
- bool [end](#) ()

Public Attributes

- si2drValuesIdT [values_](#)
- si2drValueTypeT [vtype_](#)
- si2drInt32T [int_](#)
- si2drFloat64T [float_](#)
- si2drStringT [str_](#)
- si2drBooleanT [bool_](#)
- si2drExprT * [exprp_](#)
- si2drErrorT & [err_](#)

5.14.1 Detailed Description

Definition at line 37 of file [lterators.hpp](#).

5.14.2 Constructor & Destructor Documentation

5.14.2.1 ValuesIterator()

```
ValuesIterator::ValuesIterator (  
    si2drValuesIdT values,  
    si2drErrorT & err )
```

Definition at line 24 of file [lterators.cpp](#).

5.14.2.2 ~ValuesIterator()

ValuesIterator::~~ValuesIterator ()

Definition at line 27 of file [Iterators.cpp](#).

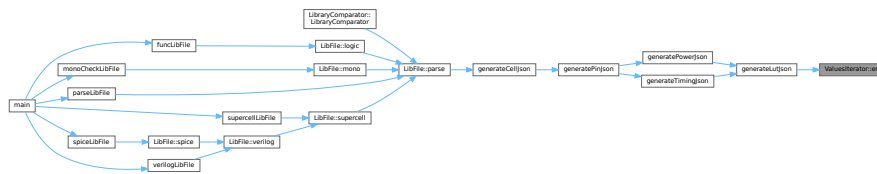
5.14.3 Member Function Documentation

5.14.3.1 end()

bool ValuesIterator::end ()

Definition at line 32 of file [Iterators.cpp](#).

Here is the caller graph for this function:

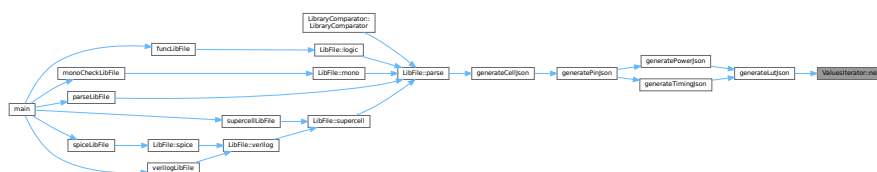


5.14.3.2 next()

void ValuesIterator::next ()

Definition at line 29 of file [Iterators.cpp](#).

Here is the caller graph for this function:



5.14.4 Member Data Documentation

5.14.4.1 bool_

si2drBooleanT ValuesIterator::bool_

Definition at line 49 of file [lterators.hpp](#).

5.14.4.2 err_

si2drErrorT& ValuesIterator::err_

Definition at line 51 of file [lterators.hpp](#).

5.14.4.3 exprp_

si2drExprT* ValuesIterator::exprp_

Definition at line 50 of file [lterators.hpp](#).

5.14.4.4 float_

si2drFloat64T ValuesIterator::float_

Definition at line 47 of file [lterators.hpp](#).

5.14.4.5 int_

si2drInt32T ValuesIterator::int_

Definition at line 46 of file [lterators.hpp](#).

5.14.4.6 str_

```
si2drStringT ValuesIterator::str_
```

Definition at line 48 of file [lterators.hpp](#).

5.14.4.7 values_

```
si2drValuesIdT ValuesIterator::values_
```

Definition at line 44 of file [lterators.hpp](#).

5.14.4.8 vtype_

```
si2drValueTypeT ValuesIterator::vtype_
```

Definition at line 45 of file [lterators.hpp](#).

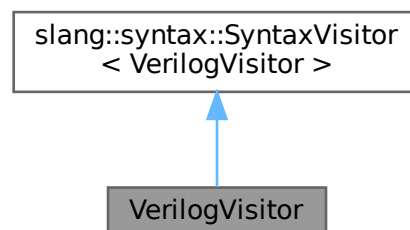
The documentation for this class was generated from the following files:

- [include/lterators.hpp](#)
- [src/lterators.cpp](#)

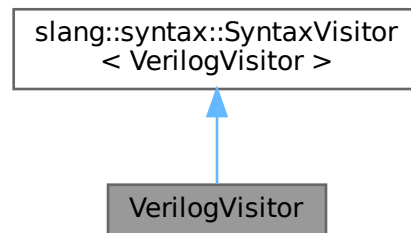
5.15 VerilogVisitor Class Reference

```
#include <verilog_utils.hpp>
```

Inheritance diagram for VerilogVisitor:



Collaboration diagram for VerilogVisitor:



Public Member Functions

- `VerilogVisitor` (const std::string &targetCell)
- void `handle` (const slang::syntax::SyntaxNode &node)
- void `handle` (const slang::syntax::ModuleDeclarationSyntax &module)
- void `handle` (const slang::syntax::PortDeclarationSyntax &portDecl)
- void `handle` (const slang::syntax::HierarchyInstantiationSyntax &hierarchyInst)
- void `handle` (const slang::syntax::SpecifyBlockSyntax &specifyBlock)

Private Attributes

- const std::string & `targetCell_`
- int `depth_`
- bool `inTargetModule_`

5.15.1 Detailed Description

Definition at line 15 of file [verilog_utils.hpp](#).

5.15.2 Constructor & Destructor Documentation

5.15.2.1 VerilogVisitor()

```
VerilogVisitor::VerilogVisitor (
    const std::string & targetCell ) [inline], [explicit]
```

Definition at line 17 of file [verilog_utils.hpp](#).

5.15.3 Member Function Documentation

5.15.3.1 handle() [1/5]

```
void VerilogVisitor::handle (
    const slang::syntax::HierarchyInstantiationSyntax & hierarchyInst )
```

Definition at line 78 of file [verilog_utils.cpp](#).

5.15.3.2 handle() [2/5]

```
void VerilogVisitor::handle (
    const slang::syntax::ModuleDeclarationSyntax & module )
```

Definition at line 17 of file [verilog_utils.cpp](#).

5.15.3.3 handle() [3/5]

```
void VerilogVisitor::handle (
    const slang::syntax::PortDeclarationSyntax & portDecl )
```

Definition at line 46 of file [verilog_utils.cpp](#).

5.15.3.4 `handle()` [4/5]

```
void VerilogVisitor::handle (  
    const slang::syntax::SpecifyBlockSyntax & specifyBlock )
```

Definition at line 258 of file [verilog_utils.cpp](#).

Here is the call graph for this function:



5.15.3.5 `handle()` [5/5]

```
void VerilogVisitor::handle (  
    const slang::syntax::SyntaxNode & node )
```

Definition at line 6 of file [verilog_utils.cpp](#).

Here is the caller graph for this function:



5.15.4 Member Data Documentation

5.15.4.1 depth_

```
int VerilogVisitor::depth_ [private]
```

Definition at line 28 of file [verilog_utils.hpp](#).

5.15.4.2 inTargetModule_

```
bool VerilogVisitor::inTargetModule_ [private]
```

Definition at line 29 of file [verilog_utils.hpp](#).

5.15.4.3 targetCell_

```
const std::string& VerilogVisitor::targetCell_ [private]
```

Definition at line 27 of file [verilog_utils.hpp](#).

The documentation for this class was generated from the following files:

- [include/verilog_utils.hpp](#)
- [src/verilog_utils.cpp](#)

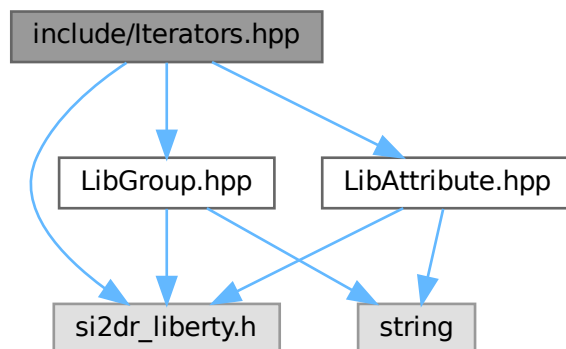
Chapter 6

File Documentation

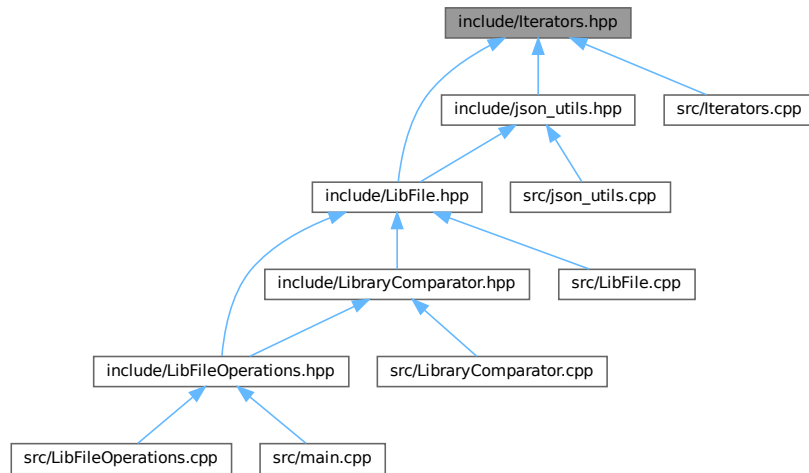
6.1 include/Iterators.hpp File Reference

```
#include "si2dr_liberty.h"  
#include "LibAttribute.hpp"  
#include "LibGroup.hpp"
```

Include dependency graph for Iterators.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class [GroupsIterator](#)
- class [AttributesIterator](#)
- class [ValuesIterator](#)

6.2 Iterators.hpp

[Go to the documentation of this file.](#)

```

00001 #ifndef ITERATORS_H
00002 #define ITERATORS_H
00003
00004 #include "si2dr_liberty.h"
00005
00006 #include "LibAttribute.hpp"
00007 #include "LibGroup.hpp"
00008
00009 class GroupsIterator {
00010 public:
00011     GroupsIterator(si2drGroupsIdT groups, si2drErrorT &err);
00012     ~GroupsIterator();
00013     void next();
00014     bool end();
00015     LibGroup get();
00016
00017 private:
00018     si2drGroupsIdT groups_;
00019     si2drGroupIdT group_;
00020     si2drErrorT &err_;
00021 };
00022
00023 class AttributesIterator {
00024 public:

```

```

00025  AttributesIterator(si2drAttrsIdT attrs, si2drErrorT &err);
00026  ~AttributesIterator();
00027  void next();
00028  bool end();
00029  LibAttribute get();
00030
00031 private:
00032  si2drAttrsIdT attrs_;
00033  si2drAttrIdT attr_;
00034  si2drErrorT &err_;
00035 };
00036
00037 class ValuesIterator {
00038 public:
00039  ValuesIterator(si2drValuesIdT values, si2drErrorT &err);
00040  ~ValuesIterator();
00041  void next();
00042  bool end();
00043
00044  si2drValuesIdT values_;
00045  si2drValueTypeT vtype_;
00046  si2drInt32T int_;
00047  si2drFloat64T float_;
00048  si2drStringT str_;
00049  si2drBooleanT bool_;
00050  si2drExprT *exprp_;
00051  si2drErrorT &err_;
00052 };
00053
00054 #endif // ITERATORS_H

```

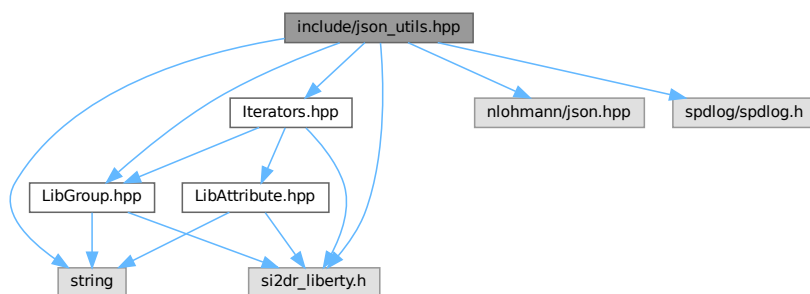
6.3 include/json_utils.hpp File Reference

```

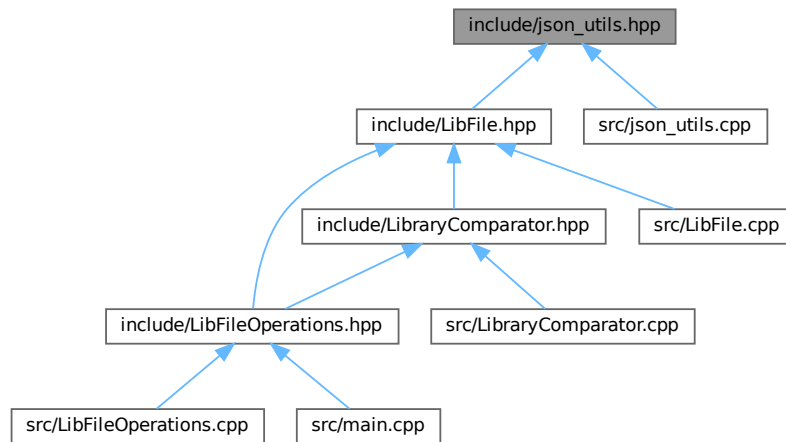
#include <string>
#include "nlohmann/json.hpp"
#include "si2dr_liberty.h"
#include "spdlog/spdlog.h"
#include "Iterators.hpp"
#include "LibGroup.hpp"

```

Include dependency graph for json_utils.hpp:



This graph shows which files directly or indirectly include this file:



Typedefs

- using `json` = `nlohmann::json`

Functions

- `json generateLutJson (LibGroup &lib_lut_group, si2drErrorT &err)`
Generates a JSON object representation of a look-up table (LUT) from a `LibGroup` object.
- `json generatePowerJson (LibGroup &lib_power_group, si2drErrorT &err)`
Converts a Liberty power group into a JSON representation.
- `std::pair< std::string, json > generatePinJson (LibGroup &lib_pin_group, si2drErrorT &err)`
Generates a JSON representation of a Liberty pin group.
- `json generateCellJson (LibGroup &lib_cell_group, si2drErrorT &err)`
Generates a JSON representation of a cell from a `LibGroup` object.

6.3.1 Typedef Documentation

6.3.1.1 json

```
using json = nlohmann::json
```

Definition at line 13 of file `json_utils.hpp`.

6.3.2 Function Documentation

6.3.2.1 generateCellJson()

```
json generateCellJson (
    LibGroup & lib_cell_group,
    si2drErrorT & err )
```

Generates a JSON representation of a cell from a [LibGroup](#) object.

This function processes a [LibGroup](#) representing a cell and converts it into a JSON object. It extracts the cell name and processes specific attributes such as area, cell_leakage_power, and cell_footprint. It also processes pins within the cell by categorizing them based on their direction (input, output, internal, inout).

Parameters

<i>lib_cell_group</i>	The LibGroup object representing the cell
<i>err</i>	Reference to a si2drErrorT object for error handling

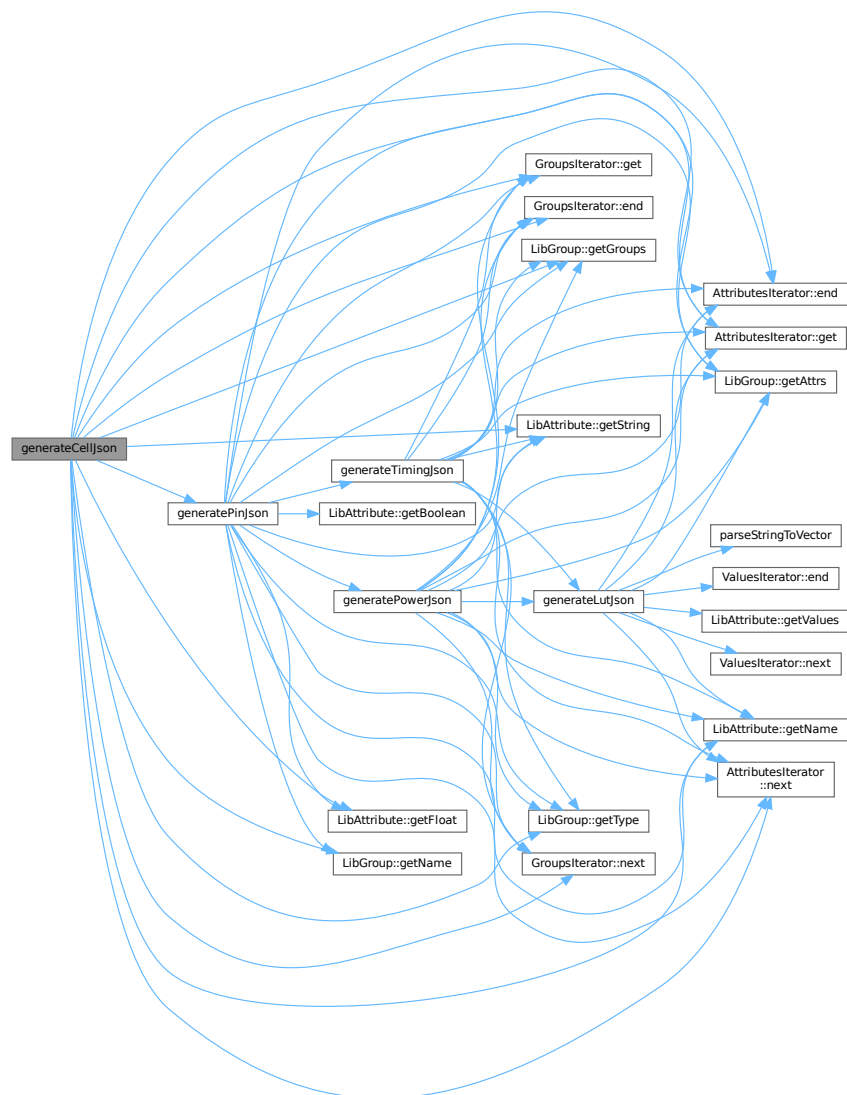
Returns

json A JSON object containing the cell's information with the following structure:

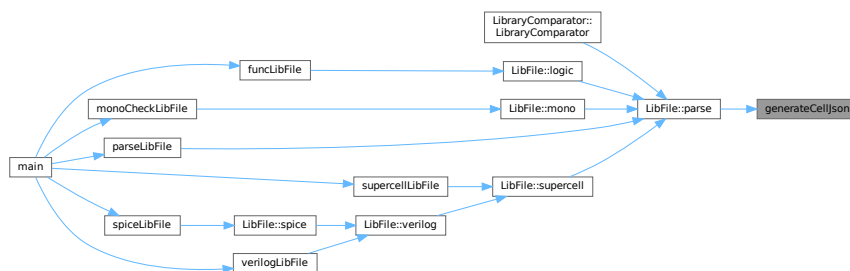
- "cell_name": string - name of the cell
- "area": float (optional) - cell area if present
- "cell_leakage_power": float (optional) - cell leakage power if present
- "cell_footprint": string (optional) - cell footprint if present
- "input_pins": array - list of input pins
- "output_pins": array - list of output pins
- "internal_pins": array - list of internal pins
- "inout_pins": array - list of inout pins

Definition at line 269 of file [json_utils.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



6.3.2.2 generateLutJson()

```
json generateLutJson (
    LibGroup & lib_lut_group,
    si2drErrorT & err )
```

Generates a JSON object representation of a look-up table (LUT) from a [LibGroup](#) object.

This function iterates through the attributes of the provided [LibGroup](#) object representing a LUT and converts them into a JSON structure. It specifically handles the following attributes:

- "index_1": Converted to a vector and stored in the JSON
- "index_2": Converted to a vector and stored in the JSON
- "values": Each value is parsed into a vector and added to an array in the JSON

Any other attributes encountered will trigger a warning message.

Parameters

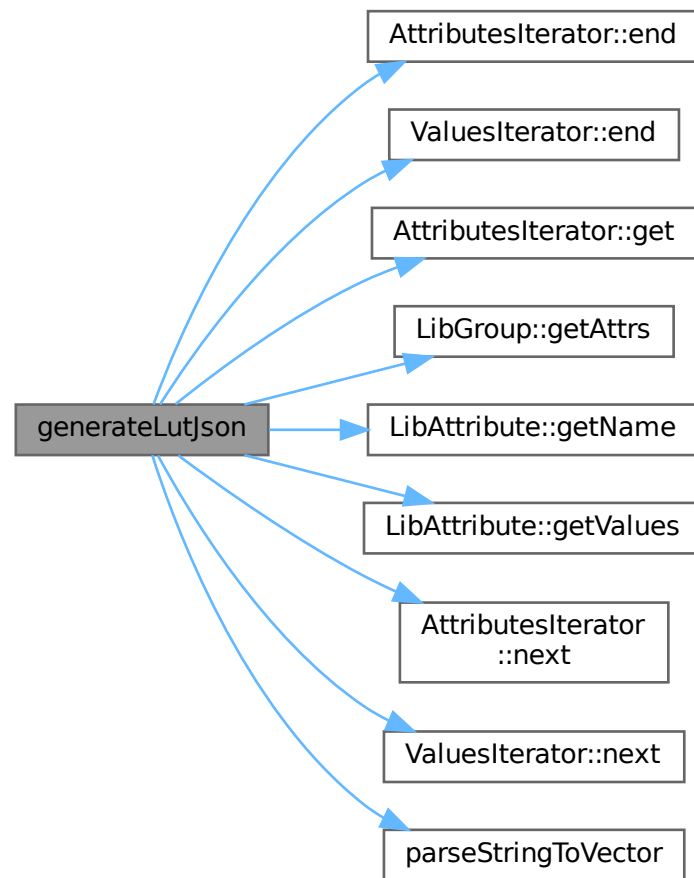
<i>lib_lut_group</i>	The LibGroup object containing the LUT data to be converted
<i>err</i>	Reference to an <code>si2drErrorT</code> object to track any errors during processing

Returns

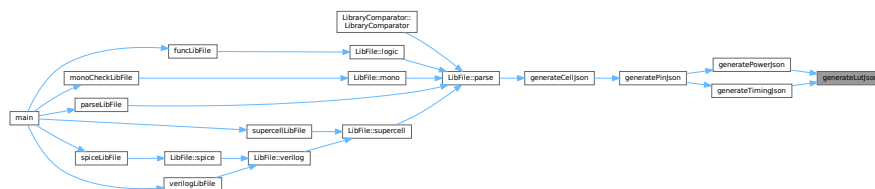
json A JSON object representing the LUT data with keys for "index_1", "index_2", and "values"

Definition at line 59 of file [json_utils.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



6.3.2.3 generatePinJson()

```
std::pair< std::string, json > generatePinJson (
```



```
LibGroup & lib_pin_group,  
si2drErrorT & err )
```

Generates a JSON representation of a Liberty pin group.

This function traverses a Liberty pin group, extracts relevant attributes and sub-groups, and converts them into a JSON object. It also determines the pin's direction.

Processes the following pin attributes:

- direction
- max_transition, capacitance, rise_capacitance, fall_capacitance, max_capacitance (float types)
- function, power_down_function, related_ground_pin, related_power_pin, three_state (string types)
- clock (boolean type)

Handles the following sub-groups:

- internal_power: Converted using [generatePowerJson\(\)](#)
- timing: Converted using [generateTimingJson\(\)](#)

Parameters

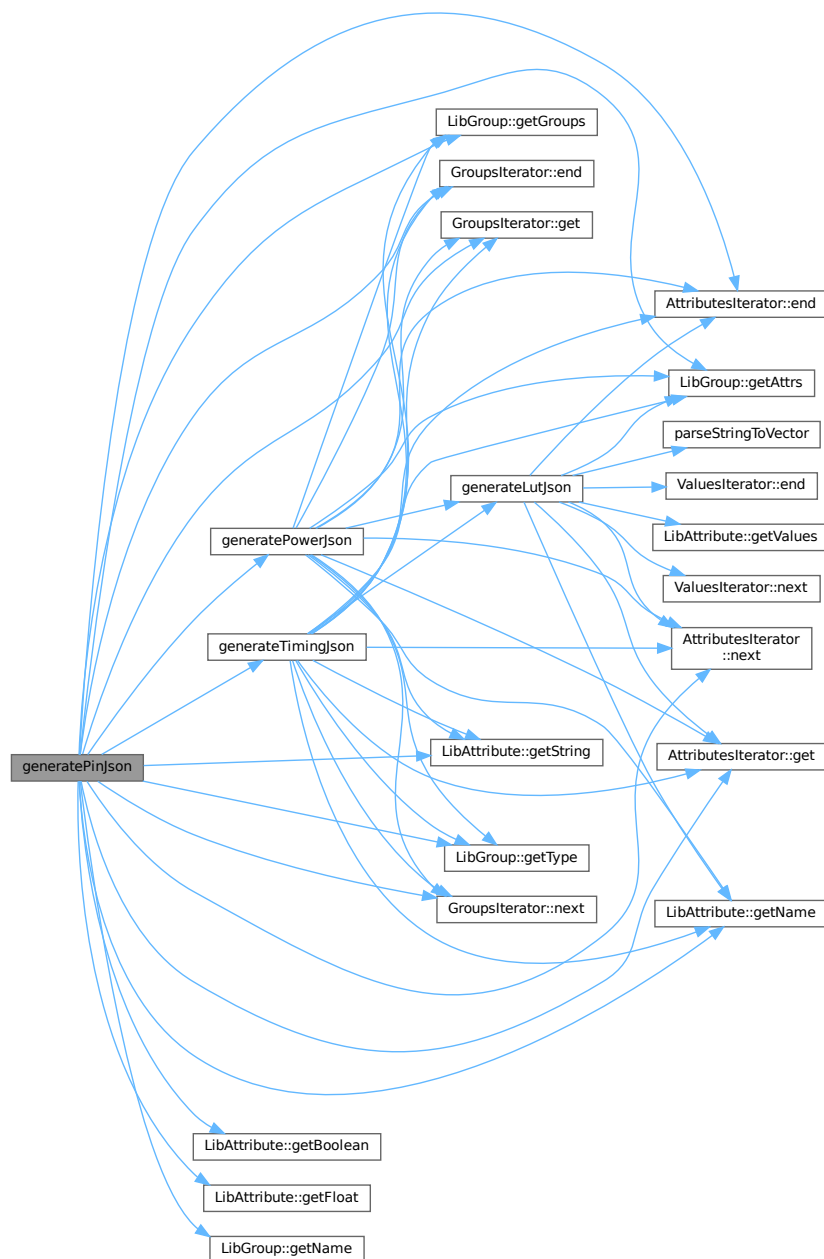
<i>lib_pin_group</i>	The Liberty pin group to process
<i>err</i>	Reference to an error object for tracking Liberty API errors

Returns

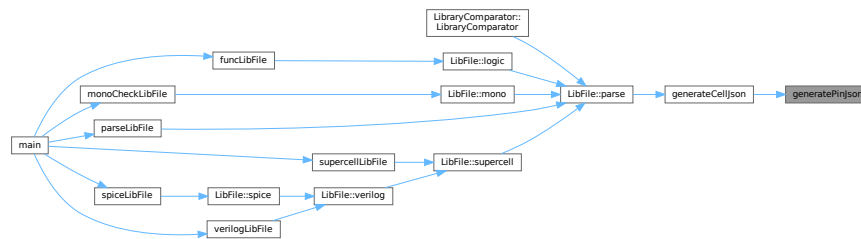
A pair containing the pin direction (string) and the JSON representation of the pin

Definition at line 203 of file [json_utils.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



6.3.2.4 generatePowerJson()

```

json generatePowerJson (
    LibGroup & lib_power_group,
    si2drErrorT & err )

```

Converts a Liberty power group into a JSON representation.

This function processes a Liberty power group and converts its attributes and subgroups into a JSON object. It handles attributes like "when", "related_pin", and "related_pg_pin", as well as "rise_power" and "fall_power" subgroups.

Parameters

<i>lib_power_group</i>	The Liberty power group to convert
<i>err</i>	Reference to an si2drErrorT object for error handling

Returns

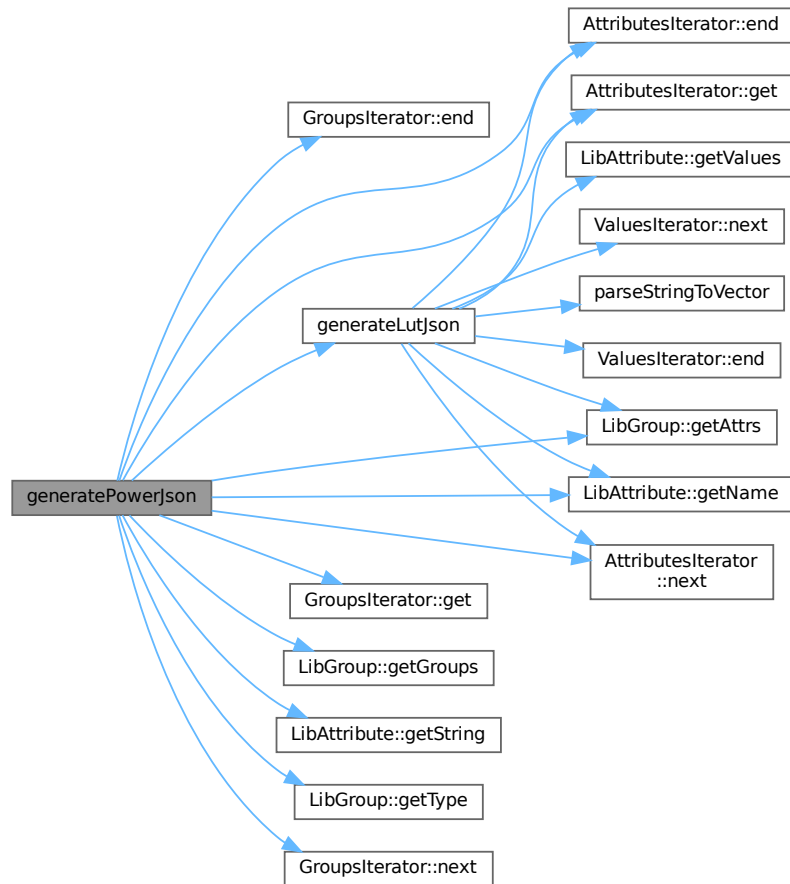
json A JSON object representing the power group data

The function:

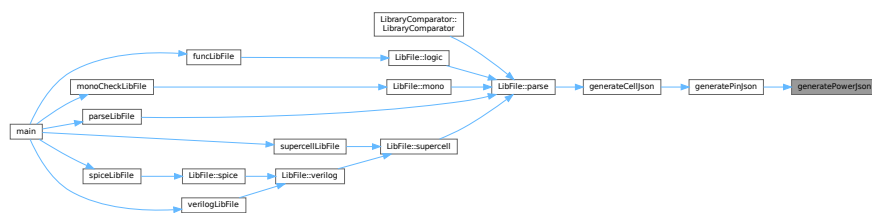
- Extracts string attributes (when, related_pin, related_pg_pin)
- Processes rise_power and fall_power subgroups by converting them to LUT JSON format
- Logs warnings for unknown power subgroup types

Definition at line 151 of file json_utils.cpp.

Here is the call graph for this function:



Here is the caller graph for this function:



6.4 json_utils.hpp

[Go to the documentation of this file.](#)

```

00001 #ifndef JSON_UTILS_HPP
00002 #define JSON_UTILS_HPP

```

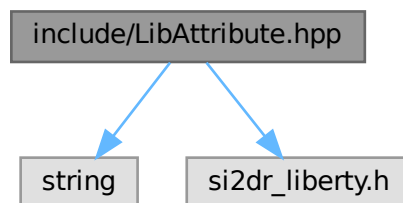
```
00003
00004 #include <string>
00005
00006 #include "nlohmann/json.hpp"
00007 #include "si2dr_liberty.h"
00008 #include "spdlog/spdlog.h"
00009
00010 #include "Iterators.hpp"
00011 #include "LibGroup.hpp"
00012
00013 using json = nlohmann::json;
00014
00015 json generateLutJson(LibGroup &lib_lut_group, si2drErrorT &err);
00016 json generatePowerJson(LibGroup &lib_power_group, si2drErrorT &err);
00017 std::pair<std::string, json> generatePinJson(LibGroup &lib_pin_group, si2drErrorT &err);
00018 json generateCellJson(LibGroup &lib_cell_group, si2drErrorT &err);
00019
00020 #endif // JSON_UTILS_HPP
```

6.5 include/LibAttribute.hpp File Reference

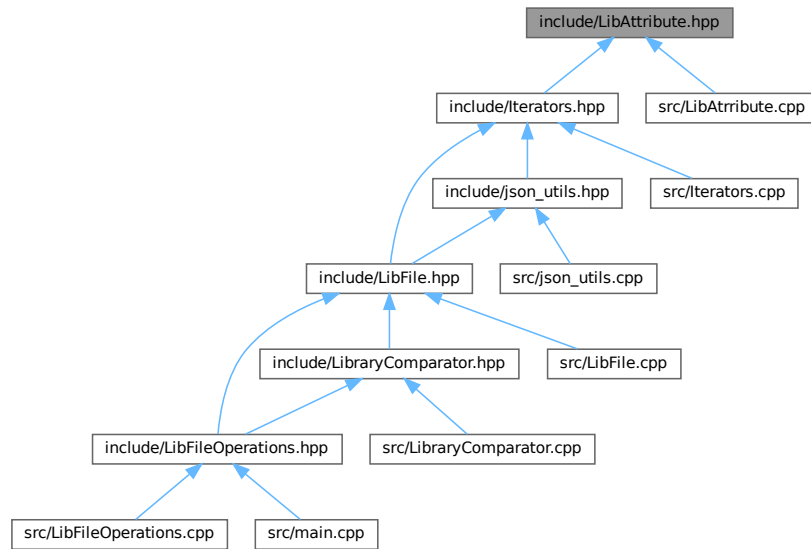
```
#include <string>
```

```
#include "si2dr_liberty.h"
```

Include dependency graph for LibAttribute.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class [LibAttribute](#)

6.6 LibAttribute.hpp

[Go to the documentation of this file.](#)

```

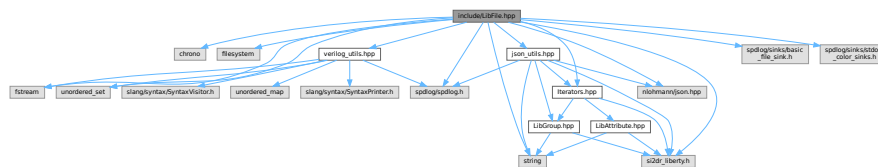
00001 #ifndef LIB_ATTRIBUTE_H
00002 #define LIB_ATTRIBUTE_H
00003
00004 #include <string>
00005
00006 #include "si2dr_liberty.h"
00007
00008 class LibAttribute {
00009 public:
00010   LibAttribute(si2drAttrIdT attr, si2drErrorT &err);
00011   ~LibAttribute();
00012   std::string getName();
00013   bool isComplex();
00014   si2drValuesIdT getValues();
00015   long int getInt();
00016   double getFloat();
00017   std::string getString();
00018   bool getBoolean();
00019
00020 private:
00021   si2drAttrIdT attr_;
00022   si2drErrorT &err_;
00023 };
00024
00025 #endif // LIB_ATTRIBUTE_H

```

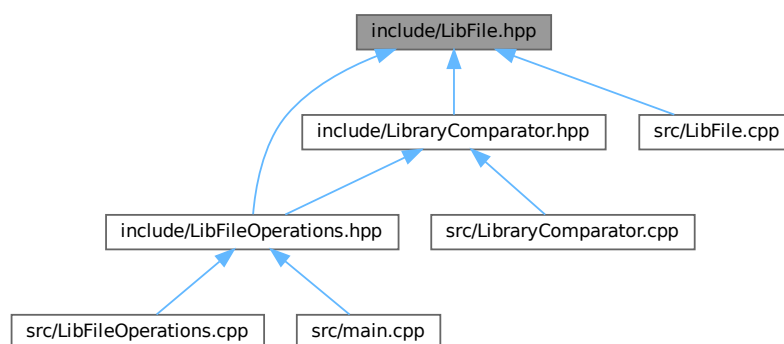
6.7 include/LibFile.hpp File Reference

```
#include <chrono>
#include <filesystem>
#include <fstream>
#include <string>
#include <unordered_set>
#include "nlohmann/json.hpp"
#include "si2dr_liberty.h"
#include "spdlog/sinks/basic_file_sink.h"
#include "spdlog/sinks/stdout_color_sinks.h"
#include "spdlog/spdlog.h"
#include "Iterators.hpp"
#include "json_utils.hpp"
#include "verilog_utils.hpp"
```

Include dependency graph for LibFile.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class [LibFile](#)

Typedefs

- using `json` = `nlohmann::json`

6.7.1 Typedef Documentation

6.7.1.1 json

using `json` = `nlohmann::json`

Definition at line 20 of file [LibFile.hpp](#).

6.8 LibFile.hpp

[Go to the documentation of this file.](#)

```

00001 #ifndef LIB_FILE_H
00002 #define LIB_FILE_H
00003
00004 #include <chrono>
00005 #include <filesystem>
00006 #include <fstream>
00007 #include <string>
00008 #include <unordered_set>
00009
00010 #include "nlohmann/json.hpp"
00011 #include "si2dr_liberty.h"
00012 #include "spdlog/sinks/basic_file_sink.h"
00013 #include "spdlog/sinks/stdout_color_sinks.h"
00014 #include "spdlog/spdlog.h"
00015
00016 #include "Iterators.hpp"
00017 #include "json_utils.hpp"
00018 #include "verilog_utils.hpp"
00019
00020 using json = nlohmann::json;
00021
00022 class LibFile {
00023 public:
00024     LibFile(const std::string &filepath, const std::string &loggername);
00025     ~LibFile();
00026     std::shared_ptr<spdlog::logger> logger_;
00027     std::filesystem::path filepath_; // full path to the file
00028     std::string basename_;          // file name without extension
00029     std::string filename_;          // full file name with extension
00030     std::string libname_ = "";      // library name obtained through parsing
00031     std::string jsonname_ = "";     // json file name to store parsed data
00032     std::string loggername_ = "";   // log file name
00033     json lib_json_ = json::object();
00034     void writeJsonToFile();
00035     void parse();
00036     void modify();
00037     void mono(const bool is_slew);
00038     void supercell(const int chain_length, const std::vector<std::string> &cell_names);

```



```

00039 void verilog(const int chain_length, const std::vector<std::string> &cell_names);
00040 void spice(const int chain_length, const std::vector<std::string> &cell_names,
00041           const std::string &verilog_lib_file, const std::string &spice_lib_file);
00042 std::map<std::string, std::string> logic(const std::string &cell_name);
00043
00044 private:
00045     si2drErrorT err_;
00046     int process_ = 0;
00047     float voltage_ = 0.0;
00048     int temperature_ = 0;
00049     void read();
00050     bool checkTimingArcMonotonicity(const json &cell, const json &pin, const json &arc,
00051                                    const std::string &type, bool is_slew);
00052 };
00053
00054 #endif // LIB_FILE_H

```

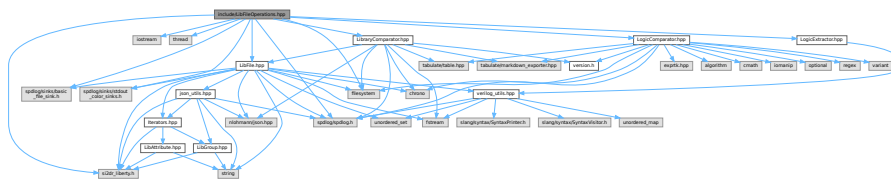
6.9 include/LibFileOperations.hpp File Reference

```

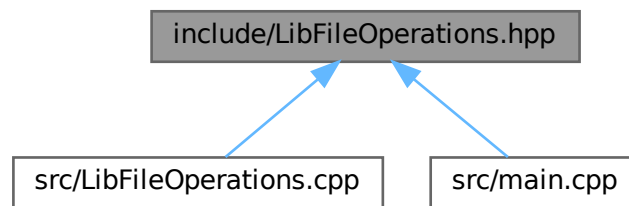
#include <filesystem>
#include <iostream>
#include <thread>
#include "si2dr_liberty.h"
#include "spdlog/sinks/basic_file_sink.h"
#include "spdlog/sinks/stdout_color_sinks.h"
#include "spdlog/spdlog.h"
#include "LibFile.hpp"
#include "LibraryComparator.hpp"
#include "LogicComparator.hpp"
#include "LogicExtractor.hpp"

```

Include dependency graph for LibFileOperations.hpp:



This graph shows which files directly or indirectly include this file:



Functions

- void [printInfo](#) ()
Sets up and configures the global logger and prints application information.
- void [parseLibFile](#) (const std::string &library_path, const std::string log_file_name)
Parses a library file and generates corresponding output.
- void [monoCheckLibFile](#) (const std::string &library_path, const std::string log_file_name, bool is↵_slew)
Performs monotonicity check on a library file.
- void [supercellLibFile](#) (const std::string &library_path, const std::string &log_file_name, int chain↵_length, const std::vector< std::string > &cell_names)
Creates supercell map structures from a Liberty library file.
- void [verilogLibFile](#) (const std::string &library_path, const std::string &log_file_name, int chain↵_length, const std::vector< std::string > &cell_names)
Generates Verilog files from a library file for specified cells.
- void [spiceLibFile](#) (const std::string &library_path, const std::string &log_file_name, int chain↵_length, const std::vector< std::string > &cell_names, const std::string &verilog_lib_file, const std::string &spice_lib_file)
- void [compareLibFiles](#) (const std::string &ref_lib, const std::string &comp_lib, const double reltol, const double abstol, std::string &report_file_name)
Compares two library files and generates a detailed comparison report.
- void [funcLibFile](#) (const std::string &ref_file, const std::string &comp_file, const std::vector< std↵::string > &cell_names, std::string &report_file_name)

6.9.1 Function Documentation

6.9.1.1 compareLibFiles()

```
void compareLibFiles (
    const std::string & ref_lib,
    const std::string & comp_lib,
    const double reltol,
    const double abstol,
    std::string & report_file_name )
```

Compares two library files and generates a detailed comparison report.

This function compares a reference library file with another library file, performing validation checks based on specified tolerance parameters. The comparison results are written to a report file in markdown or text format.

Parameters

<i>ref_lib</i>	Path to the reference library file to use as the baseline
<i>comp_lib</i>	Path to the library file to compare against the reference
<i>reltol</i>	Relative tolerance for numerical comparisons (must be ≥ 0.0)
<i>abstol</i>	Absolute tolerance for numerical comparisons
<i>report_file_name</i>	[in,out] Name of the file to write the comparison report to. If empty, defaults to [comp_lib_basename].cmp.md. If provided but doesn't end with .txt or .md, .md will be appended.

Note

The function will log an error and return without comparing if reltol is invalid.

Log files will be created with the naming pattern [library_basename].cmp.log

The function uses the [LibraryComparator](#) class to perform the actual comparison.

Definition at line 231 of file [LibFileOperations.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:

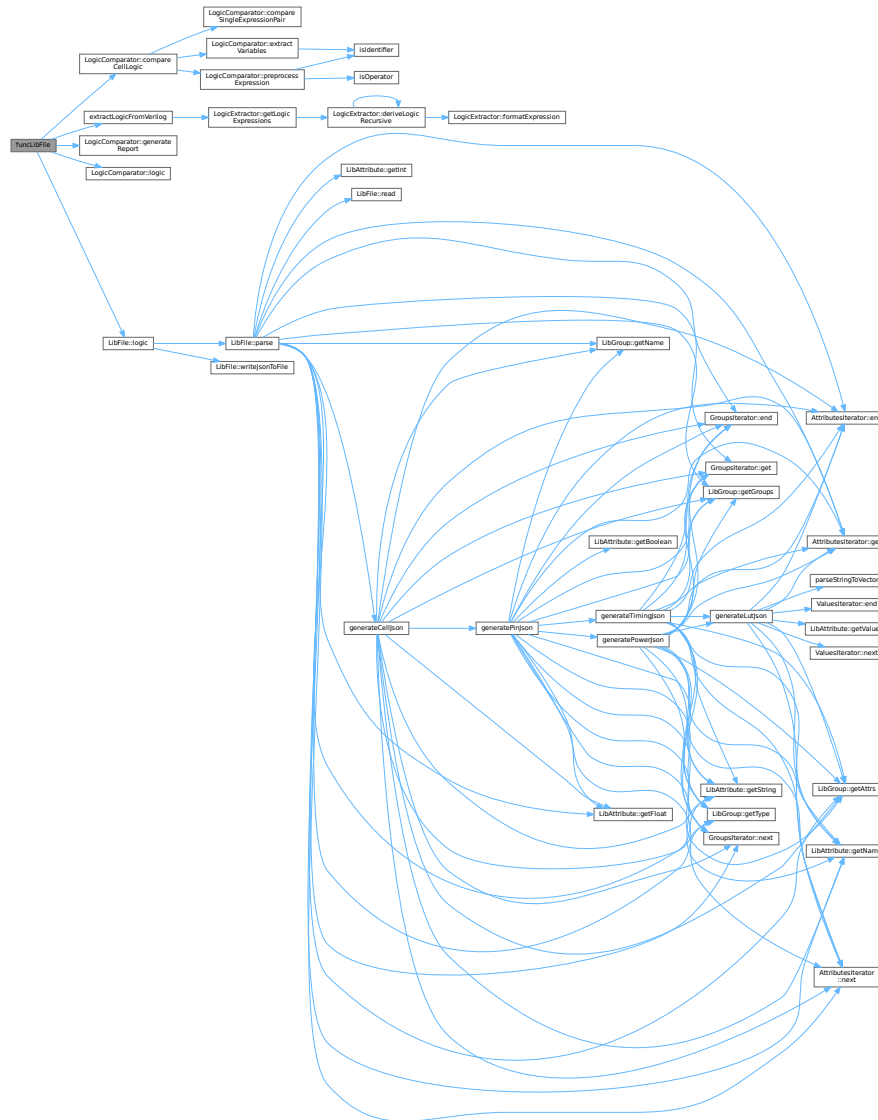


6.9.1.2 funcLibFile()

```
void funcLibFile (
    const std::string & ref_file,
    const std::string & comp_file,
    const std::vector< std::string > & cell_names,
    std::string & report_file_name )
```

Definition at line 259 of file [LibFileOperations.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



6.9.1.3 monoCheckLibFile()

```
void monoCheckLibFile (  
    const std::string & library_path,  
    const std::string log_file_name,  
    bool is_slew )
```

Performs monotonicity check on a library file.

This function validates the monotonicity of timing data in a library file. It creates a log file to record the results of the check and handles any exceptions that occur during the process.

Parameters

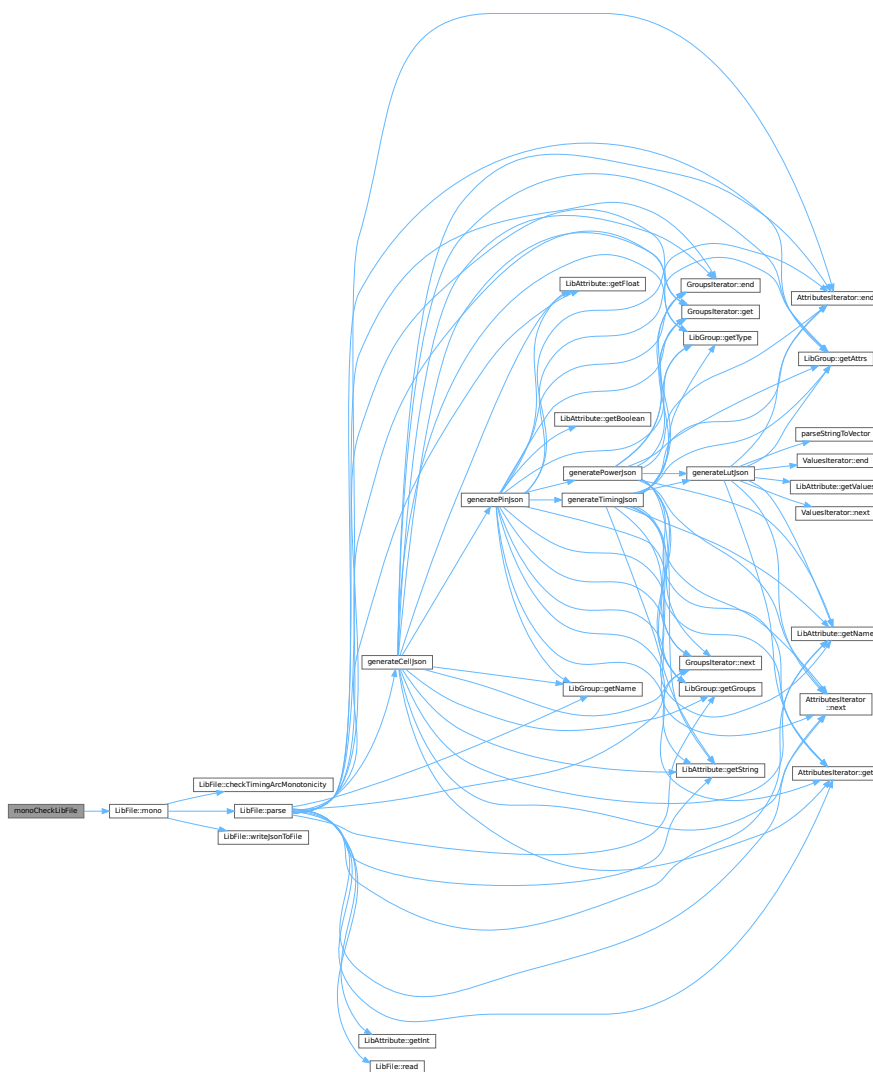
<i>library_path</i>	Path to the library file to check
<i>log_file_name</i>	Name of the log file to create (optional). If empty, a default name will be generated from the library file name
<i>is_slew</i>	Flag indicating whether to check input slew monotonicity (true) or output load monotonicity (false)

Exceptions

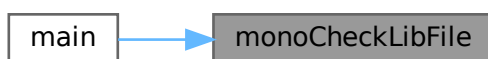
<i>The</i>	function catches and logs any exceptions but does not rethrow them
------------	--

Definition at line 81 of file [LibFileOperations.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



6.9.1.4 parseLibFile()

```
void parseLibFile (  
    const std::string & library_path,  
    const std::string log_file_name )
```

Parses a library file and generates corresponding output.

This function processes the given library file, parsing its contents and generating a JSON output. It also logs the parsing process to a specified log file or creates a default log file if none is provided.

Parameters

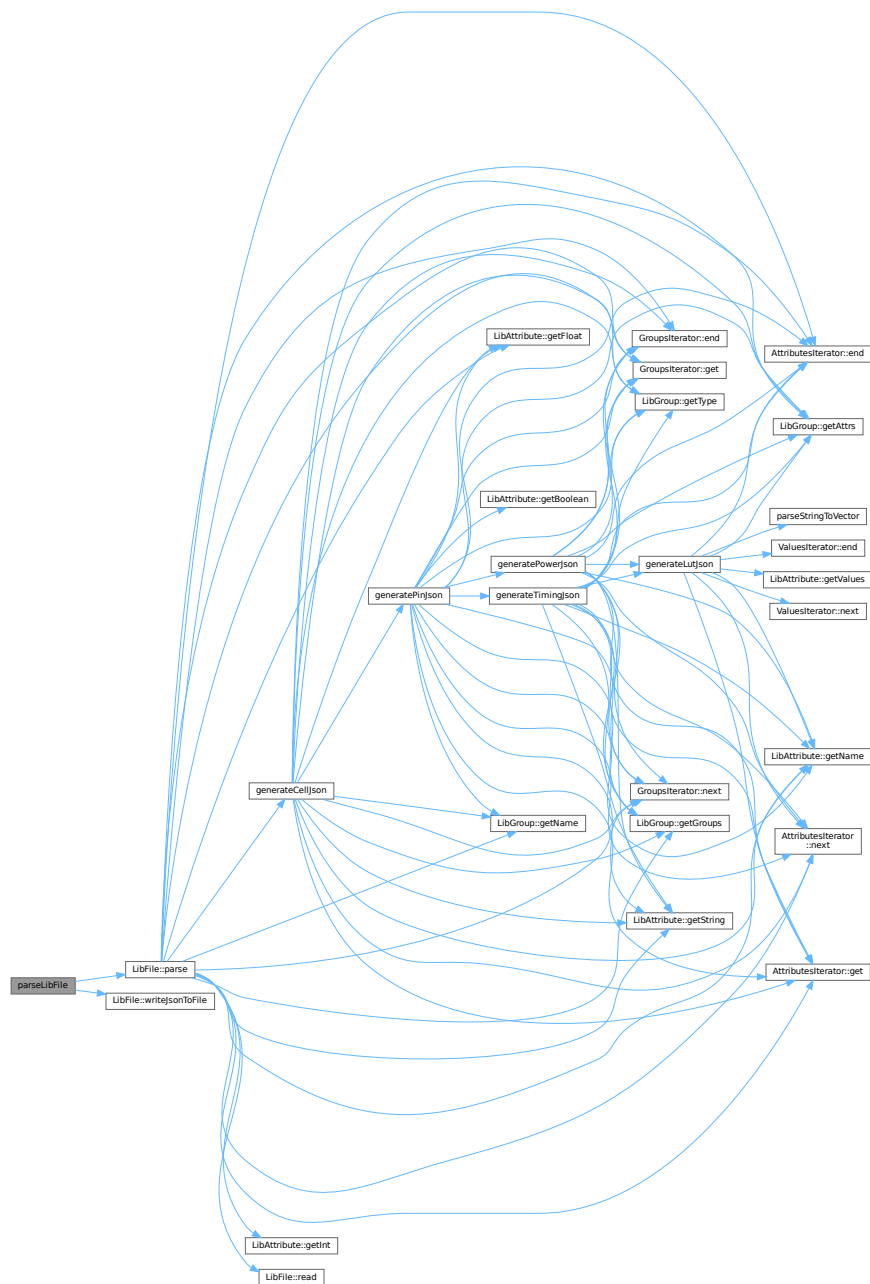
<i>library_path</i>	Path to the library file that needs to be parsed
<i>log_file_name</i>	Optional name for the log file. If empty, a default name is generated based on the library filename with ".parse.log" extension

Exceptions

<i>The</i>	function catches and logs any exceptions but doesn't propagate them
------------	---

Definition at line 49 of file [LibFileOperations.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



6.9.1.5 printInfo()

```
void printInfo ( )
```

Sets up and configures the global logger and prints application information.

This function performs the following operations:

1. Creates a console sink for logging with level set to INFO
2. Creates a file sink for logging with level set to DEBUG, saving to [APP_NAME].log
3. Configures a logger with both sinks and sets it as the default logger
4. Outputs basic application information:
 - Version and build timestamp
 - Author information
 - Log file location

Note

Uses spdlog library for logging functionality

Depends on APP_NAME, APP_VERSION, BUILD_TIMESTAMP, APP_AUTHOR, and APP↵_CONTACT macros

Definition at line 18 of file [LibFileOperations.cpp](#).

Here is the caller graph for this function:



```
void spiceLibFile (

    const std::string & library_path,

    const std::string & log_file_name,

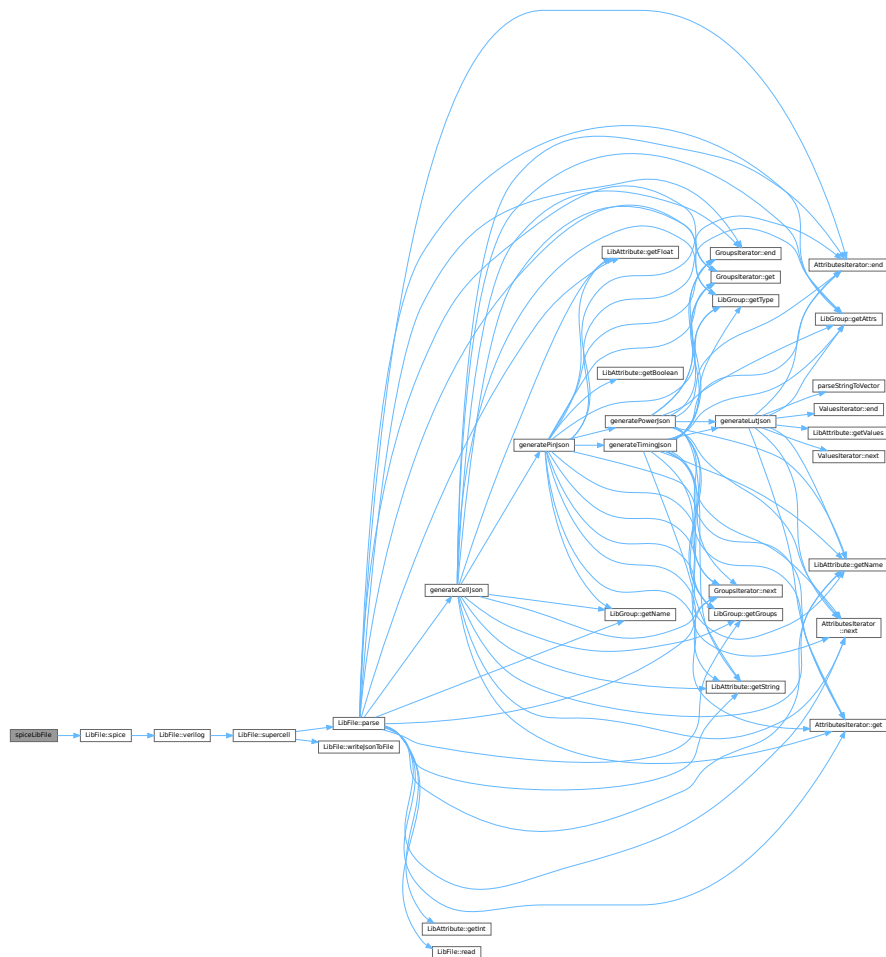
    int chain_length,

    const std::vector< std::string > & cell_names,

    const std::string & verilog_lib_file,

    const std::string & spice_lib_file )
```

Here is the call graph for this function:



Here is the caller graph for this function:



6.9.1.7 supercellLibFile()

```

void supercellLibFile (
    const std::string & library_path,
    const std::string & log_file_name,
    int chain_length,
    const std::vector< std::string > & cell_names )
  
```

Creates supercell map structures from a Liberty library file.

This function reads a Liberty file, creates supercell structures based on the specified chain length and cell names, and logs the process to a file.

Parameters

<i>library_path</i>	Path to the Liberty file to process
<i>log_file_name</i>	Name of the log file (if empty, defaults to "[library_name].supercell.log")
<i>chain_length</i>	The length of chains to create (must be ≥ 1)
<i>cell_names</i>	Vector of cell names to process for supercell generation

Exceptions

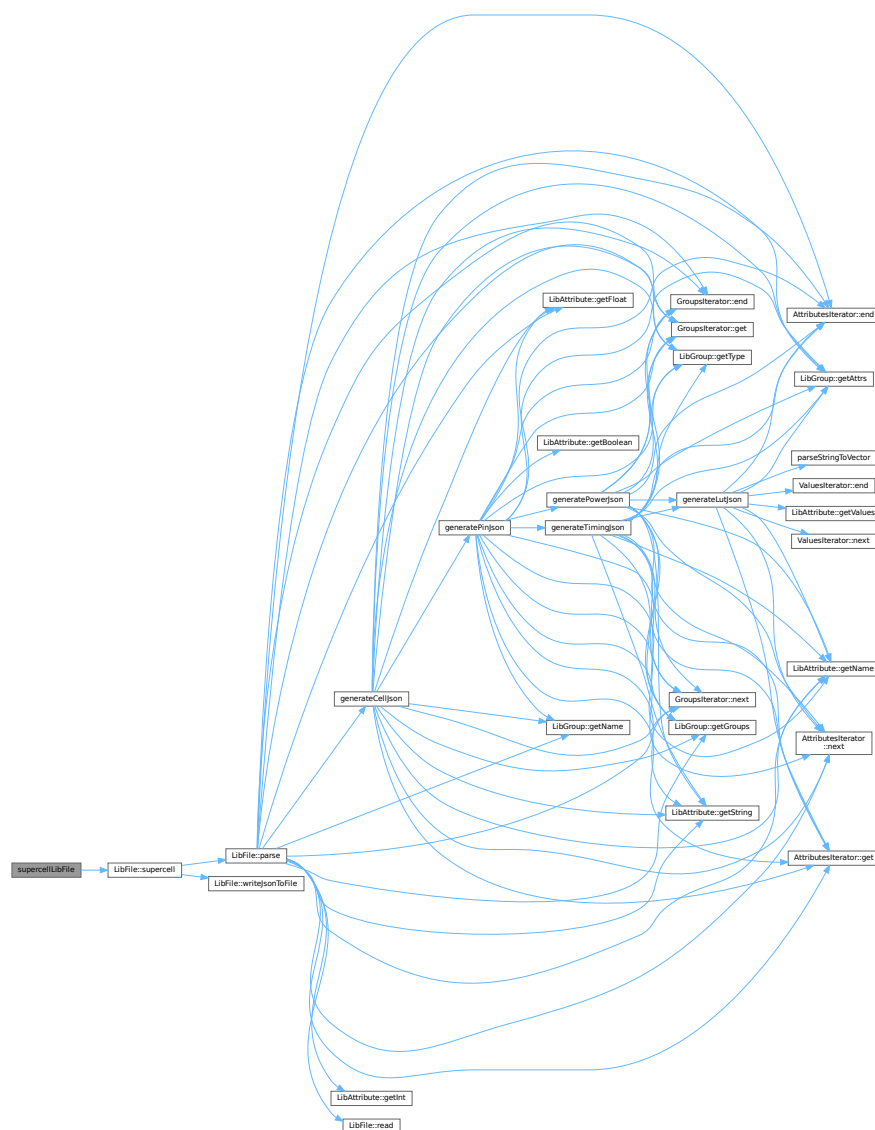
<i>May</i>	pass through exceptions from the LibFile::supercell method
------------	--

Note

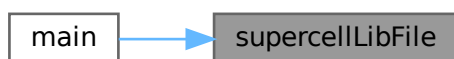
The function validates the chain length and logs all activities including errors that might occur during processing

Definition at line 116 of file [LibFileOperations.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



6.9.1.8 verilogLibFile()

```
void verilogLibFile (
    const std::string & library_path,
    const std::string & log_file_name,
    int chain_length,
    const std::vector< std::string > & cell_names )
```

Generates Verilog files from a library file for specified cells.

This function processes a library file and generates Verilog representation for the specified cell names with a given chain length. The operation results are logged to a specified or default log file.

Parameters

<i>library_path</i>	Path to the library file to process
<i>log_file_name</i>	Name for the log file (if empty, a default name will be generated)
<i>chain_length</i>	Number of cells to chain together, must be ≥ 1
<i>cell_names</i>	Vector of cell names to generate Verilog for

Exceptions

<i>Catches</i>	any exceptions from the verilog generation process and logs them
----------------	--

Definition at line 157 of file [LibFileOperations.cpp](#).


```

00001 #ifndef LIBFILEOPERATIONS_H
00002 #define LIBFILEOPERATIONS_H
00003
00004 #include <filesystem>
00005 #include <iostream>
00006 #include <thread>
00007
00008 #include "si2dr_liberty.h"
00009 #include "spdlog/sinks/basic_file_sink.h"
00010 #include "spdlog/sinks/stdout_color_sinks.h"
00011 #include "spdlog/spdlog.h"
00012
00013 #include "LibFile.hpp"
00014 #include "LibraryComparator.hpp"
00015 #include "LogicComparator.hpp"
00016 #include "LogicExtractor.hpp"
00017
00018 void printInfo();
00019 void parseLibFile(const std::string &library_path, const std::string log_file_name);
00020 void monoCheckLibFile(const std::string &library_path, const std::string log_file_name,
00021                      bool is_slew);
00022 void supercellLibFile(const std::string &library_path, const std::string &log_file_name,
00023                      int chain_length, const std::vector<std::string> &cell_names);
00024 void verilogLibFile(const std::string &library_path, const std::string &log_file_name,
00025                    int chain_length, const std::vector<std::string> &cell_names);
00026 void spiceLibFile(const std::string &library_path, const std::string &log_file_name,
00027                  int chain_length, const std::vector<std::string> &cell_names,
00028                  const std::string &verilog_lib_file, const std::string &spice_lib_file);
00029 void compareLibFiles(const std::string &ref_lib, const std::string &comp_lib, const double reltol,
00030                     const double abstol, std::string &report_file_name);
00031 void funcLibFile(const std::string &ref_file, const std::string &comp_file,
00032                 const std::vector<std::string> &cell_names, std::string &report_file_name);
00033
00034 #endif // LIBFILEOPERATIONS_H

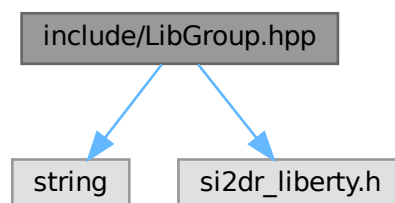
```

6.11 include/LibGroup.hpp File Reference

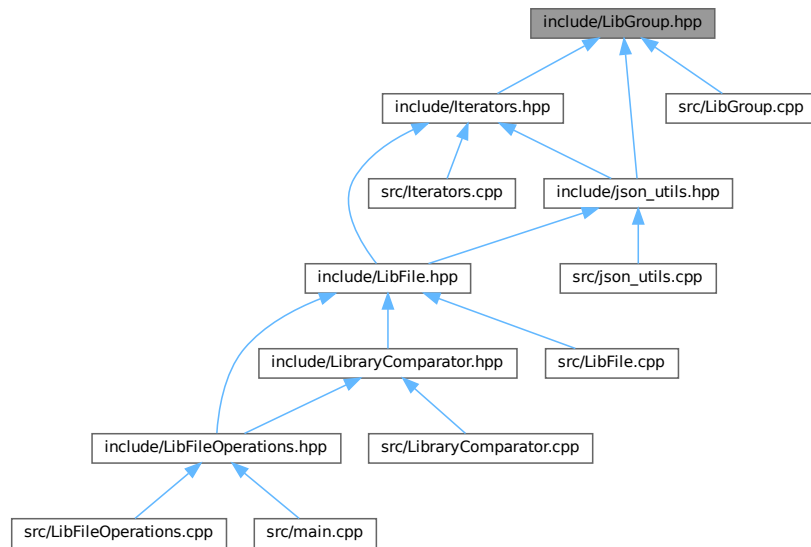
```
#include <string>
```

```
#include "si2dr_liberty.h"
```

Include dependency graph for LibGroup.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class [LibGroup](#)

6.12 LibGroup.hpp

[Go to the documentation of this file.](#)

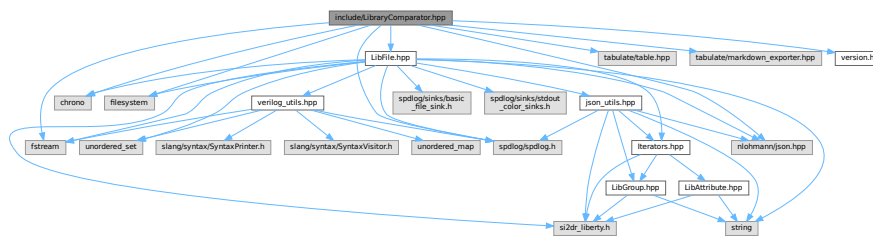
```

00001 #ifndef LIB_GROUP_H
00002 #define LIB_GROUP_H
00003
00004 #include <string>
00005
00006 #include "si2dr_liberty.h"
00007
00008 class LibGroup {
00009 public:
00010     LibGroup(si2drGroupIdT group, si2drErrorT &err);
00011     ~LibGroup();
00012     std::string getName();
00013     std::string getType();
00014     si2drAttrsIdT getAttrs();
00015     si2drGroupsIdT getGroups();
00016
00017 private:
00018     si2drGroupIdT group_;
00019     si2drErrorT &err_;
00020 };
00021
00022 #endif // LIB_GROUP_H
  
```

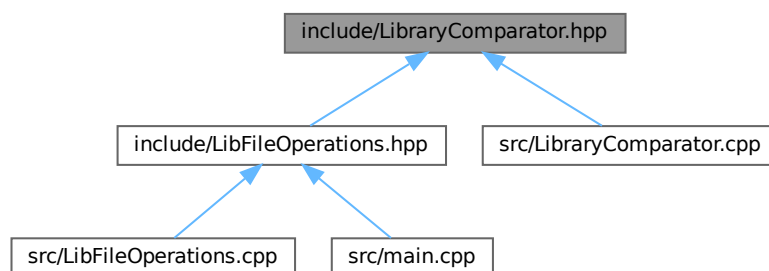
6.13 include/LibraryComparator.hpp File Reference

```
#include <chrono>
#include <filesystem>
#include <fstream>
#include "nlohmann/json.hpp"
#include "spdlog/spdlog.h"
#include "tabulate/table.hpp"
#include <tabulate/markdown_exporter.hpp>
#include "LibFile.hpp"
#include "version.h"
```

Include dependency graph for LibraryComparator.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class [LibraryComparator](#)

Typedefs

- using [json](#) = `nlohmann::json`

6.13.1 Typedef Documentation

6.13.1.1 json

using `json` = `nlohmann::json`

Definition at line 16 of file [LibraryComparator.hpp](#).

6.14 LibraryComparator.hpp

[Go to the documentation of this file.](#)

```
00001 #ifndef COMPARE_HPP
00002 #define COMPARE_HPP
00003
00004 #include <chrono>
00005 #include <filesystem>
00006 #include <fstream>
00007
00008 #include "nlohmann/json.hpp"
00009 #include "spdlog/spdlog.h"
00010 #include "tabulate/table.hpp"
00011 #include <tabulate/markdown_exporter.hpp>
00012
00013 #include "LibFile.hpp"
00014 #include "version.h"
00015
00016 using json = nlohmann::json;
00017 using namespace tabulate;
00018
00019 class LibraryComparator {
00020 public:
00021     LibraryComparator(LibFile &ref_libfile, LibFile &comp_libfile, double reltol, double abstol);
00022     std::filesystem::path ref_lib_path_;
00023     std::filesystem::path comp_lib_path_;
00024     void generateReport(const std::string &output_file);
00025
00026 private:
00027     json ref_json_;
00028     json comp_json_;
00029     double reltol_;
00030     double abstol_;
00031
00032     void compareCell(const std::string &cell_name, const json &ref_cell, const json &comp_cell,
00033                     Table &table);
00034     void comparePin(const std::string &cell_name, const std::string &pin_name, const json &ref_pin,
00035                    const json &comp_pin, Table &table);
00036     void compareTimingArc(const std::string &cell_name, const std::string &pin_name,
00037                           const std::string &timing_type, const json &ref_timing_arc,
00038                           const json &comp_timing_arc, Table &table);
00039     void compareLut(const std::string &cell_name, const std::string &pin_name,
00040                     const std::string &timing_type, const std::string &related_pin,
00041                     const std::string &arc_name, const json &ref_lut, const json &comp_lut,
00042                     Table &table);
00043     // void configureTableFormat(Table &table);
00044 };
00045
00046 #endif // COMPARE_HPP
```

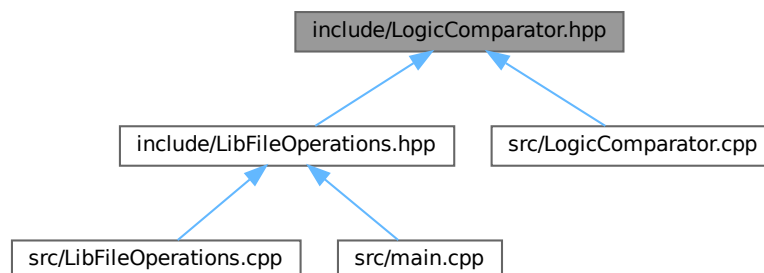
6.15 include/LogicComparator.hpp File Reference

```
#include <algorithm>
#include <chrono>
#include <cmath>
#include <iomanip>
#include <optional>
#include <regex>
#include <variant>
#include <filesystem>
#include "exprtk.hpp"
#include "tabulate/markdown_exporter.hpp"
#include "tabulate/table.hpp"
#include <spdlog/spdlog.h>
#include "version.h"
```

Include dependency graph for LogicComparator.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- struct [PinComparisonResult](#)
- class [LogicComparator](#)

6.16 LogicComparator.hpp

[Go to the documentation of this file.](#)

```

00001 #ifndef LOGIC_COMPARATOR_HPP
00002 #define LOGIC_COMPARATOR_HPP
00003
00004 #include <algorithm>
00005 #include <chrono>
00006 #include <cmath>
00007 #include <iomanip>
00008 #include <optional> // To store tables optionally
00009 #include <regex>    // For regular expressions
00010 #include <variant>
00011 #include <filesystem>
00012
00013 #include "exptrk.hpp"
00014 #include "tabulate/markdown_exporter.hpp"
00015 #include "tabulate/table.hpp"
00016 #include <spdlog/spdlog.h>
00017
00018 #include "version.h"
00019
00020 using namespace tabulate;
00021
00022 // Structure to hold results for a single pin comparison
00023 struct PinComparisonResult {
00024     std::string pin_name;
00025     std::string ref_expr_raw;
00026     std::string comp_expr_raw;
00027     std::string ref_expr_processed;
00028     std::string comp_expr_processed;
00029     bool comparison_possible = false; // Was comparison attempted?
00030     bool are_equivalent = false;
00031     bool ref_compiles = false;
00032     bool comp_compiles = false;
00033     std::optional<Table> ref_truth_table; // Store tables only if needed/successful
00034     std::optional<Table> comp_truth_table;
00035     std::string error_message; // Store any error during comparison
00036 };
00037
00038 class LogicComparator {
00039 public:
00040     LogicComparator(const std::map<std::string, std::string> &ref_outpin_map,
00041                    const std::map<std::string, std::string> &comp_outpin_map,
00042                    const std::string &cell_name);
00043
00044     // Example code from exptrk documentation
00045     void logic();
00046
00047     // Preprocessing function
00048     std::string preprocessExpression(const std::string &input_expr);
00049
00050     // Helper to extract unique sorted variables from TWO expressions
00051     bool extractVariables(const std::string &expr1, const std::string &expr2,
00052                          std::vector<std::string> &sorted_vars);
00053
00054     void compareSingleExpressionPair(const std::string &ref_expression_processed,
00055                                     const std::string &comp_expression_processed,
00056                                     const std::vector<std::string> &sorted_vars,
00057                                     PinComparisonResult &result); // Pass result struct
00058
00059     void compareCellLogic();
00060
00061     void generateReport(const std::string &output_file);
00062
00063 private:
00064     std::map<std::string, std::string> ref_outpin_map_;

```

```

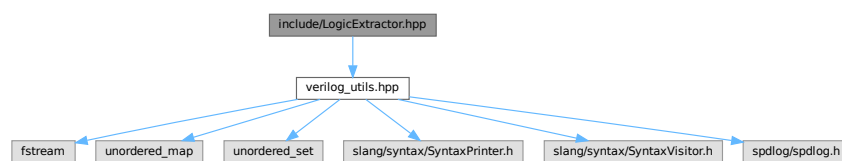
00084  std::map<std::string, std::string> comp_outpin_map_;
00085  std::string cell_name_;
00086  std::map<std::string, PinComparisonResult> all_pin_results_;
00087  };
00088
00089  #endif // LOGIC_COMPARATOR_HPP

```

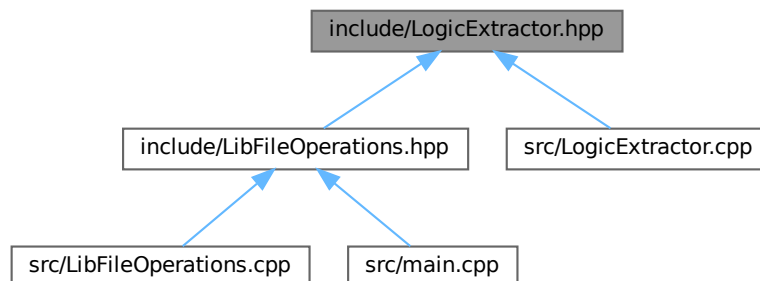
6.17 include/LogicExtractor.hpp File Reference

#include "verilog_utils.hpp"

Include dependency graph for LogicExtractor.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- struct [GateInfo](#)
- class [LogicExtractor](#)

Functions

- void [extractAndPrintNetlistInfo](#) (const std::string &verilog_file, const std::string &cell)
- std::map< std::string, std::string > [extractLogicFromVerilog](#) (const std::string &verilog_file, const std::string &cell)

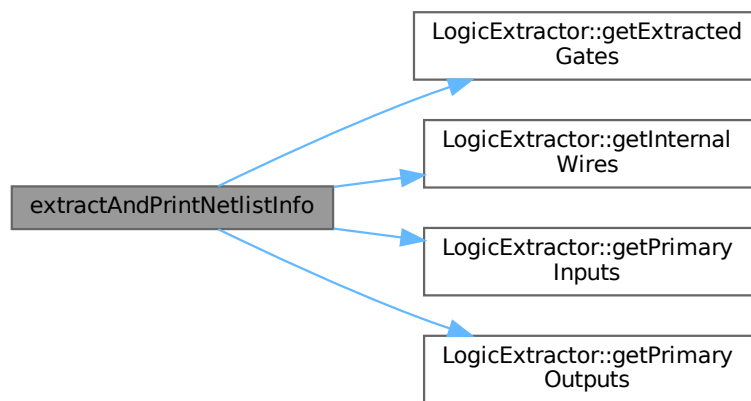
6.17.1 Function Documentation

6.17.1.1 extractAndPrintNetlistInfo()

```
void extractAndPrintNetlistInfo (
    const std::string & verilog_file,
    const std::string & cell )
```

Definition at line 514 of file [LogicExtractor.cpp](#).

Here is the call graph for this function:



6.17.1.2 extractLogicFromVerilog()

```
std::map< std::string, std::string > extractLogicFromVerilog (
    const std::string & verilog_file,
    const std::string & cell )
```

Definition at line 575 of file [LogicExtractor.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



6.18 LogicExtractor.hpp

[Go to the documentation of this file.](#)

```

00001 // include/LogicExtractor.hpp
00002
00003 #ifndef LOGIC_EXTRACTOR_HPP
00004 #define LOGIC_EXTRACTOR_HPP
00005
00006 #include "verilog_utils.hpp"
00007
00008 // Structure to represent a gate instance's information
00009 struct GateInfo {
00010     slang::parsing::TokenKind kind; // Store the TokenKind for reliable checks
00011     std::string gateTypeName;       // Store the string name like "and", "xor"
00012     std::vector<std::string> inputSignals;
00013     std::string outputSignal;
00014 };
00015
00016 // Visitor class to extract netlist information and derive logic expressions
00017 class LogicExtractor : public slang::syntax::SyntaxVisitor<LogicExtractor> {
00018 public:
00019     // Constructor takes the target cell name
00020     explicit LogicExtractor(const std::string &targetCell)
00021         : targetCell_(targetCell), inTargetModule_(false), parsingComplete_(false) {}
00022
00023     // --- Visitor Handlers ---
00024     void handle(const slang::syntax::ModuleDeclarationSyntax &module);
00025     void handle(
00026         const slang::syntax::PortDeclarationSyntax &portDecl); // Handles direction/type declaration
00027     void
00028     handle(const slang::syntax::NonAnsiPortListSyntax &portList); // Handles port names list in header
00029     void
00030     handle(const slang::syntax::NetDeclarationSyntax &netDecl); // To find explicitly declared wires
00031     void handle(const slang::syntax::PrimitiveInstantiationSyntax
00032                 &primitiveInst); // Handles gate instantiation
00033     // Potentially handle ContinuousAssignSyntax if needed later:
00034     // void handle(const slang::syntax::ContinuousAssignSyntax& assign);
00035
00036     // --- Access Extracted Info (for Step 1 debugging) ---
00037     const std::unordered_map<std::string, GateInfo> &getExtractedGates()const {
00038         return gateOutputDrivers_;
00039     }
00040     const std::unordered_set<std::string> &getPrimaryInputs()const { return primaryInputs_; }
00041     const std::unordered_set<std::string> &getPrimaryOutputs()const { return primaryOutputs_; }
00042     const std::unordered_set<std::string> &getInternalWires()const { return internalWires_; }
00043
00044     // --- Logic Derivation (Commented out for Step 1) ---
00045     std::map<std::string, std::string> getLogicExpressions();
00046
00047 private:
00048     // --- Internal State ---
  
```



```

00049  const std::string &targetCell_;
00050  bool inTargetModule_;
00051  bool parsingComplete_; // Flag to indicate AST traversal is done
00052
00053  // Netlist Information
00054  std::unordered_set<std::string> primaryInputs_;
00055  std::unordered_set<std::string> primaryOutputs_;
00056  std::unordered_set<std::string> internalWires_; // Includes gate outputs
00057
00058  // Temporary map to store directions found in PortDeclarationSyntax
00059  // Key: port name, Value: direction ("input", "output", "inout", "unknown")
00060  std::unordered_map<std::string, std::string> portDirections_;
00061
00062  // Map: output signal name -> GateInfo driving it
00063  std::unordered_map<std::string, GateInfo> gateOutputDrivers_;
00064
00065  // Map: signal name -> Logic expression string (Memoization Cache) - (Used in Step 2)
00066  std::unordered_map<std::string, std::string> logicCache_;
00067
00068  // --- Helper Methods ---
00069  // Recursive function to derive logic for a given signal (Used in Step 2)
00070  std::string deriveLogicRecursive(const std::string &signalName);
00071  // Helper to format expressions based on gate type (Used in Step 2)
00072  std::string formatExpression(const GateInfo &gateInfo,
00073                               const std::vector<std::string> &inputExprs);
00074 };
00075
00076 // --- Function Declaration ---
00077 // For Step 1, this function will just run the visitor and maybe print summary
00078 void extractAndPrintNetlistInfo(const std::string &verilog_file, const std::string &cell);
00079
00080 // Function to be implemented in Step 2
00081 std::map<std::string, std::string> extractLogicFromVerilog(const std::string &verilog_file,
00082                                                           const std::string &cell);
00083
00084 #endif // LOGIC_EXTRACTOR_HPP

```

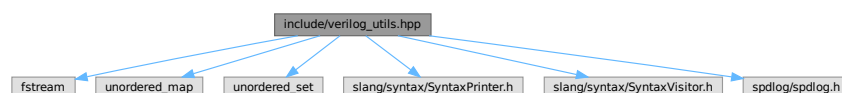
6.19 include/verilog_utils.hpp File Reference

```

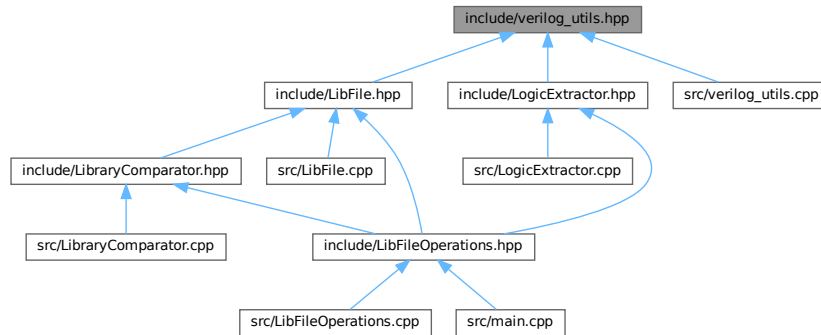
#include <fstream>
#include <unordered_map>
#include <unordered_set>
#include "slang/syntax/SyntaxPrinter.h"
#include "slang/syntax/SyntaxVisitor.h"
#include "spdlog/spdlog.h"

```

Include dependency graph for verilog_utils.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class [VerilogVisitor](#)
- class [CellExtractor](#)
- class [CellPrinter](#)
- class [ModuleRewriter](#)

Functions

- void [getAST](#) (const std::string &verilog_file, const std::string &cell)

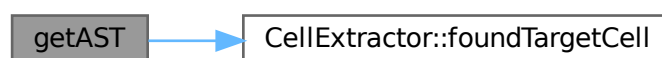
6.19.1 Function Documentation

6.19.1.1 getAST()

```
void getAST (
    const std::string & verilog_file,
    const std::string & cell )
```

Definition at line [544](#) of file [verilog_utils.cpp](#).

Here is the call graph for this function:



6.20 verilog_utils.hpp

[Go to the documentation of this file.](#)

```

00001 // verilog_utils.hpp
00002
00003 #ifndef VERILOG_UTILS_H
00004 #define VERILOG_UTILS_H
00005
00006 #include <fstream>
00007 #include <unordered_map>
00008 #include <unordered_set>
00009
00010 #include "slang/syntax/SyntaxPrinter.h"
00011 #include "slang/syntax/SyntaxVisitor.h"
00012 #include "spdlog/spdlog.h" // Include spdlog
00013
00014 // Creating custom visitor class
00015 class VerilogVisitor : public slang::syntax::SyntaxVisitor<VerilogVisitor> {
00016 public:
00017     explicit VerilogVisitor(const std::string &targetCell)
00018         : targetCell_(targetCell), depth_(0), inTargetModule_(false) {}
00019     void handle(const slang::syntax::SyntaxNode &node);
00020     void handle(const slang::syntax::ModuleDeclarationSyntax &module);
00021     void handle(const slang::syntax::PortDeclarationSyntax &portDecl);
00022     void handle(const slang::syntax::HierarchyInstantiationSyntax &hierarchyInst);
00023     // void handle(const slang::syntax::PrimitiveInstantiationSyntax &primitiveInst);
00024     void handle(const slang::syntax::SpecifyBlockSyntax &specifyBlock);
00025
00026 private:
00027     const std::string &targetCell_;
00028     int depth_;
00029     bool inTargetModule_;
00030 };
00031
00032 // Creating a Rewriter to extract a specific cell
00033 class CellExtractor : public slang::syntax::SyntaxRewriter<CellExtractor> {
00034 public:
00035     explicit CellExtractor(const std::string &targetCell)
00036         : targetCell_(targetCell), foundTarget_(false) {}
00037     void handle(const slang::syntax::ModuleDeclarationSyntax &module);
00038     bool foundTargetCell() const;
00039
00040 private:
00041     const std::string &targetCell_;
00042     bool foundTarget_;
00043 };
00044
00045 // Print specific cell when visiting SyntaxTree
00046 class CellPrinter : public slang::syntax::SyntaxVisitor<CellPrinter> {
00047 public:
00048     explicit CellPrinter(const std::string &targetCell, std::ostream &out)
00049         : targetCell_(targetCell), out_(out), foundTarget_(false) {}
00050     void handle(const slang::syntax::ModuleDeclarationSyntax &module);
00051
00052 private:
00053     const std::string &targetCell_;
00054     std::ostream &out_;
00055     bool foundTarget_;
00056 };
00057
00058 // Comprehensive module rewriter for adding ports, instances, and connections
00059 class ModuleRewriter : public slang::syntax::SyntaxRewriter<ModuleRewriter> {
00060 public:
00061     explicit ModuleRewriter(const std::vector<std::string> &inputPins,
00062                             const std::vector<std::string> &outputPins,
00063                             const std::pair<std::string, std::string> &superCellEntry,
00064                             int instanceCount, std::shared_ptr<spdlog::logger> logger)

```

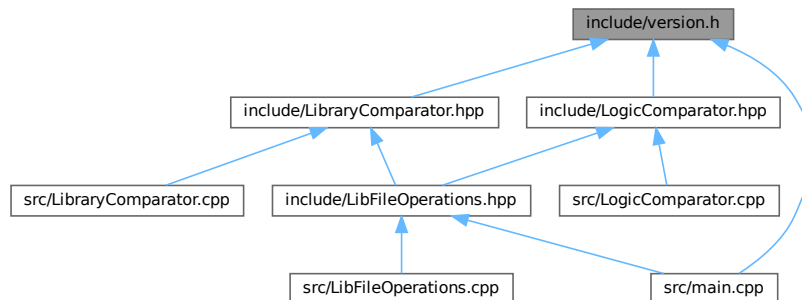
```

00065     : inputPins_(inputPins), outputPins_(outputPins), cellName_(supercell_entry.first),
00066     moduleName_(supercell_entry.second), logger_(logger), depth_(0),
00067     instance_count_(instance_count) {}
00068 std::shared_ptr<spdlog::logger> logger_;
00069 void handle(const slang::syntax::SyntaxNode &node);
00070 void handle(const slang::syntax::ModuleDeclarationSyntax &module);
00071
00072 private:
00073     const std::vector<std::string> &inputPins_;
00074     const std::vector<std::string> &outputPins_;
00075     const std::string cellName_;
00076     const std::string moduleName_;
00077     std::map<std::string, std::string> portInfoMap_; // Map from port name to direction
00078     int depth_;
00079     int instance_count_;
00080 };
00081
00082 void getAST(const std::string &verilog_file, const std::string &cell);
00083
00084 #endif // VERILOG_UTILS_H

```

6.21 include/version.h File Reference

This graph shows which files directly or indirectly include this file:



Macros

- #define APP_NAME "ZlibValidation"
- #define APP_VERSION_MAJOR 0
- #define APP_VERSION_MINOR 1
- #define APP_VERSION_PATCH 1
- #define APP_VERSION "0.1.1"
- #define APP_AUTHOR "Song Zixuan"
- #define APP_CONTACT "cedar@zju.edu.cn"
- #define BUILD_TIMESTAMP "2025-04-05 13:56:42"

6.21.1 Macro Definition Documentation

6.21.1.1 APP_AUTHOR

```
#define APP_AUTHOR "Song Zixuan"
```

Definition at line 10 of file [version.h](#).

6.21.1.2 APP_CONTACT

```
#define APP_CONTACT "cedar@zju.edu.cn"
```

Definition at line 11 of file [version.h](#).

6.21.1.3 APP_NAME

```
#define APP_NAME "ZlibValidation"
```

Definition at line 5 of file [version.h](#).

6.21.1.4 APP_VERSION

```
#define APP_VERSION "0.1.1"
```

Definition at line 9 of file [version.h](#).

6.21.1.5 APP_VERSION_MAJOR

```
#define APP_VERSION_MAJOR 0
```

Definition at line 6 of file [version.h](#).

6.21.1.6 APP_VERSION_MINOR

```
#define APP_VERSION_MINOR 1
```

Definition at line 7 of file [version.h](#).

6.21.1.7 APP_VERSION_PATCH

```
#define APP_VERSION_PATCH 1
```

Definition at line 8 of file [version.h](#).

6.21.1.8 BUILD_TIMESTAMP

```
#define BUILD_TIMESTAMP "2025-04-05 13:56:42"
```

Definition at line 12 of file [version.h](#).

6.22 version.h

[Go to the documentation of this file.](#)

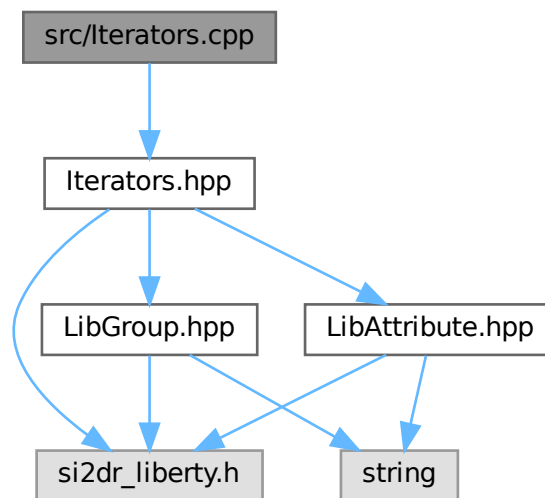
```
00001 // version.h.in
00002 #pragma once
00003
00004 // Auto-generated by CMake - DO NOT EDIT
00005 #define APP_NAME "ZlibValidation"
00006 #define APP_VERSION_MAJOR 0
00007 #define APP_VERSION_MINOR 1
00008 #define APP_VERSION_PATCH 1
00009 #define APP_VERSION "0.1.1"
00010 #define APP_AUTHOR "Song Zixuan"
00011 #define APP_CONTACT "cedar@zju.edu.cn"
00012 #define BUILD_TIMESTAMP "2025-04-05 13:56:42"
```

6.23 README.md File Reference

6.24 src/Iterators.cpp File Reference

```
#include "Iterators.hpp"
```

Include dependency graph for Iterators.cpp:



6.25 Iterators.cpp

[Go to the documentation of this file.](#)

```

00001 #include "Iterators.hpp"
00002
00003 GroupsIterator::GroupsIterator(si2drGroupsIdT groups, si2drErrorT &err) : groups_(groups), err_(err) {
00004     group_ = si2drIterNextGroup(groups_, &err_);
00005 }
00006 GroupsIterator::~GroupsIterator() { si2drIterQuit(groups_, &err_); }
00007
00008 void GroupsIterator::next() { group_ = si2drIterNextGroup(groups_, &err_); }
00009 bool GroupsIterator::end() { return si2drObjectIsNull(group_, &err_); }
00010
00011 LibGroup GroupsIterator::get() { return LibGroup(group_, err_); }
00012
00013 AttributesIterator::AttributesIterator(si2drAttrsIdT attrs, si2drErrorT &err)
00014     : attrs_(attrs), err_(err) {
00015     attr_ = si2drIterNextAttr(attrs_, &err_);
00016 }
00017 AttributesIterator::~AttributesIterator() { si2drIterQuit(attrs_, &err_); }
00018
00019 void AttributesIterator::next() { attr_ = si2drIterNextAttr(attrs_, &err_); }
00020 bool AttributesIterator::end() { return si2drObjectIsNull(attr_, &err_); }

```

```

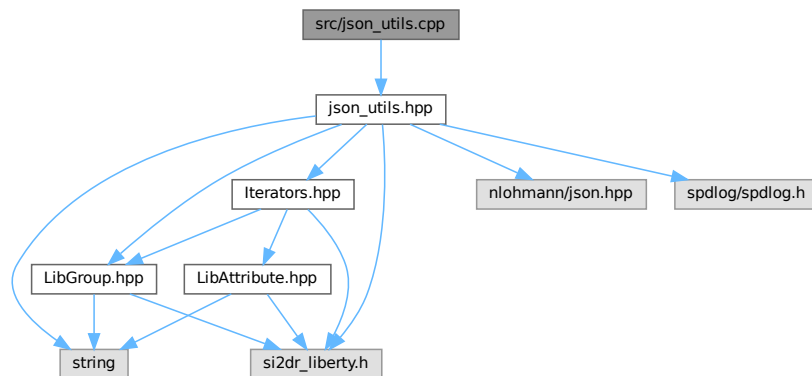
00021
00022 LibAttribute AttributesIterator::get() { return LibAttribute(attr_, err_); }
00023
00024 ValuesIterator::ValuesIterator(si2drValuesIdT values, si2drErrorT &err) : values_(values), err_(err) {
00025     si2drIterNextComplexValue(values_, &vtype_, &int_, &float_, &str_, &bool_, &exprp_, &err_);
00026 }
00027 ValuesIterator::~ValuesIterator() { si2drIterQuit(values_, &err_); }
00028
00029 void ValuesIterator::next() {
00030     si2drIterNextComplexValue(values_, &vtype_, &int_, &float_, &str_, &bool_, &exprp_, &err_);
00031 }
00032 bool ValuesIterator::end() { return vtype_ == SI2DR_UNDEFINED_VALUETYPE; }

```

6.26 src/json_utils.cpp File Reference

#include "json_utils.hpp"

Include dependency graph for json_utils.cpp:



Functions

- `std::vector< double > parseStringToVector (const std::string &str)`
Parses a string representation of a vector of doubles into a vector of doubles.
- `json generateLutJson (LibGroup &lib_lut_group, si2drErrorT &err)`
Generates a JSON object representation of a look-up table (LUT) from a [LibGroup](#) object.
- `json generateTimingJson (LibGroup &lib_timing_group, si2drErrorT &err)`
Generates a JSON representation of timing information from a library timing group.
- `json generatePowerJson (LibGroup &lib_power_group, si2drErrorT &err)`
Converts a Liberty power group into a JSON representation.
- `std::pair< std::string, json > generatePinJson (LibGroup &lib_pin_group, si2drErrorT &err)`
Generates a JSON representation of a Liberty pin group.
- `json generateCellJson (LibGroup &lib_cell_group, si2drErrorT &err)`
Generates a JSON representation of a cell from a [LibGroup](#) object.

6.26.1 Function Documentation

6.26.1.1 generateCellJson()

```
json generateCellJson (
    LibGroup & lib_cell_group,
    si2drErrorT & err )
```

Generates a JSON representation of a cell from a [LibGroup](#) object.

This function processes a [LibGroup](#) representing a cell and converts it into a JSON object. It extracts the cell name and processes specific attributes such as area, cell_leakage_power, and cell_footprint. It also processes pins within the cell by categorizing them based on their direction (input, output, internal, inout).

Parameters

<i>lib_cell_group</i>	The LibGroup object representing the cell
<i>err</i>	Reference to a si2drErrorT object for error handling

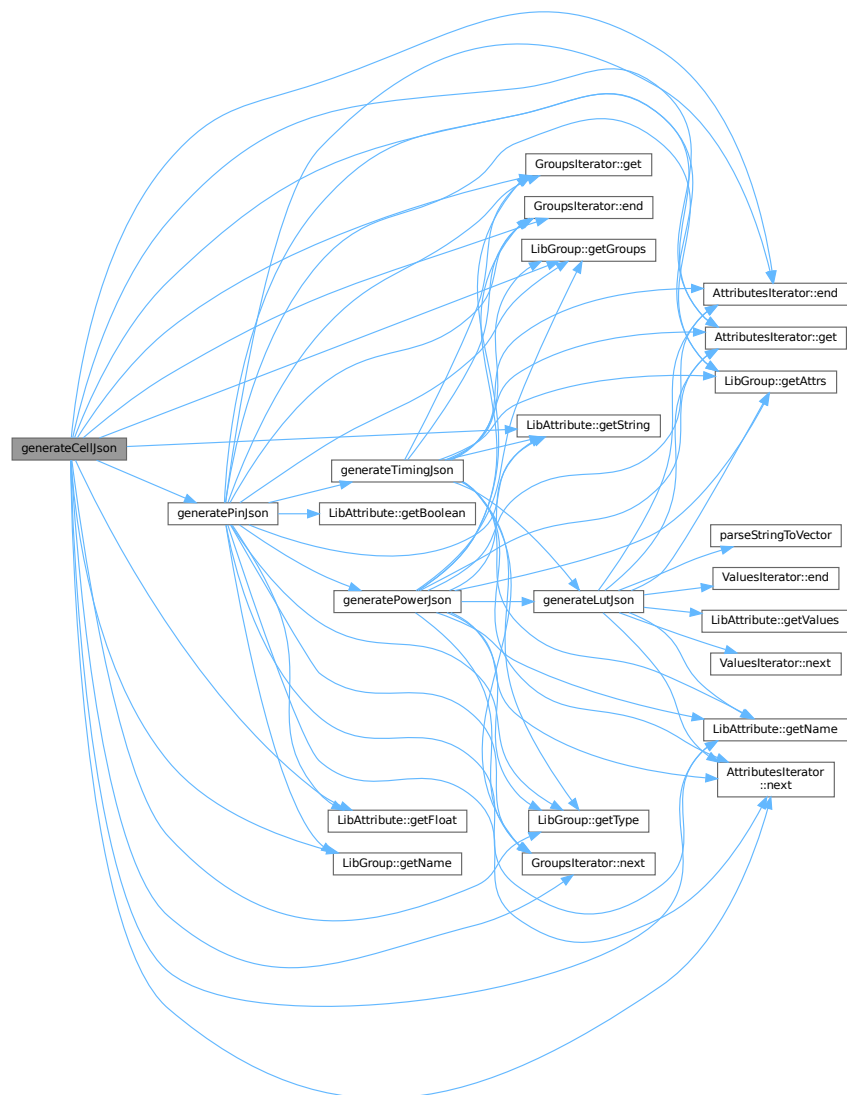
Returns

json A JSON object containing the cell's information with the following structure:

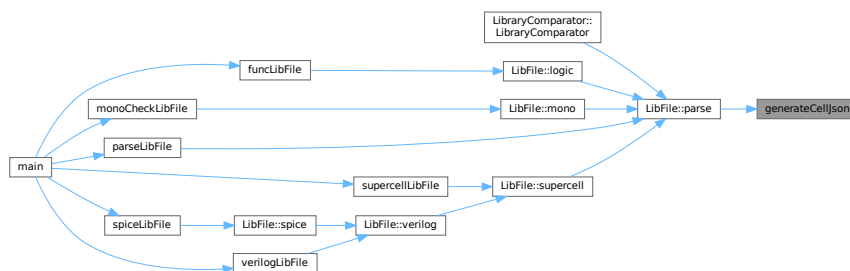
- "cell_name": string - name of the cell
- "area": float (optional) - cell area if present
- "cell_leakage_power": float (optional) - cell leakage power if present
- "cell_footprint": string (optional) - cell footprint if present
- "input_pins": array - list of input pins
- "output_pins": array - list of output pins
- "internal_pins": array - list of internal pins
- "inout_pins": array - list of inout pins

Definition at line 269 of file [json_utils.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



6.26.1.2 generateLutJson()

```
json generateLutJson (
    LibGroup & lib_lut_group,
    si2drErrorT & err )
```

Generates a JSON object representation of a look-up table (LUT) from a [LibGroup](#) object.

This function iterates through the attributes of the provided [LibGroup](#) object representing a LUT and converts them into a JSON structure. It specifically handles the following attributes:

- "index_1": Converted to a vector and stored in the JSON
- "index_2": Converted to a vector and stored in the JSON
- "values": Each value is parsed into a vector and added to an array in the JSON

Any other attributes encountered will trigger a warning message.

Parameters

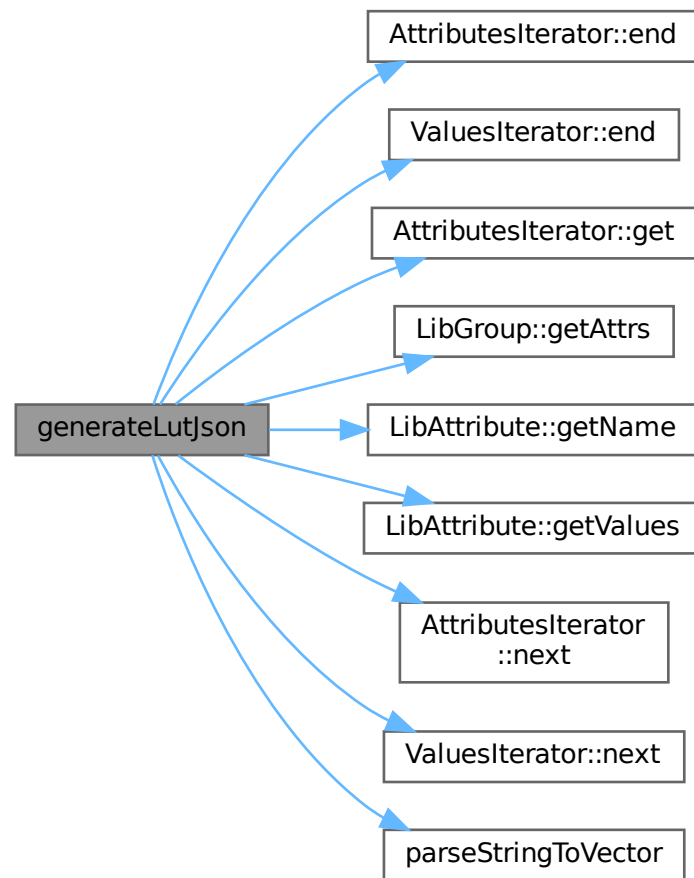
<i>lib_lut_group</i>	The LibGroup object containing the LUT data to be converted
<i>err</i>	Reference to an <code>si2drErrorT</code> object to track any errors during processing

Returns

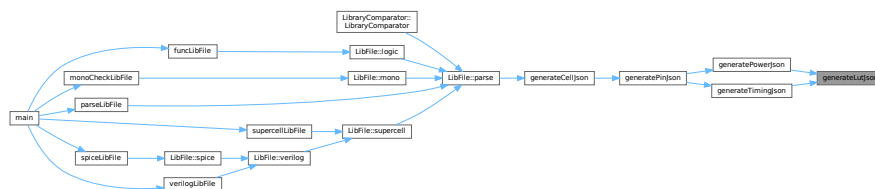
json A JSON object representing the LUT data with keys for "index_1", "index_2", and "values"

Definition at line 59 of file [json_utils.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



6.26.1.3 generatePinJson()

```
std::pair< std::string, json > generatePinJson (
```

```
LibGroup & lib_pin_group,  
si2drErrorT & err )
```

Generates a JSON representation of a Liberty pin group.

This function traverses a Liberty pin group, extracts relevant attributes and sub-groups, and converts them into a JSON object. It also determines the pin's direction.

Processes the following pin attributes:

- direction
- max_transition, capacitance, rise_capacitance, fall_capacitance, max_capacitance (float types)
- function, power_down_function, related_ground_pin, related_power_pin, three_state (string types)
- clock (boolean type)

Handles the following sub-groups:

- internal_power: Converted using [generatePowerJson\(\)](#)
- timing: Converted using [generateTimingJson\(\)](#)

Parameters

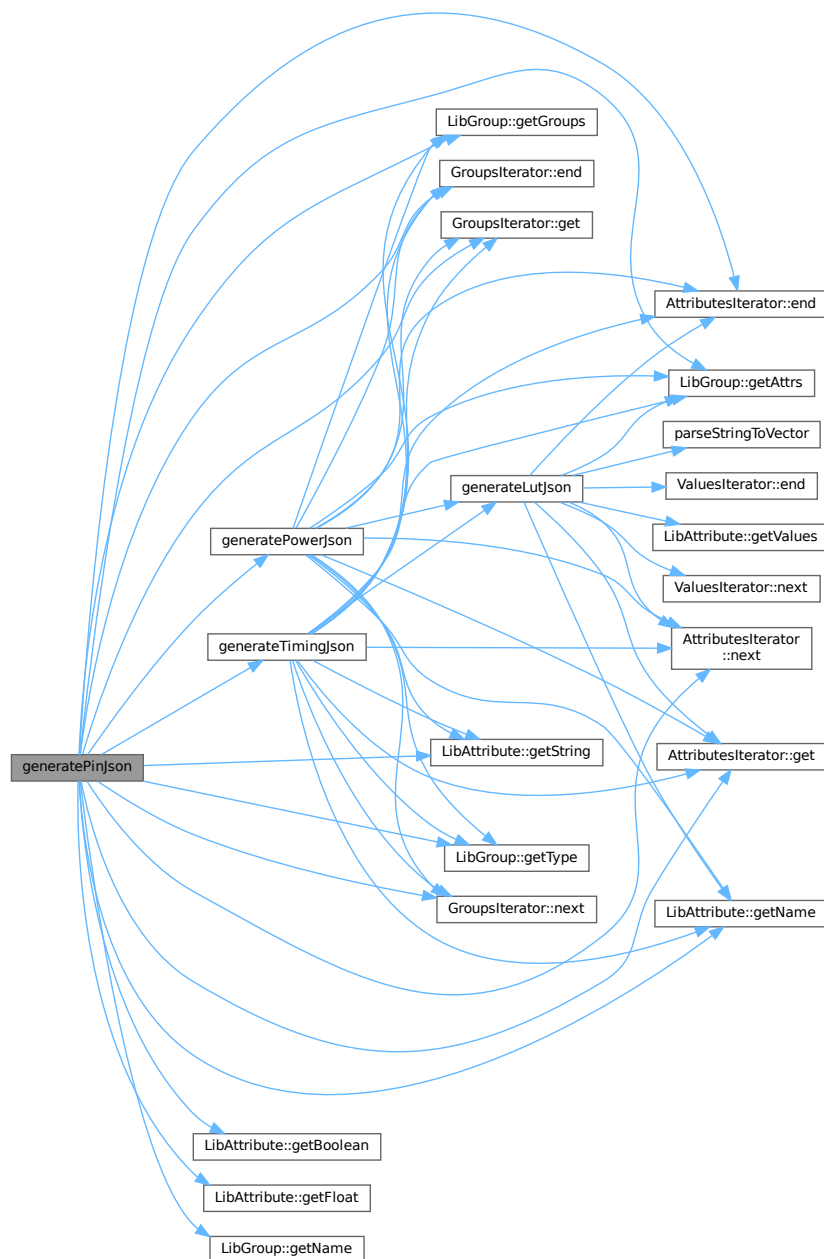
<i>lib_pin_group</i>	The Liberty pin group to process
<i>err</i>	Reference to an error object for tracking Liberty API errors

Returns

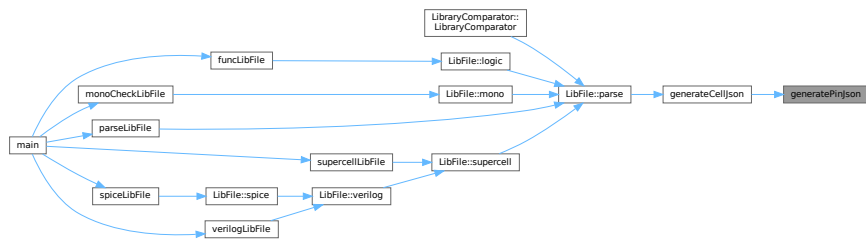
A pair containing the pin direction (string) and the JSON representation of the pin

Definition at line 203 of file [json_utils.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



6.26.1.4 generatePowerJson()

```

json generatePowerJson (
    LibGroup & lib_power_group,
    si2drErrorT & err )

```

Converts a Liberty power group into a JSON representation.

This function processes a Liberty power group and converts its attributes and subgroups into a JSON object. It handles attributes like "when", "related_pin", and "related_pg_pin", as well as "rise_power" and "fall_power" subgroups.

Parameters

<i>lib_power_group</i>	The Liberty power group to convert
<i>err</i>	Reference to an si2drErrorT object for error handling

Returns

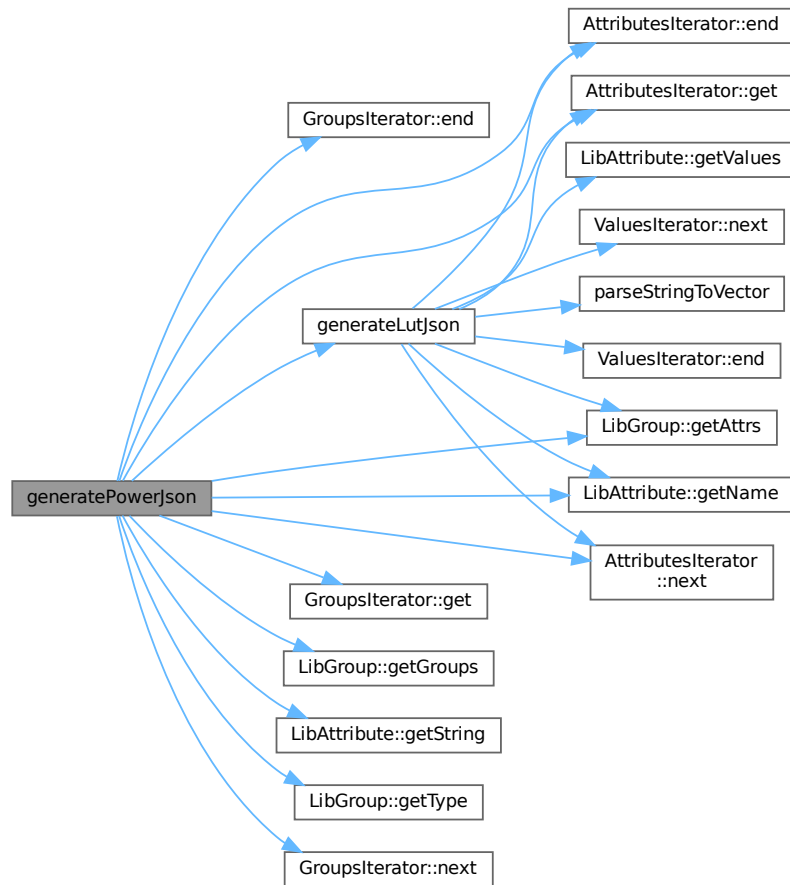
json A JSON object representing the power group data

The function:

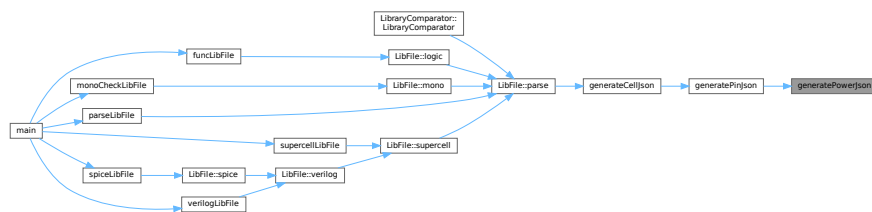
- Extracts string attributes (when, related_pin, related_pg_pin)
- Processes rise_power and fall_power subgroups by converting them to LUT JSON format
- Logs warnings for unknown power subgroup types

Definition at line 151 of file json_utils.cpp.

Here is the call graph for this function:



Here is the caller graph for this function:



6.26.1.5 generateTimingJson()

```

json generateTimingJson (
    LibGroup & lib_timing_group,
    si2drErrorT & err )
  
```


Generates a JSON representation of timing information from a library timing group.

This function extracts timing attributes and sub-groups from a Liberty timing group and organizes them into a JSON object. It processes standard timing attributes like 'related_pin', 'timing_type', 'timing_sense', and 'when', as well as timing tables such as 'cell_fall', 'cell_rise', 'fall_transition', 'rise_transition', 'fall_constraint', and 'rise_constraint'.

Parameters

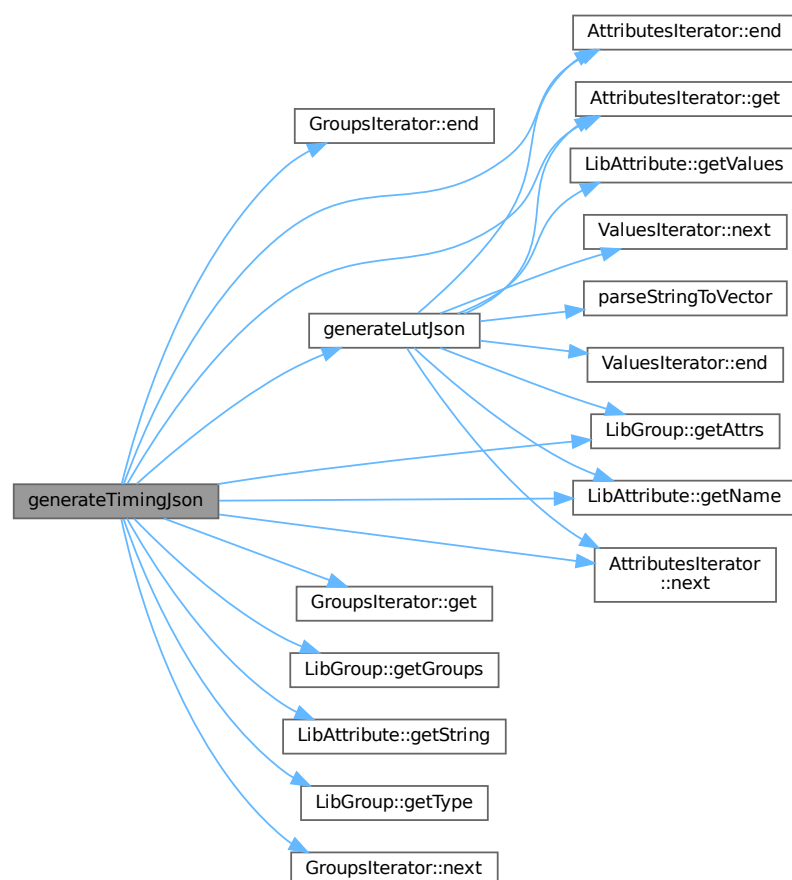
<i>lib_timing_group</i>	The Liberty timing group to process
<i>err</i>	Reference to an si2drErrorT object for error reporting

Returns

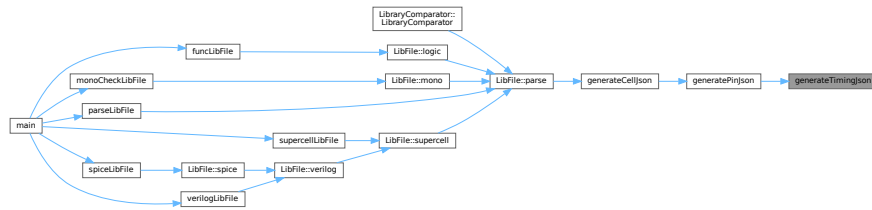
json A JSON object containing the extracted timing information

Definition at line 103 of file [json_utils.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



6.26.1.6 parseStringToVector()

```
std::vector< double > parseStringToVector (
    const std::string & str )
```

Parses a string representation of a vector of doubles into a vector of doubles.

This function takes a string containing comma-separated numbers, cleans it by removing backslashes and newline characters, and then converts each comma-separated value into a double that is added to the resulting vector.

Parameters

<i>str</i>	The input string to be parsed, containing comma-separated numbers
------------	---

Returns

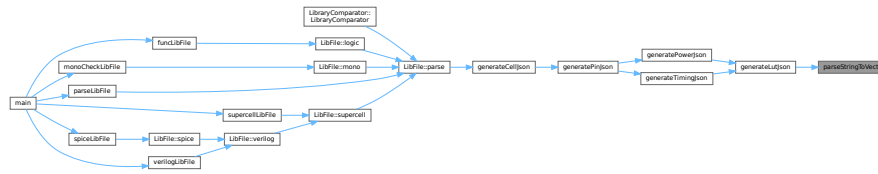
`std::vector<double>` A vector of doubles parsed from the input string

Exceptions

<code>std::invalid_argument</code>	If the string contains values that cannot be converted to double
<code>std::out_of_range</code>	If the string contains values that are out of range for double

Definition at line 25 of file `json_utils.cpp`.

Here is the caller graph for this function:



6.27 json_utils.cpp

[Go to the documentation of this file.](#)

```

00001 #include "json_utils.hpp"
00002
00003 /*
00004 json 类型的值可以是以下几种类型之一：
00005 方括号 []：用于包含一组有序的值（数组）。
00006 数组中的每个值可以是任意类型的 JSON 值，包括对象、数组、字符串、数字、布尔值或 null。
00007 花括号 {}：用于包含一组键值对（对象）。
00008 对象中的每个键都是一个字符串，值可以是任意类型的 JSON 值，包括对象、数组、字符串、数字、布尔值或
00009 null。
00010 */
00011
00025 std::vector<double> parseStringToVector(const std::string &str) {
00026     std::vector<double> result;
00027     std::string cleaned_str;
00028
00029     // Remove backslashes and newline characters
00030     for (char c : str) {
00031         if (c != '\\' && c != '\n') {
00032             cleaned_str += c;
00033         }
00034     }
00035
00036     std::stringstream ss(cleaned_str);
00037     std::string item;
00038     while (std::getline(ss, item, ',')) {
00039         result.push_back(std::stod(item));
00040     }
00041     return result;
00042 }
00043
00059 json generateLutJson(LibGroup &lib_lut_group, si2drErrorT &err) {
00060     json lut_json;
00061
00062     AttributesIterator attr_iter(lib_lut_group.getAttrs(), err);
00063     for (; !attr_iter.end(); attr_iter.next()) {
00064         LibAttribute lib_attr = attr_iter.get();
00065         std::string lut_attr_name = lib_attr.getName();
00066
00067         // spdlog::debug("LUT Attribute Name: {}", lib_attr.getName());
00068         // spdlog::debug("Is Complex? {}", lib_attr.isComplex());
00069
00070         if (lut_attr_name == "index_1" || lut_attr_name == "index_2") {
00071             ValuesIterator values_iter(lib_attr.getValues(), err);
00072             for (; !values_iter.end(); values_iter.next()) {
00073                 // spdlog::debug("Type: {}", int(values_iter.vtype_)); // 5 is string
00074                 // spdlog::debug("Str: {}", values_iter.str_);
00075                 lut_json[lut_attr_name] = parseStringToVector(std::string(values_iter.str_));
00076             }
00077         } else if (lut_attr_name == "values") {

```

```

00078     ValuesIterator values_iter(lib_attr.getValues(), err);
00079     for (; !values_iter.end(); values_iter.next()) {
00080         // spdlog::debug("{} ", values_iter.str_);
00081         lut_json["values"].push_back(parseStringToVector(std::string(values_iter.str_)));
00082     }
00083 } else {
00084     spdlog::warn("Unknown LUT attribute name: {}", lut_attr_name);
00085 }
00086 }
00087 return lut_json;
00088 }
00089
00103 json generateTimingJson(LibGroup &lib_timing_group, si2drErrorT &err) {
00104     json timing_json;
00105
00106     AttributesIterator attr_iter(lib_timing_group.getAttrs(), err);
00107     for (; !attr_iter.end(); attr_iter.next()) {
00108         LibAttribute lib_attr = attr_iter.get();
00109
00110         std::string attr_name = lib_attr.getName();
00111         if (attr_name == "related_pin" || attr_name == "timing_type" || attr_name == "timing_sense" ||
00112             attr_name == "when") {
00113             timing_json[attr_name] = lib_attr.getString();
00114         }
00115         // More timing attributes can be added here
00116     }
00117
00118     GroupsIterator timing_sub_group_iter(lib_timing_group.getGroups(), err);
00119     for (; !timing_sub_group_iter.end(); timing_sub_group_iter.next()) {
00120         LibGroup lib_timing_sub_group = timing_sub_group_iter.get();
00121
00122         std::string timing_sub_group_type = lib_timing_sub_group.getType();
00123         // std::string timing_sub_group_name = lib_timing_sub_group.getName();
00124         if (timing_sub_group_type == "cell_fall" || timing_sub_group_type == "cell_rise" ||
00125             timing_sub_group_type == "fall_transition" || timing_sub_group_type == "rise_transition" ||
00126             timing_sub_group_type == "fall_constraint" || timing_sub_group_type == "rise_constraint") {
00127             timing_json[timing_sub_group_type] = generateLutJson(lib_timing_sub_group, err);
00128         } else {
00129             spdlog::warn("Unknown timing sub group type: {}", timing_sub_group_type);
00130         }
00131     }
00132     return timing_json;
00133 }
00134
00151 json generatePowerJson(LibGroup &lib_power_group, si2drErrorT &err) {
00152     json power_json;
00153
00154     AttributesIterator attr_iter(lib_power_group.getAttrs(), err);
00155     for (; !attr_iter.end(); attr_iter.next()) {
00156         LibAttribute lib_attr = attr_iter.get();
00157
00158         std::string attr_name = lib_attr.getName();
00159         if (attr_name == "when" || attr_name == "related_pin" || attr_name == "related_pg_pin") {
00160             power_json[attr_name] = lib_attr.getString();
00161         }
00162         // More power attributes can be added here
00163     }
00164
00165     GroupsIterator power_sub_group_iter(lib_power_group.getGroups(), err);
00166     for (; !power_sub_group_iter.end(); power_sub_group_iter.next()) {
00167         LibGroup lib_power_sub_group = power_sub_group_iter.get();
00168
00169         std::string power_sub_group_type = lib_power_sub_group.getType();
00170         // std::string power_sub_group_name = lib_power_sub_group.getName();
00171         if (power_sub_group_type == "rise_power") {
00172             // spdlog::debug("Rise Power: {}", power_sub_group_name);
00173             power_json["rise_power"] = generateLutJson(lib_power_sub_group, err);
00174         } else if (power_sub_group_type == "fall_power") {
00175             power_json["fall_power"] = generateLutJson(lib_power_sub_group, err);

```

```

00176     } else {
00177         spdlog::warn("Unknown power sub group type: {}", power_sub_group_type);
00178     }
00179 }
00180 return power_json;
00181 }
00182
00203 std::pair<std::string, json> generatePinJson(LibGroup &lib_pin_group, si2drErrorT &err) {
00204     json pin_json;
00205     std::string direction;
00206     pin_json["pin_name"] = lib_pin_group.getName();
00207
00208     AttributesIterator attr_iter(lib_pin_group.getAttrs(), err);
00209     for (; !attr_iter.end(); attr_iter.next()) {
00210         LibAttribute lib_attr = attr_iter.get();
00211
00212         std::string attr_name = lib_attr.getName();
00213         if (attr_name == "direction") {
00214             direction = lib_attr.getString();
00215         } else if (attr_name == "max_transition" || attr_name == "capacitance" ||
00216             attr_name == "rise_capacitance" || attr_name == "fall_capacitance" ||
00217             attr_name == "max_capacitance") {
00218             pin_json[attr_name] = lib_attr.getFloat();
00219         } else if (attr_name == "function" || attr_name == "power_down_function" ||
00220             attr_name == "related_ground_pin" || attr_name == "related_power_pin" ||
00221             attr_name == "three_state") {
00222             pin_json[attr_name] = lib_attr.getString();
00223         } else if (attr_name == "clock") {
00224             pin_json[attr_name] = lib_attr.getBoolean();
00225         }
00226         // More pin attributes can be added here
00227     }
00228
00229     GroupsIterator pin_sub_group_iter(lib_pin_group.getGroups(), err);
00230     for (; !pin_sub_group_iter.end(); pin_sub_group_iter.next()) {
00231         LibGroup lib_pin_sub_group = pin_sub_group_iter.get();
00232
00233         std::string pin_sub_group_type = lib_pin_sub_group.getType();
00234         // std::string pin_sub_group_name = lib_pin_sub_group.getName();
00235         if (pin_sub_group_type == "internal_power") {
00236             json power_json = generatePowerJson(lib_pin_sub_group, err);
00237             pin_json["power_arcs"].push_back(power_json);
00238         } else if (pin_sub_group_type == "timing") {
00239             // spdlog::debug("Has Timing: {}", lib_pin_group.getName());
00240             json timing_json = generateTimingJson(lib_pin_sub_group, err);
00241             pin_json["timing_arcs"].push_back(timing_json);
00242         } else {
00243             spdlog::warn("Unknown pin sub group type: {}", pin_sub_group_type);
00244         }
00245     }
00246     return std::make_pair(direction, pin_json);
00247 }
00248
00269 json generateCellJson(LibGroup &lib_cell_group, si2drErrorT &err) {
00270     json cell_json;
00271     cell_json["cell_name"] = lib_cell_group.getName();
00272
00273     AttributesIterator attr_iter(lib_cell_group.getAttrs(), err);
00274     for (; !attr_iter.end(); attr_iter.next()) {
00275         LibAttribute lib_attr = attr_iter.get();
00276
00277         std::string attr_name = lib_attr.getName();
00278         if (attr_name == "area" || attr_name == "cell_leakage_power") {
00279             cell_json[attr_name] = lib_attr.getFloat();
00280         } else if (attr_name == "cell_footprint") {
00281             cell_json[attr_name] = lib_attr.getString();
00282         }
00283         // More cell attributes can be added here
00284     }

```

```

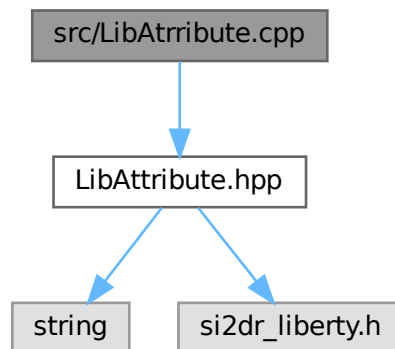
00285
00286 GroupsIterator cell_sub_group_iter(lib_cell_group.getGroups(), err);
00287 for (; !cell_sub_group_iter.end(); cell_sub_group_iter.next()) {
00288     LibGroup lib_cell_sub_group = cell_sub_group_iter.get();
00289
00290     std::string cell_sub_group_type = lib_cell_sub_group.getType();
00291     // std::string cell_sub_group_name = lib_cell_sub_group.getName();
00292
00293     // spdlog::debug("Cell Sub Group Type: {}", cell_sub_group_type);
00294     // spdlog::debug("Cell Sub Group Name: {}", cell_sub_group_name);
00295
00296     if (cell_sub_group_type == "pin") {
00297         auto [direction, pin_json] = generatePinJson(lib_cell_sub_group, err);
00298         if (direction == "input") {
00299             cell_json["input_pins"].push_back(pin_json);
00300         } else if (direction == "output") {
00301             cell_json["output_pins"].push_back(pin_json);
00302         } else if (direction == "internal") {
00303             cell_json["internal_pins"].push_back(pin_json);
00304         } else if (direction == "inout") {
00305             cell_json["inout_pins"].push_back(pin_json);
00306         } else {
00307             spdlog::warn("Unknown direction: {}", direction);
00308         }
00309     }
00310 }
00311 return cell_json;
00312 }

```

6.28 src/LibAtrribute.cpp File Reference

#include "LibAtrribute.hpp"

Include dependency graph for LibAtrribute.cpp:



6.29 LibAtrribute.cpp

[Go to the documentation of this file.](#)

```

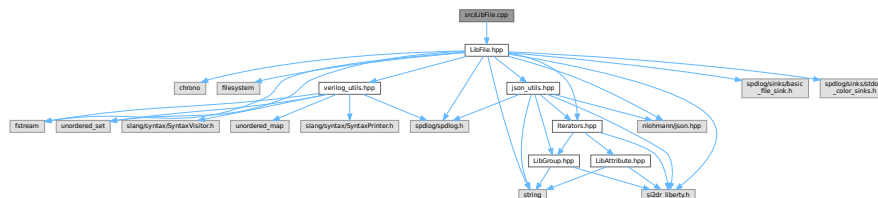
00001 #include "LibAttribute.hpp"
00002
00003 LibAttribute::LibAttribute(si2drAttrIdT attr, si2drErrorT &err) : attr_(attr), err_(err) {}
00004
00005 LibAttribute::~LibAttribute() {}
00006
00007 std::string LibAttribute::getName() {
00008     si2drStringT name = si2drAttrGetName(attr_, &err_);
00009     return name ? std::string(name) : std::string();
00010 }
00011
00012 bool LibAttribute::isComplex() { return si2drAttrGetAttrType(attr_, &err_) ? 1 : 0; }
00013
00014 si2drValuesIdT LibAttribute::getValues() { return si2drComplexAttrGetValues(attr_, &err_); }
00015
00016 long int LibAttribute::getInt() { return si2drSimpleAttrGetInt32Value(attr_, &err_); }
00017
00018 double LibAttribute::getFloat() { return si2drSimpleAttrGetFloat64Value(attr_, &err_); }
00019
00020 std::string LibAttribute::getString() {
00021     si2drStringT str = si2drSimpleAttrGetStringValue(attr_, &err_);
00022     return str ? std::string(str) : std::string();
00023 }
00024
00025 bool LibAttribute::getBoolean() { return si2drSimpleAttrGetBooleanValue(attr_, &err_) ? 1 : 0; }

```

6.30 src/LibFile.cpp File Reference

#include "LibFile.hpp"

Include dependency graph for LibFile.cpp:



6.31 LibFile.cpp

[Go to the documentation of this file.](#)

```

00001 #include "LibFile.hpp"
00002
00015 LibFile::LibFile(const std::string &filepath, const std::string &loggername)
00016     : filepath_(filepath), loggername_(loggername) {
00017     filename_ = filepath_.string();
00018     basename_ = filepath_.stem().string();
00019     jsonname_ = basename_ + ".json";
00020
00021     auto console_sink = std::make_shared<spdlog::sinks::stdout_color_sink_mt>();
00022     console_sink->set_level(spdlog::level::info);
00023
00024     auto file_sink = std::make_shared<spdlog::sinks::basic_file_sink_mt>(loggername_, true);
00025     file_sink->set_level(spdlog::level::debug);

```

```

00026
00027 std::vector<spdlog::sink_ptr> sinks{console_sink, file_sink};
00028 logger_ = std::make_shared<spdlog::logger>(loggername_, sinks.begin(), sinks.end());
00029 logger_>set_level(spdlog::level::debug);
00030
00031 logger_>info("Created LibFile object for '{}'", filename_);
00032 logger_>info("Debug log in: '{}'", loggername_);
00033 }
00034
00035 LibFile::~LibFile() { logger_>info("Closing file: '{}'", filename_); }
00036
00046 void LibFile::writeJsonToFile() {
00047     std::ofstream out(jsonname_);
00048     if (!out.is_open()) {
00049         logger_>error("Could not open file '{}' for writing", jsonname_);
00050         return;
00051     }
00052     out << lib_json_.dump(2);
00053     out.close();
00054     logger_>info("JSON data written to '{}'", jsonname_);
00055 }
00056
00077 void LibFile::read() {
00078     logger_>info("Reading '{} ...'", filename_);
00079
00080     auto start = std::chrono::high_resolution_clock::now();
00081     si2drReadLibertyFile(const_cast<char *>(filepath_.c_str()), &err_);
00082     auto end = std::chrono::high_resolution_clock::now();
00083     std::chrono::duration<double> duration = end - start;
00084
00085     if (err_ == SI2DR_INVALID_NAME) {
00086         logger_>error("Could not open file '{}' for parsing, quitting...", filename_);
00087         exit(301);
00088     } else if (err_ == SI2DR_SYNTAX_ERROR) {
00089         logger_>error("Syntax Errors were detected in the input file!");
00090         exit(401);
00091     } else {
00092         logger_>info("Done. Read time: {:.2f} seconds", duration.count());
00093     }
00094 }
00095
00116 void LibFile::parse() {
00117     si2drPIInit(&err_); // Initialize private error handler
00118     this->read();        // Read the Liberty file
00119
00120     logger_>info("Parsing '{} ...'", filename_);
00121     // Create a scope for the top-level groups iteration
00122     {
00123         GroupsIterator group_iter(si2drPIGetGroups(&err_), err_);
00124         for (; !group_iter.end(); group_iter.next()) {
00125             LibGroup lib_group = group_iter.get();
00126
00127             if (lib_group.getName() != "") {
00128                 libname_ = lib_group.getName();
00129                 lib_json_["library_name"] = this->libname_;
00130                 logger_>info("Library Name: {}", libname_);
00131             } else {
00132                 logger_>warn("Library Name: <NONAME>");
00133             }
00134             // Second level groups
00135             GroupsIterator sub_group_iter(lib_group.getGroups(), err_);
00136             for (; !sub_group_iter.end(); sub_group_iter.next()) {
00137                 LibGroup lib_sub_group = sub_group_iter.get();
00138
00139                 std::string sub_group_type = lib_sub_group.getType();
00140                 std::string sub_group_name = lib_sub_group.getName();
00141
00142                 if (sub_group_type == "cell") {
00143                     logger_>debug("Cell Name: {}", sub_group_name);

```



```

00144         // if (sub_group_name != "AN2D0") {
00145         //     continue;
00146         // }
00147         // Handle each cell
00148         json cell_json = generateCellJson(lib_sub_group, err_);
00149         lib_json["cells"].push_back(cell_json);
00150     } else if (sub_group_type == "operating_conditions") {
00151         logger_>info("Operating Conditions: {}", sub_group_name);
00152         // Get PVT from Operating Conditions
00153         AttributesIterator attr_iter(lib_sub_group.getAttrs(), err_);
00154         for (; !attr_iter.end(); attr_iter.next()) {
00155             LibAttribute lib_attr = attr_iter.get();
00156             logger_>debug("Attribute Name: {}", lib_attr.getName());
00157             logger_>debug("Int Value: {}", lib_attr.getInt());
00158             logger_>debug("Float Value: {}", lib_attr.getFloat());
00159             logger_>debug("String Value: {}", lib_attr.getString());
00160             if (lib_attr.getName() == "process") {
00161                 process_ = lib_attr.getInt();
00162                 lib_json["process"] = process_;
00163             } else if (lib_attr.getName() == "voltage") {
00164                 voltage_ = lib_attr.getFloat();
00165                 // Round to 2 decimal places to avoid floating-point precision issues
00166                 lib_json["voltage"] = std::round(voltage_ * 100) / 100.0;
00167             } else if (lib_attr.getName() == "temperature") {
00168                 temperature_ = lib_attr.getInt();
00169                 lib_json["temperature"] = temperature_;
00170             }
00171         }
00172         logger_>info("P: {}, V: {}, T: {}", process_, voltage_, temperature_);
00173     }
00174     // sub_group_iter's lifetime ends here, and the destructor is called
00175 }
00176 }
00177 // group_iter's lifetime ends here, and the destructor is called
00178 }
00179 si2drPIQuit(&err_);
00180 }
00181
00182 void LibFile::modify() { logger_>info("Modifying the file..."); }
00183
00205 bool LibFile::checkTimingArcMonotonicity(const json &cell, const json &pin, const json &arc,
00206                                           const std::string &timing_arc_name, const bool is_slew) {
00207     if (arc.contains("when") && !arc["when"].get<std::string>().empty()) {
00208         logger_>debug(
00209             "Checking cell: '{}', pin: '{}', related_pin: '{}', timing_arc: '{}', when: '{}'",
00210             cell["cell_name"].get<std::string>(), pin["pin_name"].get<std::string>(),
00211             arc["related_pin"].get<std::string>(), timing_arc_name, arc["when"].get<std::string>());
00212     } else {
00213         logger_>debug("Checking cell: '{}', pin: '{}', related_pin: '{}', timing_arc: '{}'",
00214             cell["cell_name"].get<std::string>(), pin["pin_name"].get<std::string>(),
00215             arc["related_pin"].get<std::string>(), timing_arc_name);
00216     }
00217     bool is_monotonic = true; // Assume the values are monotonic initially
00218     if (arc.contains(timing_arc_name) && arc[timing_arc_name].contains("values")) {
00219         std::vector<std::vector<double>> value_matrix;
00220         // Generate a matrix of values from the JSON array
00221         for (const auto &row_val : arc[timing_arc_name]["values"]) {
00222             std::vector<double> row_data;
00223             // Check if the value is an array
00224             if (!row_val.is_array()) {
00225                 logger_>error("Invalid format: '{}' values should be an array of arrays in cell '{}', pin "
00226                     "'{}', related_pin '{}'",
00227                     timing_arc_name, cell["cell_name"].get<std::string>(),
00228                     pin["pin_name"].get<std::string>(), arc["related_pin"].get<std::string>());
00229                 return false;
00230             }
00231             // Check for non-numeric values
00232             for (const auto &val : row_val) {
00233                 if (!val.is_number()) {

```

```

00234         row_data.push_back(val.get<double>());
00235     } else {
00236         logger_>warn("Non-numeric value found in '{}'.values, skipping value: {} in cell '{}', "
00237             "pin '{}', related_pin '{}'",
00238             timing_arc_name, val.dump(), cell["cell_name"].get<std::string>(),
00239             pin["pin_name"].get<std::string>(), arc["related_pin"].get<std::string>());
00240     }
00241 }
00242 value_matrix.push_back(row_data);
00243 }
00244 // Check the matrix for monotonicity
00245 if (!value_matrix.empty() && !value_matrix[0].empty()) {
00246     // Check the monotonic incrementality of rows (by output load capacitance, index 2)
00247     for (size_t i = 0; i < value_matrix.size(); ++i) {
00248         for (size_t j = 1; j < value_matrix[i].size(); ++j) {
00249             if ((value_matrix[i][j] < value_matrix[i][j - 1] &&
00250                 pin["pin_name"] != arc["related_pin"]) ||
00251                 (value_matrix[i][j] == 0 && value_matrix[i][j - 1] == 0)) {
00252                 // If contains "when" key, log the when value
00253                 if (arc.contains("when") && !arc["when"].get<std::string>().empty()) {
00254                     logger_>warn("Non-monotonic (by load) '{}' values: ({} , {}) {} < ({} , {}) {} "
00255                         "for Cell: {} Pin: {}->{} when: \"{}\\\"",
00256                         timing_arc_name, i, j, value_matrix[i][j], i, j - 1,
00257                         value_matrix[i][j - 1], cell["cell_name"].get<std::string>(),
00258                         arc["related_pin"].get<std::string>(),
00259                         pin["pin_name"].get<std::string>(), arc["when"].get<std::string>());
00260                 } else {
00261                     logger_>warn("Non-monotonic (by load) '{}' values: ({} , {}) {} < ({} , {}) {} "
00262                         "for Cell: {} Pin: {}->{}",
00263                         timing_arc_name, i, j, value_matrix[i][j], i, j - 1,
00264                         value_matrix[i][j - 1], cell["cell_name"].get<std::string>(),
00265                         arc["related_pin"].get<std::string>(),
00266                         pin["pin_name"].get<std::string>());
00267                 }
00268                 is_monotonic = false;
00269             }
00270         }
00271     }
00272     if (is_slew) {
00273         // Check the monotonic incrementality of columns (by input slew, index 1)
00274         for (size_t j = 0; j < value_matrix[0].size(); ++j) {
00275             for (size_t i = 1; i < value_matrix.size(); ++i) {
00276                 if ((value_matrix[i][j] < value_matrix[i - 1][j] &&
00277                     pin["pin_name"] != arc["related_pin"]) ||
00278                     (value_matrix[i][j] == 0 && value_matrix[i - 1][j] == 0)) {
00279                     // If contains "when" key, log the when value
00280                     if (arc.contains("when") && !arc["when"].get<std::string>().empty()) {
00281                         logger_>warn("Non-monotonic (by slew) '{}' values: ({} , {}) {} < ({} , {}) {} "
00282                             "for Cell: {} Pin: {}->{} when: \"{}\\\"",
00283                             timing_arc_name, i, j, value_matrix[i][j], i - 1, j,
00284                             value_matrix[i - 1][j], cell["cell_name"].get<std::string>(),
00285                             arc["related_pin"].get<std::string>(),
00286                             pin["pin_name"].get<std::string>(), arc["when"].get<std::string>());
00287                     } else {
00288                         logger_>warn("Non-monotonic (by slew) '{}' values: ({} , {}) {} < ({} , {}) {} "
00289                             "for Cell: {} Pin: {}->{}",
00290                             timing_arc_name, i, j, value_matrix[i][j], i - 1, j,
00291                             value_matrix[i - 1][j], cell["cell_name"].get<std::string>(),
00292                             arc["related_pin"].get<std::string>(),
00293                             pin["pin_name"].get<std::string>());
00294                     }
00295                     is_monotonic = false;
00296                 }
00297             }
00298         }
00299     }
00300 } else {
00301     logger_>warn(
00302         "Empty or invalid 'values' array found for cell: '{}', pin: '{}', related_pin: '{}', "

```

```

00303         "timing_arc: '{}'",
00304         cell["cell_name"].get<std::string>(), pin["pin_name"].get<std::string>(),
00305         arc["related_pin"].get<std::string>(), timing_arc_name);
00306     }
00307 }
00308 return is_monotonic;
00309 }
00310
00331 void LibFile::mono(const bool is_slew) {
00332     /*
00333     * Use this command to check the following data in the current library to ensure
00334     * the tables are monotonically increasing with respect to output load:
00335     * cell_rise retaining_rise
00336     * cell_fall retaining_fall
00337     * rise_transition retain_rise_slew
00338     * fall_transition retain_fall_slew
00339     * mpw
00340     */
00341     logger->info("Monotonicity check of '{}' starting ...", filename_);
00342
00343     if (!std::filesystem::exists(jsonname_)) {
00344         logger->info("JSON file not found. Parsing Liberty file first.");
00345         this->parse();
00346         this->writeJsonToFile();
00347     } else {
00348         // Read the JSON file into json object
00349         std::ifstream in(jsonname_);
00350         if (!in.is_open()) {
00351             logger->error("Could not open file '{}' for reading", jsonname_);
00352             return;
00353         }
00354         try {
00355             lib_json_ = json::parse(in);
00356         } catch (const json::parse_error &e) {
00357             logger->error("JSON parsing error in file '{}': {}", jsonname_, e.what());
00358             in.close();
00359             return;
00360         }
00361         in.close();
00362     }
00363
00364     std::map<std::string, bool> cell_monotonicity_status; // Track pass/fail status for each cell
00365     std::vector<std::string> failed_cells;               // List of failed cell names
00366     int total_cells = 0;
00367     int passed_cells = 0;
00368
00369     // Check the monotonicity of delay values
00370     // The index 1 (time values) must be monotonically increasing and >= 0
00371     for (const auto &cell : lib_json_["cells"]) {
00372         total_cells++;
00373         bool cell_is_monotonic = true; // Assume cell is monotonic initially
00374         std::string cell_name = cell["cell_name"].get<std::string>();
00375         cell_monotonicity_status[cell_name] =
00376             true; // Initialize to pass, will be set to false if any check fails
00377
00378         if (cell.contains("output_pins")) {
00379             for (const auto &pin : cell["output_pins"]) {
00380                 if (pin.contains("timing_arcs")) {
00381                     for (const auto &arc : pin["timing_arcs"]) {
00382                         // Check 4 timing arc names and accumulate monotonicity status
00383                         for (const auto &name :
00384                             {"cell_rise", "cell_fall", "rise_transition", "fall_transition"}) {
00385                             if (!checkTimingArcMonotonicity(cell, pin, arc, name, is_slew)) {
00386                                 cell_is_monotonic = false;
00387                             }
00388                         }
00389                     }
00390                 }
00391             }

```

```

00392     }
00393     if (cell.contains("input_pins")) {
00394         for (const auto &pin : cell["input_pins"]) {
00395             if (!pin.contains("clock")) {
00396                 continue; // Skip non-clock pins
00397             }
00398             // logger->debug("Clock pin: {}", pin["pin_name"].get<std::string>());
00399             if (pin.contains("timing_arcs")) {
00400                 for (const auto &arc : pin["timing_arcs"]) {
00401                     if (arc.contains("timing_type")) {
00402                         // logger->debug("Timing type: {}", arc["timing_type"].get<std::string>());
00403                         if (arc["timing_type"].get<std::string>() == "min_pulse_width") {
00404                             for (const auto &name : {"rise_constraint", "fall_constraint"}) {
00405                                 if (!checkTimingArcMonotonicity(cell, pin, arc, name, is_slew)) {
00406                                     cell_is_monotonic = false;
00407                                 }
00408                             }
00409                         }
00410                     }
00411                 }
00412             }
00413         }
00414     }
00415     // Update cell status based on all checks
00416     cell_monotonicity_status[cell_name] = cell_is_monotonic;
00417     if (cell_is_monotonic) {
00418         passed_cells++;
00419     } else {
00420         failed_cells.push_back(cell_name);
00421     }
00422 }
00423 logger->info("Monotonicity check of '{}' completed.", filename_);
00424
00425 // Output summary statistics
00426 logger->info("validate_monotonicity : {} out of {} cells passed", passed_cells, total_cells);
00427 logger->info("validate_monotonicity : {} out of {} cells failed", total_cells - passed_cells,
00428             total_cells);
00429 if (!failed_cells.empty()) {
00430     std::stringstream failed_cells_ss;
00431     for (size_t i = 0; i < failed_cells.size(); ++i) {
00432         failed_cells_ss << failed_cells[i];
00433         if (i < failed_cells.size() - 1) {
00434             failed_cells_ss << ", "; // Add comma if not the last element
00435         }
00436     }
00437     logger->info("Failed cell list : {}", failed_cells_ss.str());
00438 }
00439 }
00440
00440 void LibFile::supercell(const int chain_length, const std::vector<std::string> &cell_names) {
00441     /*
00442     * Supercells are named as follows:
00443     * <cellname>_X<chain_length>_<input_pin>_<output_pin>
00444     */
00445     logger->info("Creating supercells for '{}'", filename_);
00446
00447     if (!std::filesystem::exists(jsonname_)) {
00448         logger->info("JSON file not found. Parsing Liberty file first.");
00449         this->parse();
00450         this->writeJsonToFile();
00451     } else {
00452         // Read the JSON file into json object
00453         std::ifstream in(jsonname_);
00454         if (!in.is_open()) {
00455             logger->error("Could not open file '{}' for reading", jsonname_);
00456             return;
00457         }
00458         try {
00459             lib_json_ = json::parse(in);

```

```

00480     } catch (const json::parse_error &e) {
00481         logger->error("JSON parsing error in file '{}': {}", jsonname_, e.what());
00482         in.close();
00483         return;
00484     }
00485     in.close();
00486 }
00487
00488 // write to .map file
00489 std::ofstream out(basename_ + ".map");
00490 if (!out.is_open()) {
00491     logger->error("Could not open file '{}' for writing", basename_ + ".map");
00492     return;
00493 }
00494 // if cell_names is empty, create supercells for all cells
00495 // else create supercells for only the specified cells
00496
00497 // Check if we're processing specific cells or all cells
00498 bool process_all_cells = cell_names.empty();
00499
00500 if (process_all_cells) {
00501     logger->info("Creating supercells for ALL cells in '{}'", filename_);
00502 } else {
00503     logger->info("Creating supercells for {} specified cells in '{}'", cell_names.size(),
00504                 filename_);
00505 }
00506
00507 // Create a set for faster lookups if we have specific cell names
00508 std::unordered_set<std::string> cell_set;
00509 std::unordered_set<std::string> found_cells;
00510 if (!process_all_cells) {
00511     cell_set.insert(cell_names.begin(), cell_names.end());
00512 }
00513
00514 for (const auto &cell : lib_json_["cells"]) {
00515     bool is_sequential = false;
00516     std::string cell_name = cell["cell_name"].get<std::string>();
00517
00518     // Skip cells not in the specified list
00519     if (!process_all_cells && cell_set.find(cell_name) == cell_set.end()) {
00520         continue;
00521     }
00522
00523     // Mark this cell as found
00524     if (!process_all_cells) {
00525         found_cells.insert(cell_name);
00526     }
00527
00528     logger->debug("Creating supercells for cell: '{}'", cell_name);
00529
00530     std::vector<std::string> output_pins;
00531     std::vector<std::string> input_pins;
00532     // Extract input and output pins
00533     if (cell.contains("output_pins")) {
00534         for (const auto &pin : cell["output_pins"]) {
00535             output_pins.push_back(pin["pin_name"].get<std::string>());
00536         }
00537     }
00538     if (cell.contains("input_pins")) {
00539         for (const auto &pin : cell["input_pins"]) {
00540             if (pin.contains("clock")) {
00541                 is_sequential = true;
00542                 // clock pin not use for supercell
00543                 continue;
00544             }
00545             input_pins.push_back(pin["pin_name"].get<std::string>());
00546         }
00547     }
00548 }

```

```

00549 // create supercells for all combinations of input and output pins
00550 for (const auto &output_pin : output_pins) {
00551     for (const auto &input_pin : input_pins) {
00552         if (is_sequential) {
00553             const int chain_len = 1;
00554             std::string supercell_name =
00555                 cell_name + "__X" + std::to_string(chain_len) + "__" + input_pin + "__" + output_pin;
00556             out << cell_name << " " << supercell_name << std::endl;
00557         } else {
00558             std::string supercell_name = cell_name + "__X" + std::to_string(chain_length) + "__" +
00559                 input_pin + "__" + output_pin;
00560             out << cell_name << " " << supercell_name << std::endl;
00561         }
00562     }
00563 }
00564 }
00565
00566 // Check for cells that were specified but not found
00567 if (!process_all_cells) {
00568     std::vector<std::string> not_found_cells;
00569     for (const auto &requested_cell : cell_names) {
00570         if (found_cells.find(requested_cell) == found_cells.end()) {
00571             not_found_cells.push_back(requested_cell);
00572         }
00573     }
00574
00575     // Output warning for cells that weren't found
00576     if (!not_found_cells.empty()) {
00577         std::string missing_cells = not_found_cells[0];
00578         for (size_t i = 1; i < not_found_cells.size(); ++i) {
00579             missing_cells += ", " + not_found_cells[i];
00580         }
00581         logger->warn("{} specified cell{} not found in the library: {}", not_found_cells.size(),
00582             not_found_cells.size() > 1 ? "s were" : " was", missing_cells);
00583     }
00584 }
00585
00586 out.close();
00587 logger->info("Supercell creation complete in '{}', basenane_ + ".map");
00588 }
00589
00614 void LibFile::verilog(const int chain_length, const std::vector<std::string> &cell_names) {
00615     logger->info("Creating Verilog for '{}', filename_);
00616
00617     this->supercell(chain_length, cell_names);
00618
00619     // Create a temporary file for individual modules
00620     std::string temp_file = basenane_ + "_temp.v";
00621     std::ofstream out(temp_file);
00622     if (!out.is_open()) {
00623         logger->error("Could not open temp file '{}' for writing", temp_file);
00624         return;
00625     }
00626
00627     // Read the .map file into a vector to preserve all entries and their order
00628     std::vector<std::pair<std::string, std::string>> supercell_entries;
00629     std::ifstream in(basenane_ + ".map");
00630     if (!in.is_open()) {
00631         logger->error("Could not open file '{}' for reading", basenane_ + ".map");
00632         return;
00633     }
00634
00635     std::string line;
00636     while (std::getline(in, line)) {
00637         std::istringstream iss(line);
00638         std::string cell_name, supercell_name;
00639         if (!(iss >> cell_name >> supercell_name)) {
00640             logger->warn("Invalid line format in map file: '{}', line);
00641             continue;

```

```

00642     }
00643     supercell_entries.push_back({cell_name, supercell_name});
00644 }
00645 in.close();
00646 logger_>info("Read {} supercells from '{}'", supercell_entries.size(), basename_ + ".map");
00647
00648 // Store module information for top-level creation
00649 std::vector<std::string> module_names;
00650 std::map<std::string, std::vector<std::string>> module_inputs;
00651 std::map<std::string, std::vector<std::string>> module_outputs;
00652
00653 // Generate verilog module for each supercell
00654 for (const auto &supercell_entry : supercell_entries) {
00655     // Check if the cell is sequential
00656     bool is_sequential = false;
00657     // Count the number of instances will be created in verilog
00658     int instance_count = 0;
00659     // Get original cell name from pair
00660     const std::string &cell_name = supercell_entry.first;
00661     // Get supercell name(as well as module name) from pair
00662     const std::string &module_name = supercell_entry.second;
00663
00664     // Store module name for top-level creation
00665     module_names.push_back(module_name);
00666
00667     logger_>info("Creating Module: '{}' from Cell '{}'", module_name, cell_name);
00668
00669     // Get the cell JSON object
00670     json cell_json;
00671     for (const auto &cell : lib_json_["cells"]) {
00672         if (cell["cell_name"].get<std::string>() == cell_name) {
00673             cell_json = cell;
00674             break;
00675         }
00676     }
00677
00678     // Get input/output pins from the cell JSON object
00679     std::vector<std::string> input_pins;
00680     std::vector<std::string> output_pins;
00681     std::stringstream input_pins_ss;
00682     std::stringstream output_pins_ss;
00683     if (cell_json.contains("input_pins")) {
00684         for (const auto &pin : cell_json["input_pins"]) {
00685             if (pin.contains("clock")) {
00686                 is_sequential = true;
00687             }
00688             std::string pin_name = pin["pin_name"].get<std::string>();
00689             input_pins.push_back(pin_name);
00690             input_pins_ss << pin_name << " ";
00691
00692             // Store input pin name for top-level creation
00693             module_inputs[module_name].push_back(pin_name);
00694         }
00695     }
00696     if (cell_json.contains("output_pins")) {
00697         for (const auto &pin : cell_json["output_pins"]) {
00698             std::string pin_name = pin["pin_name"].get<std::string>();
00699             output_pins.push_back(pin_name);
00700             output_pins_ss << pin_name << " ";
00701
00702             // Store output pin name for top-level creation
00703             module_outputs[module_name].push_back(pin_name);
00704         }
00705     }
00706     logger_>debug("Input pins set: {}", input_pins_ss.str());
00707     logger_>debug("Output pins set: {}", output_pins_ss.str());
00708
00709     // Sequential cells will have only one instance
00710     // Combinational cells will have instances equal to chain length

```

```

00711     if (is_sequential) {
00712         instance_count = 1;
00713     } else {
00714         instance_count = chain_length;
00715     }
00716
00717     // Create ANSI port list
00718     std::string port_list_text = "(";
00719     // Add input ports
00720     bool isFirstPort = true;
00721     for (const auto &input_pin : input_pins) {
00722         if (!isFirstPort) {
00723             port_list_text += ", ";
00724         }
00725         port_list_text += "input " + input_pin;
00726         isFirstPort = false;
00727     }
00728     // Add output ports
00729     for (const auto &output_pin : output_pins) {
00730         if (!isFirstPort) {
00731             port_list_text += ", ";
00732         }
00733         port_list_text += "output " + output_pin;
00734         isFirstPort = false;
00735     }
00736     port_list_text += ")";
00737     logger->debug("ANSI Port list: {}", port_list_text);
00738
00739     // Create full module header text
00740     std::string fullModuleText = "module " + module_name + port_list_text + ";\nendmodule";
00741     logger->debug("Full module text: {}", fullModuleText);
00742
00743     // Generate the syntax tree for the module using slang library
00744     auto tree = slang::syntax::SyntaxTree::fromText(fullModuleText);
00745     if (tree) {
00746         // Add ports to the syntax tree
00747         ModuleRewriter rewriter(input_pins, output_pins, supercell_entry, instance_count, logger);
00748
00749         tree = rewriter.transform(tree);
00750         // Revisit the syntax tree to check architecture
00751         rewriter.visit(tree->root());
00752
00753         // Output the transformed syntax tree
00754         out << "timescale 1ns/10ps" << std::endl;
00755         out << slang::syntax::SyntaxPrinter::printFile(*tree) << std::endl << std::endl;
00756     } else {
00757         logger->error("Failed to create syntax tree for module '{}'", module_name);
00758         continue;
00759     }
00760 }
00761
00762 out.close();
00763
00764 // Create the top-level module
00765 std::ofstream final_out(basename_ + ".v");
00766 if (!final_out.is_open()) {
00767     logger->error("Could not open file '{}' for writing", basename_ + ".v");
00768     return;
00769 }
00770
00771 // Collect all port names separately for inputs and outputs
00772 std::vector<std::string> all_input_ports;
00773 std::vector<std::string> all_output_ports;
00774 std::stringstream top_instances;
00775
00776 // First pass: collect all port names
00777 for (const auto &module_name : module_names) {
00778     // Collect input port names
00779     for (const auto &pin : module_inputs[module_name]) {

```



```

00780     all_input_ports.push_back(module_name + "__" + pin);
00781 }
00782
00783 // Collect output port names
00784 for (const auto &pin : module_outputs[module_name]) {
00785     all_output_ports.push_back(module_name + "__" + pin);
00786 }
00787
00788 // Create instance
00789 std::string instance_name = "I_" + module_name.substr(0, module_name.find("__X"));
00790 for (size_t i = module_name.find("__X") + 3; i < module_name.length(); i++) {
00791     if (module_name[i] == '_' && module_name[i + 1] == '_') {
00792         instance_name += "__" + module_name.substr(i + 2);
00793         break;
00794     }
00795 }
00796
00797 // Generate port connections for this instance
00798 std::stringstream instance_ports;
00799 for (const auto &pin : module_inputs[module_name]) {
00800     instance_ports << "." << pin << "(" << module_name << "__" << pin << ")", ";";
00801 }
00802 for (const auto &pin : module_outputs[module_name]) {
00803     instance_ports << "." << pin << "(" << module_name << "__" << pin << ")", ";";
00804 }
00805
00806 // Remove last comma and space
00807 std::string instance_ports_str = instance_ports.str();
00808 if (!instance_ports_str.empty()) {
00809     instance_ports_str = instance_ports_str.substr(0, instance_ports_str.length() - 2);
00810 }
00811
00812 top_instances << " " << module_name << " " << instance_name << " (" << instance_ports_str
00813     << ");\n";
00814 }
00815
00816 // Now generate the port list with all inputs first, then all outputs
00817 std::stringstream top_ports;
00818
00819 // Add all input ports first
00820 for (size_t i = 0; i < all_input_ports.size(); ++i) {
00821     top_ports << all_input_ports[i];
00822     if (i < all_input_ports.size() - 1 || !all_output_ports.empty()) {
00823         top_ports << ", ";
00824     }
00825 }
00826
00827 // Then add all output ports
00828 for (size_t i = 0; i < all_output_ports.size(); ++i) {
00829     top_ports << all_output_ports[i];
00830     if (i < all_output_ports.size() - 1) {
00831         top_ports << ", ";
00832     }
00833 }
00834
00835 // Generate input and output declarations
00836 std::stringstream top_inputs;
00837 std::stringstream top_outputs;
00838
00839 // Generate input declarations
00840 for (const auto &port : all_input_ports) {
00841     top_inputs << " input " << port << ";\n";
00842 }
00843
00844 // Generate output declarations
00845 for (const auto &port : all_output_ports) {
00846     top_outputs << " output " << port << ";\n";
00847 }
00848

```

```

00849 // Write the top module
00850 final_out << "`timescale 1ns/10ps" << std::endl;
00851 final_out << "module validate_top ( " << top_ports.str() << " );" << std::endl;
00852 final_out << std::endl; // Add newline before port declarations
00853 final_out << top_inputs.str();
00854 final_out << top_outputs.str();
00855 final_out << std::endl;
00856 final_out << top_instances.str();
00857 final_out << "endmodule" << std::endl << std::endl;
00858
00859 // Append the module definitions from the temporary file
00860 std::ifstream temp_in(temp_file);
00861 if (temp_in.is_open()) {
00862     final_out << temp_in.rdbuf();
00863     temp_in.close();
00864 } else {
00865     logger_>error("Could not open temp file '{}' for reading", temp_file);
00866 }
00867
00868 final_out.close();
00869
00870 // Remove temporary file
00871 // if (std::filesystem::exists(temp_file)) {
00872 //     std::filesystem::remove(temp_file);
00873 // }
00874
00875 logger_>info("Verilog creation complete in '{}', basename_ + ".v");
00876 }
00877
00878 void LibFile::spice(const int chain_length, const std::vector<std::string> &cell_names,
00879                   const std::string &verilog_lib_file, const std::string &spice_lib_file) {
00880     logger_>info("Creating SPICE for '{}', filename_);
00881
00882     this->verilog(chain_length, cell_names);
00883
00884     // Read netlist from temporary verilog file
00885     std::ifstream in(basename_ + "_temp.v");
00886     if (!in.is_open()) {
00887         logger_>error("Could not open file {} for reading", basename_ + "_temp.v");
00888         return;
00889     }
00890
00891     // Check if V2LVS tool is available by typing "which v2lvs"
00892     std::string v2lvs_cmd = "which v2lvs";
00893     std::string result;
00894     std::array<char, 128> buffer;
00895     std::unique_ptr<FILE, decltype(&pclose)> pipe(popen(v2lvs_cmd.c_str(), "r"), pclose);
00896     if (!pipe) {
00897         logger_>error("Failed to run command: {}", v2lvs_cmd);
00898         return;
00899     }
00900     while (fgets(buffer.data(), buffer.size(), pipe.get()) != nullptr) {
00901         result += buffer.data();
00902     }
00903     if (result.empty()) {
00904         logger_>error("V2LVS tool not found. Please install it to use this feature.");
00905         return;
00906     } else {
00907         // Log the path of the V2LVS tool
00908         logger_>info("V2LVS tool found at: {}", result);
00909     }
00910
00911     // Run V2LVS command
00912     logger_>info("Running V2LVS to generate SPICE netlist...");
00913     std::string v2lvs_cmd_full = "v2lvs -v " + basename_ + "_temp.v -o " + basename_ + ".spi";
00914     // Add more options for V2LVS
00915     // TODO
00916     // Add verilog and spice library files
00917     v2lvs_cmd_full += " -l " + verilog_lib_file + " -s " + spice_lib_file;

```

```

00918 // Specifies that calls to subcircuits with pins be done in order according to traditional Spice
00919 // rather than with $PINS.
00920 v2lvs_cmd_full += " -i";
00921 int ret = system(v2lvs_cmd_full.c_str());
00922 if (ret != 0) {
00923     logger->error("Failed to run command: {}", v2lvs_cmd_full);
00924     return;
00925 }
00926
00927 // Add T-port connections
00928
00929 // Final
00930 logger->info("SPICE generation complete in '{}', basename_ + ".spi");
00931 }
00932
00950 std::map<std::string, std::string> LibFile::logic(const std::string &cell_name) {
00951     std::map<std::string, std::string> logic_map = {};
00952
00953     logger->info("Getting output pin logic function for '{}', filename_);
00954
00955     if (!std::filesystem::exists(jsonname_)) {
00956         logger->info("JSON file not found. Parsing Liberty file first.");
00957         this->parse();
00958         this->writeJsonToFile();
00959     } else {
00960         // Read the JSON file into json object
00961         std::ifstream in(jsonname_);
00962         if (!in.is_open()) {
00963             logger->error("Could not open file '{}' for reading", jsonname_);
00964             return logic_map;
00965         }
00966         try {
00967             lib_json_ = json::parse(in);
00968         } catch (const json::parse_error &e) {
00969             logger->error("JSON parsing error in file '{}': {}", jsonname_, e.what());
00970             in.close();
00971             return logic_map;
00972         }
00973         in.close();
00974     }
00975
00976     // Check if the cell name exists in the JSON object
00977     for (const auto &cell : lib_json_["cells"]) {
00978         if (cell["cell_name"].get<std::string>() == cell_name) {
00979             logger->debug("Found cell: '{}', cell_name);
00980
00981             // Tranverse all output pins
00982             if (cell.contains("output_pins")) {
00983                 for (const auto &pin : cell["output_pins"]) {
00984                     std::string pin_name = pin["pin_name"].get<std::string>();
00985                     // Check if the pin has a "timing_arcs" attribute
00986                     if (pin.contains("function")) {
00987                         std::string function = pin["function"].get<std::string>();
00988                         logic_map[pin_name] = function;
00989                         logger->debug("Pin: '{}' Logic Function: '{}', pin_name, function);
00990                     } else {
00991                         logger->warn("Pin: '{}' does not have a 'function' attribute", pin_name);
00992                     }
00993                 }
00994             }
00995         }
00996     }
00997     if (logic_map.empty()) {
00998         logger->warn("No logic functions found for cell: '{}', cell_name);
00999     } else {
01000         logger->info("Found {} Logic functions for cell: '{}', logic_map.size(), cell_name);
01001         // Print the logic functions
01002         for (const auto &pair : logic_map) {
01003             logger->info("Pin -> Logic Function: {} -> {}", pair.first, pair.second);

```


6.32.1 Function Documentation

6.32.1.1 compareLibFiles()

```
void compareLibFiles (
    const std::string & ref_lib,
    const std::string & comp_lib,
    const double reltol,
    const double abstol,
    std::string & report_file_name )
```

Compares two library files and generates a detailed comparison report.

This function compares a reference library file with another library file, performing validation checks based on specified tolerance parameters. The comparison results are written to a report file in markdown or text format.

Parameters

<i>ref_lib</i>	Path to the reference library file to use as the baseline
<i>comp_lib</i>	Path to the library file to compare against the reference
<i>reltol</i>	Relative tolerance for numerical comparisons (must be ≥ 0.0)
<i>abstol</i>	Absolute tolerance for numerical comparisons
<i>report_file_name</i>	[in,out] Name of the file to write the comparison report to. If empty, defaults to [comp_lib_basename].cmp.md. If provided but doesn't end with .txt or .md, .md will be appended.

Note

The function will log an error and return without comparing if reltol is invalid.

Log files will be created with the naming pattern [library_basename].cmp.log

The function uses the [LibraryComparator](#) class to perform the actual comparison.

Definition at line 231 of file [LibFileOperations.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:

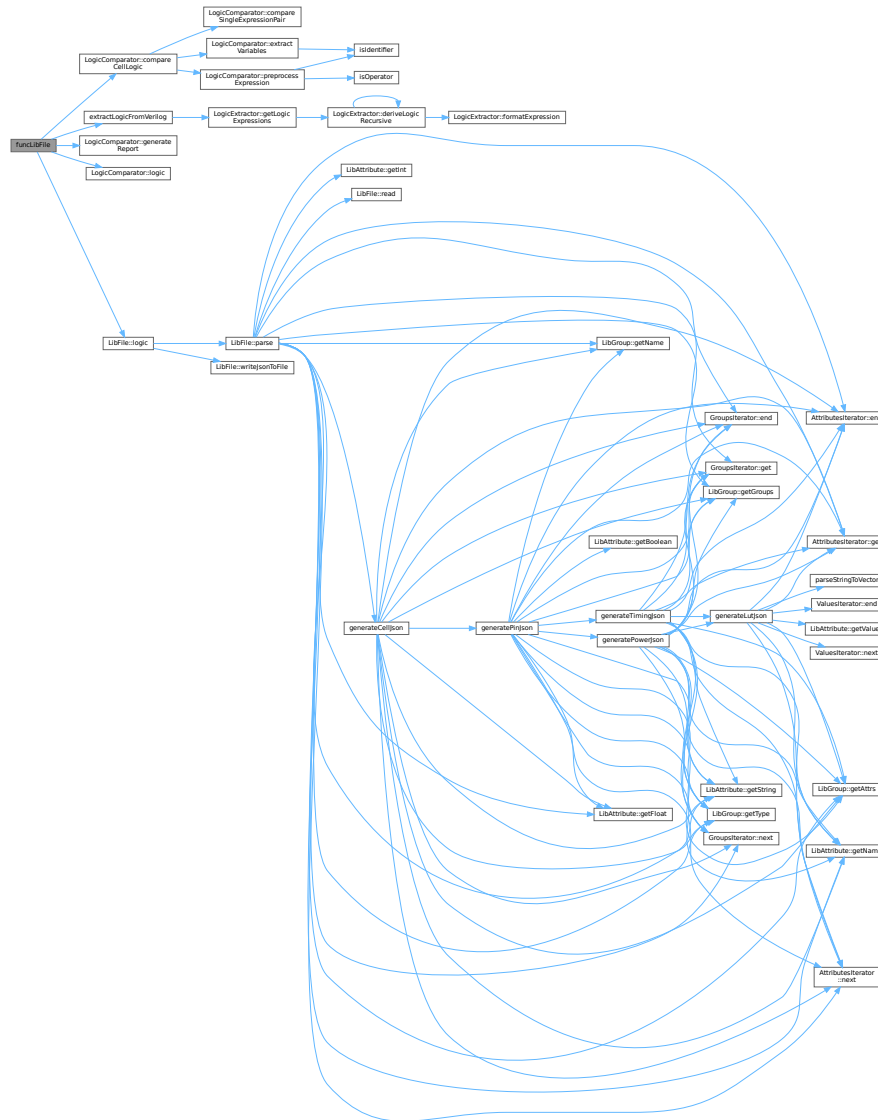


6.32.1.2 funcLibFile()

```
void funcLibFile (  
    const std::string & ref_file,  
    const std::string & comp_file,  
    const std::vector< std::string > & cell_names,  
    std::string & report_file_name )
```

Definition at line 259 of file [LibFileOperations.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



6.32.1.3 monoCheckLibFile()

```
void monoCheckLibFile (  
    const std::string & library_path,  
    const std::string log_file_name,  
    bool is_slew )
```

Performs monotonicity check on a library file.

This function validates the monotonicity of timing data in a library file. It creates a log file to record the results of the check and handles any exceptions that occur during the process.

Parameters

<i>library_path</i>	Path to the library file to check
<i>log_file_name</i>	Name of the log file to create (optional). If empty, a default name will be generated from the library file name
<i>is_slew</i>	Flag indicating whether to check input slew monotonicity (true) or output load monotonicity (false)

Exceptions

<i>The</i>	function catches and logs any exceptions but does not rethrow them
------------	--

Definition at line 81 of file [LibFileOperations.cpp](#).

[illegible]

```
graph LR; main --> monoCheckLibFile
```

6.32.1.4 parseLibFile()

```
void parseLibFile (  
    const std::string & library_path,  
    const std::string log_file_name )
```

Parses a library file and generates corresponding output.

This function processes the given library file, parsing its contents and generating a JSON output. It also logs the parsing process to a specified log file or creates a default log file if none is provided.

Parameters

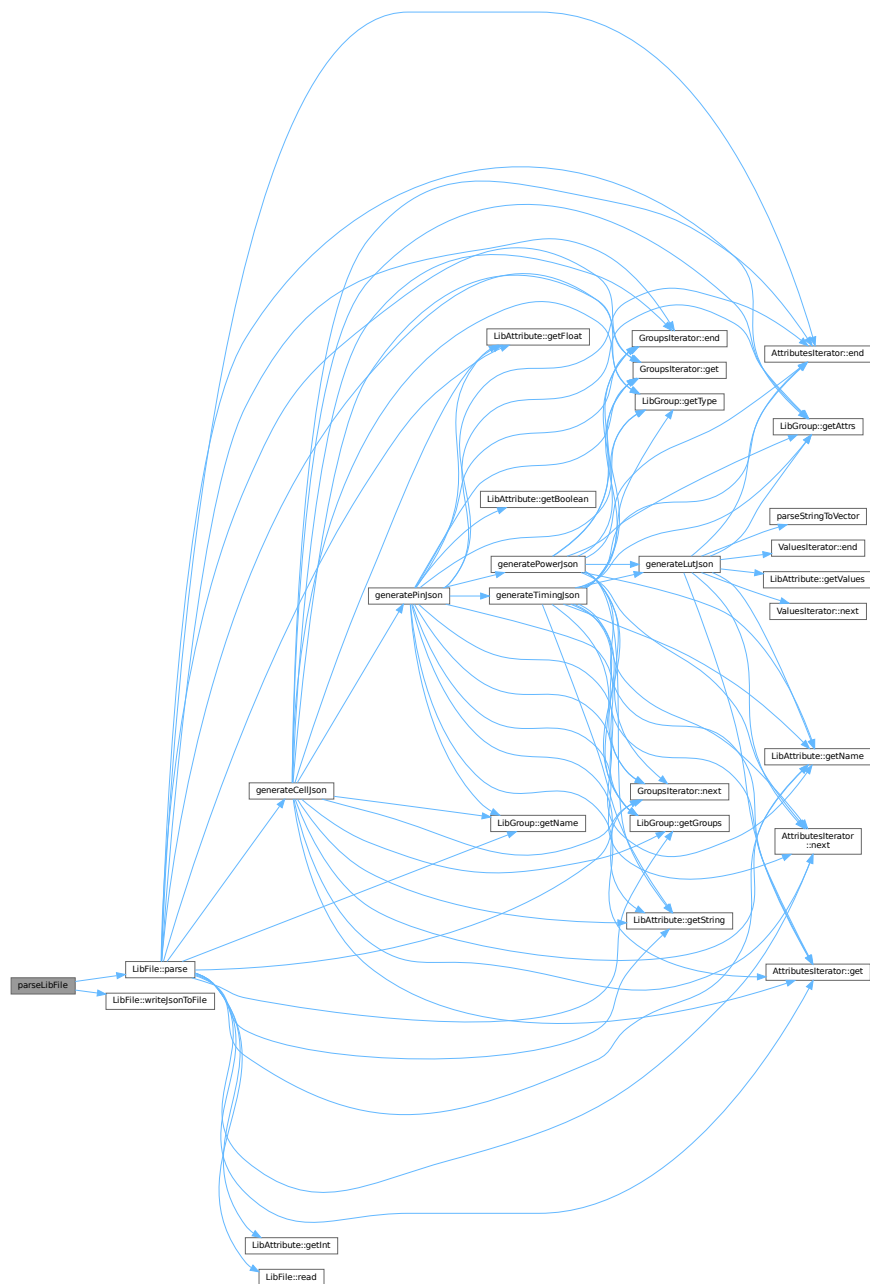
<i>library_path</i>	Path to the library file that needs to be parsed
<i>log_file_name</i>	Optional name for the log file. If empty, a default name is generated based on the library filename with ".parse.log" extension

Exceptions

<i>The</i>	function catches and logs any exceptions but doesn't propagate them
------------	---

Definition at line 49 of file [LibFileOperations.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



6.32.1.5 printInfo()

```
void printInfo ( )
```

Sets up and configures the global logger and prints application information.

This function performs the following operations:

1. Creates a console sink for logging with level set to INFO
2. Creates a file sink for logging with level set to DEBUG, saving to [APP_NAME].log
3. Configures a logger with both sinks and sets it as the default logger
4. Outputs basic application information:
 - Version and build timestamp
 - Author information
 - Log file location

Note

Uses spdlog library for logging functionality

Depends on APP_NAME, APP_VERSION, BUILD_TIMESTAMP, APP_AUTHOR, and APP↵_CONTACT macros

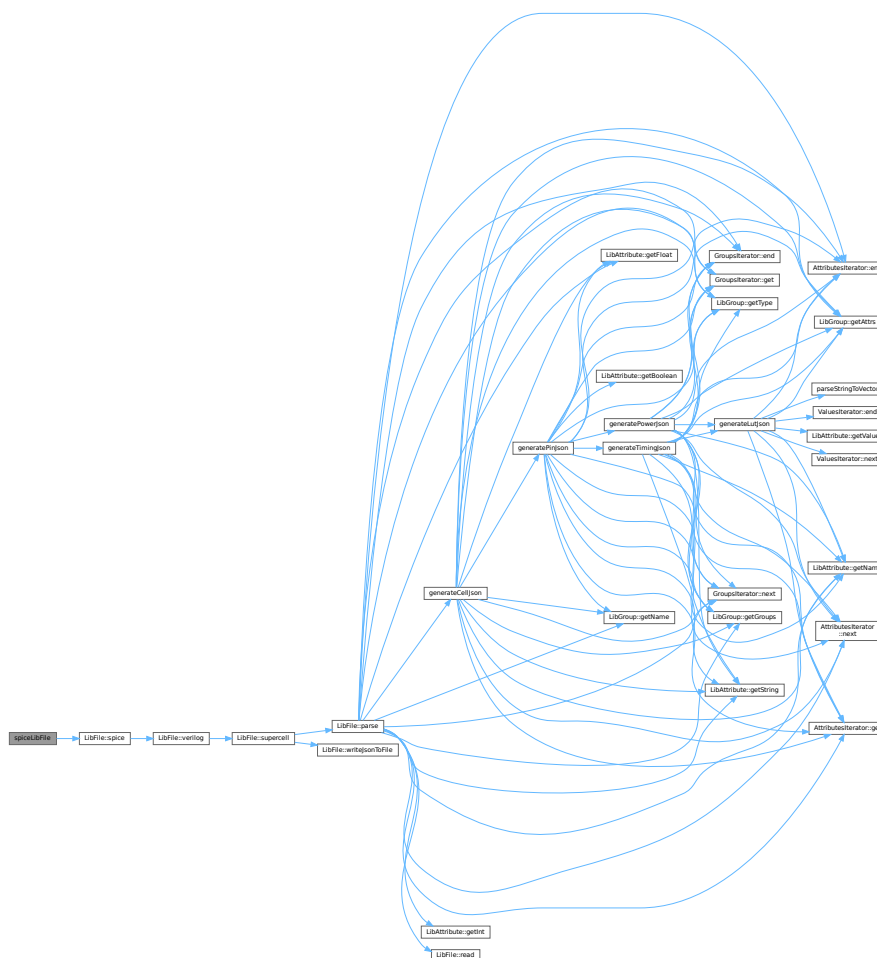
Definition at line 18 of file [LibFileOperations.cpp](#).

Here is the caller graph for this function:



```
void spiceLibFile (
    const std::string & library_path,
    const std::string & log_file_name,
    int chain_length,
    const std::vector< std::string > & cell_names,
    const std::string & verilog_lib_file,
    const std::string & spice_lib_file )
```

Here is the call graph for this function:



Here is the caller graph for this function:



6.32.1.7 supercellLibFile()

```

void supercellLibFile (
    const std::string & library_path,
    const std::string & log_file_name,
    int chain_length,
    const std::vector< std::string > & cell_names )
  
```

Creates supercell map structures from a Liberty library file.

This function reads a Liberty file, creates supercell structures based on the specified chain length and cell names, and logs the process to a file.

Parameters

<i>library_path</i>	Path to the Liberty file to process
<i>log_file_name</i>	Name of the log file (if empty, defaults to "[library_name].supercell.log")
<i>chain_length</i>	The length of chains to create (must be ≥ 1)
<i>cell_names</i>	Vector of cell names to process for supercell generation

Exceptions

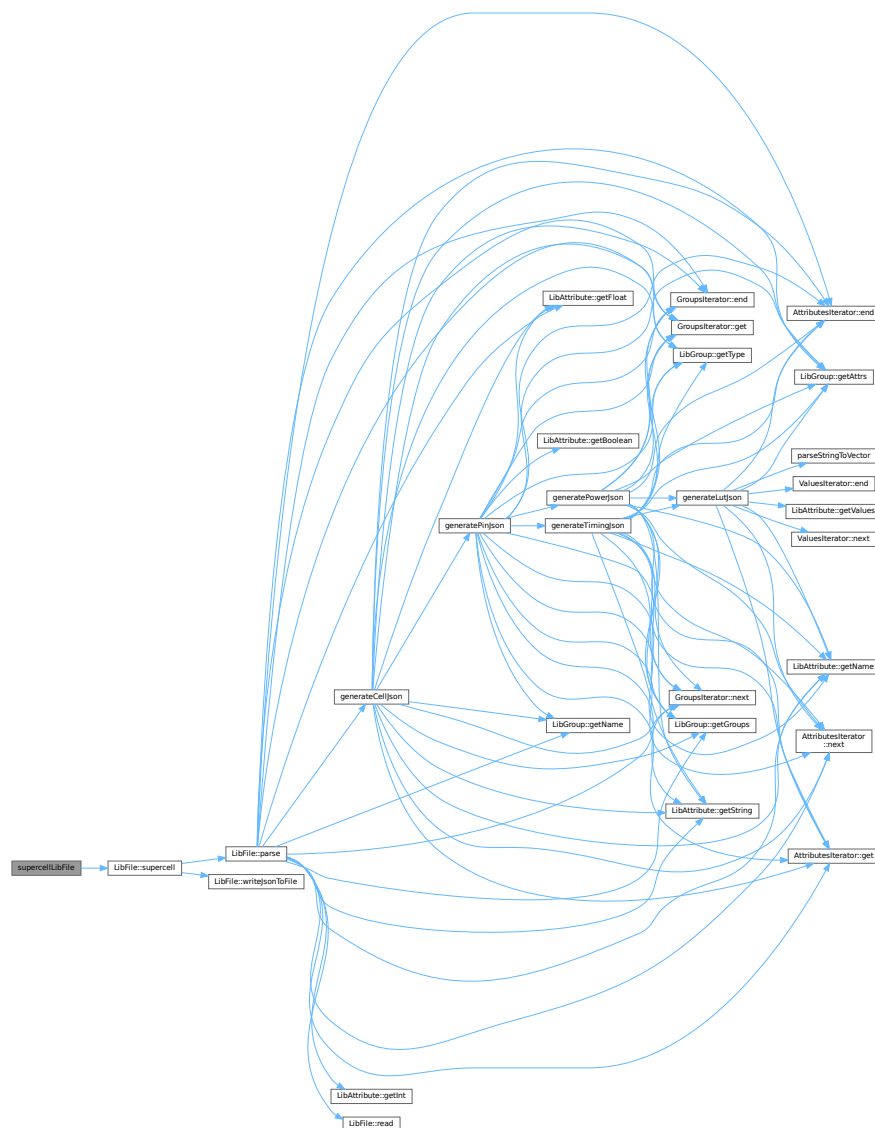
<i>May</i>	pass through exceptions from the LibFile::supercell method
------------	--

Note

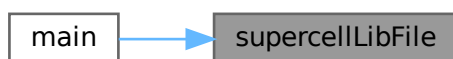
The function validates the chain length and logs all activities including errors that might occur during processing

Definition at line 116 of file [LibFileOperations.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



6.32.1.8 verilogLibFile()

```
void verilogLibFile (
    const std::string & library_path,
    const std::string & log_file_name,
    int chain_length,
    const std::vector< std::string > & cell_names )
```

Generates Verilog files from a library file for specified cells.

This function processes a library file and generates Verilog representation for the specified cell names with a given chain length. The operation results are logged to a specified or default log file.

Parameters

<i>library_path</i>	Path to the library file to process
<i>log_file_name</i>	Name for the log file (if empty, a default name will be generated)
<i>chain_length</i>	Number of cells to chain together, must be ≥ 1
<i>cell_names</i>	Vector of cell names to generate Verilog for

Exceptions

<i>Catches</i>	any exceptions from the verilog generation process and logs them
----------------	--

Definition at line 157 of file [LibFileOperations.cpp](#).


```

00001 #include "LibFileOperations.hpp"
00002
00018 void printInfo() {
00019     // Set Global Logger
00020     auto console_sink = std::make_shared<spdlog::sinks::stdout_color_sink_mt>();
00021     console_sink->set_level(spdlog::level::info);
00022     std::string app_name = APP_NAME;
00023     auto file_sink = std::make_shared<spdlog::sinks::basic_file_sink_mt>(app_name + ".log", true);
00024     file_sink->set_level(spdlog::level::trace);
00025
00026     std::vector<spdlog::sink_ptr> sinks{console_sink, file_sink};
00027     auto logger = std::make_shared<spdlog::logger>(APP_NAME, sinks.begin(), sinks.end());
00028     logger->set_level(spdlog::level::trace);
00029     spdlog::set_default_logger(logger);
00030
00031     spdlog::info("Version {}, Built: {}", APP_VERSION, BUILD_TIMESTAMP);
00032     spdlog::info("Author: {}, Email: {}", APP_AUTHOR, APP_CONTACT);
00033     spdlog::info("Global log file in: '{}'", app_name + ".log");
00034 }
00035
00049 void parseLibFile(const std::string &library_path, const std::string log_file_name) {
00050     std::string logname = log_file_name.empty()
00051         ? std::filesystem::path(library_path).stem().string() + ".parse.log"
00052         : log_file_name;
00053     LibFile libfile(library_path, logname);
00054
00055     libfile.logger->info("Starting parse for file: '{} ...", library_path);
00056
00057     try {
00058         libfile.parse();
00059         libfile.writeJsonToFile();
00060         libfile.logger->info("Successfully parsed file: '{}'", library_path);
00061     } catch (const std::exception &e) {
00062         libfile.logger->error("Error parsing file '{}': {}", library_path, e.what());
00063     }
00064 }
00065
00081 void monoCheckLibFile(const std::string &library_path, const std::string log_file_name,
00082     bool is_slew) {
00083     std::string logname = log_file_name.empty()
00084         ? std::filesystem::path(library_path).stem().string() + ".mono.log"
00085         : log_file_name;
00086     LibFile libfile(library_path, logname);
00087
00088     libfile.logger->info("Starting monotonicity check for file: '{}', input slew: {}", library_path,
00089         is_slew);
00090
00091     try {
00092         libfile.mono(is_slew);
00093         libfile.logger->info("Successfully completed monotonicity check for file: '{}'", library_path);
00094     } catch (const std::exception &e) {
00095         libfile.logger->error("Error during monotonicity check for file '{}': {}", library_path,
00096             e.what());
00097     }
00098 }
00099
00116 void supercellLibFile(const std::string &library_path, const std::string &log_file_name,
00117     int chain_length, const std::vector<std::string> &cell_names) {
00118     std::string logname = log_file_name.empty()
00119         ? std::filesystem::path(library_path).stem().string() + ".supercell.log"
00120         : log_file_name;
00121     LibFile libfile(library_path, logname);
00122
00123     // Check chain length validity
00124     if (chain_length < 1) {
00125         libfile.logger->error("Invalid chain length: {}. Chain length must be >= 1.", chain_length);
00126         return;
00127     }
00128     std::stringstream ss;

```

```

00129     for (const auto &cell : cell_names) {
00130         ss << cell << ", ";
00131     }
00132     logfile.logger->info("Starting supercell generation for file: '{}', chain length: {}, cells: {}",
00133         library_path, chain_length, ss.str());
00134     try {
00135         logfile.supercell(chain_length, cell_names);
00136         logfile.logger->info("Successfully generated supercells for file: '{}', library_path);
00137     } catch (const std::exception &e) {
00138         logfile.logger->error("Error during supercell generation for file '{}': {}", library_path,
00139             e.what());
00140     }
00141 }
00142
00157 void verilogLibFile(const std::string &library_path, const std::string &log_file_name,
00158     int chain_length, const std::vector<std::string> &cell_names) {
00159     std::string logname = log_file_name.empty()
00160         ? std::filesystem::path(library_path).stem().string() + ".verilog.log"
00161         : log_file_name;
00162     LibFile logfile(library_path, logname);
00163
00164     // Check chain length validity
00165     if (chain_length < 1) {
00166         logfile.logger->error("Invalid chain length: {}. Chain length must be >= 1.", chain_length);
00167         return;
00168     }
00169     std::stringstream ss;
00170     for (const auto &cell : cell_names) {
00171         ss << cell << ", ";
00172     }
00173     logfile.logger->info("Starting Verilog generation for file: '{}', chain length: {}, cells: {}",
00174         library_path, chain_length, ss.str());
00175     try {
00176         logfile.verilog(chain_length, cell_names);
00177         logfile.logger->info("Successfully generated Verilog for file: '{}', library_path);
00178     } catch (const std::exception &e) {
00179         logfile.logger->error("Error during Verilog generation for file '{}': {}", library_path,
00180             e.what());
00181     }
00182 }
00183
00184 void spiceLibFile(const std::string &library_path, const std::string &log_file_name,
00185     int chain_length, const std::vector<std::string> &cell_names,
00186     const std::string &verilog_lib_file, const std::string &spice_lib_file) {
00187     std::string logname = log_file_name.empty()
00188         ? std::filesystem::path(library_path).stem().string() + ".spice.log"
00189         : log_file_name;
00190     LibFile logfile(library_path, logname);
00191
00192     // Check chain length validity
00193     if (chain_length < 1) {
00194         logfile.logger->error("Invalid chain length: {}. Chain length must be >= 1.", chain_length);
00195         return;
00196     }
00197     std::stringstream ss;
00198     for (const auto &cell : cell_names) {
00199         ss << cell << ", ";
00200     }
00201     logfile.logger->info("Starting SPICE generation for file: '{}', chain length: {}, cells: {}",
00202         library_path, chain_length, ss.str());
00203     try {
00204         logfile.spice(chain_length, cell_names, verilog_lib_file, spice_lib_file);
00205         logfile.logger->info("Successfully generated SPICE for file: '{}', library_path);
00206     } catch (const std::exception &e) {
00207         logfile.logger->error("Error during SPICE generation for file '{}': {}", library_path,
00208             e.what());
00209     }
00210 }
00211

```

```

00231 void compareLibFiles(const std::string &ref_lib, const std::string &comp_lib, const double reltol,
00232                      const double abstol, std::string &report_file_name) {
00233     std::string ref_logname = std::filesystem::path(ref_lib).stem().string() + ".cmp.log";
00234     std::string comp_logname = std::filesystem::path(comp_lib).stem().string() + ".cmp.log";
00235     LibFile ref_libfile(ref_lib, ref_logname), comp_libfile(comp_lib, comp_logname);
00236
00237     // Check relative tolerance validity
00238     if (reltol < 0.0) {
00239         spdlog::error("Invalid relative tolerance: {}. Relative tolerance must be >= 0.0.", reltol);
00240         return;
00241     }
00242
00243     // Check the report file name
00244     if (report_file_name.empty()) {
00245         report_file_name = comp_libfile.basename_ + ".cmp.md";
00246     } else if (std::filesystem::path(report_file_name).extension() != ".txt" &&
00247               std::filesystem::path(report_file_name).extension() != ".md") {
00248         // report file name not end in .txt or .md, warn user and append .md
00249         report_file_name = report_file_name + ".md";
00250         spdlog::warn("Report file name does not end in .txt or .md. Report will be written to '{}'",
00251                     report_file_name);
00252     }
00253
00254     LibraryComparator comparator(ref_libfile, comp_libfile, reltol, abstol);
00255     comparator.generateReport(report_file_name);
00256     spdlog::info("Comparison completed. Report written to: '{}'", report_file_name);
00257 }
00258
00259 void funcLibFile(const std::string &ref_file, const std::string &comp_file,
00260                 const std::vector<std::string> &cell_names, std::string &report_file_name) {
00261     // Check if the files are Liberty or Verilog
00262     std::string ref_ext = std::filesystem::path(ref_file).extension();
00263     std::string comp_ext = std::filesystem::path(comp_file).extension();
00264     std::string ref_logname = std::filesystem::path(ref_file).stem().string() + ".cmp.log";
00265     std::string comp_logname = std::filesystem::path(comp_file).stem().string() + ".cmp.log";
00266
00267     // Pre cell names check
00268     if (cell_names.empty()) {
00269         spdlog::error("No cell names provided for functional equivalence check.");
00270         return;
00271     }
00272
00273     // Check the report file name
00274     if (report_file_name.empty()) {
00275         report_file_name = std::filesystem::path(comp_file).stem().string() + ".logic.cmp.md";
00276     } else if (std::filesystem::path(report_file_name).extension() != ".txt" &&
00277               std::filesystem::path(report_file_name).extension() != ".md") {
00278         // report file name not end in .txt or .md, warn user and append .md
00279         report_file_name = report_file_name + ".md";
00280         spdlog::warn("Report file name does not end in .txt or .md. Report will be written to '{}'",
00281                     report_file_name);
00282     }
00283
00284     // Clear the report file
00285     std::ofstream report_file(report_file_name);
00286     if (!report_file.is_open()) {
00287         spdlog::error("Failed to open report file: '{}'", report_file_name);
00288         return;
00289     }
00290     report_file.close();
00291     spdlog::info("Report file cleared: '{}'", report_file_name);
00292
00293     // Declare LibFile objects as pointers, initialized to nullptr
00294     LibFile *ref_libfile = nullptr;
00295     LibFile *comp_libfile = nullptr;
00296
00297     // Check the reference file extension
00298     if (ref_ext == ".v") {
00299         spdlog::info("Reference file in Verilog format.");

```

```

00300 } else if (ref_ext == ".lib") {
00301     spdlog::info("Reference file in Liberty format.");
00302     ref_libfile = new LibFile(ref_file, ref_logname);
00303 } else {
00304     spdlog::error("Unsupported reference file format: '{}'", ref_ext);
00305     return;
00306 }
00307 // Check the comparison file extension
00308 if (comp_ext == ".v") {
00309     spdlog::info("Comparison file in Verilog format.");
00310 } else if (comp_ext == ".lib") {
00311     spdlog::info("Comparison file in Liberty format.");
00312     comp_libfile = new LibFile(comp_file, comp_logname);
00313 } else {
00314     spdlog::error("Unsupported comparison file format: '{}'", comp_ext);
00315     return;
00316 }
00317
00318 // Tranverse the cell names and check functional equivalence
00319 for (const auto &cell : cell_names) {
00320     spdlog::info("Checking functional equivalence for cell: '{}'", cell);
00321     std::map<std::string, std::string> ref_outpin_map;
00322     std::map<std::string, std::string> comp_outpin_map;
00323
00324     if (ref_ext == ".v") {
00325         ref_outpin_map = extractLogicFromVerilog(ref_file, cell);
00326     } else if (ref_ext == ".lib") {
00327         ref_outpin_map = ref_libfile->logic(cell);
00328     }
00329
00330     if (comp_ext == ".v") {
00331         // getAST(comp_file, cell);
00332         // extractAndPrintNetlistInfo(comp_file, cell);
00333         comp_outpin_map = extractLogicFromVerilog(comp_file, cell);
00334     } else if (comp_ext == ".lib") {
00335         comp_outpin_map = comp_libfile->logic(cell);
00336     }
00337
00338     spdlog::info("Logic function expressions collected for cell: '{}'", cell);
00339     spdlog::info("Starting logic comparison ...");
00340
00341     LogicComparator comparator(ref_outpin_map, comp_outpin_map, cell);
00342     // Easter egg
00343     if (cell == "easteregg") {
00344         spdlog::info("Easter egg found! Running ExprTk Eample 07...");
00345         comparator.logic();
00346         return;
00347     }
00348
00349     // // Test for preprocessing
00350     // std::string ref_expr;
00351     // std::string comp_expr;
00352     // for (const auto &pin : ref_outpin_map) {
00353     //     spdlog::info("Pin -> Expression: {} -> {}", pin.first, pin.second);
00354     //     ref_expr = comparator.preprocessExpression(pin.second);
00355     // }
00356     // for (const auto &pin : comp_outpin_map) {
00357     //     spdlog::info("Pin => Expression: {} => {}", pin.first, pin.second);
00358     //     comp_expr = comparator.preprocessExpression(pin.second);
00359     // }
00360     // std::vector<std::string> sorted_vars;
00361     // comparator.extractVariables(ref_expr, comp_expr, sorted_vars);
00362     // struct PinComparisonResult pin_comparison_result;
00363     // // comp_expr = "not ((not(A1) and not(A2)) or B)";
00364     // comparator.compareSingleExpressionPair(ref_expr, comp_expr, sorted_vars,
00365     //                                         pin_comparison_result);
00366
00367     // comparator.compareCellLogic();
00368     comparator.compareCellLogic();

```

```

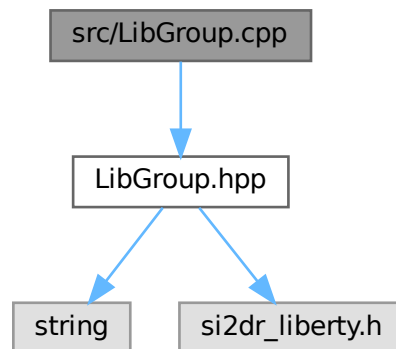
00369     comparator.generateReport(report_file_name);
00370
00371     spdlog::info("Functional equivalence check completed for cell: '{}'", cell);
00372 }
00373 spdlog::info("Functional equivalence check completed for all cells,");
00374 spdlog::info("Report written to: '{}'", report_file_name);
00375 }

```

6.34 src/LibGroup.cpp File Reference

```
#include "LibGroup.hpp"
```

Include dependency graph for LibGroup.cpp:



6.35 LibGroup.cpp

[Go to the documentation of this file.](#)

```

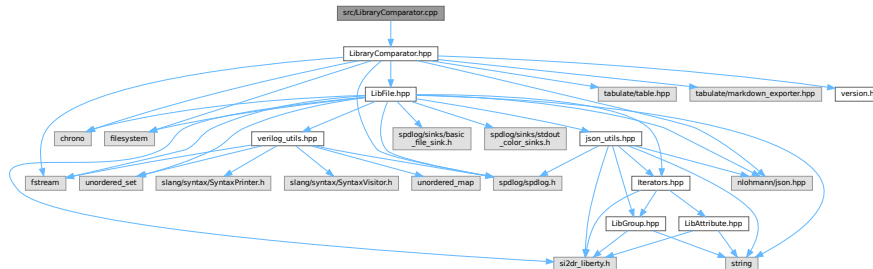
00001 #include "LibGroup.hpp"
00002
00003 LibGroup::LibGroup(si2drGroupIdT group, si2drErrorT &err) : group_(group), err_(err) {}
00004
00005 LibGroup::~LibGroup() {}
00006
00007 std::string LibGroup::getName() {
00008     si2drNamesIdT names = si2drGroupGetNames(group_, &err_);
00009     si2drStringT name = si2drIterNextName(names, &err_);
00010     si2drIterQuit(names, &err_);
00011     return name ? std::string(name) : std::string();
00012 }
00013
00014 std::string LibGroup::getType() {
00015     si2drStringT type = si2drGroupGetGroupType(group_, &err_);
00016     return type ? std::string(type) : std::string();
00017 }
00018
00019 si2drAttrsIdT LibGroup::getAttrs() { return si2drGroupGetAttrs(group_, &err_); }
00020
00021 si2drGroupsIdT LibGroup::getGroups() { return si2drGroupGetGroups(group_, &err_); }

```

6.36 src/LibraryComparator.cpp File Reference

#include "LibraryComparator.hpp"

Include dependency graph for LibraryComparator.cpp:



6.37 LibraryComparator.cpp

[Go to the documentation of this file.](#)

```
00001 #include "LibraryComparator.hpp"
00002
00021 LibraryComparator::LibraryComparator(LibFile &ref_libfile, LibFile &comp_libfile, double reltol,
00022                                     double abstol)
00023 : reltol_(reltol), abstol_(abstol) {
00024     std::string ref_json_name = ref_libfile.basename_ + ".json";
00025     if (!std::filesystem::exists(ref_json_name)) {
00026         spdlog::info("Reference JSON file {} not found. Parsing first.", ref_json_name);
00027         ref_libfile.parse();
00028         ref_libfile.writeJsonToFile();
00029         ref_json_ = ref_libfile.lib_json_;
00030     } else {
00031         // Read the JSON file into json object
00032         std::ifstream ref_in(ref_json_name);
00033         if (!ref_in.is_open()) {
00034             spdlog::error("Could not open file '{}' for reading", ref_json_name);
00035             return;
00036         }
00037         try {
00038             ref_json_ = json::parse(ref_in);
00039         } catch (const json::parse_error &e) {
00040             spdlog::error("Error parsing reference JSON file '{}': {}", ref_json_name, e.what());
00041             return;
00042         }
00043     }
00044     std::string comp_json_name = comp_libfile.basename_ + ".json";
00045     if (!std::filesystem::exists(comp_json_name)) {
00046         spdlog::info("Comparison JSON file {} not found. Parsing first.", comp_json_name);
00047         comp_libfile.parse();
00048         comp_libfile.writeJsonToFile();
00049         comp_json_ = comp_libfile.lib_json_;
00050     } else {
00051         // Read the JSON file into json object
00052         std::ifstream comp_in(comp_json_name);
00053         if (!comp_in.is_open()) {
00054             spdlog::error("Could not open file '{}' for reading", comp_json_name);
00055             return;
00056         }
00057         try {
```

```

00058     comp_json_ = json::parse(comp_in);
00059 } catch (const json::parse_error &e) {
00060     spdlog::error("Error parsing comparison JSON file '{}': {}", comp_json_name, e.what());
00061     return;
00062 }
00063 }
00064
00065 ref_lib_path_ = ref_libfile.filepath_;
00066 comp_lib_path_ = comp_libfile.filepath_;
00067 spdlog::info("Successfully loaded JSON files for comparison");
00068 }
00069
00098 void LibraryComparator::compareLut(const std::string &cell_name, const std::string &pin_name,
00099                                     const std::string &timing_type, const std::string &related_pin,
00100                                     const std::string &arc_name, const json &ref_lut,
00101                                     const json &comp_lut, Table &table) {
00102     spdlog::info("Comparing LUT of '{}' ...", arc_name);
00103     std::vector<double> ref_index_1 = ref_lut["index_1"].get<std::vector<double>>();
00104     std::vector<double> ref_index_2 = ref_lut["index_2"].get<std::vector<double>>();
00105     std::vector<double> comp_index_1 = comp_lut["index_1"].get<std::vector<double>>();
00106     std::vector<double> comp_index_2 = comp_lut["index_2"].get<std::vector<double>>();
00107     if (ref_index_1 != comp_index_1 || ref_index_2 != comp_index_2) {
00108         spdlog::warn("Mismatch in LUT indices for '{}' in cell: '{}', pin: '{}', related_pin: '{}', "
00109                     "timing_type: '{}'",
00110                     arc_name, cell_name, pin_name, related_pin, timing_type);
00111         return;
00112     } else {
00113         spdlog::debug("LUT indices match for '{}'", arc_name);
00114
00115         std::vector<std::vector<double>> comp_value_matrix;
00116         // Generate a matrix of values from the JSON array for comparison library
00117         for (const auto &row_val : comp_lut["values"]) {
00118             std::vector<double> row_data;
00119             // Check if the row value is an array
00120             if (!row_val.is_array()) {
00121                 spdlog::error(
00122                     "Invalid LUT format: '{}' values should be an array of arrays in comp_lib '{}', "
00123                     "cell '{}', pin '{}', "
00124                     "related_pin '{}', timing_type '{}'",
00125                     arc_name, this->comp_lib_path_.string(), cell_name, pin_name, related_pin, timing_type);
00126                 return;
00127             }
00128             // Check if non-numeric values
00129             for (const auto &val : row_val) {
00130                 if (val.is_number()) {
00131                     row_data.push_back(val.get<double>());
00132                 } else {
00133                     spdlog::error("Non-numeric value found in '{}'.values, skipping value: {} in comp_lib "
00134                                 "'{}', cell '{}', "
00135                                 "pin '{}', related_pin '{}', timing_type '{}'",
00136                                 arc_name, val.dump(), this->comp_lib_path_.string(), cell_name, pin_name,
00137                                 related_pin, timing_type);
00138                     return;
00139                 }
00140             }
00141             comp_value_matrix.push_back(row_data);
00142         }
00143         // Generate a matrix of values from the JSON array for reference library
00144         std::vector<std::vector<double>> ref_value_matrix;
00145         for (const auto &row_val : ref_lut["values"]) {
00146             std::vector<double> row_data;
00147             // Check if the row value is an array
00148             if (!row_val.is_array()) {
00149                 spdlog::error(
00150                     "Invalid LUT format: '{}' values should be an array of arrays in ref_lib '{}', "
00151                     "cell '{}', pin '{}', "
00152                     "related_pin '{}', timing_type '{}'",
00153                     arc_name, this->ref_lib_path_.string(), cell_name, pin_name, related_pin, timing_type);
00154                 return;

```



```

00155     }
00156     // Check if non-numeric values
00157     for (const auto &val : row_val) {
00158         if (val.is_number()) {
00159             row_data.push_back(val.get<double>());
00160         } else {
00161             spdlog::error("Non-numeric value found in '{}'.values, skipping value: {} in ref_lib "
00162                 "'{}', cell '{}', "
00163                 "pin '{}', related_pin '{}', timing_type '{}'",
00164                 arc_name, val.dump(), this->ref_lib_path_.string(), cell_name, pin_name,
00165                 related_pin, timing_type);
00166             return;
00167         }
00168     }
00169     ref_value_matrix.push_back(row_data);
00170 }
00171 // Compare the matrix for relative tolerance
00172 if (!comp_value_matrix.empty() && !comp_value_matrix[0].empty()) {
00173     for (size_t i = 0; i < comp_value_matrix.size(); ++i) {
00174         for (size_t j = 0; j < comp_value_matrix[i].size(); ++j) {
00175             double ref_val = ref_value_matrix[i][j];
00176             double comp_val = comp_value_matrix[i][j];
00177             if (std::abs(ref_val - comp_val) > reitol_ * std::abs(ref_val) &&
00178                 std::abs(ref_val - comp_val) > abstol_) {
00179                 spdlog::debug("LUT value mismatch for '{}', index_1: {}, index_2: {}", arc_name,
00180                     ref_index_1[i], ref_index_2[j]);
00181                 table.add_row({related_pin + "->" + pin_name, std::to_string(ref_val),
00182                     std::to_string(comp_val), std::to_string(comp_val - ref_val),
00183                     std::to_string((comp_val - ref_val) / ref_val * 100), timing_type,
00184                     arc_name, std::to_string(i + 1), std::to_string(ref_index_1[i]),
00185                     std::to_string(j + 1), std::to_string(ref_index_2[j]), "<"});
00186                 spdlog::debug("table size: {}", table.size());
00187             } else {
00188                 spdlog::debug("LUT value match for '{}', index_1: {}, index_2: {}", arc_name,
00189                     ref_index_1[i], ref_index_2[j]);
00190             }
00191         }
00192     }
00193 } else {
00194     spdlog::error(
00195         "Empty or invalid 'values' array found for cell: '{}', pin: '{}', related_pin: '{}', "
00196         "timing_type: '{}', arc: '{}'",
00197         cell_name, pin_name, related_pin, timing_type, arc_name);
00198 }
00199 }
00200 }
00201
00220 void LibraryComparator::compareTimingArc(const std::string &cell_name, const std::string &pin_name,
00221     const std::string &timing_type, const json &ref_timing_arc,
00222     const json &comp_timing_arc, Table &table) {
00223     spdlog::info("Comparing timing type '{}' ...", timing_type);
00224
00225     std::string related_pin = ref_timing_arc["related_pin"].get<std::string>();
00226     spdlog::debug("Related pin: '{}'", related_pin);
00227
00228     std::vector<std::string> arc_names = {"cell_rise", "cell_fall", "rise_transition",
00229         "fall_transition"};
00230     for (auto arc_name : arc_names) {
00231         if (comp_timing_arc.contains(arc_name)) {
00232             spdlog::debug("Comparing timing arc: '{}'", arc_name);
00233             // Look for the arc in the reference JSON
00234             if (ref_timing_arc.contains(arc_name)) {
00235                 // Found this arc
00236                 spdlog::debug("Found timing arc: '{}'", arc_name);
00237                 compareLut(cell_name, pin_name, timing_type, related_pin, arc_name,
00238                     ref_timing_arc[arc_name], comp_timing_arc[arc_name], table);
00239             } else {
00240                 // arc not found in reference JSON
00241                 spdlog::warn("Timing arc: '{}' not found in reference JSON", arc_name);

```

```

00242     }
00243 }
00244 }
00245 }
00246
00261 void LibraryComparator::comparePin(const std::string &cell_name, const std::string &pin_name,
00262                                     const json &ref_pin, const json &comp_pin, Table &table) {
00263     spdlog::info("Comparing pin '{}'. ...", pin_name);
00264
00265     if (comp_pin.contains("timing_arcs")) {
00266         for (const auto &comp_arc : comp_pin["timing_arcs"]) {
00267             std::string timing_type = comp_arc["timing_type"].get<std::string>();
00268             spdlog::debug("Comparing timing type: '{}'", timing_type);
00269             // Look for the timing type in the reference JSON
00270             auto ref_arc_it = std::find_if(
00271                 ref_pin["timing_arcs"].begin(), ref_pin["timing_arcs"].end(),
00272                 [&timing_type](const json &ref_arc) { return ref_arc["timing_type"] == timing_type; });
00273             if (ref_arc_it != ref_pin["timing_arcs"].end()) {
00274                 // Found this timing type
00275                 spdlog::debug("Found timing type: '{}'", timing_type);
00276                 compareTimingArc(cell_name, pin_name, timing_type, *ref_arc_it, comp_arc, table);
00277             } else {
00278                 // timing arc not found in reference JSON
00279                 spdlog::warn("Timing arc: '{}'. not found in reference JSON", timing_type);
00280             }
00281         }
00282     } else {
00283         spdlog::info("No timing arcs found for pin: '{}'.", comp_pin["pin_name"].get<std::string>());
00284     }
00285 }
00286
00301 void LibraryComparator::compareCell(const std::string &cell_name, const json &ref_cell,
00302                                     const json &comp_cell, Table &table) {
00303     spdlog::info("Comparing cell '{}'. ...", cell_name);
00304
00305     if (comp_cell.contains("output_pins")) {
00306         for (const auto &comp_pin : comp_cell["output_pins"]) {
00307             std::string pin_name = comp_pin["pin_name"].get<std::string>();
00308             spdlog::debug("Comparing pin: '{}'", pin_name);
00309             // Look for the pin in the reference JSON
00310             auto ref_pin_it = std::find_if(
00311                 ref_cell["output_pins"].begin(), ref_cell["output_pins"].end(),
00312                 [&pin_name](const json &ref_pin) { return ref_pin["pin_name"] == pin_name; });
00313             if (ref_pin_it != ref_cell["output_pins"].end()) {
00314                 // Found this pin
00315                 spdlog::debug("Found pin: '{}'", pin_name);
00316                 comparePin(cell_name, pin_name, *ref_pin_it, comp_pin, table);
00317             } else {
00318                 // pin not found in reference JSON
00319                 spdlog::warn("Pin: '{}'. not found in reference JSON", pin_name);
00320             }
00321         }
00322     } else {
00323         spdlog::info("No output pins found for cell: '{}'.", cell_name);
00324     }
00325 }
00326
00343 void LibraryComparator::generateReport(const std::string &output_file) {
00344     std::ofstream outfile(output_file);
00345     outfile << "# LIBRARY comparison\n" << std::endl;
00346     outfile << "**Reference library: " << ref_lib_path_ << "**\n" << std::endl;
00347     outfile << "**Comparison library: " << comp_lib_path_ << "**\n" << std::endl;
00348     outfile << "**Absolute tolerance: " << abstol_ << "**\n" << std::endl;
00349     outfile << "**Relative tolerance: " << reltol_ << "**\n" << std::endl;
00350     outfile << "**Performed by " << APP_NAME << " v" << APP_VERSION << " from " << APP_AUTHOR;
00351     auto now = std::chrono::system_clock::now();
00352     std::time_t now_time_t = std::chrono::system_clock::to_time_t(now);
00353     std::tm *now_tm = std::localtime(&now_time_t);
00354     outfile << ". on: " << std::put_time(now_tm, "%c") << "**\n" << std::endl;

```

```

00355 outfile << "> Legend: < outlier, * scaled, ! indices switched, ^ slews extrapolated, - loads "
00356         "extrapolated,\n";
00357 outfile << "> \n";
00358 outfile << "> Legend: + padding added, /0 divide by zero, / slews interpolated, # loads "
00359         "interpolated\n";
00360 outfile << "> \n";
00361 outfile << "> Legend: << value is less but unknown, >> value is more but unknown\n\n";
00362
00363 spdlog::info("Starting comparison report generation ...");
00364 for (const auto &comp_cell : comp_json_["cells"]) {
00365     std::string cell_name = comp_cell["cell_name"].get<std::string>();
00366     spdlog::debug("Comparing cell: '{}'", cell_name);
00367     // Look for the cell in the reference JSON
00368     auto ref_cell_it = std::find_if(
00369         ref_json_["cells"].begin(), ref_json_["cells"].end(),
00370         [&cell_name](const json &ref_cell) { return ref_cell["cell_name"] == cell_name; });
00371     if (ref_cell_it != ref_json_["cells"].end()) {
00372         // Found this cell
00373         spdlog::debug("Found cell: '{}'", cell_name);
00374         Table table;
00375         table.add_row({"Pin Name", "Reference", "Comparison", "Diff", "Diff %", "Type", "Arc Name",
00376             "Row #", "Index_1", "Col #", "Index_2", "Note"});
00377         // Header formatting
00378         for (size_t i = 0; i < table[0].size(); ++i) {
00379             table[0][i].format().font_color(Color::yellow).font_style({FontStyle::bold});
00380         }
00381         compareCell(cell_name, *ref_cell_it, comp_cell, table);
00382
00383         if (table.size() > 1) {
00384             // Data starts from row 1, row 0 is header
00385             size_t failed_count = table.size() - 1;
00386             double sum_diff = 0.0;
00387             double sum_diff_percent = 0.0;
00388             size_t outlier_count = 0;
00389             double max_diff = -1e9;
00390             double max_diff_percent = -1e9;
00391             std::vector<double> diff_values; // Store diff values for std deviation calculation
00392
00393             // Index of columns
00394             const int diff_index = 3;
00395             const int diff_percent_index = 4;
00396             const int note_index = 11;
00397
00398             for (size_t i = 1; i < table.size(); ++i) {
00399                 try {
00400                     double diff = std::stod(table[i][diff_index].get_text());
00401                     double diff_percent = std::stod(table[i][diff_percent_index].get_text());
00402
00403                     sum_diff += diff;
00404                     sum_diff_percent += diff_percent;
00405                     diff_values.push_back(diff_percent);
00406
00407                     if (table[i][note_index].get_text() == "<") {
00408                         outlier_count++;
00409                     }
00410                     max_diff = std::max(max_diff, diff);
00411                     max_diff_percent = std::max(max_diff_percent, diff_percent);
00412
00413                 } catch (const std::invalid_argument &e) {
00414                     spdlog::error("Could not convert value to double: {}", e.what());
00415                     continue; // Skip this row if there's an error
00416                 }
00417             }
00418
00419             double avg_diff = sum_diff / failed_count;
00420             double avg_diff_percent = sum_diff_percent / failed_count;
00421
00422             outfile << "## " << cell_name << "\n\n";
00423             outfile << "Delay Comparison in ns\n\n";

```

```

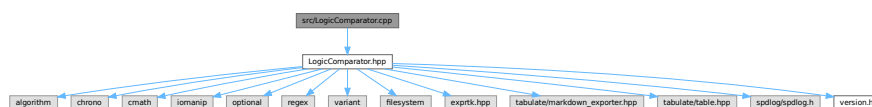
00424     if (output_file.substr(output_file.size() - 3) == ".md") {
00425         MarkdownExporter exporter;
00426         auto markdown = exporter.dump(table);
00427         outfile << markdown << std::endl;
00428     } else {
00429         outfile << table << std::endl;
00430     }
00431     outfile << std::endl;
00432     std::cout << table << std::endl;
00433     outfile << cell_name << " delay SUMMARY (abstol: " << abstol_ << ", reitol: " << reitol_
00434         << ")\n";
00435
00436     // Create summary table
00437     Table summary_table;
00438     summary_table.add_row({"Cell Name", "Data Type", "Failed Count", "Avg Diff", "Avg Diff%",
00439         "Max Diff", "Max Diff%", "Outliers"});
00440     summary_table.add_row({cell_name, "delay(ns)", std::to_string(failed_count),
00441         std::to_string(avg_diff), std::to_string(avg_diff_percent) + "%",
00442         std::to_string(max_diff), std::to_string(max_diff_percent) + "%",
00443         std::to_string(outlier_count)});
00444
00445     // Output summary table
00446     // Use std::filesystem to reliably get the extension
00447     std::filesystem::path file_path(output_file);
00448     if (file_path.has_extension() && file_path.extension() == ".md") {
00449         MarkdownExporter exporter;
00450         auto markdown = exporter.dump(summary_table);
00451         outfile << markdown << std::endl;
00452     } else {
00453         outfile << summary_table << std::endl;
00454     }
00455     outfile << "Worst delay outlier: Max Abs: " << max_diff << ", Max Rel: " << max_diff_percent
00456         << "%, Outliers: " << outlier_count << "\n\n";
00457 }
00458 } else {
00459     // cell not found in reference JSON
00460     spdlog::warn("Cell: '{}' not found in reference JSON", cell_name);
00461 }
00462 }
00463
00464 outfile.close();
00465 }

```

6.38 src/LogicComparator.cpp File Reference

#include "LogicComparator.hpp"

Include dependency graph for LogicComparator.cpp:



Functions

- bool `isIdentifier` (const std::string &token)

Checks if a given string is a valid identifier.

- bool `isOperator` (const std::string &token)

Checks if a given token is a logical operator.

6.38.1 Function Documentation

6.38.1.1 isIdentifier()

```
bool isIdentifier (
    const std::string & token )
```

Checks if a given string is a valid identifier.

A valid identifier must:

- Not be empty.
- Consist of alphanumeric characters and underscores.
- Start with an uppercase alphabetic character.
- Contain at least one alphabetic character.

Parameters

<code>token</code>	The string to check.
--------------------	----------------------

Returns

True if the string is a valid identifier, false otherwise.

Definition at line 74 of file [LogicComparator.cpp](#).

Here is the caller graph for this function:



6.38.1.2 isOperator()

```
bool isOperator (
    const std::string & token )
```

Checks if a given token is a logical operator.

This function determines whether the input string token is one of the supported logical operators: "and", "or", "xor", or "not".

Parameters

<i>token</i>	The string to check.
--------------	----------------------

Returns

True if the token is a logical operator, false otherwise.

Definition at line 99 of file [LogicComparator.cpp](#).

Here is the caller graph for this function:



6.39 LogicComparator.cpp

[Go to the documentation of this file.](#)

```
00001 #include "LogicComparator.hpp"
00002
00003 LogicComparator::LogicComparator(const std::map<std::string, std::string> &ref_outpin_map,
00004                                 const std::map<std::string, std::string> &comp_outpin_map,
00005                                 const std::string &cell_name)
00006     : ref_outpin_map_(ref_outpin_map), comp_outpin_map_(comp_outpin_map), cell_name_(cell_name) {}
00024 void LogicComparator::logic() {
00025     typedef exprtk::symbol_table<double> symbol_table_t;
00026     typedef exprtk::expression<double> expression_t;
00027     typedef exprtk::parser<double> parser_t;
00028
00029     const std::string expression_string = "not(A and B) or C";
00030
00031     symbol_table_t symbol_table;
00032     symbol_table.create_variable("A");
00033     symbol_table.create_variable("B");
00034     symbol_table.create_variable("C");
00035
00036     expression_t expression;
00037     expression.register_symbol_table(symbol_table);
```

Generated on Sat Apr 5 2025 14:07:22 for ZlibValidation by Doxygen

```

00128 // Replace logical NOT '!' with " not ", ensuring not to replace '!='.
00129 std::string temp_processed;
00130 temp_processed.reserve(processed.length() * 1.2); // Pre-allocate a bit more space
00131 for (size_t i = 0; i < processed.length(); ++i) {
00132     if (processed[i] == '!') {
00133         if (i + 1 < processed.length() && processed[i + 1] == '=') {
00134             temp_processed += "!="; // Keep inequality operator
00135             i++;                     // Skip the '='
00136         } else {
00137             temp_processed += " not "; // Replace logical NOT with spaced operator
00138         }
00139     } else {
00140         temp_processed += processed[i];
00141     }
00142 }
00143 processed = temp_processed;
00144 spdlog::trace("After '!' to 'not' conversion: {}", processed);
00145
00146 // 2. Add spaces around other operators and parentheses that need them
00147 processed = std::regex_replace(processed, std::regex("\\("), " ( ");
00148 processed = std::regex_replace(processed, std::regex("\\)"), " ) ");
00149 processed = std::regex_replace(processed, std::regex("\\+"), " + "); // OR
00150 processed = std::regex_replace(processed, std::regex("\\^"), " ^ "); // XOR
00151 processed = std::regex_replace(processed, std::regex("\\*"), " * "); // AND
00152 // Consolidate multiple spaces into one and trim again
00153 processed = std::regex_replace(processed, std::regex("\\s+"), " ");
00154 processed = std::regex_replace(processed, std::regex("^\\s+|\\s+$"), "");
00155 spdlog::trace("After adding spaces: {}", processed);
00156
00157 // 3. Replace symbolic operators with their keyword equivalents
00158 // Note: 'not' is already handled.
00159 processed = std::regex_replace(processed, std::regex("\\|+ "), " or ");
00160 processed = std::regex_replace(processed, std::regex("\\|^ "), " xor ");
00161 processed = std::regex_replace(processed, std::regex("\\* "), " and ");
00162 spdlog::trace("After operator keyword replacement: {}", processed);
00163
00164 // 4. Tokenize based on spaces
00165 std::stringstream ss(processed);
00166 // Read tokens skipping whitespace
00167 std::vector<std::string> tokens{std::istream_iterator<std::string>(ss),
00168                                std::istream_iterator<std::string>()};
00169
00170 if (tokens.empty()) {
00171     spdlog::debug("No tokens found after tokenization.");
00172     return "";
00173 }
00174 spdlog::trace("Tokens after initial processing: [{}]", build_log_string(tokens));
00175
00176 // 5. Insert implied 'and'
00177 std::vector<std::string> tokens_with_implied_and;
00178 if (!tokens.empty()) {
00179     tokens_with_implied_and.push_back(tokens[0]);
00180     for (size_t i = 0; i < tokens.size() - 1; ++i) {
00181         const std::string &current_token = tokens[i];
00182         const std::string &next_token = tokens[i + 1];
00183
00184         // An operand ends if it's an identifier or a closing parenthesis.
00185         bool current_ends_operand = isIdentifier(current_token) || current_token == ")";
00186         // An operand starts if it's an identifier, an opening parenthesis, or 'not'.
00187         bool next_starts_operand =
00188             isIdentifier(next_token) || next_token == "(" || next_token == "not";
00189
00190         // We should insert 'and' if the current token ends an operand AND the next token starts one,
00191         // UNLESS the current token is already an operator/opening bracket OR the next token is an
00192         // operator/closing bracket. Explicit check: Don't insert 'and' if an operator already exists
00193         // between them.
00194         bool current_is_op_or_open =
00195             isOperator(current_token) || current_token == "not" || current_token == "(";
00196         bool next_is_op_or_close = isOperator(next_token) || next_token == "not" || next_token == ")";

```



```

00197
00198     if (current_ends_operand && next_starts_operand && !current_is_op_or_open &&
00199         !next_is_op_or_close) {
00200         spdlog::trace("Inserting implied 'and' between '{}' and '{}'", current_token, next_token);
00201         tokens_with_implied_and.push_back("and");
00202     }
00203
00204     tokens_with_implied_and.push_back(next_token); // Add the next token regardless
00205 }
00206 } else {
00207     // Handle case where initial tokenization yielded no tokens
00208     return "";
00209 }
00210
00211 spdlog::trace("Tokens after implied 'and': [{}]", build_log_string(tokens_with_implied_and));
00212
00213 // 6. Handle 'not Identifier' ONLY during reconstruction (NEW LOGIC)
00214 std::vector<std::string> final_tokens;
00215 for (size_t i = 0; i < tokens_with_implied_and.size(); ++i) {
00216     const std::string &current_token = tokens_with_implied_and[i];
00217
00218     if (current_token == "not") {
00219         // Check if the *next* token is an Identifier
00220         if (i + 1 < tokens_with_implied_and.size() && isIdentifier(tokens_with_implied_and[i + 1])) {
00221             // Found 'not Identifier', transform to 'not(Identifier)'
00222             final_tokens.push_back("not");
00223             final_tokens.push_back("(");
00224             final_tokens.push_back(tokens_with_implied_and[i + 1]); // The identifier
00225             final_tokens.push_back(")");
00226             i++; // Increment i to skip the identifier we just processed
00227         } else {
00228             // 'not' is followed by '(', another operator, or is at the end.
00229             // Add 'not' as is, let ExprTk parse 'not(...)' or handle errors.
00230             final_tokens.push_back("not");
00231         }
00232     } else {
00233         // Token is not 'not', just add it to the final list
00234         final_tokens.push_back(current_token);
00235     }
00236 }
00237 spdlog::trace("Tokens after 'not' parenthesis handling: [{}]", build_log_string(final_tokens));
00238
00239 // 7. Reconstruct the final expression string from tokens
00240 std::string final_expr;
00241 if (!final_tokens.empty()) {
00242     final_expr = final_tokens[0];
00243     for (size_t i = 1; i < final_tokens.size(); ++i) {
00244         const std::string &prev = final_tokens[i - 1];
00245         const std::string &curr = final_tokens[i];
00246
00247         // Add space smartly: No space after '(', no space before ')',
00248         // no space after 'not' if followed by '(', no space before ',' (in function args, if
00249         // applicable)
00250         bool add_space = true;
00251         if (prev == "(" || curr == ")" || curr == ",") {
00252             add_space = false;
00253         }
00254         if (prev == "not" && curr == "(") {
00255             add_space = false; // Already handled by 'not(...)' structure
00256         }
00257         // Potentially add more rules if needed for other operators/functions
00258
00259         if (add_space) {
00260             final_expr += " ";
00261         }
00262         final_expr += curr;
00263     }
00264 }
00265

```

```

00266 // 8. Final cleanup (consolidate any remaining multiple spaces and trim)
00267 final_expr = std::regex_replace(final_expr, std::regex("\\s+"), " ");
00268 final_expr = std::regex_replace(final_expr, std::regex("^\\s+|\\s+$"), "");
00269
00270 spdlog::debug("Preprocessed expression result: {}", final_expr);
00271 return final_expr;
00272 }
00273
00274 // --- Variable Extraction Helper ---
00303 bool LogicComparator::extractVariables(const std::string &expr1_raw, const std::string &expr2_raw,
00304                                       std::vector<std::string> &sorted_vars) {
00305     // Define the set of ExprTk keywords/function names to filter out (lowercase)
00306     // Even though isIdentifier checks for uppercase, keep this filter as a safety measure
00307     static const std::set<std::string> keywords = {"abs",
00308                                                    "acos",
00309                                                    "acosh",
00310                                                    "and",
00311                                                    "asin",
00312                                                    "asinh",
00313                                                    "assert",
00314                                                    "atan",
00315                                                    "atan2",
00316                                                    "atanh",
00317                                                    "avg",
00318                                                    "break",
00319                                                    "case",
00320                                                    "ceil",
00321                                                    "clamp",
00322                                                    "continue",
00323                                                    "cosh",
00324                                                    "cos",
00325                                                    "cot",
00326                                                    "csc",
00327                                                    "default",
00328                                                    "deg2grad",
00329                                                    "deg2rad",
00330                                                    "else",
00331                                                    "equal",
00332                                                    "erfc",
00333                                                    "erf",
00334                                                    "exp",
00335                                                    "expm1",
00336                                                    "false",
00337                                                    "floor",
00338                                                    "for",
00339                                                    "frac",
00340                                                    "grad2deg",
00341                                                    "hypot",
00342                                                    "iclamp",
00343                                                    "if",
00344                                                    "ilike",
00345                                                    "in",
00346                                                    "inrange",
00347                                                    /*"in",*/ "like",
00348                                                    "log",
00349                                                    "log10",
00350                                                    "log1p",
00351                                                    "log2",
00352                                                    "logn",
00353                                                    "mand",
00354                                                    "max",
00355                                                    "min",
00356                                                    "mod",
00357                                                    "mor",
00358                                                    "mul",
00359                                                    "nand",
00360                                                    "ncdf",
00361                                                    "nor",
00362                                                    "not",

```

```

00363         "not_equal",
00364         /*"not",*/ "null",
00365         "or",
00366         "pow",
00367         "rad2deg",
00368         "repeat",
00369         "return",
00370         "root",
00371         "roundn",
00372         "round",
00373         "sec",
00374         "sgn",
00375         "shl",
00376         "shr",
00377         "sinc",
00378         "sinh",
00379         "sin",
00380         "sqrt",
00381         "sum",
00382         "swap",
00383         "switch",
00384         "tanh",
00385         "tan",
00386         "true",
00387         "trunc",
00388         "until",
00389         "var",
00390         "while",
00391         "xnor",
00392         "xor"};
00393
00394 // Regular expression to find potential identifiers (unchanged)
00395 const std::regex identifier_regex("[a-zA-Z_][a-zA-Z0-9_]*");
00396
00397 // Store validated and filtered variable names
00398 std::set<std::string> actual_vars1_set;
00399 std::set<std::string> actual_vars2_set;
00400
00401 // --- Helper function: build log string ---
00402 auto build_log_string = [](const auto &container) {
00403     std::ostringstream oss;
00404     oss << "[";
00405     for (auto it = container.begin(); it != container.end(); ++it) {
00406         oss << *it;
00407         if (std::next(it) != container.end()) {
00408             oss << ", ";
00409         }
00410     }
00411     oss << "]";
00412     return oss.str();
00413 };
00414
00415 // --- Process the first expression ---
00416 spdlog::debug("Extracting and validating identifiers from raw expr1: {}", expr1_raw);
00417 auto words_begin1 = std::sregex_iterator(expr1_raw.begin(), expr1_raw.end(), identifier_regex);
00418 auto words_end1 = std::sregex_iterator();
00419 for (std::sregex_iterator i = words_begin1; i != words_end1; ++i) {
00420     std::string potential_var = i->str();
00421     spdlog::trace("Regex found in expr1: {}", potential_var);
00422
00423     // 1. Validate using isIdentifier (now checks for uppercase etc.)
00424     if (isIdentifier(potential_var)) {
00425         // 2. Check if it's a keyword (still use lowercase comparison as a precaution)
00426         std::string lower_var = potential_var;
00427         std::transform(lower_var.begin(), lower_var.end(), lower_var.begin(),
00428             [](unsigned char c) { return std::tolower(c); });
00429
00430         if (keywords.find(lower_var) == keywords.end()) {
00431             actual_vars1_set.insert(potential_var); // Keep original case

```

```

00432         spdlog::trace("Kept variable from expr1: {}", potential_var);
00433     } else {
00434         spdlog::trace("Filtered keyword from expr1: {}", potential_var);
00435     }
00436 } else {
00437     spdlog::trace("Filtered invalid identifier from expr1: {}", potential_var);
00438 }
00439 }
00440 spdlog::debug("Validated variables found in expr1: {}", build_log_string(actual_vars1_set));
00441
00442 // --- Process the second expression ---
00443 spdlog::debug("Extracting and validating identifiers from raw expr2: {}", expr2_raw);
00444 auto words_begin2 = std::sregex_iterator(expr2_raw.begin(), expr2_raw.end(), identifier_regex);
00445 auto words_end2 = std::sregex_iterator();
00446 for (std::sregex_iterator i = words_begin2; i != words_end2; ++i) {
00447     std::string potential_var = i->str();
00448     spdlog::trace("Regex found in expr2: {}", potential_var);
00449
00450     // 1. Validate using isIdentifier
00451     if (isIdentifier(potential_var)) {
00452         // 2. Check if it's a keyword (still use lowercase comparison)
00453         std::string lower_var = potential_var;
00454         std::transform(lower_var.begin(), lower_var.end(), lower_var.begin(),
00455             [](unsigned char c) { return std::tolower(c); });
00456
00457         if (keywords.find(lower_var) == keywords.end()) {
00458             actual_vars2_set.insert(potential_var); // Keep original case
00459             spdlog::trace("Kept variable from expr2: {}", potential_var);
00460         } else {
00461             spdlog::trace("Filtered keyword from expr2: {}", potential_var);
00462         }
00463     } else {
00464         spdlog::trace("Filtered invalid identifier from expr2: {}", potential_var);
00465     }
00466 }
00467 spdlog::debug("Validated variables found in expr2: {}", build_log_string(actual_vars2_set));
00468
00469 // --- Compare the two variable sets ---
00470 if (actual_vars1_set == actual_vars2_set) {
00471     // Sets are equal, extract the variable list
00472     sorted_vars.assign(actual_vars1_set.begin(), actual_vars1_set.end());
00473     std::sort(sorted_vars.begin(), sorted_vars.end()); // Sort alphabetically
00474     spdlog::info("Variable sets match. Found unique variables for comparison: {}",
00475         build_log_string(sorted_vars));
00476     spdlog::info("Extracted variables done !");
00477     return true; // Variable sets are consistent
00478 } else {
00479     // Sets are not equal, report an error and return false
00480     spdlog::warn("Variable sets do not match between the two expressions!");
00481     // Calculate the differences for more detailed logging
00482     std::vector<std::string> diff1, diff2;
00483     std::set_difference(actual_vars1_set.begin(), actual_vars1_set.end(), actual_vars2_set.begin(),
00484         actual_vars2_set.end(),
00485         std::back_inserter(diff1)); // Vars only in expr1
00486     std::set_difference(actual_vars2_set.begin(), actual_vars2_set.end(), actual_vars1_set.begin(),
00487         actual_vars1_set.end(),
00488         std::back_inserter(diff2)); // Vars only in expr2
00489
00490     if (!diff1.empty()) {
00491         spdlog::warn("Variables only in reference expression: {}", build_log_string(diff1));
00492     }
00493     if (!diff2.empty()) {
00494         spdlog::warn("Variables only in comparison expression: {}", build_log_string(diff2));
00495     }
00496     sorted_vars.clear(); // Clear the output variable list
00497     return false; // Variable sets are inconsistent
00498 }
00499 }
00500

```

```

00501 // --- Implementation of compareSingleExpressionPair ---
00529 void LogicComparator::compareSingleExpressionPair(const std::string &ref_expression_processed,
00530                                                    const std::string &comp_expression_processed,
00531                                                    const std::vector<std::string> &sorted_vars,
00532                                                    PinComparisonResult &result) {
00533     // Store processed expressions in the result struct
00534     result.ref_expr_processed = ref_expression_processed;
00535     result.comp_expr_processed = comp_expression_processed;
00536     result.comparison_possible = true; // Assume comparison is possible initially
00537
00538     spdlog::debug("Comparing processed expressions for pin '{}':", result.pin_name);
00539     spdlog::debug("  Ref : {}", ref_expression_processed);
00540     spdlog::debug("  Comp: {}", comp_expression_processed);
00541
00542     // --- ExprTk Setup ---
00543     typedef exprtk::symbol_table<double> symbol_table_t;
00544     typedef exprtk::expression<double> expression_t;
00545     typedef exprtk::parser<double> parser_t;
00546
00547     // Setup for Reference Expression
00548     symbol_table_t ref_symbol_table;
00549     expression_t ref_expression;
00550     parser_t ref_parser;
00551
00552     for (const auto &var_name : sorted_vars) {
00553         // Create variable, ExprTk manages its memory within the table
00554         if (!ref_symbol_table.create_variable(var_name)) {
00555             result.error_message = "Failed to create variable '" + var_name + "' in ref symbol table.";
00556             spdlog::error(result.error_message);
00557             result.comparison_possible = false;
00558             return; // Cannot proceed
00559         }
00560     }
00561     ref_expression.register_symbol_table(ref_symbol_table);
00562
00563     // Compile Reference Expression
00564     result.ref_compiles = ref_parser.compile(ref_expression_processed, ref_expression);
00565     if (!result.ref_compiles) {
00566         result.error_message = "Reference expression compilation failed: " + ref_parser.error();
00567         spdlog::warn("Pin '{}': {}", result.pin_name, result.error_message);
00568         result.comparison_possible = false;
00569         // Don't return yet, maybe the comparison one also fails
00570     } else {
00571         spdlog::debug("Pin '{}': Reference expression compiled successfully.", result.pin_name);
00572     }
00573
00574     // Setup for Comparison Expression
00575     symbol_table_t comp_symbol_table;
00576     expression_t comp_expression;
00577     parser_t comp_parser;
00578
00579     for (const auto &var_name : sorted_vars) {
00580         if (!comp_symbol_table.create_variable(var_name)) {
00581             // Append error if ref also failed, otherwise set it
00582             std::string comp_err = "Failed to create variable '" + var_name + "' in comp symbol table.";
00583             result.error_message += (result.error_message.empty() ? "" : "\n") + comp_err;
00584             spdlog::error(comp_err);
00585             result.comparison_possible = false; // Mark as impossible now
00586             return; // Cannot proceed if variable creation fails
00587         }
00588     }
00589     comp_expression.register_symbol_table(comp_symbol_table);
00590
00591     // Compile Comparison Expression
00592     result.comp_compiles = comp_parser.compile(comp_expression_processed, comp_expression);
00593     if (!result.comp_compiles) {
00594         std::string comp_err = "Comparison expression compilation failed: " + comp_parser.error();
00595         result.error_message += (result.error_message.empty() ? "" : "\n") + comp_err;
00596         spdlog::warn("Pin '{}': {}", result.pin_name, comp_err);

```

```

00597     result.comparison_possible = false; // Mark as impossible if not already
00598 } else {
00599     spdlog::debug("Pin '{}': Comparison expression compiled successfully.", result.pin_name);
00600 }
00601
00602 // If either failed to compile, comparison is not possible
00603 if (!result.ref_compiles || !result.comp_compiles) {
00604     result.comparison_possible = false;
00605     return;
00606 }
00607
00608 // --- Truth Table Generation and Comparison ---
00609 size_t num_vars = sorted_vars.size();
00610 // Use unsigned long long for potentially large number of combinations
00611 unsigned long long num_combinations = 1ULL << num_vars;
00612 // Prevent excessively large tables (e.g., > 20 variables is 1M+ rows)
00613 const unsigned long long MAX_COMBINATIONS = 1ULL << 20; // Limit to 2^20 combinations
00614 if (num_vars > 20) { // Check against var count for clarity
00615     result.error_message = "Too many variables (" + std::to_string(num_vars) +
00616         "). Max supported for truth table is 20.";
00617     spdlog::error("Pin '{}': {}", result.pin_name, result.error_message);
00618     result.comparison_possible = false;
00619     return;
00620 }
00621 if (num_combinations == 0 && num_vars > 0) { // Overflow check
00622     result.error_message = "Number of combinations calculation overflowed for " +
00623         std::to_string(num_vars) + " variables.";
00624     spdlog::error("Pin '{}': {}", result.pin_name, result.error_message);
00625     result.comparison_possible = false;
00626     return;
00627 }
00628
00629 std::vector<bool> ref_results;
00630 std::vector<bool> comp_results;
00631 ref_results.reserve(static_cast<size_t>(num_combinations)); // Avoid reallocations
00632 comp_results.reserve(static_cast<size_t>(num_combinations));
00633
00634 // --- Prepare Tabulate Tables ---
00635 Table ref_table;
00636 Table comp_table;
00637 Table::Row_t header_row;
00638 header_row.push_back("#"); // Row number column
00639 for (const auto &var_name : sorted_vars) {
00640     header_row.push_back(var_name);
00641 }
00642 header_row.push_back(ref_expression_processed); // Reference expression as last column header
00643 ref_table.add_row(header_row);
00644
00645 // Adjust header for comparison table
00646 header_row.back() = comp_expression_processed; // Change last header to comparison expression
00647 comp_table.add_row(header_row);
00648
00649 // Format header rows
00650 for (size_t i = 0; i < ref_table[0].size(); ++i) {
00651     ref_table[0][i].format().font_color(Color::yellow).font_style({FontStyle::bold});
00652     comp_table[0][i].format().font_color(Color::yellow).font_style({FontStyle::bold});
00653 }
00654
00655 spdlog::debug("Pin '{}': Generating truth table with {} variables ({} combinations)...",
00656     result.pin_name, num_vars, num_combinations);
00657
00658 bool evaluation_error = false;
00659 for (unsigned long long i = 0; i < num_combinations; ++i) {
00660     Table::Row_t ref_data_row;
00661     Table::Row_t comp_data_row;
00662     ref_data_row.push_back(std::to_string(i));
00663     comp_data_row.push_back(std::to_string(i));
00664
00665     // Set variable values for this combination

```

```

00666     for (size_t j = 0; j < num_vars; ++j) {
00667         // The j-th bit of i determines the value of the j-th variable
00668         bool bit_value = ((i >> j) & 1ULL);
00669         double var_value = bit_value ? double(1.0) : double(0.0); // ExprTk uses floating point
00670
00671         // Get variable reference and assign value
00672         // Need error checking here in theory, but create_variable should have caught issues
00673         ref_symbol_table.get_variable(sorted_vars[j])->ref() = var_value;
00674         comp_symbol_table.get_variable(sorted_vars[j])->ref() = var_value;
00675
00676         // Add input value to table rows (as string "0" or "1")
00677         std::string bit_str = bit_value ? "1" : "0";
00678         ref_data_row.push_back(bit_str);
00679         comp_data_row.push_back(bit_str);
00680     }
00681
00682     // Evaluate expressions
00683     double ref_val, comp_val;
00684     try {
00685         ref_val = ref_expression.value();
00686     } catch (const std::exception &e) {
00687         result.error_message +=
00688             "\nReference evaluation failed at combination " + std::to_string(i) + ": " + e.what();
00689         spdlog::error("Pin '{}': {}", result.pin_name, result.error_message);
00690         result.comparison_possible = false;
00691         evaluation_error = true;
00692         break; // Stop evaluation
00693     }
00694     try {
00695         comp_val = comp_expression.value();
00696     } catch (const std::exception &e) {
00697         result.error_message +=
00698             "\nComparison evaluation failed at combination " + std::to_string(i) + ": " + e.what();
00699         spdlog::error("Pin '{}': {}", result.pin_name, result.error_message);
00700         result.comparison_possible = false;
00701         evaluation_error = true;
00702         break; // Stop evaluation
00703     }
00704
00705     // Store boolean results (commonly, non-zero is true in logic contexts)
00706     bool ref_bool_result = (ref_val != double(0.0));
00707     bool comp_bool_result = (comp_val != double(0.0));
00708
00709     ref_results.push_back(ref_bool_result);
00710     comp_results.push_back(comp_bool_result);
00711
00712     // Add output results to table rows (as string "0" or "1")
00713     ref_data_row.push_back(ref_bool_result ? "1" : "0");
00714     comp_data_row.push_back(comp_bool_result ? "1" : "0");
00715
00716     // Add rows to tables
00717     ref_table.add_row(ref_data_row);
00718     comp_table.add_row(comp_data_row);
00719
00720 } // End of combination loop
00721
00722 if (evaluation_error) {
00723     return; // Don't proceed if evaluation failed
00724 }
00725
00726 // --- Final Comparison ---
00727 if (result.comparison_possible) {
00728     result.are_equivalent = (ref_results == comp_results);
00729     if (result.are_equivalent) {
00730         spdlog::info("Pin '{}': Expressions ARE logically equivalent.", result.pin_name);
00731     } else {
00732         spdlog::warn("Pin '{}': Expressions ARE NOT logically equivalent.", result.pin_name);
00733         result.error_message +=
00734             "\nTruth table outputs differ."; // Add specific reason for non-equivalence

```

```

00735     }
00736
00737     // Store the generated tables in the result struct
00738     result.ref_truth_table = ref_table;
00739     result.comp_truth_table = comp_table;
00740 }
00741 }
00742
00788 void LogicComparator::compareCellLogic() {
00789
00790     // 1. Collect all unique pin names from both maps
00791     std::set<std::string> unique_pin_names;
00792     for (const auto &pair : ref_outpin_map_) {
00793         unique_pin_names.insert(pair.first);
00794     }
00795     for (const auto &pair : comp_outpin_map_) {
00796         unique_pin_names.insert(pair.first);
00797     }
00798
00799     spdlog::info("Starting logic comparison for cell '{}' with {} unique output pins...", cell_name_,
00800                 unique_pin_names.size());
00801
00802     // 2. Iterate through each unique pin name
00803     for (const std::string &pin_name : unique_pin_names) {
00804         PinComparisonResult pin_result;
00805         pin_result.pin_name = pin_name;
00806         pin_result.comparison_possible = true; // Assume possible initially
00807
00808         // 3. Get raw expressions
00809         auto ref_it = ref_outpin_map_.find(pin_name);
00810         auto comp_it = comp_outpin_map_.find(pin_name);
00811
00812         if (ref_it == ref_outpin_map_.end()) {
00813             pin_result.error_message = "Pin not found in reference map.";
00814             spdlog::warn("Cell '{}', Pin '{}': {}", cell_name_, pin_name, pin_result.error_message);
00815             pin_result.comparison_possible = false;
00816             // Still try to get the comparison expression for reporting
00817             if (comp_it != comp_outpin_map_.end()) {
00818                 pin_result.comp_expr_raw = comp_it->second;
00819             }
00820         } else {
00821             pin_result.ref_expr_raw = ref_it->second;
00822             spdlog::debug("Pin -> Expression: {} -> {}", pin_name, pin_result.ref_expr_raw);
00823         }
00824
00825         if (comp_it == comp_outpin_map_.end()) {
00826             std::string comp_err = "Pin not found in comparison map.";
00827             pin_result.error_message += (pin_result.error_message.empty() ? "" : "\n") + comp_err;
00828             spdlog::warn("Cell '{}', Pin '{}': {}", cell_name_, pin_name, comp_err);
00829             pin_result.comparison_possible = false;
00830             // If ref existed, store it
00831             if (ref_it != ref_outpin_map_.end()) {
00832                 pin_result.ref_expr_raw = ref_it->second; // Already stored if ref exists
00833             }
00834         } else {
00835             pin_result.comp_expr_raw = comp_it->second;
00836             spdlog::debug("Pin => Expression: {} => {}", pin_name, pin_result.comp_expr_raw);
00837         }
00838
00839         // If pin missing in either, we can't compare, but store result and continue
00840         if (!pin_result.comparison_possible) {
00841             all_pin_results_[pin_name] = pin_result;
00842             continue;
00843         }
00844
00845         // 4. Preprocess expressions
00846         std::string ref_processed = preprocessExpression(pin_result.ref_expr_raw);
00847         std::string comp_processed = preprocessExpression(pin_result.comp_expr_raw);
00848

```



```

00849     if (ref_processed.empty()) {
00850         pin_result.error_message += "\nReference expression became empty after preprocessing.";
00851         spdlog::warn("Cell '{}', Pin '{}': {}", cell_name_, pin_name,
00852             "Reference expression became empty after preprocessing.");
00853         pin_result.comparison_possible = false;
00854     }
00855     if (comp_processed.empty()) {
00856         std::string comp_err = "\nComparison expression became empty after preprocessing.";
00857         pin_result.error_message += (pin_result.error_message.empty() ? "" : "\n") + comp_err;
00858         spdlog::warn("Cell '{}', Pin '{}': {}", cell_name_, pin_name,
00859             "Comparison expression became empty after preprocessing.");
00860         pin_result.comparison_possible = false; // Mark as impossible if not already
00861     }
00862     // If preprocessing failed for either, store and continue
00863     if (ref_processed.empty() || comp_processed.empty()) {
00864         pin_result.ref_expr_processed = ref_processed; // Store possibly empty strings
00865         pin_result.comp_expr_processed = comp_processed;
00866         all_pin_results_[pin_name] = pin_result;
00867         continue;
00868     }
00869
00870     // 5. Extract and validate variables
00871     std::vector<std::string> sorted_vars;
00872     // Pass RAW expressions to extractVariables as it uses regex on original format
00873     bool variables_match =
00874         extractVariables(pin_result.ref_expr_raw, pin_result.comp_expr_raw, sorted_vars);
00875
00876     if (!variables_match) {
00877         pin_result.error_message = "Variable sets do not match between expressions.";
00878         // extractVariables already logs details
00879         pin_result.comparison_possible = false;
00880         pin_result.ref_expr_processed = ref_processed; // Store processed even if vars mismatch
00881         pin_result.comp_expr_processed = comp_processed;
00882         all_pin_results_[pin_name] = pin_result;
00883         continue;
00884     }
00885     // --- Helper function: build log string ---
00886     auto build_log_string = [](const auto &container) {
00887         std::ostringstream oss;
00888         for (auto it = container.begin(); it != container.end(); ++it) {
00889             oss << *it;
00890             if (std::next(it) != container.end()) {
00891                 oss << ", ";
00892             }
00893         }
00894         return oss.str();
00895     };
00896     spdlog::info("Pin '{}': Variable sets match. Found unique variables: {}", pin_name,
00897         build_log_string(sorted_vars));
00898
00899     // 6. Compare the single pair using the processed expressions
00900     // Pass pin_result by reference - it will be populated by the function
00901     compareSingleExpressionPair(ref_processed, comp_processed, sorted_vars, pin_result);
00902
00903     // 7. Store the detailed result for this pin
00904     all_pin_results_[pin_name] = pin_result;
00905
00906 } // End of pin loop
00907
00908 spdlog::info("Logic comparison finished for cell '{}'.", cell_name_);
00909 }
00910
00911 void LogicComparator::generateReport(const std::string &output_file) {
00912     // --- 0. Start Info ---
00913     spdlog::info("Generating report for cell '{}' to '{}'....", cell_name_, output_file);
00914
00915     // --- 1. Determine Output Format ---
00916     bool output_markdown = false;
00917     try {

```

```

00918 // Use std::filesystem to reliably get the extension
00919 std::filesystem::path file_path(output_file);
00920 if (file_path.has_extension() && file_path.extension() == ".md") {
00921     output_markdown = true;
00922 }
00923 } catch (const std::exception &e) {
00924     spdlog::warn("Could not determine file extension for '{}': {}. Defaulting to plain text.",
00925         output_file, e.what());
00926     output_markdown = false; // Default to plain text on error
00927 }
00928
00929 // --- 2. Open Output File ---
00930 std::ofstream outfile(output_file,
00931     std::ios::app); // Use app to append if file exists, or create new
00932 if (!outfile.is_open()) {
00933     spdlog::error("Failed to open output report file: {}", output_file);
00934     return;
00935 }
00936 spdlog::info("Generating report for cell '{}' to '{}' (Format: {})...", cell_name_, output_file,
00937     output_markdown ? "Markdown" : "Plain Text");
00938
00939 // --- 3. Report Header ---
00940 outfile << "# Logic Equivalence Comparison Report\n\n";
00941 outfile << "**Cell Name: " << cell_name_ << "**\n\n";
00942 // Reference Pin Functions Table
00943 Table ref_pin_table;
00944 ref_pin_table.add_row({"Reference Pin", "Function"});
00945 ref_pin_table[0][0].format().font_style({FontStyle::bold});
00946 ref_pin_table[0][1].format().font_style({FontStyle::bold});
00947 for (const auto &[pin, function] : ref_outpin_map_) {
00948     ref_pin_table.add_row({pin, "\"" + function + "\""}); // Add backticks to function
00949 }
00950 // Comparison Pin Functions Table
00951 Table comp_pin_table;
00952 comp_pin_table.add_row({"Comparison Pin", "Function"});
00953 comp_pin_table[0][0].format().font_style({FontStyle::bold});
00954 comp_pin_table[0][1].format().font_style({FontStyle::bold});
00955 for (const auto &[pin, function] : comp_outpin_map_) {
00956     comp_pin_table.add_row({pin, "\"" + function + "\""}); // Add backticks to function
00957 }
00958
00959 // --- 4. Report Metadata ---
00960 outfile << "**Performed by " << APP_NAME << " v" << APP_VERSION << " from " << APP_AUTHOR;
00961 auto now = std::chrono::system_clock::now();
00962 std::time_t now_time_t = std::chrono::system_clock::to_time_t(now);
00963 std::tm *now_tm = std::localtime(&now_time_t);
00964 outfile << ". on: " << std::put_time(now_tm, "%c") << "**\n" << std::endl;
00965
00966 // --- 5. Legend Generation ---
00967 outfile << "## Legend\n\n";
00968 Table legend_table;
00969 legend_table.add_row({"Symbol", "Meaning"});
00970 legend_table[0][0].format().font_style({FontStyle::bold});
00971 legend_table[0][1].format().font_style({FontStyle::bold});
00972 legend_table.add_row({"[OK]", "Logically Equivalent"});
00973 legend_table.add_row({"[NO]", "Not Logically Equivalent"});
00974 legend_table.add_row({"[NA]", "Comparison Not Applicable (General)"}); // Changed from possible
00975 legend_table.add_row({"[VM]", "Variable Mismatch (Cannot Compare)"});
00976 legend_table.add_row({"[CE]", "Compilation Error (Cannot Compare)"});
00977 legend_table.add_row({"[EE]", "Evaluation Error (Cannot Compare)"});
00978 legend_table.add_row({"[PE]", "Preprocessing/Pin/Setup Error (Cannot Compare)"});
00979
00980 // --- 6. Export Legend Table ---
00981 if (output_markdown) {
00982     MarkdownExporter exporter;
00983     outfile << exporter.dump(ref_pin_table) << "\n"; // Export ref pin table
00984     outfile << exporter.dump(comp_pin_table) << "\n"; // Export comp pin table
00985     outfile << exporter.dump(legend_table) << "\n"; // Export legend table
00986 } else {

```

```

00987     outfile << ref_pin_table << "\n"; // Export ref pin table
00988     outfile << comp_pin_table << "\n"; // Export comp pin table
00989     outfile << legend_table << "\n"; // Export legend table (plain text)
00990 }
00991
00992 // --- 7. Process Pin Results ---
00993 for (const auto &[pin_name, result] : all_pin_results_) { // Use passed argument
00994     outfile << "## Pin: `" << pin_name << "`\n\n"; // Use backticks for pin name
00995
00996     // --- Output Truth Tables (Always if available) ---
00997     if (result.ref_truth_table.has_value()) {
00998         outfile << "### Reference Truth Table\n\n";
00999         Table ref_table = result.ref_truth_table.value();
01000         if (output_markdown) {
01001             MarkdownExporter exporter;
01002             outfile << exporter.dump(ref_table) << "\n";
01003         } else {
01004             outfile << ref_table << "\n";
01005         }
01006     } else {
01007         outfile << "*Reference truth table not generated (e.g., due to compilation error).*\n\n";
01008     }
01009
01010     if (result.comp_truth_table.has_value()) {
01011         outfile << "### Comparison Truth Table\n\n";
01012         Table comp_table = result.comp_truth_table.value();
01013         if (output_markdown) {
01014             MarkdownExporter exporter;
01015             outfile << exporter.dump(comp_table) << "\n";
01016         } else {
01017             outfile << comp_table << "\n";
01018         }
01019     } else {
01020         outfile << "*Comparison truth table not generated (e.g., due to compilation error).*\n\n";
01021     }
01022
01023     // --- Generate Summary Table ---
01024     outfile << "### Comparison Summary\n\n";
01025     Table summary_table;
01026     summary_table.add_row({"Property", "Value"});
01027     summary_table[0][0].format().font_style({FontStyle::bold});
01028     summary_table[0][1].format().font_style({FontStyle::bold});
01029     summary_table[0][0].format().font_color(Color::yellow);
01030     summary_table[0][1].format().font_color(Color::yellow);
01031
01032     // Determine Status Symbol using ASCII codes
01033     std::string status_symbol;
01034     if (!result.comparison_possible) {
01035         status_symbol = "[NA]"; // Default Not Applicable
01036     } else if (!result.error_message.empty()) {
01037         if (result.error_message.find("Variable sets do not match") != std::string::npos) {
01038             status_symbol = "[VM]";
01039         } else if (!result.ref_compiles || !result.comp_compiles ||
01040             result.error_message.find("compilation failed") != std::string::npos) {
01041             // Check flags first, then error message as fallback
01042             status_symbol = "[CE]";
01043         } else if (result.error_message.find("evaluation failed") != std::string::npos) {
01044             status_symbol = "[EE]";
01045         } else {
01046             // If comparison_possible is false for other reasons (e.g., pin missing, setup errors)
01047             status_symbol = "[PE]";
01048         }
01049     } else if (!result.ref_compiles || !result.comp_compiles) {
01050         // Handle cases where comparison_possible might be true but compilation failed (should
01051         // ideally not happen if logic is sound)
01052         status_symbol = "[CE]";
01053     } else {
01054         status_symbol = "[PE]"; // Fallback if no error message but still not possible
01055     }

```

```

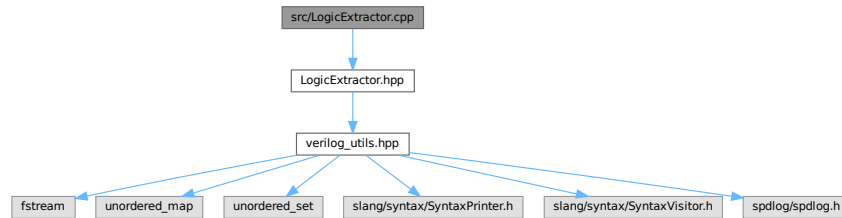
01056     } else if (result.are_equivalent) {
01057         status_symbol = "[OK]";
01058     } else {
01059         status_symbol = "[NO]";
01060     }
01061
01062     summary_table.add_row({"Status", status_symbol});
01063     summary_table.add_row({"Reference (Raw)", "" + result.ref_expr_raw + ""});
01064     summary_table.add_row({"Comparison (Raw)", "" + result.comp_expr_raw + ""});
01065     summary_table.add_row({"Reference (Processed)", "" + result.ref_expr_processed + ""});
01066     summary_table.add_row({"Comparison (Processed)", "" + result.comp_expr_processed + ""});
01067     summary_table.add_row({"Ref Expression Compiled", result.ref_compiles ? "Yes" : "No"});
01068     summary_table.add_row({"Comp Expression Compiled", result.comp_compiles ? "Yes" : "No"});
01069     summary_table.add_row({"Logically Equivalent", result.comparison_possible
01070                             ? (result.are_equivalent ? "Yes" : "No")
01071                             : "N/A"});
01072     summary_table[1][0].format().font_style({FontStyle::bold});
01073     summary_table[2][0].format().font_style({FontStyle::bold});
01074     summary_table[3][0].format().font_style({FontStyle::bold});
01075     summary_table[4][0].format().font_style({FontStyle::bold});
01076     summary_table[5][0].format().font_style({FontStyle::bold});
01077     summary_table[6][0].format().font_style({FontStyle::bold});
01078     summary_table[7][0].format().font_style({FontStyle::bold});
01079     summary_table[8][0].format().font_style({FontStyle::bold});
01080
01081     if (!result.error_message.empty()) {
01082         std::string formatted_error = result.error_message;
01083         if (output_markdown) {
01084             // Basic newline replacement for Markdown
01085             size_t pos = 0;
01086             while ((pos = formatted_error.find('\n', pos)) != std::string::npos) {
01087                 formatted_error.replace(pos, 1, "<br>");
01088                 pos += 4; // Length of "<br>"
01089             }
01090         }
01091         summary_table.add_row({"Details/Error", formatted_error});
01092     }
01093
01094     // Output Summary Table
01095     if (output_markdown) {
01096         MarkdownExporter exporter;
01097         outfile << exporter.dump(summary_table) << "\n";
01098     } else {
01099         outfile << summary_table << "\n";
01100     }
01101     std::cout << summary_table << "\n"; // Print to console for immediate feedback
01102
01103     outfile << "---\n\n"; // Separator between pins
01104
01105 } // End of pin results loop
01106
01107 outfile.close();
01108 spdlog::info("Report generation complete for cell '{}'.", cell_name_);
01109 }

```

6.40 src/LogicExtractor.cpp File Reference

```
#include "LogicExtractor.hpp"
```

Include dependency graph for LogicExtractor.cpp:



Functions

- void [extractAndPrintNetlistInfo](#) (const std::string &verilog_file, const std::string &cell)
- std::map< std::string, std::string > [extractLogicFromVerilog](#) (const std::string &verilog_file, const std::string &cell)

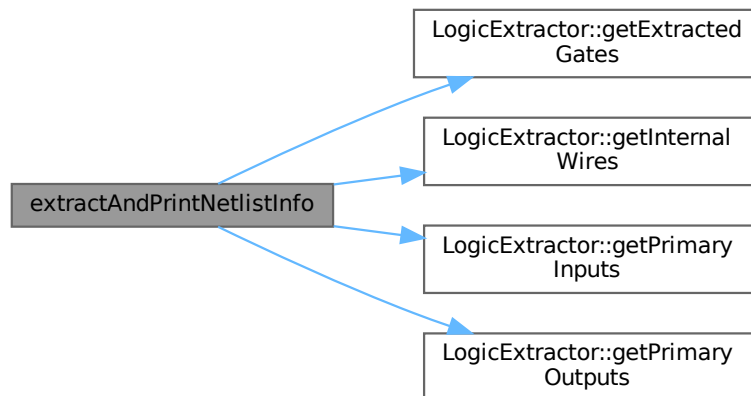
6.40.1 Function Documentation

6.40.1.1 extractAndPrintNetlistInfo()

```
void extractAndPrintNetlistInfo (
    const std::string & verilog_file,
    const std::string & cell )
```

Definition at line 514 of file [LogicExtractor.cpp](#).

Here is the call graph for this function:



6.40.1.2 extractLogicFromVerilog()

```

std::map< std::string, std::string > extractLogicFromVerilog (
    const std::string & verilog_file,
    const std::string & cell )
  
```

Definition at line 575 of file [LogicExtractor.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



6.41 LogicExtractor.cpp

[Go to the documentation of this file.](#)

```

00001 #include "LogicExtractor.hpp"
00002
00003 // --- Implementation for LogicExtractor ---
00004
00005 void LogicExtractor::handle(const slang::syntax::ModuleDeclarationSyntax &module) {
00006     if (parsingComplete_)
00007         return; // Don't re-process if called again
00008
00009     if (module.header && module.header->name.valid()) {
00010         std::string_view moduleName = module.header->name.valueText();
00011         if (!targetCell_.empty() && moduleName == targetCell_) {
00012             spdlog::info("LogicExtractor: Found target module: {}", targetCell_);
00013             inTargetModule_ = true;
00014
00015             // Reset state for the target module
00016             primaryInputs_.clear();
00017             primaryOutputs_.clear();
00018             internalWires_.clear();
00019             portDirections_.clear(); // Clear temporary direction map
00020             gateOutputDrivers_.clear();
00021             // logicCache_.clear(); // For Step 2
00022
00023             // Visit children (ports, declarations, instances) IN ORDER
00024             // It's often better to visit declarations first, then the port list
00025             // Slang's default visitDefault might handle this, but be aware.
00026             visitDefault(module);
00027
00028             // --- Finalize Ports after visiting declarations and port list ---
00029             // This part might be better placed *after* visiting NonAnsiPortList,
00030             // assuming the list visit confirms the names from the header.
00031             // Let's move the finalization logic to handle(NonAnsiPortListSyntax)
00032
00033             // Mark that we finished processing the target module
00034             inTargetModule_ = false; // Important to prevent processing other modules
00035             parsingComplete_ = true; // Stop processing after finding the target
00036             spdlog::info("LogicExtractor: Finished visiting target module '{}'.", targetCell_);
00037
00038         } else if (!parsingComplete_) {
00039             // If not the target module yet, continue searching
00040             visitDefault(module);
00041         }
00042     } else if (!parsingComplete_) {
00043         // Handle modules without valid headers if necessary, or just traverse
00044         visitDefault(module);
00045     }
00046 }
00047
00048 // Handle Port Declarations (defines direction and type, usually inside module body)
00049 void LogicExtractor::handle(const slang::syntax::PortDeclarationSyntax &portDecl) {
00050     if (!inTargetModule_)
00051         return;
00052     spdlog::debug("LogicExtractor: Handling Port Declaration");
00053
00054     std::string direction = "unknown";
00055     if (portDecl.header) {
00056         // Determine direction (input/output/inout)
00057         // Important: Use valueText() as Slang typically represents keywords as text tokens
00058         if (auto varHeader = portDecl.header->as_if<slang::syntax::VariablePortHeaderSyntax>()) {
00059             if (varHeader->direction.valid()) {
00060                 direction = std::string(varHeader->direction.valueText());
00061             }
00062         } else if (auto netHeader = portDecl.header->as_if<slang::syntax::NetPortHeaderSyntax>()) {
00063             if (netHeader->direction.valid()) {
00064                 direction = std::string(netHeader->direction.valueText());

```

```

00065     }
00066 }
00067 // Add InterfacePortHeaderSyntax if needed
00068 } else {
00069     spdlog::warn("LogicExtractor: Port declaration without header found.");
00070 }
00071
00072 for (const auto &declarator : portDecl.declarators) {
00073     if (declarator->name.valid()) {
00074         std::string portName = std::string(declarator->name.valueText());
00075         if (portDirections_.count(portName) && portDirections_[portName] != "unknown") {
00076             spdlog::warn("LogicExtractor: Multiple direction declarations for port '{}'. Keeping first "
00077                 "one '{}'. New direction: '{}'",
00078                 portName, portDirections_[portName], direction);
00079         } else {
00080             spdlog::info("LogicExtractor: Storing direction '{}' for port '{}'", direction, portName);
00081             portDirections_[portName] = direction;
00082             if (direction == "input") {
00083                 primaryInputs_.insert(portName);
00084                 internalWires_.insert(portName); // Inputs are also technically 'wires' usable internally
00085             } else if (direction == "output") {
00086                 primaryOutputs_.insert(portName);
00087                 internalWires_.insert(portName); // Outputs are also wires driven by something
00088             } else if (direction == "inout") {
00089                 spdlog::info("LogicExtractor: Inout port '{}' found. Adding to both inputs and outputs.",
00090                     portName);
00091                 primaryInputs_.insert(portName);
00092                 primaryOutputs_.insert(portName);
00093                 internalWires_.insert(portName);
00094             } else {
00095                 spdlog::warn("LogicExtractor: Port '{}' found in declaration but has unknown or missing "
00096                     "direction '{}'. Treating as internal wire.",
00097                     portName, direction);
00098                 internalWires_.insert(portName);
00099             }
00100             // logicCache_[portName] = portName; // For Step 2
00101         }
00102     } else {
00103         spdlog::warn(
00104             "LogicExtractor: Port declarator without a name found in PortDeclarationSyntax.");
00105     }
00106 }
00107 // Don't call visitDefault here, as it might revisit things unexpectedly.
00108 // Let the main module visitor handle traversing into children if necessary.
00109 }
00110
00111 // Handle Non-ANSI Port List (defines names, usually in module header)
00112 void LogicExtractor::handle(const slang::syntax::NonAnsiPortListSyntax &portList) {
00113     if (!inTargetModule_)
00114         return;
00115     spdlog::debug("LogicExtractor: Handling NonAnsi Port List");
00116
00117     for (const auto portSyntax : portList.ports) {
00118         if (!portSyntax)
00119             continue;
00120
00121         std::string portName = "";
00122         // Most common case: ImplicitNonAnsiPortSyntax contains the expression (usually just the name)
00123         if (auto implicitPort = portSyntax->as_if<slang::syntax::ImplicitNonAnsiPortSyntax>()) {
00124             if (implicitPort->expr) {
00125                 // The expression itself might be complex, try to get the simple name
00126                 if (auto portRef = implicitPort->expr->as_if<slang::syntax::PortReferenceSyntax>()) {
00127                     if (portRef->name.valid()) {
00128                         portName = std::string(portRef->name.valueText());
00129                     }
00130                 } else if (auto portConcat =
00131                     implicitPort->expr->as_if<slang::syntax::PortConcatenationSyntax>()) {
00132                     // Handle concatenations if necessary - complex
00133                     spdlog::warn("LogicExtractor: Port concatenation found in NonAnsi port list - currently "

```



```

00134         "not fully handled for logic extraction. Port: {}",
00135         implicitPort->toString());
00136     continue; // Skip complex ports for now
00137 } else {
00138     // Fallback: try getting name from the expression directly (might be just identifier)
00139     portName = implicitPort->toString();
00140 }
00141 }
00142 }
00143 // Handle ExplicitNonAnsiPortSyntax (e.g., .A(A)) if needed
00144 else if (auto explicitPort = portSyntax->as_if<slang::syntax::ExplicitNonAnsiPortSyntax>()) {
00145     if (explicitPort->name.valid()) {
00146         portName = std::string(explicitPort->name.valueText());
00147         // Note: explicitPort->expr is the internal signal it connects to.
00148         // We are primarily interested in the port's own name (explicitPort->name) here.
00149     }
00150 }
00151 // Handle EmptyNonAnsiPortSyntax (commas for placeholders) if necessary
00152 else if (portSyntax->kind == slang::syntax::SyntaxKind::EmptyNonAnsiPort) {
00153     spdlog::debug("LogicExtractor: Skipping empty non-ANSI port placeholder.");
00154     continue;
00155 } else {
00156     spdlog::warn("LogicExtractor: Unhandled NonAnsi port syntax kind: {}",
00157         slang::syntax::toString(portSyntax->kind));
00158     continue;
00159 }
00160
00161 // Now we (hopefully) have the portName.
00162 if (!portName.empty()) {
00163     if (portDirections_.count(portName)) {
00164         const std::string &direction = portDirections_[portName];
00165         spdlog::debug("LogicExtractor: Finalizing port '{}' with direction '{}'", portName,
00166             direction);
00167
00168         if (direction == "input") {
00169             primaryInputs_.insert(portName);
00170             internalWires_.insert(portName); // Inputs are also technically 'wires' usable internally
00171             // logicCache_[portName] = portName; // For Step 2
00172         } else if (direction == "output") {
00173             primaryOutputs_.insert(portName);
00174             internalWires_.insert(portName); // Outputs are also wires driven by something
00175         } else if (direction == "inout") {
00176             spdlog::info("LogicExtractor: Inout port '{}' found. Adding to both inputs and outputs.",
00177                 portName);
00178             primaryInputs_.insert(portName);
00179             primaryOutputs_.insert(portName);
00180             internalWires_.insert(portName);
00181             // logicCache_[portName] = portName; // For Step 2
00182         } else {
00183             spdlog::warn("LogicExtractor: Port '{}' found in list but has unknown or missing "
00184                 "direction '{}'. Treating as internal wire.",
00185                 portName, direction);
00186             internalWires_.insert(portName);
00187         }
00188     } else {
00189         spdlog::debug("LogicExtractor: Port '{}' found in NonAnsi list, no direction declaration ",
00190             portName);
00191     }
00192 } else {
00193     spdlog::warn("LogicExtractor: Could not determine port name from NonAnsi port list item: {}",
00194         portSyntax->toString());
00195 }
00196 }
00197 // Don't call visitDefault here
00198 }
00199
00200 // Handle explicit wire declarations
00201 void LogicExtractor::handle(const slang::syntax::NetDeclarationSyntax &netDecl) {
00202     if (!inTargetModule_)

```

```

00203     return;
00204
00205     for (const auto &declarator : netDecl.declarators) {
00206         if (declarator->name.valid()) {
00207             std::string wireName = std::string(declarator->name.valueText());
00208             spdlog::debug("LogicExtractor: Found wire declaration: {}", wireName);
00209             // Avoid adding ports again if they were also declared as nets (common)
00210             if (!primaryInputs_.count(wireName) && !primaryOutputs_.count(wireName)) {
00211                 internalWires_.insert(wireName);
00212             } else {
00213                 spdlog::trace("LogicExtractor: Wire '{}' is already known as a port.", wireName);
00214             }
00215         }
00216     }
00217     // Don't call visitDefault here
00218 }
00219
00220 // Handle primitive gate instantiations (MOST IMPORTANT PART)
00221 void LogicExtractor::handle(const slang::syntax::PrimitiveInstantiationSyntax &primitiveInst) {
00222     if (!inTargetModule_)
00223         return;
00224
00225     if (!primitiveInst.type.valid()) {
00226         spdlog::warn("LogicExtractor: Primitive instance without a type token found.");
00227         return;
00228     }
00229
00230     // Use token kind for reliable checking, text for name storage
00231     slang::parsing::TokenKind gateKind = primitiveInst.type.kind;
00232     std::string gateTypeName = std::string(primitiveInst.type.valueText());
00233
00234     spdlog::debug("LogicExtractor: Found Primitive Instance of Type: {} (Kind: {})", gateTypeName,
00235                 slang::parsing::toString(gateKind));
00236
00237     for (const auto &instance : primitiveInst.instances) {
00238         // if (!instance || !instance->connections.isInitialized()) { // Check if connections are valid
00239         //     spdlog::warn("LogicExtractor: Skipping primitive instance of type {} due to null pointer or
00240         //         "
00241         //             "uninitialized connections.",
00242         //             gateTypeName);
00243         //     continue;
00244         // }
00245
00246         GateInfo currentGateInfo;
00247         currentGateInfo.gateTypeName = gateTypeName;
00248         currentGateInfo.kind = gateKind;
00249
00250         // Primitives usually have ordered connections.
00251         // The FIRST connection is typically the OUTPUT.
00252         // The REST are INPUTS.
00253         if (instance->connections.empty()) {
00254             spdlog::warn("LogicExtractor: Gate instance of type {} has no connections.", gateTypeName);
00255             continue;
00256         }
00257
00258         // Extract Output Signal
00259         // Ensure the connection itself is not null
00260         if (!instance->connections[0]) {
00261             spdlog::error("LogicExtractor: First connection (output) is null for primitive {}",
00262                         gateTypeName);
00263             continue;
00264         }
00265         if (auto firstConn =
00266             instance->connections[0]->as_if<slang::syntax::OrderedPortConnectionSyntax>()) {
00267             currentGateInfo.outputSignal =
00268                 firstConn->expr->toString(); // Get the output signal name from the connection
00269             // remove extra spaces in the signal name
00270             currentGateInfo.outputSignal.erase(
00271                 std::remove_if(currentGateInfo.outputSignal.begin(), currentGateInfo.outputSignal.end(),

```

```

00272         [](unsigned char x) { return std::isspace(x); }},
00273         currentGateInfo.outputSignal.end());
00274     if (!currentGateInfo.outputSignal.empty()) {
00275         spdlog::debug(" Output Signal: {}", currentGateInfo.outputSignal);
00276         // Gate outputs are internal signals (unless they are module outputs)
00277         // Add to internal wires if not already a primary output.
00278         if (!primaryOutputs_.count(currentGateInfo.outputSignal)) {
00279             internalWires_.insert(currentGateInfo.outputSignal);
00280         }
00281     } else {
00282         spdlog::error(
00283             "LogicExtractor: Could not determine output signal name for gate instance of type {}",
00284             gateTypeName);
00285         continue; // Skip this instance if output is unknown
00286     }
00287 } else {
00288     spdlog::error("LogicExtractor: Expected OrderedPortConnectionSyntax for output of primitive "
00289         "{}", but got kind: {}. Conn: {}",
00290         gateTypeName, slang::syntax::toString(instance->connections[0]->kind),
00291         instance->connections[0]->toString());
00292     continue; // Skip this instance
00293 }
00294
00295 // Extract Input Signals
00296 for (size_t i = 1; i < instance->connections.size(); ++i) {
00297     // Ensure the connection itself is not null
00298     if (!instance->connections[i]) {
00299         spdlog::warn("LogicExtractor: Input connection {} is null for primitive {}", i,
00300             gateTypeName);
00301         continue;
00302     }
00303     if (auto conn =
00304         instance->connections[i]->as_if<slang::syntax::OrderedPortConnectionSyntax>()) {
00305         std::string inputSig =
00306             conn->expr->toString(); // Get the input signal name from the connection
00307         // remove extra spaces in the signal name
00308         inputSig.erase(std::remove_if(inputSig.begin(), inputSig.end(),
00309             [](unsigned char x) { return std::isspace(x); }},
00310             inputSig.end());
00311         if (!inputSig.empty()) {
00312             currentGateInfo.inputSignals.push_back(inputSig);
00313             spdlog::debug(" Input Signal {}: {}", i, inputSig);
00314         } else {
00315             spdlog::warn("LogicExtractor: Could not determine input signal name for input {} of gate "
00316                 "instance type {}. Conn: {}",
00317                 i, gateTypeName, conn->toString());
00318             // Decide whether to skip or continue with partial inputs
00319         }
00320     } else {
00321         spdlog::warn("LogicExtractor: Expected OrderedPortConnectionSyntax for input {} of "
00322             "primitive {}, but got kind: {}. Conn: {}",
00323             i, gateTypeName, slang::syntax::toString(instance->connections[i]->kind),
00324             instance->connections[i]->toString());
00325     }
00326 }
00327
00328 // Store the gate information, mapping the output signal to its driving gate
00329 if (!currentGateInfo.outputSignal.empty()) {
00330     if (gateOutputDrivers_.count(currentGateInfo.outputSignal)) {
00331         // This is a critical warning - indicates multiple drivers for the same net!
00332         spdlog::error("LogicExtractor: Multiple drivers found for signal '{}'. Previous driver: "
00333             "{}", New driver: {}. Netlist is likely invalid.",
00334             currentGateInfo.outputSignal,
00335             gateOutputDrivers_[currentGateInfo.outputSignal].gateTypeName, gateTypeName);
00336         // Keep the first one found for now, or decide on error handling
00337     } else {
00338         spdlog::info(" Storing driver for '{}': Gate Type '{}'", currentGateInfo.outputSignal,
00339             gateTypeName);
00340         gateOutputDrivers_[currentGateInfo.outputSignal] = currentGateInfo;

```

```

00341     }
00342 }
00343 }
00344 // Don't call visitDefault here
00345 }
00346
00347 // --- Logic Derivation Implementation ---
00348
00349 // Public method called after visiting the tree
00350 std::map<std::string, std::string> LogicExtractor::getLogicExpressions() {
00351     std::map<std::string, std::string> result_map;
00352     if (!parsingComplete_) {
00353         spdlog::error("LogicExtractor: AST parsing did not complete or target module '{}' not found. "
00354             "Cannot extract logic.",
00355             targetCell_);
00356         return result_map; // Return empty map
00357     }
00358
00359     spdlog::info("LogicExtractor: Deriving logic expressions for {} output ports...",
00360         primaryOutputs_.size());
00361
00362     for (const std::string &outputPort : primaryOutputs_) {
00363         spdlog::debug("LogicExtractor: Deriving logic for output: {}", outputPort);
00364         try {
00365             result_map[outputPort] = deriveLogicRecursive(outputPort);
00366             spdlog::info("  Output: {} => {}", outputPort, result_map[outputPort]);
00367         } catch (const std::runtime_error &e) {
00368             spdlog::error("LogicExtractor: Error deriving logic for output '{}': {}", outputPort,
00369                 e.what());
00370             result_map[outputPort] = "/* Error deriving logic */";
00371         } catch (...) {
00372             spdlog::error("LogicExtractor: Unknown error deriving logic for output '{}'", outputPort);
00373             result_map[outputPort] = "/* Unknown error deriving logic */";
00374         }
00375     }
00376     return result_map;
00377 }
00378
00379 // Recursive function with memoization
00380 std::string LogicExtractor::deriveLogicRecursive(const std::string &signalName) {
00381     // 1. Check Cache (Memoization)
00382     if (logicCache_.count(signalName)) {
00383         return logicCache_.at(signalName);
00384     }
00385
00386     // 2. Base Case: Is it a primary input?
00387     if (primaryInputs_.count(signalName)) {
00388         // Already cached during port handling, but double-check
00389         if (!logicCache_.count(signalName)) {
00390             logicCache_[signalName] = signalName;
00391         }
00392         return signalName;
00393     }
00394
00395     // 3. Recursive Step: Is it driven by a gate?
00396     if (gateOutputDrivers_.count(signalName)) {
00397         const GateInfo &driverGate = gateOutputDrivers_.at(signalName);
00398         std::vector<std::string> inputExpressions;
00399         spdlog::debug("  Tracing signal '{}', driven by {} gate", signalName,
00400             driverGate.gateTypeName());
00401
00402         // Recursively find expressions for all inputs of this gate
00403         for (const std::string &inputSig : driverGate.inputSignals) {
00404             if (inputSig.empty()) {
00405                 throw std::runtime_error("Empty input signal name encountered for gate driving " +
00406                     signalName);
00407             }
00408             spdlog::debug("    Recursing for input: {}", inputSig);
00409             inputExpressions.push_back(deriveLogicRecursive(inputSig));

```

```

00410     }
00411
00412     // Format the expression based on gate type and input expressions
00413     std::string currentExpr = formatExpression(driverGate, inputExpressions);
00414
00415     // Cache the result
00416     logicCache_[signalName] = currentExpr;
00417     return currentExpr;
00418 }
00419
00420 // 4. Handle Assign statements (if implemented)
00421 // if (assignDrivers_.count(signalName)) { ... }
00422
00423 // 5. Error Case: Signal not found or not driven by known element
00424 // Check if it's just an internal wire that wasn't driven?
00425 if (internalWires_.count(signalName)) {
00426     throw std::runtime_error(
00427         "Signal '" + signalName +
00428         "' is an internal wire but has no identified driver (gate or assign).");
00429 } else {
00430     throw std::runtime_error(
00431         "Signal '" + signalName +
00432         "' is not a primary input, known wire, or driven by a recognized gate/assignment.");
00433 }
00434 }
00435
00436 // Helper to format the expression string based on gate type
00437 std::string LogicExtractor::formatExpression(const GateInfo &gateInfo,
00438                                             const std::vector<std::string> &inputExprs) {
00439     if (inputExprs.empty() && gateInfo.kind != slang::parsing::TokenKind::NotKeyword &&
00440         gateInfo.kind != slang::parsing::TokenKind::BufKeyword) {
00441         // Gates like AND/OR/XOR need inputs
00442         spdlog::warn("Gate type {} requires inputs, but none were provided/derived for output {}",
00443             gateInfo.gateTypeName, gateInfo.outputSignal);
00444         return "/*Error: Missing Inputs for " + gateInfo.gateTypeName + ">*/";
00445     }
00446
00447     std::string result = "";
00448
00449     // Use gateInfo.type (enum) for reliable checking
00450     switch (gateInfo.kind) {
00451     case slang::parsing::TokenKind::AndKeyword:
00452         result = "(" + inputExprs[0];
00453         for (size_t i = 1; i < inputExprs.size(); ++i)
00454             result += " * " + inputExprs[i]; // Use * for AND
00455         result += ")";
00456         break;
00457     case slang::parsing::TokenKind::NandKeyword:
00458         result = "!(" + inputExprs[0];
00459         for (size_t i = 1; i < inputExprs.size(); ++i)
00460             result += " * " + inputExprs[i];
00461         result += ")";
00462         break;
00463     case slang::parsing::TokenKind::OrKeyword:
00464         result = "(" + inputExprs[0];
00465         for (size_t i = 1; i < inputExprs.size(); ++i)
00466             result += " + " + inputExprs[i]; // Use + for OR
00467         result += ")";
00468         break;
00469     case slang::parsing::TokenKind::NorKeyword:
00470         result = "!(" + inputExprs[0];
00471         for (size_t i = 1; i < inputExprs.size(); ++i)
00472             result += " + " + inputExprs[i];
00473         result += ")";
00474         break;
00475     case slang::parsing::TokenKind::XorKeyword:
00476         result = "(" + inputExprs[0];
00477         for (size_t i = 1; i < inputExprs.size(); ++i)
00478             result += " ^ " + inputExprs[i]; // Use ^ for XOR

```

```

00479     result += ")";
00480     break;
00481 case slang::parsing::TokenKind::XnorKeyword:
00482     result = "!(" + inputExprs[0];
00483     for (size_t i = 1; i < inputExprs.size(); ++i)
00484         result += " ^ " + inputExprs[i];
00485     result += ")";
00486     break;
00487 case slang::parsing::TokenKind::NotKeyword: // NOT gate
00488     if (inputExprs.size() != 1) {
00489         spdlog::warn("NOT gate expects 1 input, got {}", inputExprs.size());
00490         return "/*<Error: Incorrect Inputs for NOT>*/";
00491     }
00492     result = "!" + inputExprs[0]; // Use ! for NOT
00493     break;
00494 case slang::parsing::TokenKind::BufKeyword: // BUF gate
00495     if (inputExprs.size() != 1) {
00496         spdlog::warn("BUF gate expects 1 input, got {}", inputExprs.size());
00497         return "/*<Error: Incorrect Inputs for BUF>*/";
00498     }
00499     result = inputExprs[0]; // Output is the same as input
00500     break;
00501 // Add cases for bufif0, bufif1, notif0, notif1, pullup, pulldown, cmos, nmos, pmos, tran etc. if
00502 // needed
00503 default:
00504     spdlog::warn("Unsupported primitive gate type for logic expression generation: {}",
00505                 gateInfo.gateTypeName);
00506     result = "/*<Unsupported Gate: " + gateInfo.gateTypeName + ">*/";
00507     break;
00508 }
00509 return result;
00510 }
00511
00512 // --- New Function Implementation (Step 1: Visit and Print) ---
00513
00514 void extractAndPrintNetlistInfo(const std::string &verilog_file, const std::string &cell) {
00515     spdlog::info("--- Step 1: Starting Netlist Info Extraction ---");
00516     spdlog::info("Verilog file: '{}', Target cell: '{}'", verilog_file, cell);
00517
00518     try {
00519         auto result = slang::syntax::SyntaxTree::fromFile(verilog_file);
00520         if (!result) { // Check if result is valid
00521             spdlog::error("Error parsing Verilog file '{}'. Cannot extract info.", verilog_file);
00522             return;
00523         }
00524
00525         spdlog::info("Successfully parsed Verilog file.");
00526         std::shared_ptr<slang::syntax::SyntaxTree> tree = result.value();
00527
00528         LogicExtractor extractor(cell);
00529         tree->root().visit(extractor); // Populate extractor's internal state
00530
00531         // --- Print Summary (Debug for Step 1) ---
00532         spdlog::info("--- Extraction Summary for Cell '{}': ---", cell);
00533
00534         const auto &inputs = extractor.getPrimaryInputs();
00535         spdlog::info("Found {} Primary Inputs:", inputs.size());
00536         for (const auto &name : inputs) {
00537             spdlog::info("  - {}", name);
00538         }
00539
00540         const auto &outputs = extractor.getPrimaryOutputs();
00541         spdlog::info("Found {} Primary Outputs:", outputs.size());
00542         for (const auto &name : outputs) {
00543             spdlog::info("  - {}", name);
00544         }
00545
00546         const auto &wires = extractor.getInternalWires();
00547         spdlog::info("Found {} Internal Wires:", wires.size());

```

```

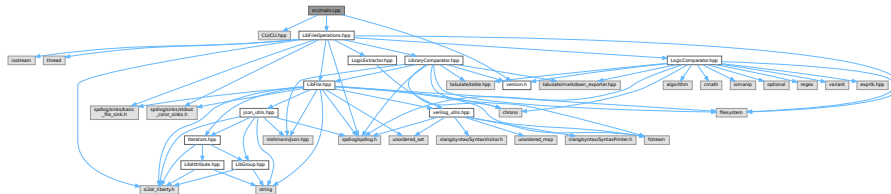
00548     for (const auto &name : wires) {
00549         spdlog::info(" - {}", name);
00550     }
00551
00552     const auto &gates = extractor.getExtractedGates();
00553     spdlog::info("Found {} Gate Drivers:", gates.size());
00554     for (const auto &pair : gates) {
00555         const std::string &outputNet = pair.first;
00556         const GateInfo &info = pair.second;
00557         std::string inputsStr = "";
00558         for (size_t i = 0; i < info.inputSignals.size(); ++i) {
00559             inputsStr += info.inputSignals[i] + (i == info.inputSignals.size() - 1 ? " : " : ", ");
00560         }
00561         spdlog::info(" - Output: {} <= Driven by: {} ({} Inputs: [{}]", outputNet,
00562             info.gateTypeName, slang::parsing::toString(info.kind), inputsStr);
00563     }
00564     spdlog::info("--- End Netlist Info Extraction ---");
00565
00566 } catch (const std::exception &e) {
00567     spdlog::error("Exception during Verilog parsing or info extraction: {}", e.what());
00568 } catch (...) {
00569     spdlog::error("Unknown exception during Verilog parsing or info extraction.");
00570 }
00571 }
00572
00573 // --- (Step 2: Extract Logic Expressions) ---
00574
00575 std::map<std::string, std::string> extractLogicFromVerilog(const std::string &verilog_file,
00576     const std::string &cell) {
00577     spdlog::info("--- Starting Logic Expression Extraction for cell: '{}' ---", cell);
00578     std::map<std::string, std::string> logicMap; // Default empty map
00579
00580     try {
00581         auto result = slang::syntax::SyntaxTree::fromFile(verilog_file);
00582         if (!result) {
00583             spdlog::error("Error parsing Verilog file '{}'. Cannot extract logic.", verilog_file);
00584             return logicMap;
00585         }
00586
00587         spdlog::info("Successfully parsed Verilog file.");
00588         std::shared_ptr<slang::syntax::SyntaxTree> tree = result.value();
00589
00590         LogicExtractor extractor(cell);
00591         tree->root().visit(extractor); // Populate extractor's internal state
00592
00593         // Now, call the method to derive and get the expressions
00594         logicMap = extractor.getLogicExpressions();
00595
00596     } catch (const std::exception &e) {
00597         spdlog::error("Exception during Verilog parsing or logic extraction: {}", e.what());
00598     } catch (...) {
00599         spdlog::error("Unknown exception during Verilog parsing or logic extraction.");
00600     }
00601
00602     if (logicMap.empty()) {
00603         spdlog::warn("Logic extraction finished, but no expressions were derived for cell '{}'. Check "
00604             "if cell exists and is correctly defined.",
00605             cell);
00606     } else {
00607         spdlog::info("Logic extraction completed for cell '{}'. Found {} output expression strings",
00608             cell, logicMap.size());
00609     }
00610
00611     spdlog::info("--- End Logic Expression Extraction ---");
00612     return logicMap;
00613 }

```

6.42 src/main.cpp File Reference

```
#include "CLI/CLI.hpp"
#include "LibFileOperations.hpp"
#include "version.h"
```

Include dependency graph for main.cpp:



Functions

- int [main](#) (int argc, char *argv[])

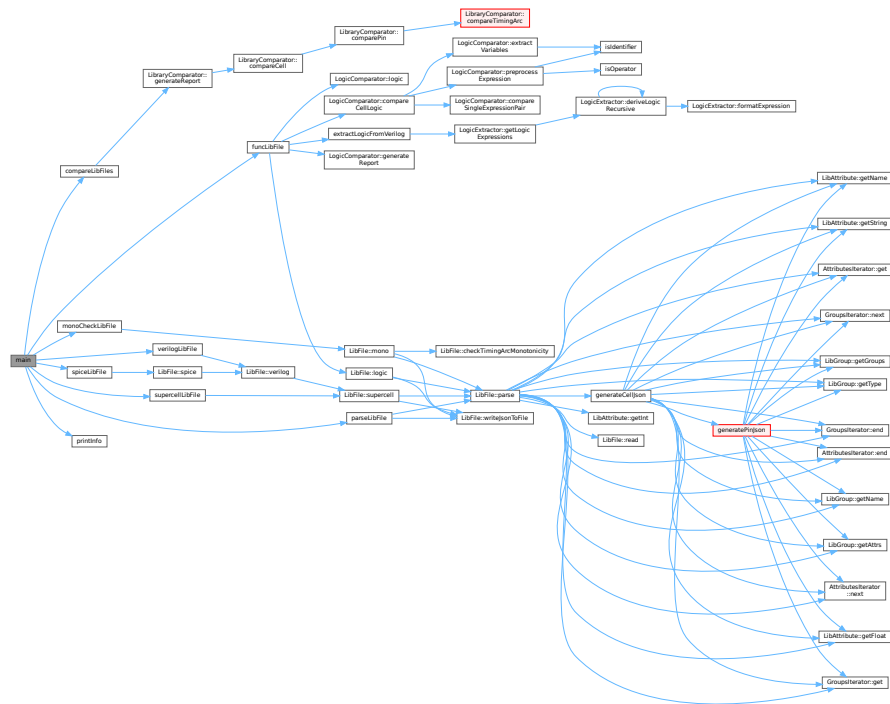
6.42.1 Function Documentation

6.42.1.1 main()

```
int main (
    int argc,
    char * argv[] )
```

Definition at line 8 of file [main.cpp](#).

Here is the call graph for this function:



6.43 main.cpp

[Go to the documentation of this file.](#)

```
00001 // ./src/main.cpp
00002
00003 #include "CLI/CLI.hpp"
00004
00005 #include "LibFileOperations.hpp"
00006 #include "version.h"
00007
00008 int main(int argc, char *argv[]) {
00009     // Parse command line arguments
00010     std::vector<std::string> library_paths; // Support multiple files
00011     std::string log_file_name = "";
00012
00013     // Command line parameter parsing using CLI11
00014     CLI::App app{APP_NAME};
00015     app.set_version_flag("-v,--version", APP_VERSION);
00016
00017     // Add subcommand for parse mode
00018     CLI::App *parse_cmd =
00019         app.add_subcommand("parse", "Parse the Liberty file and write JSON to a file");
00020     parse_cmd->add_option("library_path", library_paths, "Specify the library file to process")
00021         ->check(CLI::ExistingFile)
00022         ->required();
00023     parse_cmd->add_option("-l,--log", log_file_name,
00024         "Specify the log file name. Default: <basename>.parse.log");
00025     parse_cmd->callback([&] {
00026         printInfo();
00027         // Check if multi files
00028         if (library_paths.size() > 1) {
00029             spdlog::info("Running sequential parsing for {} files.", library_paths.size());
```

```

00030     spdlog::info("Each library will write to its own log file.");
00031     // Sequential parsing
00032     for (const auto &library_path : library_paths) {
00033         parseLibFile(library_path, log_file_name = "");
00034     }
00035 } else {
00036     parseLibFile(library_paths[0], log_file_name);
00037 }
00038 });
00039
00040 // Add subcommand for mono check mode
00041 bool is_slew = false;
00042 CLI::App *mono_cmd = app.add_subcommand("mono", "Check the monotonicity of timing arc values");
00043 mono_cmd->add_option("library_path", library_paths, "Specify the library file to process")
00044     ->check(CLI::ExistingFile)
00045     ->required();
00046 mono_cmd->add_option("-l,--log", log_file_name,
00047     "Specify the log file name. Default: <basename>.mono.log");
00048 mono_cmd->add_flag("-s,--slew", is_slew,
00049     "Specify that monotonicity checks also include input slew.");
00050 mono_cmd->callback([&] {
00051     printInfo();
00052     // Check if multi files
00053     if (library_paths.size() > 1) {
00054         spdlog::info("Running multi-threaded monotonicity check for {} files.", library_paths.size());
00055         spdlog::info("Each thread will write to its own log file.");
00056
00057         // Sequential json file check
00058         for (const auto &library_path : library_paths) {
00059             if (!std::filesystem::exists(std::filesystem::path(library_path).stem().string() +
00060                 ".json")) {
00061                 spdlog::info("JSON file not found for '{}'. Parsing Liberty file first.", library_path);
00062                 parseLibFile(library_path, log_file_name = "");
00063             }
00064         }
00065         spdlog::info("All JSON files prepared.");
00066         // Parallel monotonicity check
00067         std::vector<std::thread> threads;
00068         for (const auto &library_path : library_paths) {
00069             threads.emplace_back(monoCheckLibFile, library_path, log_file_name = "", is_slew);
00070         }
00071         for (auto &thread : threads) {
00072             thread.join();
00073         }
00074         spdlog::info("All threads completed.");
00075     } else {
00076         monoCheckLibFile(library_paths[0], log_file_name, is_slew);
00077     }
00078 });
00079
00080 // Add subcommand for compare mode
00081 std::string ref_lib, comp_lib, report_file_name;
00082 CLI::App *compare_cmd = app.add_subcommand(
00083     "compare",
00084     "Compare the comparison library against the reference library and report differences");
00085 compare_cmd->add_option("--ref", ref_lib, "Specify the reference library file")
00086     ->check(CLI::ExistingFile)
00087     ->required();
00088 compare_cmd->add_option("--comp", comp_lib, "Specify the comparison library file")
00089     ->check(CLI::ExistingFile)
00090     ->required();
00091 double abstol = 0.002;
00092 compare_cmd->add_option("--abstol", abstol,
00093     "Specify the absolute tolerance for comparison. Default: 0.002ns");
00094 double reltol = 0.02;
00095 compare_cmd->add_option("--reltol", reltol,
00096     "Specify the relative tolerance for comparison. Default: 0.02/2.0%");
00097 compare_cmd->add_option("--report", report_file_name,
00098     "Specify the report file name. Default: <comp_lib>.cmp.md");

```

```

00099 compare_cmd->callback([&] {
00100     printInfo();
00101     compareLibFiles(ref_lib, comp_lib, reltol, abstol, report_file_name);
00102 });
00103
00104 // Add subcommand for supercell generation
00105 int chain_length = 1;
00106 CLI::App *supercell_cmd =
00107     app.add_subcommand("supercell", "Generate supercells for the given Liberty file");
00108 supercell_cmd->add_option("library_path", library_paths, "Specify the library file to process")
00109     ->check(CLI::ExistingFile)
00110     ->required();
00111 supercell_cmd->add_option("-l,--log", log_file_name,
00112     "Specify the log file name. Default: <basename>.supercell.log");
00113 supercell_cmd->add_option("-c,--chain", chain_length,
00114     "Specify the chain length for supercell generation. Default: 1");
00115 std::vector<std::string> cell_names = {}; // "CMPE42D1" "AN2D0", "DFQD1"
00116 supercell_cmd->add_option("--cells", cell_names,
00117     "Specify the cell names to generate supercells for");
00118 supercell_cmd->callback([&] {
00119     printInfo();
00120     // Check if multi files
00121     if (library_paths.size() > 1) {
00122         spdlog::info("Running multi-threaded supercell generation for {} files.",
00123             library_paths.size());
00124         spdlog::info("Each library will write to its own log file.");
00125
00126         // Sequential json file check
00127         for (const auto &library_path : library_paths) {
00128             if (!std::filesystem::exists(std::filesystem::path(library_path).stem().string() +
00129                 ".json")) {
00130                 spdlog::info("JSON file not found for '{}'. Parsing Liberty file first.", library_path);
00131                 parseLibFile(library_path, log_file_name = "");
00132             }
00133         }
00134         spdlog::info("All JSON files prepared.");
00135         // Parallel supercell generation
00136         std::vector<std::thread> threads;
00137         for (const auto &library_path : library_paths) {
00138             threads.emplace_back(supercellLibFile, library_path, log_file_name = "", chain_length,
00139                 cell_names);
00140         }
00141         for (auto &thread : threads) {
00142             thread.join();
00143         }
00144         spdlog::info("All threads completed.");
00145     } else {
00146         supercellLibFile(library_paths[0], log_file_name, chain_length, cell_names);
00147     }
00148 });
00149
00150 // Add subcommand for zlibboost
00151 CLI::App *zlibboost_cmd =
00152     app.add_subcommand("zlibboost", "ZlibBoost - Multi-threaded Library Processing Tool");
00153 std::string config_dir = "/home/songzx/Projects/zlibboost/config.tcl";
00154 std::string python_dir = "/home/guocj/anaconda3/envs/myenv/bin/python";
00155 std::string main_py_dir = "/home/songzx/Projects/zlibboost/zlibboost.py";
00156 zlibboost_cmd->add_option(
00157     "-c, --config", config_dir,
00158     "Specify the configuration TCL file. Default: /home/songzx/Projects/zlibboost/config.tcl");
00159 zlibboost_cmd->add_option(
00160     "--python", python_dir,
00161     "Specify the python directory. Default: /home/guocj/anaconda3/envs/myenv/bin/python");
00162 zlibboost_cmd->add_option("--main", main_py_dir,
00163     "Specify the main python script directory. Default: "
00164     "/home/songzx/Projects/zlibboost/zlibboost.py");
00165 zlibboost_cmd->callback([&] {
00166     printInfo();
00167     // Run the ZlibBoost tool

```

```

00168     std::string command = python_dir + " " + main_py_dir + " -c " + config_dir;
00169     spdlog::info("Running ZlibBoost with command: '{}'", command);
00170     int ret = std::system(command.c_str());
00171     if (ret == 0) {
00172         spdlog::info("ZlibBoost completed successfully.");
00173     } else {
00174         spdlog::error("ZlibBoost failed with return code: {}", ret);
00175     }
00176 });
00177
00178 // Add subcommand for clear
00179 CLI::App *clear_cmd = app.add_subcommand(
00180     "clear", "Clear the log, JSON, map, markdown, Verilog, SPICE files in this directory");
00181 clear_cmd->callback([&] {
00182     printInfo();
00183     std::filesystem::path current_dir = std::filesystem::current_path();
00184     for (const auto &entry : std::filesystem::directory_iterator(current_dir)) {
00185         if (entry.path().extension() == ".log" || entry.path().extension() == ".json" ||
00186             entry.path().extension() == ".map" || entry.path().extension() == ".md" ||
00187             entry.path().extension() == ".v" || entry.path().extension() == ".spi") {
00188             spdlog::info("Removing file: '{}'", entry.path().string());
00189             std::filesystem::remove(entry.path());
00190         }
00191     }
00192     spdlog::info("All log, JSON, map, markdown files cleared.");
00193 });
00194
00195 // Add subcommand for Verilog generation
00196 CLI::App *verilog_cmd =
00197     app.add_subcommand("verilog", "Generate Verilog file for given Liberty file");
00198 verilog_cmd->add_option("library_path", library_paths, "Specify the library file to process")
00199     ->check(CLI::ExistingFile)
00200     ->required();
00201 verilog_cmd->add_option("-l,--log", log_file_name,
00202     "Specify the log file name. Default: <basename>.verilog.log");
00203 verilog_cmd->add_option("-c,--chain", chain_length,
00204     "Specify the chain length for verilog generation. Default: 1");
00205 verilog_cmd->add_option("--cells", cell_names, "Specify the cell names to generate Verilog for");
00206 verilog_cmd->callback([&] {
00207     printInfo();
00208     // Check if multi files
00209     if (library_paths.size() > 1) {
00210         spdlog::info("Running multi-threaded Verilog generation for {} files.", library_paths.size());
00211         spdlog::info("Each library will write to its own log file.");
00212
00213         // Sequential json file check
00214         for (const auto &library_path : library_paths) {
00215             if (!std::filesystem::exists(std::filesystem::path(library_path).stem().string() +
00216                 ".json")) {
00217                 spdlog::info("JSON file not found for '{}'. Parsing Liberty file first.", library_path);
00218                 parseLibFile(library_path, log_file_name = "");
00219             }
00220         }
00221         spdlog::info("All JSON files prepared.");
00222         // Parallel Verilog generation
00223         std::vector<std::thread> threads;
00224         for (const auto &library_path : library_paths) {
00225             threads.emplace_back(verilogLibFile, library_path, log_file_name = "", chain_length,
00226                 cell_names);
00227         }
00228         for (auto &thread : threads) {
00229             thread.join();
00230         }
00231         spdlog::info("All threads completed.");
00232     } else {
00233         verilogLibFile(library_paths[0], log_file_name, chain_length, cell_names);
00234     }
00235 });
00236

```

```

00237 // Add subcommand for SPICE generation
00238 CLI::App *spice_cmd = app.add_subcommand("spice", "Generate SPICE file for given Liberty file");
00239 spice_cmd->add_option("library_path", library_paths, "Specify the library file to process")
00240     ->check(CLI::ExistingFile)
00241     ->required();
00242 spice_cmd->add_option("-l,--log", log_file_name,
00243     "Specify the log file name. Default: <basename>.spice.log");
00244 spice_cmd->add_option("-c,--chain", chain_length,
00245     "Specify the chain length for SPICE generation. Default: 1");
00246 spice_cmd->add_option("--cells", cell_names, "Specify the cell names to generate SPICE for");
00247 std::string verilog_lib_file =
00248     "/home/songzx/examples/mydpk/TSMC65/TSMC65NM_CLN65LP_STDIO_STDCELL/tcbn65lp_220a/"
00249     "0396011_20170308/TSMCHOME/digital/Front_End/verilog/tcbn65lp.v";
00250 std::string spice_lib_file =
00251     "/home/songzx/examples/mydpk/TSMC65/TSMC65NM_CLN65LP_STDIO_STDCELL/tcbn65lp_220a/"
00252     "0396011_20170308/TSMCHOME/digital/Back_End/lpe_spice/tcbn65lp_200a/tcbn65lp_200a_lpe.spi";
00253 spice_cmd->add_option("--vl", verilog_lib_file,
00254     "Specify the location of the Verilog primitive library file");
00255 spice_cmd->add_option(
00256     "--sl", spice_lib_file,
00257     "Specify the location of the SPICE library file to be included in the output");
00258 spice_cmd->callback([&] {
00259     printInfo();
00260     // Check if multi files
00261     if (library_paths.size() > 1) {
00262         spdlog::info("Running multi-threaded SPICE generation for {} files.", library_paths.size());
00263         spdlog::info("Each library will write to its own log file.");
00264
00265         // Sequential json file check
00266         for (const auto &library_path : library_paths) {
00267             if (!std::filesystem::exists(std::filesystem::path(library_path).stem().string() +
00268                 ".json")) {
00269                 spdlog::info("JSON file not found for '{}'. Parsing Liberty file first.", library_path);
00270                 parseLibFile(library_path, log_file_name = "");
00271             }
00272         }
00273         spdlog::info("All JSON files prepared.");
00274         // Parallel SPICE generation
00275         std::vector<std::thread> threads;
00276         for (const auto &library_path : library_paths) {
00277             threads.emplace_back(spiceLibFile, library_path, log_file_name = "", chain_length,
00278                 cell_names, verilog_lib_file, spice_lib_file);
00279         }
00280         for (auto &thread : threads) {
00281             thread.join();
00282         }
00283         spdlog::info("All threads completed.");
00284     } else {
00285         spiceLibFile(library_paths[0], log_file_name, chain_length, cell_names, verilog_lib_file,
00286             spice_lib_file);
00287     }
00288 });
00289
00290 // Add subcommand for funtional equivalence check
00291 CLI::App *func_cmd = app.add_subcommand(
00292     "func", "Check functional equivalence of two Liberty files or Verilog files");
00293 std::string ref_file, comp_file;
00294 func_cmd->add_option("--ref", ref_file, "Specify the reference Liberty or Verilog file")
00295     ->check(CLI::ExistingFile)
00296     ->required();
00297 func_cmd->add_option("--comp", comp_file, "Specify the comparison Liberty or Verilog file")
00298     ->check(CLI::ExistingFile)
00299     ->required();
00300 func_cmd->add_option("--cells", cell_names,
00301     "Specify the cell names to check functional equivalence for");
00302 func_cmd->add_option("--report", report_file_name,
00303     "Specify the report file name. Default: <comp_lib>.cmp.md");
00304
00305 func_cmd->callback([&] {

```

```

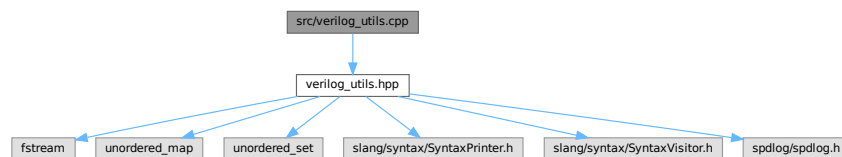
00306     printInfo();
00307     funcLibFile(ref_file, comp_file, cell_names, report_file_name);
00308 };
00309
00310 CLI11_PARSE(app, argc, argv);
00311
00312 // End of program
00313 char hostname[256];
00314 gethostname(hostname, sizeof(hostname));
00315
00316 auto now = std::chrono::system_clock::now();
00317 auto time_now = std::chrono::system_clock::to_time_t(now);
00318 std::stringstream ss;
00319 ss << std::put_time(std::localtime(&time_now), "%c");
00320
00321 spdlog::info("ZlibValidation exited on '{}' at {}", hostname, ss.str());
00322 return 0;
00323 }

```

6.44 src/verilog_utils.cpp File Reference

```
#include "verilog_utils.hpp"
```

Include dependency graph for verilog_utils.cpp:



Functions

- void [getAST](#) (const std::string &verilog_file, const std::string &cell)

6.44.1 Function Documentation

6.44.1.1 getAST()

```

void getAST (
    const std::string & verilog_file,
    const std::string & cell )

```

Definition at line [544](#) of file [verilog_utils.cpp](#).

Here is the call graph for this function:



6.45 verilog_utils.cpp

[Go to the documentation of this file.](#)

```

00001 // verilog_utils.cpp
00002
00003 #include "verilog_utils.hpp"
00004
00005 // Adding a generic handler method to record all visited nodes and increment the depth counter
00006 void VerilogVisitor::handle(const slang::syntax::SyntaxNode &node) {
00007     std::string indent(depth_ * 2, ' ');
00008     spdlog::debug("{}Node: {}", indent, slang::syntax::toString(node.kind));
00009
00010     // Increase depth, visit child nodes, then decrease depth
00011     depth_++;
00012     visitDefault(node);
00013     depth_--;
00014 }
00015
00016 // Handle module declarations
00017 void VerilogVisitor::handle(const slang::syntax::ModuleDeclarationSyntax &module) {
00018     std::string indent(depth_ * 2, ' ');
00019
00020     if (module.header && module.header->name.valid()) {
00021         std::string_view moduleName = module.header->name.valueText();
00022         spdlog::debug("{}Module Name: {}", indent, moduleName);
00023
00024         inTargetModule_ = !targetCell_.empty() && moduleName == targetCell_;
00025         if (!inTargetModule_) {
00026             return;
00027         }
00028
00029         // If a specific module name is specified, check if it matches
00030         if (inTargetModule_) {
00031             spdlog::info("{}Found target module: {}", indent, targetCell_);
00032             // Print the module ports
00033             if (module.header->ports) {
00034                 spdlog::info("{}Ports: {}", indent, module.header->ports->toString());
00035             }
00036         }
00037     }
00038
00039     // Continue processing child nodes
00040     depth_++;
00041     visitDefault(module);
00042     depth_--;
00043 }
00044
00045 // Handle port declarations
00046 void VerilogVisitor::handle(const slang::syntax::PortDeclarationSyntax &portDecl) {
00047     if (!inTargetModule_) {
  
```

```

00048     return;
00049 }
00050
00051 std::string indent = std::string(depth_ * 2, ' ');
00052
00053 // Get port direction
00054 std::string direction = "unknown";
00055 if (portDecl.header) {
00056     // Try to convert the header to different types of port headers
00057     if (auto *varPort = portDecl.header->as_if<slang::syntax::VariablePortHeaderSyntax>()) {
00058         if (varPort->direction) {
00059             direction = varPort->direction.valueText();
00060         }
00061     } else if (auto *netPort = portDecl.header->as_if<slang::syntax::NetPortHeaderSyntax>()) {
00062         if (netPort->direction) {
00063             direction = netPort->direction.valueText();
00064         }
00065     }
00066
00067     // Get port name
00068     for (const auto &declarator : portDecl.declarators) {
00069         if (declarator->name.valid()) {
00070             std::string_view portName = declarator->name.valueText();
00071             spdlog::info("{}Port Name: {} ({})", indent, portName, direction);
00072         }
00073     }
00074 }
00075 }
00076
00077 // Handle hierarchical instantiation (module instance)
00078 void VerilogVisitor::handle(const slang::syntax::HierarchyInstantiationSyntax &hierarchyInst) {
00079     if (!inTargetModule_) {
00080         return;
00081     }
00082
00083     std::string indent(depth_ * 2, ' ');
00084
00085     if (hierarchyInst.type.valid()) {
00086         std::string_view instType = hierarchyInst.type.valueText();
00087         spdlog::info("{}Hierarchy Instance Type: {}", indent, instType);
00088
00089         // Safely print parameter values (if any)
00090         try {
00091             if (hierarchyInst.parameters) {
00092                 spdlog::info("{}Parameters:", indent);
00093                 for (const auto &param : hierarchyInst.parameters->parameters) {
00094                     if (!param)
00095                         continue; // Null pointer check
00096
00097                     if (auto *orderedParam = param->as_if<slang::syntax::OrderedParamAssignmentSyntax>()) {
00098                         if (orderedParam->expr) {
00099                             spdlog::info("{} Parameter: {}", indent, orderedParam->expr->toString());
00100                         }
00101                     } else if (auto *namedParam = param->as_if<slang::syntax::NamedParamAssignmentSyntax>()) {
00102                         if (namedParam->name.valid() && namedParam->expr) {
00103                             spdlog::info("{} Parameter {}: {}", indent, namedParam->name.valueText(),
00104                                     namedParam->expr->toString());
00105                         }
00106                     }
00107                 }
00108             }
00109         } catch (const std::exception &e) {
00110             spdlog::warn("{}Exception while processing parameters: {}", indent, e.what());
00111         }
00112
00113         // Safely handle instances
00114         try {
00115             for (const auto &instance : hierarchyInst.instances) {
00116                 if (!instance) {

```



```

00117         spdlog::warn("{}Invalid instance", indent);
00118         continue;
00119     } else if (!instance->decl) {
00120         spdlog::warn("{}Invalid instance declaration", indent);
00121         // continue;
00122     } else {
00123         try {
00124             if (instance->decl->name.valid()) {
00125                 std::string_view instName = instance->decl->name.valueText();
00126                 spdlog::info("{}Instance Name: {}", indent, instName);
00127             }
00128         } catch (...) {
00129             spdlog::debug("{}Error printing name", indent);
00130         }
00131
00132         // Safely print dimensions
00133         try {
00134             if (!instance->decl->dimensions.empty()) {
00135                 spdlog::info("{} Dimensions: {}", indent, instance->decl->dimensions.toString());
00136             }
00137         } catch (...) {
00138             spdlog::debug("{}Error printing dimensions", indent);
00139         }
00140     }
00141
00142     // Safely print port connections
00143     spdlog::info("{} Port connections:", indent);
00144     for (const auto &conn : instance->connections) {
00145         if (!conn) {
00146             spdlog::warn("{} Null connection", indent);
00147             continue; // Null pointer check
00148         }
00149
00150         try {
00151             // Use as_if method for safe type conversion
00152             if (auto *ordered = conn->as_if<slang::syntax::OrderedPortConnectionSyntax>()) {
00153                 if (ordered->expr) {
00154                     spdlog::info("{} Ordered connection: {}", indent, ordered->expr->toString());
00155                 } else {
00156                     spdlog::info("{} Ordered connection: <empty>", indent);
00157                 }
00158             } else if (auto *named = conn->as_if<slang::syntax::NamedPortConnectionSyntax>()) {
00159                 if (named->name.valid()) {
00160                     std::string exprStr = named->expr ? named->expr->toString() : "<empty>";
00161                     spdlog::info("{} .{}({})", indent, named->name.valueText(), exprStr);
00162                 }
00163             } else if (conn->as_if<slang::syntax::EmptyPortConnectionSyntax>()) {
00164                 spdlog::info("{} <empty connection>", indent);
00165             } else if (conn->as_if<slang::syntax::WildcardPortConnectionSyntax>()) {
00166                 spdlog::info("{} .* (wildcard connection)", indent);
00167             } else {
00168                 spdlog::info("{} Unknown connection type: {}", indent,
00169                     slang::syntax::toString(conn->kind));
00170             }
00171         } catch (const std::exception &e) {
00172             spdlog::warn("{}Exception processing connection: {}", indent, e.what());
00173         } catch (...) {
00174             spdlog::warn("{}Unknown exception processing connection", indent);
00175         }
00176     }
00177 }
00178 } catch (const std::exception &e) {
00179     spdlog::warn("{}Exception while processing instances: {}", indent, e.what());
00180 }
00181
00182 // Continue processing child nodes
00183 depth_++;
00184 visitDefault(hierarchyInst);
00185 depth--;

```

```

00186 }
00187 }
00188 /*
00189 // Handle primitive gate instantiation
00190 void VerilogVisitor::handle(const slang::syntax::PrimitiveInstantiationSyntax &primitiveInst) {
00191 if (!inTargetModule_) {
00192 return;
00193 }
00194
00195 std::string indent(depth_ * 2, ' ');
00196
00197 // Get primitive gate type
00198 if (primitiveInst.type.valid()) {
00199 std::string_view gateType = primitiveInst.type.valueText();
00200 spdlog::info("{}Primitive Gate: {}", indent, gateType);
00201
00202 // Print delay information (if any)
00203 if (primitiveInst.delay) {
00204 spdlog::info("{}Delay: {}", indent, primitiveInst.delay->toString());
00205 }
00206
00207 // Print strength information (if any)
00208 if (primitiveInst.strength) {
00209 spdlog::info("{}Strength: {}", indent, primitiveInst.strength->toString());
00210 }
00211
00212 // Print each instance
00213 for (const auto &instance : primitiveInst.instances) {
00214 // Primitive gates usually don't have names, but if they do, print them
00215 if (instance->decl && instance->decl->name.valid()) {
00216 std::string_view instName = instance->decl->name.valueText();
00217 spdlog::info("{} Gate instance: {}", indent, instName);
00218 }
00219
00220 // Print connections
00221 spdlog::info("{} Gate connections:", indent);
00222 for (size_t i = 0; i < instance->connections.size(); ++i) {
00223 const auto &conn = instance->connections[i];
00224 // For gate-level instantiation, connections are usually ordered
00225 if (auto *ordered = conn->as_if<slang::syntax::OrderedPortConnectionSyntax>()) {
00226 if (ordered->expr) {
00227 // The first is usually the output, the rest are inputs
00228 std::string portType = (i == 0) ? "output" : "input";
00229 spdlog::info("{} {} {}: {}", indent, portType, i, ordered->expr->toString());
00230 }
00231 }
00232 }
00233 }
00234
00235 // Continue to traverse deeper when specific standard gate type
00236 // Create a set of safe standard gate types
00237 static const std::unordered_set<std::string_view> safeGateTypes = {
00238 "and", "or", "nand", "nor", "xor", "xnor", "not",
00239 "buf", "bufif0", "bufif1", "notif0", "notif1", "pullup", "pulldown",
00240 "cmos", "rcmos", "nmos", "pmos", "rnmos", "rpmos", "tran",
00241 "rtran", "tranif0", "tranif1", "rtranif0", "rtranif1"};
00242
00243 if (safeGateTypes.find(gateType) != safeGateTypes.end()) {
00244 // Continue processing child nodes
00245 depth_++;
00246 visitDefault(primitiveInst);
00247 depth_--;
00248 } else {
00249 spdlog::warn("{}Skipping deeper traversal of primitive: {}", indent, gateType);
00250 }
00251
00252 } else {
00253 spdlog::warn("{}Primitive instantiation missing gate type", indent);
00254 }

```

```

00255 }
00256 */
00257 // Handle specify block
00258 void VerilogVisitor::handle(const slang::syntax::SpecifyBlockSyntax &specifyBlock) {
00259     if (!inTargetModule_) {
00260         return;
00261     }
00262
00263     std::string indent(depth_ * 2, ' ');
00264     spdlog::info("{}Specify Block:", indent);
00265
00266     // Iterate through all path declarations
00267     for (const auto &item : specifyBlock.items) {
00268         if (auto *pathDecl = item->as_if<slang::syntax::PathDeclarationSyntax>()) {
00269             if (pathDecl->desc) {
00270                 std::string pathSrc;
00271                 std::string pathDst;
00272
00273                 // Get path source
00274                 if (!pathDecl->desc->inputs.empty()) {
00275                     // Get the first input
00276                     if (auto *identifier =
00277                         pathDecl->desc->inputs[0]->as_if<slang::syntax::IdentifierNameSyntax>()) {
00278                         if (identifier->identifier.valid()) {
00279                             pathSrc = identifier->identifier.valueText();
00280                         }
00281                     }
00282                 }
00283
00284                 // Get path destination
00285                 if (pathDecl->desc->suffix) {
00286                     if (auto *simpleSuffix =
00287                         pathDecl->desc->suffix->as_if<slang::syntax::SimplePathSuffixSyntax>()) {
00288                         if (!simpleSuffix->outputs.empty()) {
00289                             if (auto *identifier =
00290                                 simpleSuffix->outputs[0]->as_if<slang::syntax::IdentifierNameSyntax>()) {
00291                                 if (identifier->identifier.valid()) {
00292                                     pathDst = identifier->identifier.valueText();
00293                                 }
00294                             }
00295                         } else if (auto *edgeSuffix =
00296                             pathDecl->desc->suffix
00297                                 ->as_if<slang::syntax::EdgeSensitivePathSuffixSyntax>()) {
00298                             if (!edgeSuffix->outputs.empty()) {
00299                                 if (auto *identifier =
00300                                     edgeSuffix->outputs[0]->as_if<slang::syntax::IdentifierNameSyntax>()) {
00301                                     if (identifier->identifier.valid()) {
00302                                         pathDst = identifier->identifier.valueText();
00303                                     }
00304                                 }
00305                             }
00306                         }
00307                     }
00308
00309                     // Construct path string
00310                     std::string pathStr = pathSrc + " => " + pathDst;
00311
00312                     // Get path delay
00313                     std::string delayStr = "(";
00314                     for (size_t i = 0; i < pathDecl->delays.size(); ++i) {
00315                         if (i > 0)
00316                             delayStr += ", ";
00317                         delayStr += pathDecl->delays[i]->toString();
00318                     }
00319                     delayStr += ")";
00320
00321                     spdlog::info("{} Path: {} = {}", indent, pathStr, delayStr);
00322                 }
00323             }

```

```

00324     // Can add handling for ConditionalPathDeclarationSyntax and IfNonePathDeclarationSyntax
00325     else if (auto *condPath = item->as_if<slang::syntax::ConditionalPathDeclarationSyntax>()) {
00326         if (condPath->path && condPath->predicate) {
00327             spdlog::info("{} Conditional Path (if {})", indent, condPath->predicate->toString());
00328             // Recursively handle internal path
00329             handle(*condPath->path);
00330         }
00331     } else if (auto *ifNonePath = item->as_if<slang::syntax::IfNonePathDeclarationSyntax>()) {
00332         if (ifNonePath->path) {
00333             spdlog::info("{} If-None Path", indent);
00334             // Recursively handle internal path
00335             handle(*ifNonePath->path);
00336         }
00337     }
00338 }
00339
00340 // Continue processing child nodes
00341 depth_++;
00342 visitDefault(specifyBlock);
00343 depth--;
00344 }
00345 }
00346
00347 // Handle module declarations, only keep the target module
00348 void CellExtractor::handle(const slang::syntax::ModuleDeclarationSyntax &module) {
00349     if (module.header && module.header->name.valid()) {
00350         std::string_view moduleName = module.header->name.valueText();
00351
00352         // If it is the target module, mark as found, otherwise remove it
00353         if (!targetCell_.empty() && moduleName == targetCell_) {
00354             foundTarget_ = true;
00355             // Do not modify, keep this module
00356         } else {
00357             // Remove non-target modules
00358             remove(module);
00359         }
00360     }
00361 }
00362
00363 // Get result
00364 bool CellExtractor::foundTargetCell()const { return foundTarget_; }
00365
00366 // Handle module declarations
00367 void CellPrinter::handle(const slang::syntax::ModuleDeclarationSyntax &module) {
00368     if (module.header && module.header->name.valid()) {
00369         std::string_view moduleName = module.header->name.valueText();
00370
00371         // Check if it is the target module
00372         if (!targetCell_.empty() && moduleName == targetCell_) {
00373             foundTarget_ = true;
00374
00375             // Print module definition
00376             out_ << "timescale 1ns/10ps\n";
00377             out_ << module.toString();
00378
00379             return; // Do not traverse further
00380         }
00381     }
00382
00383     // Continue traversing other modules
00384     visitDefault(module);
00385 }
00386
00396 void ModuleRewriter::handle(const slang::syntax::SyntaxNode &node) {
00397     std::string indent(depth_ * 2, ' ');
00398     logger->debug("{}Node: {}", indent, slang::syntax::toString(node.kind));
00399
00400     // Continue processing child nodes
00401     depth_++;

```

```

00402 visitDefault(node);
00403 depth--;
00404 }
00405
00406 void ModuleRewriter::handle(const slang::syntax::ModuleDeclarationSyntax &module) {
00407     logger->debug("Processing module: {}", module.header->name.valueText());
00408
00409     // Create intermediate wires for connections between instances
00410     for (int i = 0; i < instance_count_ - 1; i++) {
00411         auto &newNetNode = parse("\n wire OP_" + std::to_string(i) + ";");
00412         insertAtBack(module.members, newNetNode);
00413         logger->debug("Added intermediate wire: {}", newNetNode.toString());
00414     }
00415
00416     // Get critical input port & output port from
00417     // the module name pattern: CELLNAME_#__CRITICALPORT__OUTPUTPORT
00418     std::string criticalInputPort = "";
00419     std::string criticalOutputPort = "";
00420     if (!this->moduleName_.empty()) {
00421         size_t firstSep = this->moduleName_.find("__");
00422         if (firstSep != std::string::npos) {
00423             size_t secondSep = this->moduleName_.find("__", firstSep + 2);
00424             if (secondSep != std::string::npos) {
00425                 size_t thirdSep = this->moduleName_.find("__", secondSep + 2);
00426                 if (thirdSep != std::string::npos) {
00427                     criticalInputPort = this->moduleName_.substr(secondSep + 2, thirdSep - (secondSep + 2));
00428                     criticalOutputPort = this->moduleName_.substr(thirdSep + 2);
00429                     logger->debug("Extracted critical input port: {}", criticalInputPort);
00430                     logger->debug("Extracted output port: {}", criticalOutputPort);
00431                 } else {
00432                     logger->warn("Can't find thirdSep. Invalid module name pattern: {}", this->moduleName_);
00433                     return;
00434                 }
00435             } else {
00436                 logger->warn("Can't find secondSep. Invalid module name pattern: {}", this->moduleName_);
00437                 return;
00438             }
00439         } else {
00440             logger->warn("Can't find firstSep. Invalid module name pattern: {}", this->moduleName_);
00441             return;
00442         }
00443     } else {
00444         logger->warn("Module name is empty. Can't extract critical input port.");
00445         return;
00446     }
00447
00448     // Add additional wires for intermediate outputs that aren't part of the chain
00449     if (this->outputPins_.size() > 1) { // Only add if there are multiple output pins
00450         for (int i = 0; i < instance_count_ - 1; i++) { // Skip the first and last instances
00451             for (const auto &outputPin : this->outputPins_) {
00452                 if (outputPin != criticalOutputPort) { // Found an intermediate output pin
00453                     auto &newNetNode = parse("\n wire P_" + std::to_string(i) + "__" + outputPin + ";");
00454                     insertAtBack(module.members, newNetNode);
00455                     logger->debug("Added intermediate wire: {}", newNetNode.toString());
00456                 }
00457             }
00458         }
00459     }
00460
00461     std::string portList_str = module.header->ports->toString();
00462     logger->debug("Port list: {}", portList_str);
00463
00464     std::vector<std::string> allPorts;
00465     auto ansiPortList = module.header->ports->as_if<slang::syntax::AnsiPortListSyntax>();
00466     if (!ansiPortList) {
00467         logger->warn("Port list is not ANSI style.");
00468         return;
00469     }
00470     portInfoMap_.clear(); // Clear the port info map

```

```

00471 for (const auto portMember : ansiPortList->ports) {
00472     if (portMember->kind == slang::syntax::SyntaxKind::ImplicitAnsiPort) {
00473         const auto implicitPort = portMember->as_if<slang::syntax::ImplicitAnsiPortSyntax>();
00474         const auto &directionToken =
00475             implicitPort->header->as_if<slang::syntax::VariablePortHeaderSyntax>()->direction;
00476         const auto &nameToken = implicitPort->declarator->name;
00477
00478         std::string_view portName = nameToken.valueText();
00479         std::string_view direction = directionToken.valueText();
00480         portInfoMap_[std::string(portName)] = std::string(direction); // Store port info in the map
00481         logger_->debug("Port Name: {}, Direction: {}", portName, direction);
00482     } else {
00483         logger_->warn("Port member is not ImplicitAnsiPort.");
00484     }
00485 }
00486
00487 for (int i = 0; i < instance_count_; i++) {
00488     std::string instanceName = "I_" + this->cellName_ + "__X" + std::to_string(i) + "__" +
00489         criticalInputPort + "__" + criticalOutputPort;
00490     std::string instanceCode = "\n " + this->cellName_ + " " + instanceName + " (";
00491     std::vector<std::string> portConnections;
00492
00493     for (const auto &pair : portInfoMap_) {
00494         std::string portName = pair.first;
00495         std::string direction = pair.second;
00496         std::string connectionName;
00497
00498         if (portName == criticalInputPort) {
00499             if (i == 0) {
00500                 connectionName = portName; // First instance: connect to module input port
00501             } else {
00502                 connectionName =
00503                     "OP_" + std::to_string(i - 1); // Intermediate instances: connect to previous OP wire
00504             }
00505         } else if (portName == criticalOutputPort) {
00506             if (i == instance_count_ - 1) {
00507                 connectionName = portName; // Last instance: connect to module output port
00508             } else {
00509                 connectionName = "OP_" + std::to_string(i); // Intermediate instances: connect to OP wire
00510             }
00511         } else if (direction == "output") { // Handle other output ports
00512             if (i < instance_count_ - 1) {
00513                 connectionName = "P_" + std::to_string(i) + "__" +
00514                     portName; // Connect to intermediate P_i_portName wire
00515             } else {
00516                 connectionName = portName; // Last instance: connect to module output port
00517             }
00518         } else { // For input ports (and inout?), connect directly to module port
00519             connectionName = portName;
00520         }
00521         portConnections.push_back("." + portName + "(" + connectionName + ")");
00522     }
00523
00524     // Connect all ports with comma and space
00525     if (!portConnections.empty()) {
00526         instanceCode += portConnections[0];
00527         for (size_t i = 1; i < portConnections.size(); ++i) {
00528             instanceCode += ", " + portConnections[i];
00529         }
00530     }
00531     instanceCode += ");";
00532
00533     // Blank line before the first instance
00534     if (i == 0) {
00535         instanceCode = "\n" + instanceCode;
00536     }
00537
00538     // Insert the instance code
00539     auto &instanceNode = parse(instanceCode);

```

```

00540     insertAtBack(module.members, instanceNode);
00541 }
00542 }
00543
00544 void getAST(const std::string &verilog_file, const std::string &cell) {
00545     try {
00546         spdlog::info("Starting get AST from Verilog file: '{}'", verilog_file);
00547         auto result = slang::syntax::SyntaxTree::fromFile(verilog_file);
00548         if (result) {
00549             spdlog::info("Successfully parsed Verilog file.");
00550
00551             try {
00552                 // First, use the visitor to print basic information
00553                 VerilogVisitor visitor(cell);
00554                 visitor.visit(result.value()->root());
00555
00556                 // If a target cell is specified, use the rewriter to extract the relevant code
00557                 if (!cell.empty()) {
00558                     // Create a rewriter to only keep the target cell related code
00559                     CellExtractor extractor(cell);
00560                     auto extractedTree = extractor.transform(result.value());
00561
00562                     if (extractor.foundTargetCell()) {
00563                         // Use SyntaxPrinter to print the extracted code
00564                         std::string extractedCode = slang::syntax::SyntaxPrinter::printFile(*extractedTree);
00565
00566                         // Save to a file named after the cell name
00567                         std::string outputFile = cell + ".v";
00568                         std::ofstream cellOut(outputFile);
00569                         if (cellOut) {
00570                             cellOut << extractedCode;
00571                             cellOut.close();
00572                             spdlog::info("Extracted '{}' cell code to '{}'", cell, outputFile);
00573                         } else {
00574                             spdlog::error("Failed to write extracted cell code to '{}'", outputFile);
00575                         }
00576                     } else {
00577                         spdlog::warn("Target cell '{}' not found in the Verilog file", cell);
00578                     }
00579                 }
00580
00581                 // spdlog::info("Print full source code to 'full_source_code.v'");
00582                 // // Optionally save the entire syntax tree
00583                 // std::string fullOutput = slang::syntax::SyntaxPrinter::printFile(*result.value());
00584                 // std::ofstream out("full_source_code.v");
00585                 // out << fullOutput;
00586                 // out.close();
00587
00588                 spdlog::info("Print target cell code to 'cell_code.v using toString()");
00589                 // Use CellPrinter to print the target cell's code
00590                 std::ofstream cellOut("cell_code.v");
00591                 CellPrinter cellPrinter(cell, cellOut);
00592                 cellPrinter.visit(result.value()->root());
00593                 cellOut.close();
00594
00595             } catch (const std::exception &e) {
00596                 spdlog::error("Exception during AST traversal: {}", e.what());
00597             } catch (...) {
00598                 spdlog::error("Unknown exception during AST traversal");
00599             }
00600         } else {
00601             spdlog::error("Error parsing Verilog file.");
00602         }
00603     } catch (const std::exception &e) {
00604         spdlog::error("Exception during Verilog parsing: {}", e.what());
00605     } catch (...) {
00606         spdlog::error("Unknown exception during Verilog parsing");
00607     }
00608 }

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